# 10.0 STEAM AND POWER CONVERSION SYSTEM

## 10.1 SUMMARY DESCRIPTION

The steam and power conversion system is designed to remove heat energy from the reactor coolant system via the four steam generators and to convert it to electrical power in the turbine-generator. The main condenser deaerates the condensate and transfers the heat, which is unusable in the cycle, to the circulating water system. The regenerative turbine cycle heats the feedwater, and the main feedwater system returns it to the steam generators.

Drawing AX6DD403 is an overall flow diagram of the steam and power conversion system. Table 10.1-1 gives the significant design and performance data for the major system components based on 1X4DC103, As-Built Heat Balance Diagram (100% reactor power). Drawings 1X4DC101 and 1X4DC103 provide heat balances for valves wide open (VWO), and 100% reactor power conditions, respectively.

### 10.1.1 GENERAL DESCRIPTION

The steam generated in the four steam generators is supplied to the high-pressure turbine element by the main steam system. After expansion through the high-pressure turbine element, the steam passes through the moisture separator/reheaters (MSR) and then is admitted to the three low-pressure turbine elements. A portion of the steam is extracted from the turbine for six stages of feedwater heating. Above approximately 10% of full power, hot reheat steam is supplied to the turbine-driven main feedwater pumps; below 10% of full power, main steam is used for this purpose.

Exhaust steam from the low-pressure turbines is condensed and deaerated in the main condenser. The heat rejected in the main condenser is removed by a closed loop circulating water system utilizing a natural draft cooling tower. The condensate pumps take suction from the condenser hotwell and deliver the condensate through five stages of low-pressure feedwater heaters to the suction of the steam generator feedwater pumps. The steam generator feedwater pumps discharge the feedwater through one stage of high-pressure feedwater heaters to the four steam generators.

The moisture separator drains, reheater drains, and the drains from the high-pressure feedwater heaters are pumped forward into the condensate stream between the fourth and fifth stages of low pressure feedwater heaters by the heater drain pump. The drains from the low-pressure feedwater heaters are cascaded through successively lower pressure feedwater heaters to the main condenser.

The General Electric turbine-generator has a guaranteed rating of 1,249,721 kW (1,247,083-kW nameplate rating) for a nuclear steam supply system (NSSS) thermal output of 3430 MWt. The principal turbine-generator conditions for the turbine guarantee rating are listed in table 10.1-1. The Westinghouse NSSS has a rating of 3643 MWt consisting of 3626 MWt net reactor core power plus 17 MWt reactor coolant pump input heat. The warranty rating system conditions for the NSSS are listed in table 10.1-1. The systems of the turbine cycle have been designed to meet the turbine-generator VWO, maximum calculated, not guarantee condition.

Instrumentation systems for the normal operating conditions of the steam and condensate systems are designed in accordance with accepted secondary cycle design for safe and reliable control, requirements for performance calculations, and periodic heat balances. Instrumentation

for the secondary cycle is also provided to meet recommendations by the turbine supplier and American Society of Mechanical Engineers (ASME) Standard TWDPS-1, Part 2, 1973, Recommended Practices for the Prevention of Water Damage to Steam Turbines. Other recommendations by the nuclear steam system supplier for prevention of water hammer in the steam generator are also part of the design bases. Continuous sampling system instrumentation and grab sample points are provided for maintaining acceptable limits of water chemistry in the secondary cycle, as required by the nuclear steam system and turbine suppliers. Condenser conductivity sampling is provided for tube/tube sheet leakage detection to meet requirements to identify and promptly isolate leakage in the condenser.

Criteria and bases for safety-related instrumentation for the main steam isolation and auxiliary feedwater systems are discussed in section 7.3.

## PROTECTIVE AND SAFETY-RELATED FEATURES

Note: All references to Unit 1 digital turbine controls are effective after the Unit 1 outage in 2017.

A. Loss of External Electrical Load and/or Turbine Trip

In the event of turbine trip (unless below approximately 50% power), the integrated control system provides for automatic reactor scram. Steam is relieved to the condenser via the turbine bypass valves and, if required, to the atmosphere via the atmospheric relief valves. Steam relief permits energy removal from the reactor coolant system. Load rejection capability is discussed in paragraph 10.3.2.3.1. Loss of external electrical load is discussed in section 15.2.2.

B. Overpressure Protection

Spring-loaded safety valves are provided on all four main steam lines, in accordance with the ASME Boiler and Pressure Vessel Code, Section III. The pressure relief capacity of all the safety valves is such that the energy generated at the high-flux reactor trip setting can be dissipated through this system. The design capacity of the main steam safety valves equals or exceeds 105% of the NSSS engineered safeguards design steamflow at an accumulation pressure not exceeding 110% of the main steam system design pressure.

In addition, the shell sides of the feedwater heaters and moisture separator/reheaters are provided with overpressure protection in accordance with ASME Boiler and Pressure Vessel Code, Section VIII, or equivalent standards.

C. Loss of Main Feedwater Flow

The auxiliary feedwater system is designed to provide feedwater to the steam generators for the removal of sensible and decay heat whenever main feedwater flow is interrupted, including loss of normal electric power. This system is described in subsection 10.4.9.

D. Turbine Overspeed Protection

Unit 1

During normal operations, turbine overspeed is precluded by the governing action of the digital electrohydraulic control system. Two additional redundant protective systems are provided for overspeed trip: primary overspeed trip and emergency overspeed trip. These overspeed trip systems close the main and

intermediate turbine stop valves and the control and intercept valves by electrohydraulic means. The electric trip devices (ETDs) in the trip manifold assembly (TMA) are independently testable under load. During testing, overpseed protection remains fully functional. The primary overspeed trip actuates at approximately 110% of rated speed, while the emergency overspeed trip actuates at approximately 110.5%. Turbine overspeed protection is further discussed in subsection 10.2.2.

Unit 2

During normal operations, turbine overspeed is precluded by the governing action of the electrohydraulic control system. Two additional independent protective systems are provided for overspeed trip: mechanical overspeed trip and electrical trip. These overspeed trip systems close the main and intermediate turbine stop valves by mechanical hydraulic means and the control and intercept valves by electrohydraulic means. Each of the two means of overspeed tripping is independently testable under load. During testing, overspeed protection is provided by the device not being tested. The mechanical trip valve actuates at 110% of rated speed, while the backup electrical device actuates at 111.5%. Turbine overspeed protection is further discussed in subsection 10.2.2.

E. Turbine Missile Protection

Turbine disk integrity to minimize turbine missiles is discussed in subsection 10.2.3.

Turbine missile protection is discussed in paragraph 3.5.1.3.

F. Radioactivity

Under normal operating conditions, there are no significant radioactive contaminants present in the steam and power conversion system. However, it is possible for the system to become contaminated through steam generator tube leakage. In this event, radiological monitoring of the main condenser air removal system and the steam generator blowdown system will detect contamination and alarm high radioactivity concentrations. A full discussion of the radiological aspects of primary-to-secondary system leakage and limiting conditions for operation are contained in chapter 11. The steam generator blowdown system, described in subsection 10.4.8, and the condensate demineralizer system, described in subsection 10.4.6, serve to limit the radioactivity level in the secondary cycle.

TABLE 10.1-1 (SHEET 1 OF 3)

SIGNIFICANT DESIGN FEATURES AND
PERFORMANCE CHARACTERISTICS FOR MAJOR
STEAM AND POWER CONVERSION SYSTEM COMPONENTS

| | |
|---|---|
| **Nuclear Steam Supply System, Full Power Operation** | |
| Rated NSSS power (MWt) | 3643 |
| Steam generator outlet pressure (psig) | 972 |
| Steam generator inlet feedwater temperature (°F) | 449 |
| Steam generator outlet steam moisture (%) | 0.27 |
| Steam generator outlet steam temperature (°F) | 543 |
| Quantity of steam generators per unit | 4 |
| Flowrate per steam generator (lb/h) | $4.06 \times 10^6$ |
| **Turbine-Generator** | |
| Guarantee output (kW) | 1,249,721 |
| Nameplate rating (kW) | 1,247,083 |
| Turbine type | Tandem-compound, 6-flow, 38-in. last-stage bucket |
| Turbine elements per unit | 1 high pressure<br>3 low pressure |
| Operating speed (rpm) | 1800 |
| **Moisture Separator/Reheater** | |
| Stages of reheat | 1 |
| Stages of moisture separation | 1 |
| Quantity of MSR vessels per unit | 4 |
| **Main Condenser** | |
| Type | 2-pass, single-pressure |
| Quantity, per unit | 1 (3 shells) |
| Condensing capacity (Btu/h) | $8.09 \times 10^9$ |
| Circulating water flowrate (gal/min) | 484,600 |
| Circulating water temperature rise (°F) | 33.1 |

TABLE 10.1-1 (SHEET 2 OF 3)

| | |
|---|---|
| **Condenser Vacuum Pumps** | |
| Type | Rotary motor-driven, water-sealed |
| Hogging capacity, each ($sf^3$/min) | 800 @ 10.0 in. HgA |
| Holding capacity, each ($sf^3$/min) | 36 @ 1.0 in. HgA |
| Motor (hp) each | 150 |
| Speed (rpm) | 1800 |
| Quantity per unit | 2 |
| **Steam Jet Air Ejectors** | |
| Quantity per unit | 2 |
| Type | 2-stage |
| Holding capacity, each ($sf^3$/min) | 60 @ 1.0 in. HgA |
| Steam consumption, each (lb/h) | 3500 |
| **Condensate Pumps** | |
| Type | Vertical centrifugal, motor-driven |
| Design conditions | |
| Flow (gal/min) | 10,810 |
| Total head (ft) | 1280 |
| Motor (hp) | 4500 |
| Quantity per unit | 3 |
| **Low Pressure Feedwater Heaters** | |
| No. 1 | |
| Quantity per unit | 3 |
| Operating pressure (psia) | 9.17 |
| Duty (Btu/h) per heater | $233 \times 10^6$ |
| No. 2 | |
| Quantity per unit | 3 |
| Operating pressure (psia) | 20.0 |
| Duty (Btu/h) per heater | $141 \times 10^6$ |
| No. 3 | |
| Quantity per unit | 3 |
| Operating pressure (psia) | 60.5 |
| Duty (Btu/h) per heater | $241 \times 10^6$ |
| No. 4 | |
| Quantity per unit | 2 |
| Operating pressure (psia) | 167.1 |
| Duty (Btu/h) per heater | $411 \times 10^6$ |

TABLE 10.1-1 (SHEET 3 OF 3)

| | |
|---|---|
| No. 5 | |
| Quantity per unit | 2 |
| Operating pressure (psia) | 270.2 |
| Duty (Btu/h) per heater | $329 \times 10^6$ |

High Pressure Feedwater Heater

| | |
|---|---|
| No. 6 | |
| Quantity per unit | 2 |
| Operating pressure (psia) | 424.7 |
| Duty (Btu/h) per heater | $358 \times 10^6$ |

Steam Generator Feedwater Pumps

| | |
|---|---|
| Pump type | Horizontal, centrifugal, single-stage |
| Turbine type | Six-stage, noncondensing |
| Quantity per unit | 2 |
| Design conditions per pump | |
| Flow (gal/min) | 18,650 |
| Total head (ft) | 2650 |
| Turbine hp @ 5950 rpm | 15,700 |

Heater Drain Pumps

| | |
|---|---|
| Type | Vertical centrifugal, motor-driven |
| Design conditions | |
| Flow (gal/min) | 6700 |
| Total head (ft) | 940 |
| Motor (hp) | 1800 |
| Quantity per unit | 2 |

Steam Generator Blowdown Heat Exchanger

| | |
|---|---|
| Duty (Btu/h) | $22.2 \times 10^6$ |
| Quantity per unit | 4 |

Steam Generator Blowdown Trim Heat Exchanger

| | |
|---|---|
| Duty (Btu/h) | $5.4 \times 10^6$ |
| Quantity per unit | 1 |

Heater Drain Tank

| | |
|---|---|
| Quantity per unit | 2 |
| Operating pressure (psia) | 168.1 |

## 10.2 TURBINE-GENERATOR

The function of the turbine-generator is to convert thermal energy into electric power.

### 10.2.1 DESIGN BASES

#### 10.2.1.1 Safety Design Bases

The turbine-generator serves no safety function and has no safety design bases.

#### 10.2.1.2 Power Generation Design Bases

The following is a list of the principal design bases:

A. The turbine-generator is intended for baseload operation and also has load following characteristics consistent with the requirements of the Westinghouse nuclear steam supply system (NSSS).

B. The turbine-generator system (TGS) is designed in accordance with all applicable interface requirements and system design requirements of the Westinghouse NSSS.

C. The turbine-generator is designed to trip automatically under abnormal conditions and to concurrently initiate a reactor trip.

D. The system is designed to ensure proper drainage of all related piping and components in order to eliminate water induction into the main turbine during all phases of operation.

E. The TGS satisfies the recommendations of Nuclear Regulatory Commission Branch Technical Position ASB 3-1 as related to breaks in high- and moderate energy piping systems outside containment. In this regard the TGS is considered a high-energy system.

F. The system is designed to supply hot reheat steam to the feedwater pump turbines during normal power operation (above about 40 percent load) and at pump runout conditions with one pump out of service.

G. The system provides extraction steam for six stages of regenerative feedwater heating.

### 10.2.2 DESCRIPTION

The General Electric turbine-generator (drawings 1X4DB160-1, 1X4DB160-2, 1X4DB160-3, 1X4DB166, 1X4DB167-1, 1X4DB167-2, 1X4DB191, 1X4DB193, and 1X4DB196) is designated as a TC6F 38-in. last-stage buckets unit. It consists of turbines, a generator, external moisture-separator reheaters, exciter, controls, and auxiliary subsystems. The major design parameters of the turbine-generator and auxiliaries are presented in table 10.2.2-1.

The turbine-generator and associated piping, valves, and controls are located completely within the turbine building. The effects of turbine missiles are described in paragraph 3.5.1.3. Failure of turbine-generator equipment does not preclude safe shutdown of the reactor coolant system. There is unlimited access to turbine-generator components and instrumentation associated with turbine-generator overspeed protection, under all operating conditions.

### 10.2.2.1 Turbine-Generator Description

The turbine is an 1800-rpm, tandem-compound, 6-flow, reheat unit with 38-in. last-stage buckets (TC6F 38-in. LSB). The high-pressure turbine element includes one double-flow high-pressure turbine. The low-pressure turbine elements include three double-flow low-pressure turbines and four external moisture separator/reheaters (MSRs) with one stage of reheating. The single direct-driven generator is gas cooled and rated at 1350 MVA at 25 kV, 3-phase, 60 Hz. Other related system components include a complete turbine-generator bearing lubrication oil system, a digital electrohydraulic control system (Unit 1) or electrohydraulic control system (Unit 2) with supervisory instrumentation, a turbine steam sealing system (refer to subsection 10.4.3), overspeed protective devices, turning gear, a generator hydrogen and seal oil system, a stator cooling system, a rectifier section, and a voltage adjuster.

### 10.2.2.2 Turbine-Generator Cycle Description

Steam from each of four steam generators enters the high- pressure turbine through four 28-in. stop valves and governing control valves; one stop valve and one control valve form a single assembly. Crossties are provided upstream of the turbine stop valves to provide pressure equalization with one or more stop valves closed. After expanding through the high-pressure turbine, the high-pressure turbine exhaust steam flows through four external MSR vessels. The external moisture separators reduce the moisture content of the high-pressure exhaust steam from approximately 14 percent to dry saturated steam conditions. The reheater uses a portion of the main steam supply to the main turbine to reheat the steam to superheat conditions. Next, the reheated steam flows through a combined intermediate stop and intercept valve in each of six 42-in. reheat steam lines leading to the inlets of the three low-pressure turbines. A portion of the hot reheat steam leaving the MSR is used to power the two feedwater pump turbine drivers. Steam from each low-pressure turbine is then exhausted into the shell of the condenser serving each low-pressure turbine.

Turbine steam extraction connections are provided for six stages of feedwater heating. Steam from the turbine third stage extraction point of the high-pressure turbine is supplied to high-pressure feedwater heater No. 6. The turbine fifth stage extraction point supplies steam to the low-pressure feedwater heater No. 5. Steam is taken from the high-pressure turbine exhaust crossover piping to the low-pressure feedwater heater No. 4. The low-pressure turbine ninth, eleventh, and twelfth stage extraction points supply steam to the low-pressure feedwater heaters No. 3, 2, and 1, respectively.

The rotating low-pressure turbine blades located just upstream of the eleventh and twelfth stage turbine extraction points are provided with grooved moisture separator extraction buckets to partially remove moisture from the remainder of the steam passing through the turbine and to improve thermal efficiency of the cycle. The extraction steam lines from these turbine extraction points to the feedwater heaters thus serve the dual purpose of routing the required extraction steam for feedwater preheating and the moisture extracted from the steam flowing through the turbine.

The moisture removed by the external MSR is drained to moisture separator drain tanks, from there to the No. 4 feedwater heater drain tanks, and subsequently is pumped into the feedwater system. Similarly, the condensed steam in the reheaters is drained to the reheater drain tanks, drains into the shell side of the No. 6 feedwater heater, and is eventually pumped into the feedwater system.

### 10.2.2.3 Automatic Controls

Automatic controls regulate turbine speed and acceleration through the entire speed range, with several discrete speed and acceleration rate settings. The automatic control system includes control of load and loading rate from no load to full load, with continuous load adjustments and discrete loading rates. Should it become necessary to remove the generating unit from the primary automatic controls, the standby manual control (Unit 2 only) of speed and load takes over, thus allowing continued operation of the turbine-generator.

#### 10.2.2.3.1 Electrohydraulic Control Systems

Unit 1

The turbine-generator is equipped with a digital electrohydraulic control (DEHC) system (GE MARK Vle) that combines the principles of triple channel redundancy and high-pressure hydraulics to regulate steam flow through the turbine. The control system has three major subsystems: speed control logic, load control logic, and valve flow control logic all contained within the main controller processing units. The control system contains the logic that controls the turbine speed, load, acceleration rates, loading rates, and valve position limits by providing the appropriate control signals to the turbine steam flow control valves.

The DEHC system employs six electric speed inputs. Signals are triple-redundantly processed in the DEHC system with three independent processors. Valve opening actuation is provided by a 1600-psig hydraulic system that is totally independent of the bearing lubrication system; valve closing actuation is provided by springs and steam forces upon the reduction or relief of fluid pressure. the system is designed so that loss of fluid pressure, for any reason, leads to valve closing and consequent shutdown.

All steam valves are provided in series pairs. A stop valve is actuated by either of two overspeed (primary and emergency) trip systems, followed by a control valve modulated by the speed governing logic, and tripped by either overspeed trip system.

Drawing 1X4DB194 is a schematic of the EHC hydraulic fluid system.

Unit 2

The turbine-generator is equipped with an electrohydraulic control (EHC) system that combines the principles of solid-state electronics and high-pressure hydraulics to regulate steamflow through the turbine. The control system has three major subsystems: a speed control unit, a load control unit, and valve flow control units.

The EHC system employs three electric inputs and one mechanical speed input. Signals are redundantly processed in both electronic and hydraulic logic channels. Valve opening actuation is provided by a 1600-psig hydraulic system that is totally independent of the bearing lubrication system; valve closing actuation is provided by springs and steam forces upon the reduction or relief of fluid pressure. The system is designed so that loss of fluid pressure, for any reason, leads to valve closing and consequent shutdown.

All steam valves are provided in series pairs. A stop valve is actuated by either of two overspeed trip systems, followed by a control valve modulated by the speed-governing system, and tripped by either overspeed trip system.

Drawing 2X4DB194 is a schematic of the EHC hydraulic fluid system.

10.2.2.3.1.1 Speed Control. The speed control logic provides speed control, acceleration, and overspeed protection functions. The speed control logic produces a speed error signal, which is fed to the load control logic. The speed error signal is derived by comparing the desired speed with the actual speed of the turbine at steady-state conditions or by comparing the desired acceleration rate with the actual acceleration rate during startup.

There are six passive magnetic-type probes, used in conjunction with a multi-toothed wheel on the steam turbine shaft to sense the turbine's rotational speed. A primary set of three probes, voted two out of three, is used for speed control, speed indication,zero speed detection, and primary overspeed protection. The other set of three, also voted two out of three but in separate and triple-redundant protection processors, is used exclusively for emergency overspeed protection. Probe channel failures are detected when any single channel signal differs from voted value by a preset amount. Failure of any channel will generate an alarm. Failure of any two probe channels out of the three in a set will trip the turbine.

Because of the importance in safeguarding against turbine overspeed, the speed control logic uses the median select speed value. If one speed input fails, the logic continues to provide control. If two speed inputs fail, the turbine will trip.

A close signal being sent to the control valves and intercept valves will be sent to a fast acting solenoid valve on each control valve and intercept valve. Energizing these solenoid valves will release the hydraulic fluid pressure in the valve actuators, allowing springs to close each valve.

Logic is incorporated in the speed control to slowly vary the rotor speed above and below critical frequencies. This will prevent the turbine from running at a constant speed near critical bucket resonances.

Unit 2 Speed Control Unit

The speed control unit provides speed control, acceleration, and overspeed protection functions. The speed control unit produces a speed error signal, which is fed to the load control unit. The speed error signal is derived by comparing the desired speed with the actual speed of the turbine at steady-state conditions or by comparing the desired acceleration rate with the actual acceleration rate during startup.
Three separate error signals are derived by the speed control circuits: two are the result of individual comparisons of a speed reference signal with each of the two analog speed signals (primary and backup) that are proportional to turbine speed; the other is the integral of the acceleration error, which is derived by comparing a signal proportional to rotor acceleration with an acceleration reference signal. A low value gate receives these three signals and transmits the signal demanding the smallest control valve opening.

Because of the importance in safeguarding against turbine overspeed, the speed control unit uses two redundant channels, a primary and a backup. If the primary channel fails, the backup channel takes over automatically. If the backup channel fails, the primary will maintain control. In the event that both channels are lost, the turbine will trip.

A trip signal being sent to the control valves and intercept valves will be sent to a fast acting solenoid valve on each control valve and intercept valve. Energizing these solenoid valves will release the hydraulic fluid pressure in the valve actuators, allowing springs to close each valve.

A circuit is incorporated in the speed control unit to slowly vary the rotor speed above and below critical frequencies. This will prevent the turbine from running at a constant speed near critical bucket resonances.

10.2.2.3.1.2 Load Control

Unit 1 Load Control Logic

The load control logic develops signals that are used to proportion the steam flow to the stop valves, control valves, and intercept valves. Signal outputs are based on a proper combination of the speed error signals and load reference signals.

Unit 2 Load Control Unit

The load control unit develops signals that are used to proportion the steam flow to the stop valves, control valves, and intercept valves. Signal outputs are based on a proper combination of the speed error signals and load reference signals.

10.2.2.3.1.3 Valve Flow Control

Unit 1 Valve Flow Control Logic

The valve flow control logic regulates the steam flows as directed by the load control logic. Compensation is used to ensure linear steam flow response with respect to steam flow signals. The main stop valve, the control valve, and the intercept valve position loop consists, in each case, of DEHC system servo output, an electrohydraulic servo valve, a hydraulic actuator, and linear position transducers. By use of valve position feedback, the valve control logic positions the main stop valves, the control valves, and the intercept valves according to the flow demand signal from the load control logic, or from the valve test logic.

The flow of the main steam entering the high-pressure turbine is controlled by four stop valves and four governing control valves. Each 28-in. stop valve is controlled by an electrohydraulic actuator, so that the stop valve is either fully open or fully closed. The function of the stop valves is to shut off the steam flow to the turbine, when required. The stop valves are closed within 0.3.s by actuation of the emergency trip system devices. These devices are independent of the electronic flow control logic (paragraph 10.2.2.3.1.5).

The turbine control valves are positioned by electrohydraulic servo actuators in response to signals from the flow control logic. The flow control signal positions the control valves for long-range speed control through the normal turbine operating range and for load control after the turbine-generator unit is synchronized.

The 34-in. combined intermediate stop valves, located in the hot reheat lines at the inlet to the low-pressure turbines, are stop and intercept valves in one casing and control steam flow to the low-pressure turbines. During normal operation of the turbine, the stop and intercept valves are wide open. The intercept valve flow control logic positions the valve during startup

and normal operations and closes the valve rapidly on loss of turbine load. The intermediate stop valves close completely on turbine overspeed and trip.

Unit 2 Valve Flow Control Units

The valve flow control unit regulates the steamflows as directed by the load control unit. Compensation circuits are introduced to ensure linear steamflow response with respect to steamflow signals. The main stop valve, the control valve, and the intercept valve position loop consists, in each case, of electronic circuitry, an electrohydraulic servo valve, a hydraulic actuator, and a linear position transducer. By use of valve position feedback control, the valve control units position the main stop valves, the control valves, and the intercept valves according to the flow demand signal from the load control unit, from the standby control unit, or directly from the control panel (valve test).

The flow of the main steam entering the high-pressure turbine is controlled by four stop valves and four governing control valves. Each 28-in. stop valve is controlled by an electrohydraulic actuator, so that the stop valve is either fully open or fully closed. The function of the stop valves is to shut off the steamflow to the turbine, when required. The stop valves are closed within 0.3 s by actuation of the emergency trip system devices. These devices are independent of the electronic flow control unit (paragraph 10.2.2.3.1.5).

The turbine control valves are positioned by electrohydraulic servo actuators in response to signals from their respective flow control unit. The flow control unit signal positions the control valves for long-range speed control through the normal turbine operating range and for load control after the turbine- generator unit is synchronized.

The 34-in. combined intermediate stop valves, located in the hot reheat lines at the inlet to the low-pressure turbines, are stop and intercept valves in one casing and control steamflow to the low-pressure turbines. During normal operation of the turbine, the stop and intercept valves are wide open. The intercept valve flow control unit positions the valve during startup and normal operations and closes the valve rapidly on loss of turbine load. The intermediate stop valves close completely on turbine overspeed and trip.

10 2.2.3.1.4 Power/Load Unbalance.

Unit 1 Power/Load Unbalance

Associated with the load control logic is a rate sensitive power/load unbalance circuit, the purpose of which is to initiate control valve fast closing action under load rejection conditions that might lead to rapid rotor acceleration and consequent overspeed.

Valve action will occur when the power exceeds the load by 40 percent or more, and generator current is lost in a time span of 35 ms or less. Cold reheat pressure is used as a measure of power. Generator current is used as a measure of load to provide discrimination between loss of load incidents and occurrences of electric system faults.

When DEHC detects a signal indicating a power/load unbalance condition, the load reference signal is set to zero, and the load setpoint begins to run back toward the no-load flow point. Should the condition disappear quickly, the power/load unbalance circuit will reset automatically, and the load reference signal will be re-established near its value prior to the loss of load. Should the condition persist and the load not return within approximately 45 s, the load setpoint runback will be completed. The power/load unbalance circuit will clear automatically when the cold reheat pressure drops below 40 percent.

Unit 2 Power/Load Unbalance

Associated with the load control unit is a rate sensitive power/load unbalance circuit, the purpose of which is to initiate control valve fast closing action under load rejection conditions that might lead to rapid rotor acceleration and consequent overspeed.

Valve action will occur when the power exceeds the load by 40 percent or more, and generator current is lost in a time span of 35 ms or less. Cold reheat pressure is used as a measure of power. Generator current is used as a measure of load to provide discrimination between loss of load incidents and occurrences of electric system faults.

When the detection circuitry provides a signal indicating a power/load unbalance condition, the load reference signal is grounded, and the load reference motor begins to run back toward the no-load flow point. Should the condition disappear quickly, the power/load unbalance circuit will reset automatically, and the load reference signal will be re-established near its value prior to the loss of load. Should the condition persist and the load not return within approximately 45 s, the load reference runback will be completed. The power/load unbalance circuit will clear automatically when the cold reheat pressure drops below 40 percent.

10.2.2.3.1.5 Overspeed Protection.

Unit 1 Overspeed Protection

Three separate systems have been provided to protect the turbine against overspeed. The first is the normal overspeed protection system, which is an integral part of the speed control as discussed in 10.2.2.3.1.1. The second is the primary overspeed trip protection, which will operate if the normal overspeed protection should fail. The third is the emergency overspeed trip protection, which will operate should the primary overspeed trip fail. Refer to table 10.2.2-2 for a description of the sequence of events following a full-load rejection. A schematic of the overspeed protection systems is shown in figure 10.2.2-1a.

The primary overspeed trip subsystem will trip the turbine at the primary overspeed trip point. The primary set of speed probe signals are voted two out of three and compared to the primary overspeed trip point. Values equal to or greater than the trip point will de-energize the primary trip relays (PTRs). The PTRs, in turn, will de-energize the electrical trip devices (ETDs). Actuation of the ETDs, which are located in the TMA, will release the hydraulic fluid pressure in the steam valve actuator, allowing springs to close the steam valves. The hydraulic piping system through which the pressure signal is propagated is called the emergency trip system (ETS).

The emergency overspeed trip subsystem set to operate at a slightly higher speed than the primary pvers[eed trip subsystem. The emergency set of speed probe signals are voted two out of three and compared to the emergency overspeed trip point. Values equal to or greater than the trip point will de-energize the emergency trip relays (ETRs). The ETRs, in turn, de-energize the ETDs which will release hydraulic fluid pressure in the steam valve actuators, allowing springs to close the steam valves. The emergency overspeed trip subsystem is a component of the protective system which is a completely independent set of triple-redundant processors.

Both the primary overspeed trip and the emergency overspeed trip subsystems may be independently tested online at any desired load. During a test, overspeed protection will remain fully functional.

The hydraulic piping of the primary overspeed trip and the emergency overspeed trip subsystems is also connected to an air relay dump valve. When the pressure in the hydraulic piping to the air relay dump valve is high, an air signal is supplied to the nonreturn valves

located in the extraction lines. Loss of pressure in the hydraulic piping causes a loss of the air signal, which allows the spring-assisted nonreturn valves to close.

Due to the redundancy that is built into the overspeed protection systems, the failure of a single valve will not disable the trip functions. The following component redundancies are used to guard against overspeed:

- Main stop valves/control valves.
- Intermediate stop valves/intercept valves.
- Redundant speed sensors (triple redundant).
- Fast acting solenoid valves/emergency trip system.
- Speed control/primary overspeed trip/emergency overspeed trip (triple redundant).

The overspeed protection systems are also designed for fail-safe operation. Loss of any two redundant speed signals will initiate a trip. Loss of the hydraulic pressure in the ETS will also cause a trip. Therefore, damage to the overspeed protection systems, whether at the front standard of the turbine, at the TMS, or at the steam valves, will result in the closure of the valves and the interruption of steam flow to the turbine.

Quick closure of the steam valves is an important part of preventing turbine overspeed. The following values account for valve closing time:

| Valve | Closings/(s) |
|---|---|
| Main stop valves | 0.3 |
| Control valves | 0.3 |
| Intercept valves | 0.3 |
| Intermediate stop valves | 0.3 |
| Extraction nonreturn valves | 2.0 |

Unit 2 Overspeed Protection

Two separate basic systems have been provided to protect the turbine against overspeed. The first is the normal overspeed protection system, which is an integral part of the speed control unit as discussed in 10.2.2.3.1.1. The second is the emergency overspeed protection system, which will operate if the normal overspeed protection (as a part of speed control) should fail. Refer to table 10.2.2-2 for a description of the sequence of events following a full-load rejection. A schematic of the overspeed protection systems is shown in figure 10.2.2-1b.

The emergency overspeed protection system is a mechanical hydraulic system, which uses an overspeed trip (OST) device. The OST is a mechanical ring mounted on the front end of the turbine shaft; this ring will remain concentric with the turbine shaft as long as the shaft is rotating below a certain speed. If the shaft rotates above this speed, the ring will move to an eccentric position and actuate a trip latch mechanism. This mechanism converts the mechanical trip signal to a hydraulic (low-pressure) signal in the mechanical trip valve, which passes this signal through a hydraulic piping system to the steam valve actuators. Actuation of the mechanical trip valve will release the hydraulic fluid pressure in the steam valve actuator, allowing springs to close the steam valves. The hydraulic piping system through which the pressure signal is propagated is called the emergency trip system (ETS).

Supplemental to the OST is the backup overspeed trip (BOST), an electrical trip normally set to operate at a slightly higher speed than the OST. Three independent speed signals are generated by magnetic pickups from a toothed wheel on the turbine shaft. The speed signals

are amplified through electronic circuitry and are compared to a trip speed reference signal. If a speed signal exceeds its reference setting, its master trip relay is energized. The master trip relays, through two-out-of-three logic, deenergize the electrical trip solenoid valve. This will actuate the electrical trip valve which will release hydraulic fluid pressure in the steam valve actuators, allowing springs to close the steam valves.

The BOST electric trip, through two-out-of-three logic, also sends a mechanical close-trip signal to the trip latch assembly through the mechanical trip solenoid valve and the mechanical trip piston.

Both the OST and BOST may be independently tested online at any desired load. During a test, overspeed protection will be provided by the device not being tested.

The hydraulic piping of the emergency overspeed protection system is also connected to an air relay dump valve. When the pressure in the hydraulic piping to the air relay dump valve is high, an air signal is supplied to the nonreturn valves located in the extraction lines. Loss of pressure in the hydraulic piping causes a loss of the air signal, which allows the spring-assisted nonreturn valves to close.

Due to the redundancy that is built into the overspeed protection systems, the failure of a single valve will not disable the trip functions. The following component redundancies are used to guard against overspeed:

- Main stop valves/control valves.
- Intermediate stop valves/intercept valves.
- Primary speed control/backup speed control.
- Fast acting solenoid valves/emergency trip system.
- Mechanical overspeed trip/electrical overspeed trip.
- Speed control/overspeed trip/backup overspeed trip.

The overspeed protection systems also are designed for fail-safe operation. Loss of speed signals will initiate a trip. Loss of the hydraulic pressure in the ETS will also cause a trip. Therefore, damage to the overspeed protection systems, whether at the front standard of the turbine or at the steam valves, will result in the closure of the valves and the interruption of steamflow to the turbine.

Quick closure of the steam valves is an important part of preventing turbine overspeed. The following values account for valve closing time:

| Valve | Closings/(s) |
|---|---|
| Main stop valves | 0.3 |
| Control valves | 0.3 |
| Intercept valves | 0.3 |
| Intermediate stop valves | 0.3 |
| Extraction nonreturn valves | 2.0 |

### 10.2.2.4 Turbine Protective Trips

Unit 1

Turbine protective trips operate through the DEHC but independently of the speed control logic and, when initiated, cause tripping of all turbine stop and control valves. The turbine trips are:

- Primary Overspeed (110 percent of normal).
- Emergency overspeed (110.5 percent of normal).
- Low vacuum.
- Excessive thrust bearing wear.
- Reactor trip.
- Generator trip.
- Manual trip push-buttons from the main control room.
- Manual trip push-buttons located at the turbine front standard.
- Excessive vibration.
- Moisture separator drain system high level.
- Prolonged loss of stator coolant.
- Low hydraulic fluid pressure.
- Loss of two redundant speed signals.
- Low bearing oil pressure.
- Loss of main shaft oil pump discharge pressure.

Drawing 1X5DN203-1 illustrates the turbine and generator trip logics.

Unit 2

Turbine protective trips are independent of the electronic control system and, when initiated, cause tripping of all turbine stop and control valves. The protective trips are:

- Overspeed (mechanical): 110 percent of normal.
- Backup overspeed (electrical): 112 percent of normal.
- Low vacuum.
- Excessive thrust bearing wear.
- Electric solenoid trip actuated by:
  - Reactor trip.
  - Generator trip.
  - Manual trip from control room.
- Excessive vibration.

- Manual trip handle located at the turbine front standard.
- Moisture separator drain system high level.
- Prolonged loss of stator coolant.
- Low hydraulic fluid pressure.
- Loss of both speed signals or backup overspeed trip.
- Low bearing oil pressure.
- Loss of main shaft oil pump discharge pressure.

Drawing 2X5DN203-1 illustrates the turbine and generator trip logics.

### 10.2.2.5 Protective System

The purpose of the protective system is to detect undesirable or dangerous operating conditions of the turbine-generator, take appropriate trip actions, and provide information to the operator about the detected conditions and the consequent actions. In addition, means are provided for testing all testable equipment and circuits.

The protective system consists of two major subsystems:

A. The mechanical-hydraulic trip system (Unit 2 only).

B. The electrical trip and monitoring system.

A signal flow diagram of the protective system is shown in figure 10.2.2-2a (Unit 1) and figure 10.2.2-2b (Unit 2).

The ETS is the high pressure fluid system that, when in the reset or pressurized state, permits all steam valves to open in the presence of opening signals from the DEHC (Unit 1) or EHC (Unit 2). When in the tripped or depressurized state, it overrides all opening signals; trips the main and reheat stop valves, the control valves, and the intercept valves directly by way of their disc-dump valves; and trips the extractor check valves through the air relay dump valve. The principal output function of the protective system is to control the state of the ETS.

#### 10.2.2.5.1 Mechanical-Hydraulic Trip System (Unit 2 Only)

The ETS is pressurized from the high pressure hydraulic fluid supply, through the following chain of devices, all components of the mechanical-hydraulic trip system:

A. Mechanical shutoff valve (MSOV).

B. Mechanical trip valve (MTV).

C. Mechanical lockout solenoid valve (MLV).

D. Electrical trip valve (ETV).

E. Electrical lockout solenoid valve (ELV).

The MSOV and MTV are controlled hydraulically by the mechanical trip pilot valve (MTPV), and when their pilot lines are depressurized, these valves shut off their input line and drain their output line, tripping the ETS.

The MLV is controlled electrically by the electrical trip and monitoring system, and when energized, it bypasses the MSOV and MTV, permitting these two valves and two of the three signal paths that actuate them to be tested without tripping the ETS.

The ETV is controlled hydraulically by the electrical trip solenoid valve (ETSV), and when its pilot line is depressurized, this valve shuts off its input line and drains its output line, tripping the ETS.

The ELV is controlled electrically by the electrical trip and monitoring system and, when energized, bypasses the ETV, permitting this valve and the signal path that actuates it to be tested without tripping the ETS.

In order to trip the ETS, any tripping signal has to actuate one or both of the MTPV or the ETSV. Each of these two cases will be examined separately.

10.2.2.5.1.1 MTPV Actuation. This valve is operated mechanically by the trip latch rod, which is tripped (i.e., allowed to move under the influence of a charged spring to a position where the MTPV is tripped) by the trip finger.

The trip finger is operated by:

A. Mechanical OST

This is actuated during an overspeed of the turbine exceeding the OST setting or at rated speed during a mechanical OST test. During this test the electrical trip and monitoring system energizes the oil trip solenoid valve (OTSV) which admits lubrication oil to the OST, causing it to trip. A coordinated actuation of the MLV prevents the ETS from tripping.

B. Mechanical Trip Piston

This is held in the reset position by turbine lube oil pressure. The piston is allowed to trip by action of a spring when the oil pressure is lost or when the oil is shut off by the mechanical trip solenoid valve (MTSV). This valve is energized by the electrical trip and monitoring system during a 125-V trip, as it will be defined later, or during a mechanical trip piston test. In the latter case a coordinated actuation of the MLV prevents the ETS from tripping.

C. Manual Trip Handle

There is no provision for testing the manual trip handle under lockout conditions. A manual trip handle test will result in an actual trip.

The trip latch rod, once tripped, latches mechanically and remains in the tripped position even after the condition that caused it to trip (items A, B, or C) has been cleared. It is reset by the reset mechanism, consisting of:

The oil reset solenoid valve, which actuates the oil reset piston when energized by the electrical trip and monitoring system.

The oil reset piston, which resets the trip latch rod and the MTPV, which in turn resets the MSOV and MTV.

10.2.2.5.1.2 ETSV Actuation. This valve has a 24-V-dc solenoid which is normally energized when the ETSV is in the reset state. The valve trips when the solenoid is deenergized. Failure of this solenoid will cause a trip. The solenoid is connected to the electrical trip and monitoring system and is deenergized during a 24-V trip, as will be defined later, or during an electrical trip test. In the latter case, a coordinated actuation of the ELV prevents the ETS from tripping.

### 10.2.2.5.2 Electrical Trip and Monitoring System

Unit 1

The principal function of this part of the protective system is to connect all external trip signals (except the tripping signals from the emergency overspeed trip, which act directly on the ETRs to deenergize the ETDs) to the PTRs to deenergize the ETDs after suitable modifications by logic circuits. Deenergizing two-out-of-three ETDs in each TMA manifold is capable of tripping the ETS. To allow online maintainability, valves permit isolation of a TMA manifold. All protective trips remain functional with a manifold isolated.

The incoming trip signals are arranged into two groups, signals external or internal to the DEHC system.

Unit 2

The principal function of this part of the protective system is to connect all external trip signals (except the tripping signals from the OST and manual trip handle, which act directly on the mechanical-hydraulic trip system) to one or both of the MTSV and ETSV after suitable modifications by logic circuits. Each of these valves is independently capable of tripping the ETS.

The incoming trip signals are arranged into two groups, signals external or internal to the EHC cabinet.

10.2.2.5.2.1 External Signals.

Unit 1 Signals External to the EHC System

These cause the PTRs to deenergize, in turn, deenergizing the electrical trip devices (ETDs).

Unit 2 Signals External to the EHC Cabinet

These cause 125-V trips; i.e., they activate the 125-V trip bus and energize the MTSV directly. In addition, the 24-V trip circuit is indirectly operated through a set of relays (cross-trip), and the 125-V trip bus is locked up after a short time delay.
When the generator circuit breaker is open and the MTV and ETS are tripped, an additional lockup circuit is established through pressure switch contacts.

During a mechanical trip piston test, the MTSV is energized without activating the 125-V trip bus, the cross-trip circuit, and the lockup circuits.

10.2.2.5.2.2 Internal Signals.

Unit 1 Signals Internal to the DEHC System.

These cause the PTRs to deenergize, in turn, deenergizing the electrical trip devices (ETDs).

A loss of the Input/Output Network (IONet) will result in a turbine trip. The electrical trip and monitoring system receives feedback signals from each component in the IONet that allow it to determine network health.

Loss of power to two-out-of-three control processors will result in a turbine trip. The electrical trip and monitoring system receives feedback signals from each power source that allow it to determine system health.

The electrical trip and monitoring system also contains the logic for testing the ETDs. When one of these tests is initiated, the electrical trip and monitoring system logic deenergizes the selected PTR or ETR and receives feedback signals that allow it to sense the status of the PTRs, ETRs, and ETDs after each step and whether the test was successful.

Each of the three PTR or ETR circuits for each ETD can be separately tested without causing an actual trip. These circuits, as well as many other trip circuits of the electrical trip and monitoring system, are arranged in a two-out-of-three logic system.

Unit 2 Signals Internal to the EHC Cabinet

These cause a 24-V trip; i.e., they deenergize the ETSV solenoid through a set of relay contacts and lock up the 24-V trip circuit through another set of relays contacts.

The first set of relays is also operated during an electrical trip test; in this case, the lockup circuit is not activated and the situation is cleared once the cause producing it is removed, without necessitating any positive resetting action.

Of the signals internal to the EHC cabinet, the loss of both speed signal and the BOST signal energize, in addition to the tripping and the locking relays, a third set of relays which cross-trip the 125-V trip bus. The other three signals that cause a 24-V trip do not energize these cross-trip relays; however, of these, the master trip button causes a 125-V trip through a separate contact.

A loss of 24 V dc deenergizes the ETSV solenoid, causing a 24-V trip and cross-trips the 125-V trip bus.

The electrical trip and monitoring system also contains the logic for testing the trip devices (mechanical overspeed, mechanical trip piston, and electrical trip tests.) When one of these tests is initiated, the electrical trip and monitoring system logic provides a sequence of signals to the appropriate lockout, trip, and reset valve solenoids and receives feedback signals that allow it to sense the status of the mechanical-hydraulic trip system after each step and whether the test was successful.

Each of the three BOST circuits can be separately tested without causing an actual trip. These circuits, as well as many other trip circuits of the electrical trip and monitoring system, are arranged in a two-out-of-three logic system.

### 10.2.2.6 Other Protective Systems

Additional protective features of the turbine and steam system are:

A. Safety valves on the MSR to protect the high-pressure turbine cylinder from overpressure in the event of a turbine trip.

B. Extraction line nonreturn valves (with the exception of the last two low-pressure heaters) to protect the turbine from overspeed due to reverse flow in case of a turbine trip.

C. Exhaust casing rupture diaphragms to protect the low- pressure turbine cylinders from overpressure in case of a loss of condenser vacuum.

### 10.2.2.7 Plant Loading and Load Following

The turbine-generator is intended to be base loaded but is designed to match or exceed the transient load-following capabilities of the nuclear steam supply system. The reactor regulating system (RRS) automatically adjusts reactor power to follow turbine load transients. The RRS senses turbine first- stage pressure as a linear indication of load and generates signals that regulate control element assembly drive direction and speed. As a combined unit consisting of turbine-generator and reactor, the system accepts step-load changes of ±10 percent and ramp-load changes of ±5 percent/min over the range of 20- to 100-percent full power. It also accepts, with the aid of the turbine bypass system, a load rejection of 50 percent full-load power without reactor trip and without steam release to the atmosphere through the power-operated atmospheric steam relief or steam safety valves. For load rejections greater than 50 percent of full load, a portion of the steam generated during the reactor runback will be released to the atmosphere. The turbine bypass system is described in subsection 10.4.4.

The turbine control system is designed to provide protection to the turbine by tripping the turbine for certain predetermined conditions as discussed in paragraph 10.2.2.4. The turbine is tripped upon reactor trip. The reactor protective system provides two separate signals of reactor trip to the turbine control system.

### 10.2.2.8 Inspection and Testing Requirements

Major system components are readily accessible for inspection and are available for testing during normal plant operation. Controls and protective devices associated with each turbine-generator component will be tested on a regularly scheduled basis. Turbine trip circuitry will be tested prior to unit startup. The main steam and reheat valves are nonsafety related, and the testing requirements are detailed in the Technical Requirements Manual.

## 10.2.3 TURBINE INTEGRITY (UNIT 1)

### 10.2.3.1 Materials Selection

Vogtle Unit 1 low-pressure turbine rotors are made from individual single forgings (monoblock) which eliminate the shrunk-on disks. The rotors are made from vacuum-degassed, Ni-Cr-Mo-V alloy steel. The susceptibility to stress corrosion cracking (SCC) is most pronounced in materials with yield strength above 105 ksi. The threshold of SCC susceptibility is in material with yield strength in the range of 85 to 100 ksi. The material (similar to ASTM A470 Class 6) used for the monoblock rotor, which has a yield strength in the range of 80 ksi, was selected to optimize load-carrying capability and resistance to SCC. Hence, the material yield strength is less than the SCC threshold, and stresses are reduced to provide maximum resistance to SCC.

### 10.2.3.2 High-Temperature Properties

The operating temperatures of the high-pressure rotors are below the creep rupture range. Creep rupture is, therefore, not considered to be a significant factor in ensuring rotor integrity over the lifetime of the turbine. Basic data are obtained from laboratory creep rupture tests.

### 10.2.3.3 Turbine Disk Design

The turbine assembly is designed to withstand normal conditions and anticipated transients, including those resulting in turbine trip, without loss of structural integrity. The design of the turbine assembly meets the following criteria:

A. Turbine shaft bearings are designed to retain their structural integrity under normal operating loads and anticipated transients, including those leading to turbine trips.

B. The multitude of natural critical frequencies of the turbine shaft assemblies existing between zero speed and 20-percent overspeed are controlled in the design and operation so as to cause no distress to the unit during operation.

### 10.2.3.4 Preservice Tests and Inspections

Preservice inspections for turbine rotors include the following:

A. Rotor forgings are rough machined, with minimum stock allowance prior to heat treatment.

B. Each rotor forging is subjected to a 100-percent volumetric (ultrasonic) examination. Each finish-machined rotor is subjected to a surface magnetic particle and visual examination. Results of the above examination are evaluated by use of General Electric acceptance criteria. The criteria are more restrictive than those specified for Class 1 components in the American Society of Mechanical Engineers (ASME) Boiler and Pressure Vessel Code, Sections III and V. These criteria include the requirement that subsurface sonic indications are either removed or evaluated to ensure that they do not grow to a size which compromises the integrity of the unit during the service life of the unit.

C. Each fully bucketed turbine rotor assembly is spin tested at 20 percent overspeed, the maximum speed anticipated following a load rejection from full load.

Additional preservice tests include air leakage tests to determine whether the hydrogen cooling system is tight, hydrogen purity tests, generator windings and motor megger tests, vibration tests, hydrostatic testing of coolers, pressure testing of piping, and in-place testing of motor-operated valves.

### 10.2.3.5 Inservice Inspection

Inspections for the turbine assembly and valves include the following:

A. Disassembly of the turbine is conducted during plant shutdown. Inspection of all parts that are normally inaccessible when the turbine is assembled for operation (couplings, coupling bolts, turbine shafts, low-pressure turbine buckets, and high-pressure rotors) is conducted. The turbine is inspected in sections so that over a 10-year period, the entire turbine is inspected once.

   This inspection consists of visual, surface, and volumetric examinations, as indicated below:

   1. Ultrasonic inspection of the tangential entry dovetails and pins of the finger dovetails is conducted. This inspection should be conducted at intervals of about 10 years.

2. A thorough volumetric ultrasonic examination of the high-pressure rotor is conducted. In addition, all accessible rotor surfaces are inspected visually and by magnetic particle testing. This inspection should be conducted at intervals of about 10 years.
3. Visual and surface examination of all low pressure buckets is conducted.
4. A thorough visual inspection of couplings and coupling bolts is performed.

B. Each main steam stop valve, control valve, intermediate stop valve, and intercept valve is disassembled approximately every 120 months during scheduled refueling or maintenance shutdowns. A visual and surface examination of valve seats, wheels, and stems is conducted. Valve bushings are inspected and cleaned, and bore diameters are checked for proper clearance.

C. All main stop valves, control valves, and combined intercept valves will be tested with the turbine online through the DEHC (Unit 1). Pushbuttons on the electrohydraulic control test panel will be used to stroke main stop, control, and combined intercept valves from full-open to full-closed (Unit 2).

Turbine valve testing will be performed at intervals as specified in subsection 16.3.8.

D. All extraction nonreturn valves will be tested prior to each startup.

Extraction nonreturn valves are tested locally by stroking the valve full-open with air, then equalizing air pressure, allowing the spring closure mechanism to close the valve. Closure of each valve is verified by direct observation of the valve arm movement.

#### 10.2.3.6 References

1. Begley, J. A., and Logsdon, W. A., Westinghouse Scientific Paper 71-1E7-MSLRF-P1.
2. Spencer, R. C., and Timo, D. P., Starting and Loading of Turbines, General Electric Company, Presented at the 36th Annual Meeting of the American Power Conference, Chicago, Illinois, April 29 – May 1, 1974.
3. General Electric letter dated March 28, 1979, from Richard V. Dufresna to H. L. Welch, Jr.

### 10.2.4 TURBINE INTEGRITY (UNIT 2)

#### 10.2.4.1 Materials Selection

Turbine wheels and rotors are made from vacuum-melted, or vacuum-degassed, Ni-Cr-Mo-V alloy steel by processes which minimize flaw occurrence and provide adequate fracture toughness. Tramp elements are controlled to the lowest practical concentrations consistent with good scrap selection and melting practices and consistent with obtaining adequate initial and long-life fracture toughness for the environment in which the parts operate. The turbine wheel and rotor materials have the lowest fracture appearance transition temperatures (FATT) and the highest Charpy V-notch energies obtainable, on a consistent basis, from water-quenched Ni-Cr-Mo-V material at the sizes and strength levels used. Since actual levels of

FATT and Charpy V-notch energy vary, depending upon the size of the part, the location within the part, etc., these variations will be taken into account in accepting specific forgings for use in turbines for nuclear application. Charpy tests, essentially in accordance with American Society of Testing Materials (ASTM) Specification A370, are included.

### 10.2.4.2 Fracture Toughness

Suitable material toughness is obtained through the use of materials described in paragraph 10.2.4.1, to produce a balance of adequate material strength and toughness to ensure safety while simultaneously providing high reliability, availability, and efficiency during operation. Bore stress calculations include components due to centrifugal loads, interference fit, and thermal gradients where applicable. The ratio of material fracture toughness, KIC (as derived from material tests on each wheel or rotor) to the maximum tangential stress for wheels and rotors, at speeds from normal to 115 percent of rated speed[a], will be at least 2 in.$^{(1/2)}$ Adequate material fracture toughness needed to maintain this ratio is ensured by destructive tests on material taken from the wheel or rotor, using correlation methods which are more conservative than that presented in reference 1.

Operating procedures for the turbine are employed to preclude brittle fracture at startup by ensuring that the metal temperature of the wheels and rotors is adequately above the FATT and is sufficient to maintain the fracture toughness to tangential stress ratio at or above 2 in.$^{(1/2)}$ as defined above.

Details of these startup procedures are contained in reference 2.

### 10.2.4.3 High-Temperature Properties

The operating temperatures of the high-pressure rotors are below the creep rupture range. Creep rupture is, therefore, not considered to be a significant factor in ensuring rotor integrity over the lifetime of the turbine. Basic data are obtained from laboratory creep rupture tests.

### 10.2.4.4 Turbine Disk Design

The turbine assembly is designed to withstand normal conditions and anticipated transients, including those resulting in turbine trip, without loss of structural integrity. The design of the turbine assembly meets the following criteria:

A. The maximum tangential stress in wheels and rotors, resulting from centrifugal forces, interference fit, and thermal gradients, does not exceed 0.75 of the yield strength of the materials at 115 percent of rated speed.(3)

B. Turbine shaft bearings are designed to retain their structural integrity under normal operating loads and anticipated transients, including those leading to turbine trips.

C. The multitude of natural critical frequencies of the turbine shaft assemblies existing between zero speed and 20-percent overspeed are controlled in the design and operation so as to cause no distress to the unit during operation.

a. The highest anticipated speed resulting from a loss of load is 110 percent.

The turbine disk design facilitates inspection of all high stress regions including bores and keyways without the need for removing the disks from the shafts.

### 10.2.4.5 Preservice Tests and Inspections

Preservice inspections for turbine disk and rotor include the following:

A. Wheel and rotor forgings are rough machined, with minimum stock allowance prior to heat treatment.

B. Each rotor and wheel forging is subjected to a 100-percent volumetric (ultrasonic) examination. Each finish-machined rotor and wheel is subjected to a surface magnetic particle and visual examination. Results of the above examination are evaluated by use of General Electric acceptance criteria. The criteria are more restrictive than those specified for Class 1 components in the American Society of Mechanical Engineers (ASME) Boiler and Pressure Vessel Code, Sections III and V. These criteria include the requirement that subsurface sonic indications are either removed or evaluated to ensure that they do not grow to a size which compromises the integrity of the unit during the service life of the unit.

C. Finish-machined surfaces are subjected to a magnetic particle examination. No magnetic particle flaw indications are permissible in bores, holes, keyways, and other highly stressed regions.

D. Each fully bucketed turbine rotor assembly is spin tested at 20 percent overspeed, the maximum speed anticipated following a load rejection from full load.

Additional preservice tests include air leakage tests to determine whether the hydrogen cooling system is tight, hydrogen purity tests, generator windings and motor megger tests, vibration tests, hydrostatic testing of coolers, pressure testing of piping, and in-place testing of motor-operated valves.

### 10.2.4.6 Inservice Inspection

Inspections for the turbine assembly and valves include the following:

A. Disassembly of the turbine is conducted during plant shutdown. Inspection of all parts that are normally inaccessible when the turbine is assembled for operation (couplings, coupling bolts, turbine shafts, low-pressure turbine buckets, low-pressure wheels, and high-pressure rotors) is conducted. The turbine is inspected in sections so that over a 10-year period, the entire turbine is inspected once.

   This inspection consists of visual, surface, and volumetric examinations, as indicated below:

   1. The bore and keyway region of each wheel receives an ultrasonic examination. In addition, each wheel is inspected visually and by magnetic particle testing on all accessible surfaces. Also, ultrasonic inspection of the tangential entry dovetails and pins of the finger dovetails is conducted. This inspection should be conducted at intervals of about 6 years.

2. A thorough volumetric ultrasonic examination of the high-pressure rotor is conducted. In addition, all accessible rotor surfaces are inspected visually and by magnetic particle testing. This inspection should be conducted at intervals of about 10 years.
3. Visual and surface examination of all low pressure buckets is conducted.
4. A thorough visual inspection of couplings and coupling bolts is performed.

B. Each main steam stop valve, control valve, intermediate stop valve, and intercept valve is disassembled approximately every 120 months during scheduled refueling or maintenance shutdowns. A visual and surface examination of valve seats, wheels, and stems is conducted. Valve bushings are inspected and cleaned, and bore diameters are checked for proper clearance.

C. All main stop valves, control valves, and combined intercept valves will be tested with the turbine online through the DEHC (Unit 1). Pushbuttons on the electrohydraulic control test panel will be used to stroke main stop, control, and combined intercept valves from full-open to full-closed (Unit 2).

Turbine valve testing will be performed at intervals as specified in subsection 16.3.1.

D. All extraction nonreturn valves will be tested prior to each startup.

### 10.2.5 EVALUATION

Components of the turbine-generator are conventional and are types that have been extensively used in other nuclear power plants. Instruments, controls, and protective devices are provided to ensure reliable and safe operation. Redundant, fast actuating controls are installed to prevent any damage resulting from overspeed and/or full-load rejection. The control system ensures turbine trip upon reactor trip. Automatic low pressure exhaust hood water sprays prevent excessive hood temperatures. Exhaust casing rupture diaphragms prevent low-pressure cylinder overpressure in the event of loss of condenser vacuum.

Since the steam generated in the steam generators is not normally radioactive, no radiation shielding is provided for the turbine-generator and associated components. Thus radiological considerations do not affect access to system components during normal conditions. In the event of a primary-to-secondary system leak due to a steam generator tube leak, it is possible for the main steam to become radioactively contaminated. Discussions of the radiological aspects of primary-to-secondary leakage are presented in chapters 11 and 12.

### 10.2.6 INSTRUMENTATION APPLICATIONS

The turbine-generator is provided with a full complement of turbine supervisory instruments mounted in the control room or are part of the DEHC on Unit 1. It is complete with sensors and/or transmitters mounted on the associated equipment, which indicate and record the following:

- Speed.
- Stop valve position.

- Control valve position.
- Combined intermediate valve (CIV) position.
- Temperatures as required for controlled starting, including:
  - External valve chest inner surface.
  - External valve chest outer surface.
  - First-stage shell lower inner surface.
  - Crossaround pipe upstream of CIV No. 1.
  - Crossaround pipe upstream of CIV No. 2.
  - Crossaround pipe upstream of CIV No. 3.
  - Crossaround pipe upstream of CIV No. 4.
  - Crossaround pipe upstream of CIV No. 5.
  - Crossaround pipe upstream of CIV No. 6.
- Casing and shaft differential expansion.
- Vibration of each bearing.
- Shaft eccentricity turbine standard.
- Bearing metal temperatures.

Control room alarms are provided to warn the operators of the following abnormal conditions:

- High vibration trip (Unit 2 only).
- High-high exhaust hood temperature (Unit 2 only).
- Low emergency trip system pressure trip (Unit 2 only).
- Low vacuum trip (Unit 2 only).
- Thrust bearing wear trip (Unit 2 only).
- Low shaft pump discharge pressure trip (Unit 2 only).
- Low bearing oil pressure trip (Unit 2 only).
- Steam seal pressure trouble (low steam seal pressure, high steam seal bypass pressure).
- Steam packing exhauster water level, high or low vacuum.
- High-low level in moisture separator drain tank.
- Turbine condenser vacuum low.
- Turbine exhaust hood temperature high.
- Shaft driven oil pump low pressure (Unit 1 only).

- Main turbine oil temperature Hi/Lo.
- Main turbine tripped (Unit 1 only).
- Turbine control miscellaneous alarm (Unit 1 only).
- Turbine control major alarm (Unit 1 only).
- Turbine control minor alarm (Unit 1 only).
- Vibration monitor trouble.

Local and control room indication of the following miscellaneous parameters are provided. Many of the parameters are available on the DEHC displays on Unit 1.

- Main steam throttle pressure.
- Steam seal supply header pressure.
- Steam seal condenser vacuum.
- Bearing oil header pressure.
- Bearing oil coolers coolant temperature.
- Electrohydraulic control fluid header pressure.
- Electrohydraulic control fluid temperature.
- Crossover pressure.
- Moisture separator drain tank level.
- First-stage pressure.
- High-pressure turbine exhaust pressure.
- Extraction steam pressure, each extraction point (via computer).
- Low-pressure turbine exhaust hood pressure.
- Exhaust hood temperature for each exhaust.

Instrumentation and controls are provided in the control room for the generator equipment as follows:

A. Generator supervisory instruments with sensors and/or transmitters mounted on the associated equipment, indicating or recording the following:

1. Multiple generator stator winding temperatures (via computer). The detectors are built into the generator, fully protected from the cooling medium, and suitably distributed around the circumference in positions having the highest temperature.

2. Multiple stator winding cooling coil outlet temperature detectors.
3. Stator coolant inlet and discharge temperatures.
4. Hydrogen cooler inlet gas temperature (two detectors at each point).
5. Field ampere and voltage.
6. Hydrogen gas pressure.
7. Hydrogen gas purity.
8. Generator winding overtemperature.
9. Generator ampere, voltage, and power.

B. Alarms are provided for high stator, hydrogen, stator coil coolant, and field temperature. An alarm is provided from a core monitoring system to indicate a local core overheating condition.

TABLE 10.2.2-1

TURBINE-GENERATOR AND AUXILIARIES DESIGN PARAMETERS

| | |
|---|---|
| Manufacturer | General Electric |
| **Turbine** | |
| Type | TC6F 38-in. LSB |
| No. of elements | 1 high pressure; 3 low pressure |
| Last-stage bucket length (in.) | 38 |
| Operating speed (rpm) | 1800 |
| Condensing pressure (in. HgA) | 3.50 |
| Guaranteed turbine cycle heat rate (Btu/kWh) | 10,115 |
| **Generator** | |
| Guaranteed generator rated output (Kw) | 1,156,622 |
| Nameplate rating (Kw) | 1,160,000 |
| Power factor | 0.90 |
| Generator rating (Kva) | 1,350,000 |
| Voltage (V) | 25,000 |
| Hydrogen pressure (psig) | 75 |
| **Moisture separator/reheater** | |
| Type | Shell and tube |
| Number | 4 |
| Stages of reheating | 1 |

TABLE 10.2.2-2

EVENTS FOLLOWING LOSS OF TURBINE LOAD WITH POSTULATED EQUIPMENT FAILURES

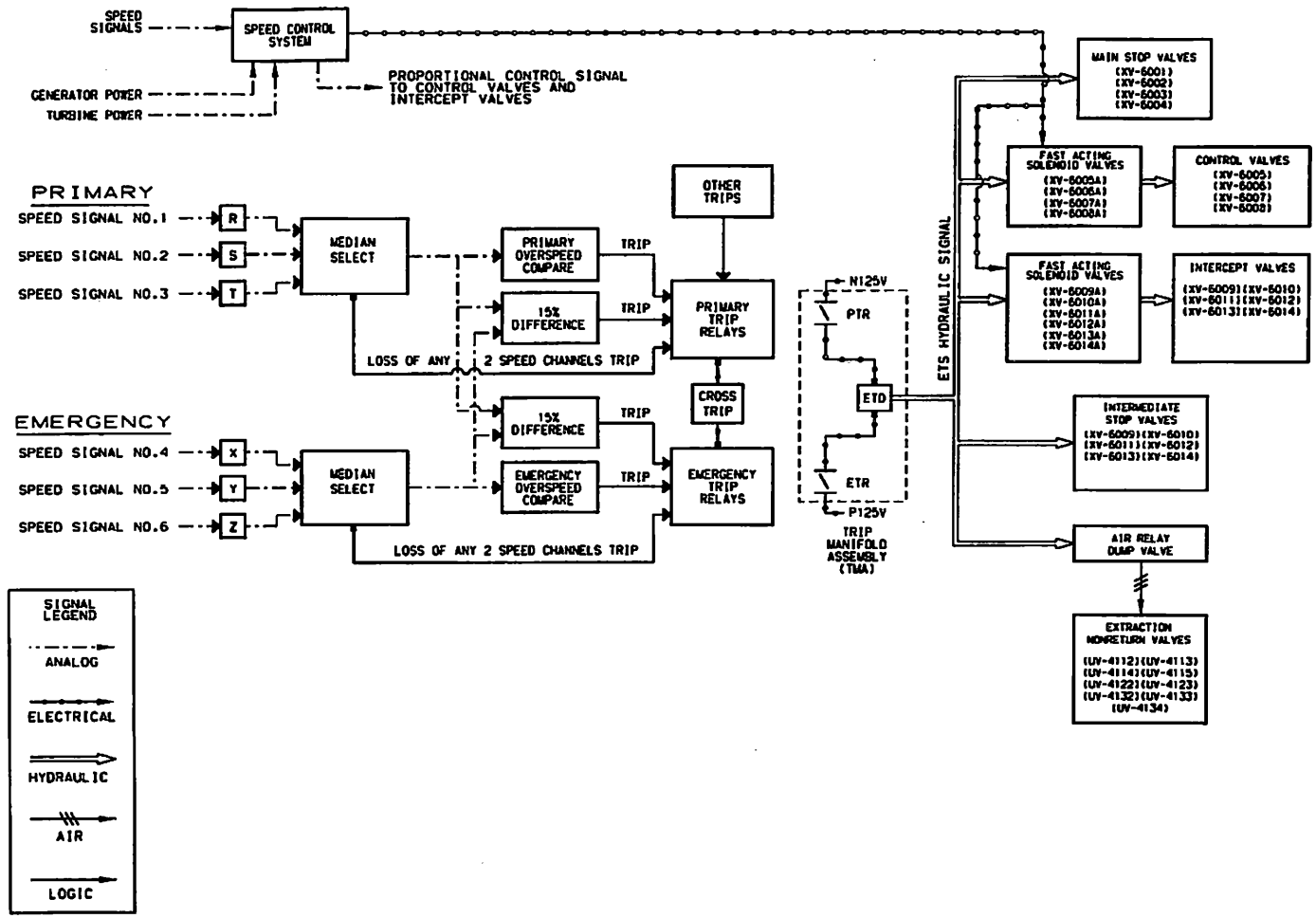
**Unit 1 Only**

| Percent of Rated Speed (Approximate) | Event |
|---|---|
| 100 | Turbine is initially at valves wide open. Full load is lost. Speed begins to rise. |
| 101 | Control and intercept valves begin to close. |
| < 110 | Peak transient speed with normally operating speed control system.<br><br>Assume that power/load unbalance and speed control systems had failed prior to loss of load. |
| 110 | Primary (Electrical) overspeed trip signal is generated to depressurize the emergency trip header and cause all of the turbine valves to close. |
| 110.5 | Emergency (Electrical) overspeed trip signal is generated to depressurize the emergency trip header and cause all of the turbine valves to close. |

**Unit 2 Only**

| Percent of Rated Speed (Approximate) | Event |
|---|---|
| 100 | Turbine is initially at valves wide open. Full load is lost. Speed begins to rise. |
| 101 | Control and intercept valves begin to close. |
| 108 | Peak transient speed with normally operating speed control system.<br><br>Assume that power/load unbalance and speed control systems had failed prior to loss of load. |
| 110-111 | Mechanical overspeed trip signals all valves to close. Operation of air relay dump valves releases spring closure mechanisms of extraction nonreturn valves. |
| 111-112 | Backup overspeed trip signals all valves to close. |

Following the above sequence of events in either tube, the turbine will approach but not exceed 120 percent of rated speed.



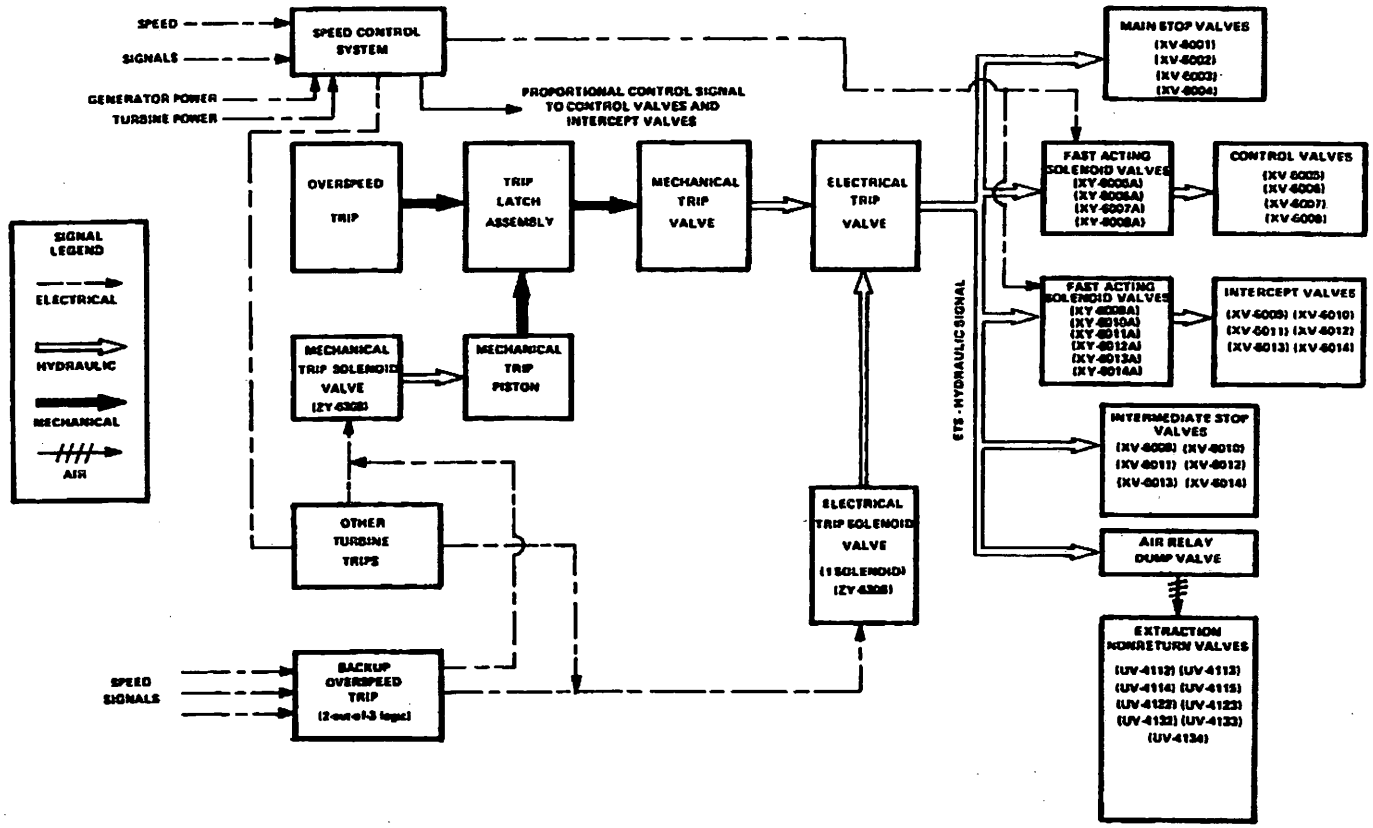

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**ELECTRIC GENERATING PLANT**
**UNIT 1 AND UNIT 2**

UNIT 1 TURBINE OVERSPEED PROTECTION SCHEMATIC

FIGURE 10.2.2–1a



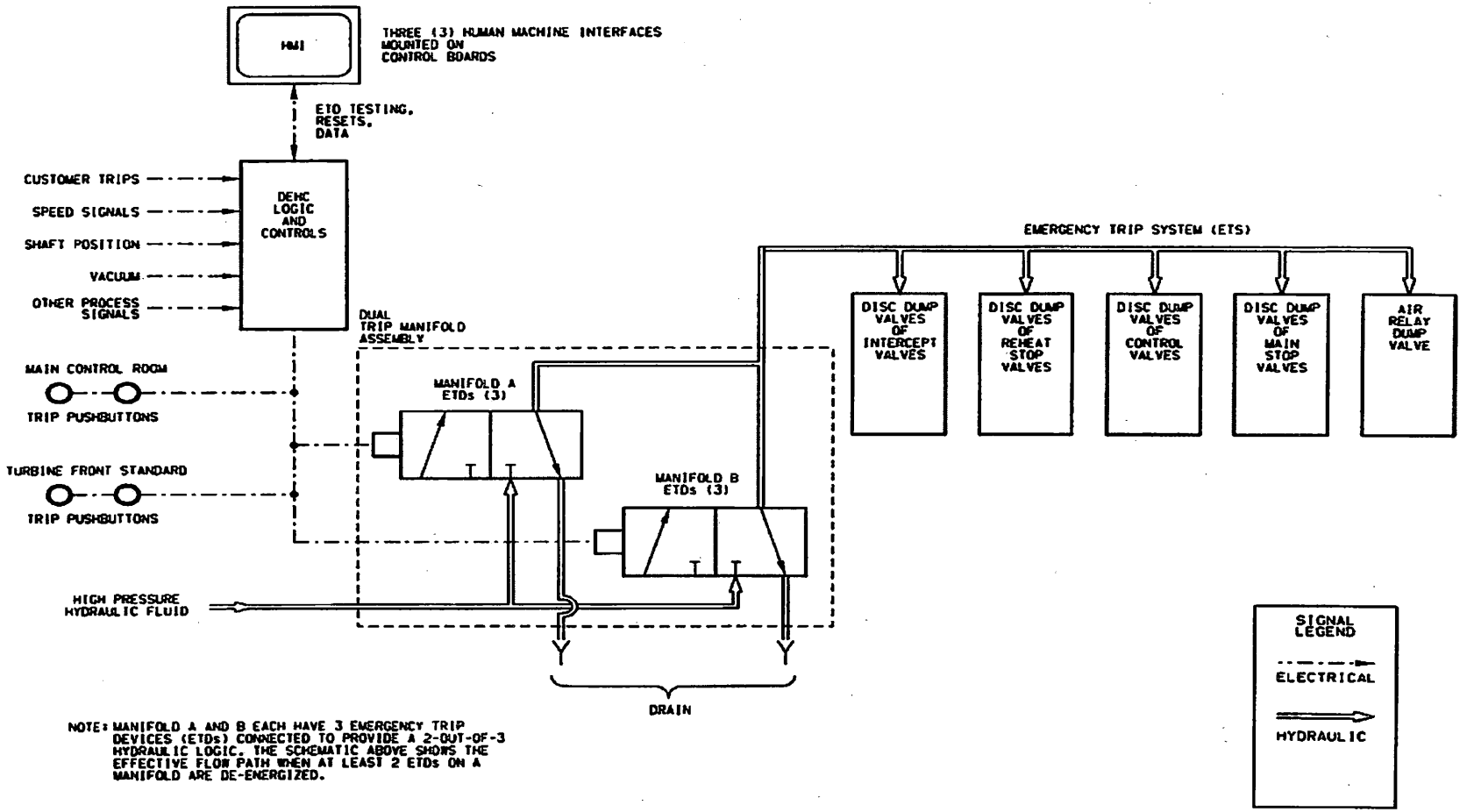

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UNIT 2 TURBINE OVERSPEED PROTECTION SCHEMATIC

FIGURE 10.2.2–1b




NOTE: MANIFOLD A AND B EACH HAVE 3 EMERGENCY TRIP DEVICES (ETDs) CONNECTED TO PROVIDE A 2-OUT-OF-3 HYDRAULIC LOGIC. THE SCHEMATIC ABOVE SHOWS THE EFFECTIVE FLOW PATH WHEN AT LEAST 2 ETDs ON A MANIFOLD ARE DE-ENERGIZED.

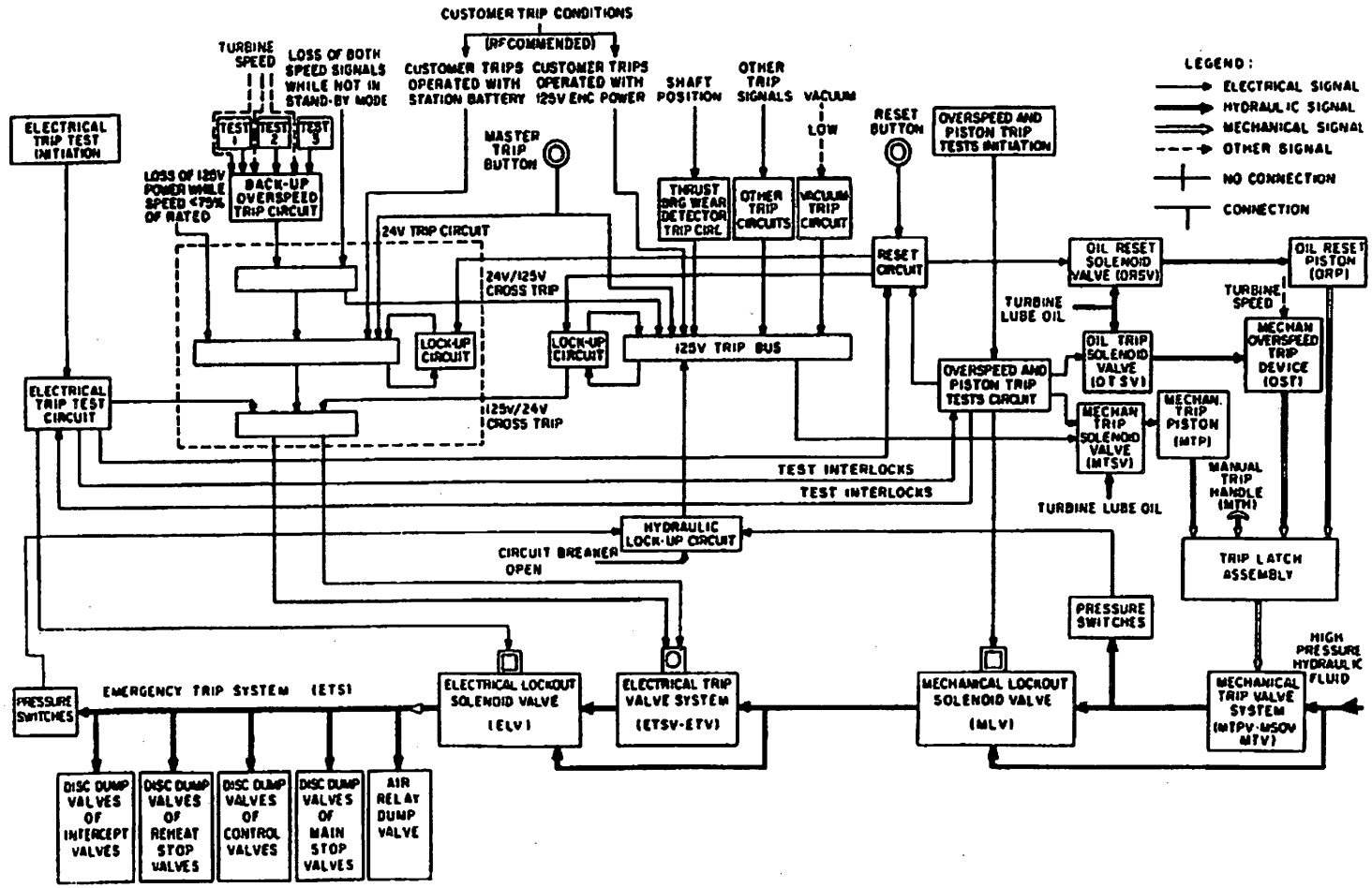
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UNIT 1 PROTECTIVE SYSTEMS SIGNAL FLOW DIAGRAM

FIGURE 10.2.2–2a




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UNIT 2 PROTECTIVE SYSTEMS SIGNAL FLOW DIAGRAM

FIGURE 10.2.2–2b

## 10.3 MAIN STEAM SUPPLY SYSTEM

The function of the main steam supply system is to supply steam from the steam generators to the high-pressure turbine over a range of flows and pressures covering the entire operating range from system warmup to valves wide open (VWO) turbine conditions.

The system also provides steam to the moisture separator/ reheaters, the auxiliary feedwater pump turbine drive, the steam seal system for the main turbine, the feedwater pump turbine drives, the condenser spargers, and the steam jet air ejectors. The system also dissipates heat generated by the nuclear steam supply system (NSSS) by means of steam dump valves to the condenser or to the atmosphere through power-operated atmospheric relief valves or spring-loaded main steam safety valves when either the turbine-generator or condenser is unavailable.

### 10.3.1 DESIGN BASES

#### 10.3.1.1 Safety Design Bases

Pertinent safety design bases are as follows:

A. The system provides the means of dissipating residual and sensible heat generated from the NSSS during hot shutdown and cooldown even when the main condenser is not available. Power-operated atmospheric relief valves are provided to allow controlled cooldown of the steam generator and the reactor coolant system when the condenser is not available.

B. The system is provided with two main steam isolation valves (MSIVs) and two associated MSIV bypass valves in parallel with the MSIVs in each main steam line. These valves isolate the secondary side of the steam generators to deal with leakage and malfunction and to prevent the uncontrolled blowdown of two steam generators and isolate nonsafety-related portions of the system.

C. The codes and standards utilized in the design of the main steam supply system are identified in table 10.3.2-1. The following components are designed in accordance with American Society of Mechanical Engineers (ASME) Boiler and Pressure Vessel Code Section III, Class 2, and Seismic Category 1 requirements and are considered the safety-related portion of the system.

   1. All piping and valves from the steam generators to and including the pipe restraints provided downstream of each outboard main steam isolation valve (MSIV) to maintain piping loads upstream of the restraint in accordance with Branch Technical Position MEB 3-1.

   2. Branch lines from the above portions of the main steam lines up to and including the first valve (including a safety or relief valve) that is either normally closed or capable of automatic/remote manual closure during all modes of normal reactor operation.

   All other components located downstream of the outboard MSIV restraint assembly are designed in accordance with the Power Piping Code, American National Standards Institute B31.1.

D. The safety-related portion of the system is designed to withstand the effects of a safe shutdown earthquake. The safety-related portion of the system is capable of withstanding the effects of natural phenomena and of performing its intended function following postulated hazards of fire, internal missile, and pipe break.

E. The safety-related portion of the system is designed so that a single failure in the main steam supply system will have no contributory effects on:

1. Initiation of a loss-of-coolant accident.
2. Integrity of other steam lines.
3. The capability of the engineered safety features system to effect a safe reactor shutdown.
4. Transmission of excessive loading to the containment pressure boundary.

Component redundancy is provided so that safety functions can be performed assuming a single active failure coincident with loss of offsite power.

F. The portion of the main steam supply system that is constructed in accordance with ASME III, Class 2 requirements is provided with access to welds and removable insulation, as required for inservice inspection in accordance with ASME Section XI. (See paragraph 10.3.4.4.)

G. The main steam supply system provides steam to the auxiliary feedwater turbine-driven pump during emergency conditions and for shutdown operations.

H. The main steam supply system is designed to function in the normal and accident environments identified in subsection 3.11.1.

I. The main steam supply system is designed in accordance with Branch Technical Positions ASB 3-1 and MEB 3-1 with regard to high-energy pipe break location and evaluation.

### 10.3.1.2 Power Generation Design Basis

The following is a list of the principal power generation design bases:

A. The main steam supply system is designed to deliver steam from the steam generators to the turbine- generator for the range of flowrates, temperatures, and pressures existing from warmup to rated conditions.

B. Each main steam line is sized to provide balanced steam pressures at the turbine stop valves, using the turbine VWO (maximum calculated - not guaranteed) steam flowrate and conditions in accordance with Westinghouse Steam System Design Manual.

C. The main steam supply system provides the capacity to dump 40 percent of VWO (maximum calculated) steamflow to the condenser during plant step-load reductions.

## 10.3.2 DESCRIPTION

### 10.3.2.1 General Description

The main steam supply system, shown in drawings 1X4DB159-1, 2X4DB159-1, 1X4DB159-2, 1X4DB159-3, 1X4DB160-1, 1X4DB160-2, 1X4DB160-3, 1X4DB166, 1X4DB167-1, 1X4DB167-2, 1X4DB191, 1X4DB193, and 1X4DB196, includes the following major components:

A. Main steam piping from the steam generator outlet steam nozzles to the main turbine stop valves.

B. Two main steam isolation valves (MSIVs) and two main steam isolation valve bypass valves per main steam line.

C. Main steam safety valves.

D. Power-operated atmospheric relief valves.

Table 10.3.2-1 lists the design data covering the major components of the main steam supply system. Table 10.3.2-2 lists the design data covering the main steam safety valve design.

### 10.3.2.2 Component Description

#### 10.3.2.2.1 Main Steam Piping

The main steam lines deliver a total steamflow of approximately 16.25 x $10^6$ lb/h from the secondary side of the four steam generators. Each of the main steam lines from the steam generators is anchored at the containment wall and has sufficient flexibility to accommodate thermal expansion. Design of the main steam piping from the steam generators includes design considerations that incorporate the allowable steam generator nozzle loading moments and stresses with all steam generators operating.

The design of the piping and supports takes into consideration all static and dynamic loadings, stresses, and moments arising from normal operation, pressure transients, or pipe rupture. The design of Seismic Category 1 piping and supports takes into consideration the loads discussed in subsection 3.9.3.

The main steam lines between the containment penetration and the first restraint downstream of the outboard MSIV are designed to meet the no-break zone criteria of Nuclear Regulatory Commission Branch Technical Position MEB 3-1 (as described in section 3.6.) so that the piping failures need not be postulated.

A description of the main steam piping from the steam generators to the turbine stop valves is presented in table 10.3.2-3.

The sizing of the 38-in. and the 36-in. common runs hydraulically balances the steam line pressure drops from the respective pairs of steam generators to the inlet of each turbine stop valve. The 36-in. and 38-in. main steam lines are cross-connected with two 20-in. lines into a common 30-in. header, just before branching into the four 28-in. lines to each turbine stop valve, to equalize flow and pressure to the inlet of the turbine stop valves. This permits online testing of each turbine stop valve without exceeding the allowable limit on steam generator pressure differential during this testing.

Each steam generator outlet nozzle contains an internal flow restrictor arrangement of about 1.4 $ft^2$ to limit flow in the event of a main steam line break. A further discussion is provided in subsection 5.4.4.

Two American Society of Mechanical Engineers (ASME) standard steam sampling nozzles are installed in the nonsafety-related portion of each main steam line downstream of the outboard MSIV to sample for conductivity, pH, and steam generator outlet moisture carryover.

The power operated atmospheric relief, safety, main steam line isolation valves, and MSIV bypass valves are located outside the containment and are installed as close as possible to the containment wall. (Containment penetrations are discussed in subsection 6.2.4.)

Turbine bypass valves are provided between the MSIVs and turbine-generator stop valves, as discussed under the turbine bypass system. (Refer to subsection 10.4.4.)

Main steam line branch connections are made downstream of the isolation valves in the American National Standards Institute (ANSI) B31.1 run of piping with the exception of the takeoffs for the power-operated atmospheric relief valve, connections for main steam safety valves, lines to the auxiliary feedwater pump turbine, low point drains, and high point vents.

Branch piping downstream of the main steam line isolation valves provides steam to the single stage reheaters, steam seal system, main feedwater pump turbines, turbine bypass system, auxiliary steam system, steam jet air ejectors, and condenser spargers.

The main steam piping in the main steam pipe tunnel downstream of the anchor forging at the end of the MSIV/main feedwater isolation valve area is designed in accordance with ANSI B31.1. Anchor forgings are provided at both terminal points for main steam and feedwater piping located in the tunnel and this portion of the piping is designed to withstand the effects of a safe shutdown earthquake (SSE). In addition, pipe whip restraints are provided as necessary to mitigate the effects of postulated pipe breaks discussed in appendix 3F.

#### 10.3.2.2.2 Main Steam Safety Valves

Main steam safety valves are provided with sufficient rated capacity to prevent the steam pressure from exceeding 110 percent of the main steam system design pressure:

A. Following a turbine trip without a reactor trip and with main feedwater flow maintained.

B. Following a turbine trip with a delayed reactor trip and with the loss of main feedwater flow.

The second situation is assumed to occur following a turbine trip resulting from high condenser pressure, which would also cause the loss of the turbine-driven main feedwater pumps. A total main steam safety valve rated capacity of 105 percent of the engineered safeguards design steamflow, at an accumulation pressure not exceeding 110 percent of the main steam system design pressure, meets this requirement. This represents a minimum relief capacity of 4,213,125 lb/h for each main steam line. At the same time, the individual safety valves are limited to the maximum allowable steam relief valve capacity of 970,000 lb/h for a main steam system design pressure of 1200 psia. This limit is imposed on safety valves connecting to the main steam system wherever a single valve failure in the open position would result in the uncontrolled release of steam from one or more steam generators. The purpose of this requirement is to limit the potential uncontrolled blowdown flowrate and the ensuing reactor transient should a single safety valve inadvertently fail or stick in the open position.

Five safety valves are provided per main steam line for the VEGP. The lowest set pressure is 1185 psig and the highest set pressure is 1235 psig. The relieving capacity per main steam line is 4,651,805 lb/h; the total relieving capacity per nuclear unit for four main steam lines is 18,607,220 lb/h. Table 10.3.2-2 lists the performance data for the VEGP main steam safety valves.

The main steam supply system safety valves are located in the safety-related portion of the main steam piping upstream of the MSIVs and outside the containment. Adequate provision is made in the steam piping for the installation and support of the valves. Particular consideration is given to the static and dynamic loads when operating or being subjected to seismic shock.

Each safety valve is connected to its vent stack by an open umbrella-type transition piece.

The vent stacks are designed to:

A. Direct the relieved steam from adjoining structures.

B. Ensure that no backflow of relieved steam can escape through the umbrella-type transition section.

C. Ensure that the total steamflow plus a small quantity of ambient air leaves the vent stack outlet.

D. Minimize the backpressure on the valve outlet in order not to restrict the valve rated capacity.

The vent stacks are not required for safety, but are structurally designed to withstand SSE and operating basis earthquake (OBE) loads in order not to jeopardize the performance of safety-related components such as the safety valves, power-operated atmospheric relief valves, or MSIVs.

#### 10.3.2.2.3 Power-Operated Atmospheric Relief Valves

A power-operated atmospheric relief valve is installed on the outlet piping from each steam generator. The four valves are installed to provide for controlled removal of reactor decay heat during normal reactor cooldown when the MSIVs are closed or the turbine bypass system is not available. The valves will pass sufficient flow at all pressures to achieve a 50°F/h plant cooldown rate. The valves are sized to provide a minimum flow of 64,000 lb/h at an inlet pressure of 100 psia. The maximum actual capacity of the relief valve at design pressure is limited to reduce the magnitude of a reactor transient if one valve would inadvertently open and remain open.

Each power-operated relief valve is located outside the containment and upstream of the MSIV, in the safety-related portion of the main steam line associated with each steam generator, to permit valve operation following all accident conditions, including those which could result in closure of the MSIVs.

The operation of the power-operated relief valves is automatically controlled by steam line pressure during plant operations. The power-operated relief valves will automatically modulate open and exhaust to atmosphere whenever the steam line pressure exceeds a predetermined setpoint. As steam line pressure decreases, the relief valves will modulate closed, eventually reseating at a pressure at least 10 psi below the opening pressure. Typically the setpoint is selected between zero-load steam pressure and the set pressure of the lowest set safety valves.

The steam generator power-operated atmospheric relief valves provide a means for plant cooldown by discharging steam to the atmosphere when either the condenser, the condenser circulating water pumps, or the steam dump is not available. Under such circumstances, the relief valves (in conjunction with the auxiliary feedwater system) allow the plant to be cooled down at a cooldown rate of 50°F/h from the pressure setpoint of the lowest set safety valves down to the point where the residual heat removal (RHR) system can assume the burden of heat removal. Residual heat removal operations are initiated when the reactor coolant system hot leg temperature has reached 350°F and primary coolant pressure is less than or equal to 400 psig. The 350°F RHR cut-in point corresponds to a steam generator steam pressure of 125 psia if reactor coolant pumps are operating or to 100 psia if only natural circulation conditions exist in the primary system.

For their use during plant cooldown, the power-operated atmospheric relief valves are automatically controlled by steam line pressure, with remote manual adjustment of the pressure setpoint from the control room or the shutdown panels. In order to effect a plant cooldown, the operator manually adjusts the pressure setpoint downward in a step-wise fashion. As the pressure setpoint of the relief valves is adjusted downward, the relief valves will initially open wide to reduce the steam generator saturation pressure. As the pressure and temperature begin to decrease, the relief valve opening will decrease to an area sufficient to pass the decay heat load and maintain constant steam pressure until the pressure setpoint is again manually reduced. The frequency of these setpoint adjustments and the magnitude of the step reductions (together with the auxiliary feedwater flowrate) determine the average cooldown rate. The maximum cooldown rate achievable is ultimately limited; however, by the flow-passing capability of the relief valves, the number of steam generators (and hence the number of relief valves) in service, and the available auxiliary feedwater pumping capacity and by the desire to either maintain or recover steam generator water levels during the cooldown.

The power-operated atmospheric relief valves also serve to prevent operation of the safety valves during relatively mild transients and, following safety valve actuation, act to assist the safety valves to positively reseat by automatically reducing and regulating steam pressure to a value below the safety valve reseating pressure. The operation of each power-operated atmospheric relief valve is controlled from a pressure tap on the steam generator steam line with which it is associated. This piping connection is separate from the other steam piping pressure taps which are used for reactor protection, to satisfy the requirement for separation between control and protection systems.

The power-operated atmospheric relief valve consists of a Control Components, Inc., pressurized seat offset globe drag valve (8-in. inlet-offset, 10-in. outlet, 900-lb valve rating) with electrohydraulic actuator and a restrictor. The nuclear Class 2 globe drag valve is located inside the MSIV areas; the restrictor (silencer) is placed on a roof outside the control and auxiliary buildings to vent to atmosphere. The noise attenuation is divided into two stages: one by the disk stack inside the main globe valve and the second by the disk stack in the restrictor. This design approach reduces the size of the main valve and the size of the restrictor. The combined design has the capability of controlling the distant field noise as well as the local field noise. It is sized for a sound pressure level of 90 dBa at 3 ft near field noise and less than 55 dBa at 4000 ft for far field noise.

The atmospheric relief valves are electrohydraulically operated and are controlled by Class 1E sources. The capability for remote manual valve operations is provided in the main control room and at the shutdown panels. Local manual operators are provided to permit operation of the valves in the event of a complete loss of automatic or remote manual control.

The valve operator is a self-contained linear modulating electro-hydraulic valve operator for use with an 8-in. x 10-in. globe drag valve. On loss of power and/or signal, the actuator will extend

the operator and close the valve. The operator is mounted on the valve by attaching the base plate to the gland of the valve. The operator rod is attached to the line valve stem.

The primary function of the operator is modulation. The actuator recognizes 4- to 20-mA command signals; four mA represents full extension (valve closed) and with increasing signal the actuator retracts (valve opens). The incoming signal is compared to the feedback signal coming from the position transducer by an onboard servo amplifier. If there is a change from the previous level greater than the deadband, the actuator is set in motion until the corresponding position is reached. Loss of command signal will extend the operator and close the valve.

The system stores energy in a pneumatic accumulator and gas bottle pressurized by an electrically driven pump which is controlled by two fluid pressure switches; one to turn pump motor on (decreasing) and the other to turn pump motor off (increasing).

Opening or closing during normal operation is accomplished by either energizing both solenoids or de-energizing both solenoids, to either retract (open valve) or extend (close valve) the operator cylinder rod. Speed or retracting (opening valve) or extending (closing valve) is controlled by a flow control valve which meters the flow of hydraulic fluid from the cylinder to return. The operator can be positioned by means of a permanently installed local hand pump in the event of pump failure.

The electrical system consists of servo valve, three pressure switches (one for gas, two for oil), and a reservoir low fluid level indicator. The system also features a servo amplifier which compares incoming command signals with actuator position feedback transducer and then energizes the corresponding solenoid valve to comply with the command.

The hydraulic system stores sufficient energy to perform its intended functions. The energy is stored in an onboard piston-type accumulator. Charging is initiated by a "fluid" pressure switch which is set to indicate the minimum system pressure necessary for proper operation. When the minimum is reached the system's hydraulic power supply is turned on to restore the system to its peak pressure. Upon reaching maximum, system pressure as indicated by another pressure switch, the power supply is turned off. As the operator modulates, the system pressure drops. When the minimum pressure switch set point is reached, the above process repeats and the actuator is recharged.

For modulation, the dual coil servo valves in the system are operated. This is automatically done by the servo amplifier when there is a change in the incoming signal. To extend the actuator (and close the valve), solenoids A and B are deenergized. Upon reaching the set point when extending, solenoid B is energized to stop the actuator. If a command signal is received to retract the actuator, the servo amplifier energizes solenoids A and B. At the set point, solenoid A is deenergized and solenoid B remains energized (as extending).

The hydraulic circuit is protected from extreme pressure transients by two thermal relief valves. These valves become active when their relief setting is exceeded. Reseat is automatic.

Stroking speed is controlled by a cartridge-type flow control valve which is preadjusted to provide the specified stroking speed.

#### 10.3.2.2.4 Main Steam Isolation Valve System

The function of the main steam isolation system is to limit blowdown to one steam generator in the event of a steam line break in order to:

A. Limit the related effect upon the reactor core within specified fuel design limits.

B. Limit containment pressure to a value less than 90 percent of design pressure.

The main steam isolation system consists of two quick-acting gate valves in series in each main steam line and two associated quick-acting globe type MSIV bypass valves with associated actuators and instrumentation. These valves are located outside the containment building, as close to the containment as practical, and downstream of the steam generator safety valves and the atmospheric relief valve. The isolation system provides positive shutoff with minimum leakage during postulated line severance conditions either upstream or downstream of the valves.

The main steam isolation valves close fully and remain fully closed within 5 s of the receipt of a manual or automatic signal. Upon receipt of the closing signal, the main steam isolation valves complete the closing cycle despite loss of normally required utility services for actuator and instrumentation. The actuator has sufficient self-contained capacity for two full closures without restoration of utility services. In loss of actuating power, the valves fail to the closed position. Position indication and remote manual operation of the isolation valves are provided in the control room. Separation of redundant control, power, and indication channels is provided for the isolation valves. Precautions are taken to prevent accidental closure on temporary loss of energy (e.g., electrical power, instrument air, etc.). Additionally, provisions are made for inservice inspection of the isolation valves.

Closure of the MSIVs and MSIV bypass valves is initiated by the following:

A. Low steam line pressure in one of four loops.

B. High containment pressure.

C. High steam pressure rate in one of four loops.

D. Manual actuation; one control (momentary actuation) per loop for single-loop isolation or manual actuation of either of two controls for system level isolation.

Each MSIV is a bidirectional, wedge-type gate valve composed of a valve body which is welded into the system pipeline. The valve body is a straight-through T-pattern with the stem and upper works vertically upright above the flowline. The valve's flow passage is sealed by closing the gate into the seat rings. The gate is made up of two gate halves, which are tapered to match with the angle of the seat rings. The gate halves are guided throughout the stroke by tongues on the gate sides which fit into grooves in the body. The gate is positioned (open or closed) by the stem which is captured between the gate halves in a T-head slot in the gate halves. The body-to-bonnet joint is sealed against leakage by a pressure seal gasket. As the differential pressure across the disk increases, the seating load also increases, thus providing a tight seal throughout the entire range of operating differential pressures. Since the disks are completely independent of each other and the design is essentially symmetrical, positive sealing can be maintained in either direction. This feature eliminates the necessity of installing a check valve to stop reverse flow.

The valve actuator is supported by the yoke, which is attached to the top of the body. The valve actuator consists of a hydraulic cylinder with a stored energy system to provide emergency closure of the isolation valve. The energy to operate the valve is stored in the form of compressed nitrogen contained in a spheroidal volume on one end of the actuator cylinder. The pressure of the nitrogen is 2500 psig maximum at 110°F with the piston fully retracted. The stored energy volume communicates directly with the actuator piston, which has a redundant sealing system to ensure that there is no leakage of nitrogen while the unit is in the standby mode. The piston is designed to protect the backseats of the valve. This is done by having a piston within a piston, which allows the main piston to seat on the piston stop ring. The floating piston (actually the top end of the actuator rod) regulates backseating force because its

pressurized area is significantly smaller than the main piston. The lower volume of the cylinder, on the other side of the piston, is filled with hydraulic fluid. This serves three basic purposes: to be throttled through a control system to control closing speed; to provide a simple, reliable method of reopening the valve; and, most importantly, to hold the unit in the standby position. The latter is accomplished by holding the fluid under high pressure, also with the hydraulic control system. The operating pressure in the oil side of the cylinder is 4300 psig maximum at 110°F. When the actuator extends to close the valve, hydraulic fluid flows from the cylinder into the hydraulic control system through cavitating venturi orifice inserts that can limit closing speed in the event of a catastrophe which causes one or both control manifolds to be severed from the body of the cylinder. The fluid flows through the control systems into a reservoir on the front of the actuator. The pump which reopens the actuator is mounted on this reservoir and is connected directly to one of the control manifolds. There is no external piping which would prevent emergency closing of the actuator should the piping be damaged.

The MSIV bypass valves are used when the MSIVs are closed to permit warming of the main steam lines prior to startup. The bypass valves are air-operated globe valves. For emergency closure, the valve solenoid, when deenergized, will result in valve closure. Electrical solenoids are energized from a separate Class IE source.

### 10.3.2.3 System Operation

#### 10.3.2.3.1 Normal Operation

At low plant operating loads, the main steam supply system provides steam to the steam generator feedwater pump turbines, the main and steam generator feedwater pump turbines steam-seal systems, the condenser spargers, the steam jet air ejectors, the single-stage reheaters, and the auxiliary steam system. At high plant operating loads, a portion of the hot reheat steam leaving the single-stage reheater is used to drive the steam generator feedwater pump turbines. The transition from using all main steam at low loads to all hot reheat steam at high loads starts at below 10-percent load. No steam is required for the condenser spargers at high loads.

The main steam supply system is capable of accepting a ±10-percent step change in load followed by a ±5-percent/min ramp change without discharging steam to the atmosphere through the main steam safety valves or to the main condenser through the turbine bypass system. For large step change load reductions, steam is bypassed (40 percent of VWO) directly to the condenser via the turbine bypass system. The system is capable of accepting a 50-percent load rejection without reactor trip and a full-load rejection with turbine trip without lifting safety valves. If the turbine bypass system is not available, steam is vented to the atmosphere via the power-operated atmospheric relief valves and the main steam safety valves, as required.

#### 10.3.2.3.2 Emergency Operation

In the event that the plant must be shut down and offsite power is lost, the MSIVs with associated MSIV bypass valves and other valves (except to the auxiliary feedpump turbine) associated with the main steam lines are closed. The power-operated atmospheric relief valves are then used to remove reactor decay and primary system sensible heat at a cooldown rate averaging 50°F/h from an average primary system temperature of 550°F to 350°F, whereupon

the RHR system performs the remaining cooldown function to 140°F to achieve cold shutdown. If the power- operated atmospheric relief valve for an individual main steam line is unavailable because of the loss of its control or power supply, the respective safety valves will provide overpressure protection. The remaining power-operated atmospheric relief valves are sufficient to achieve cold shutdown.

In the event that a design basis accident occurs which results in a large steam line break, the MSIVs with associated MSIV bypass valves automatically close.

Steam is automatically provided to the auxiliary feedwater pump turbine from one of two steam lines upon low-low level in two steam generators or loss of offsite power. Check valves are installed in the lines to the auxiliary feedwater pump turbine to ensure that only one steam generator will feed a ruptured main steam line.

The closure of either isolation valve in each of the four pairs of MSIVs and associated MSIV bypass valves will ensure that no more than one steam generator can supply a postulated break. Coordinated operation of the auxiliary feedwater system (subsection 10.4.9) and power-operated atmospheric relief valves or safety valves may be employed to remove sensible and reactor decay heat.

### 10.3.3 EVALUATION

A. Each main steam line is provided with safety valves that limit the pressure in the line to preclude over-pressurization and remove stored energy. Each line is provided with a power-operated atmospheric relief valve to permit reduction of the main steam line pressure and remove stored energy to achieve an orderly shutdown. The auxiliary feedwater system, described and evaluated in subsection 10.4.9, provides makeup to the steam generators consistent with the steaming rate.

B. Redundant power supplies and power trains operate the main steam isolation valves (MSIVs) and MSIV bypass valves to isolate safety- and nonsafety-related portions of the system. Branch lines upstream of the MSIVs contain normally closed, power-operated atmospheric relief valves which modulate open and closed on steam line pressure. In the event the atmospheric relief valves fail closed, the safety valves provide the overpressure protection.

   Accidental releases of radioactivity from the main steam supply system are minimized by the negligible amount of radioactivity in the system under normal operating conditions. Additionally, the main steam isolation system provides controls for reducing accidental releases, as discussed in chapter 15, following a steam generator tube rupture.

   Detection of radioactive leakage into the system, which is characteristic of a tube leak or rupture, is facilitated by inline radiation monitors on each steam header, the condenser air ejector radiation monitor, and steam generator blowdown sampling. Two additional primary-to-secondary leak detection systems are also provided: a noble gas detector and a system utilizing N16 as the detection medium. The N16 detector is installed in the turbine building main steam pipe chase, between the two main steam pipes. The noble gas detector is located in the condenser steam jet air ejector discharge header immediately prior to the filtration unit. Remote readout is provided at the integrated plant computer in the control room. For details of these radiation monitors see subsection 11.5.2.

C. Section 3.2 delineates the quality group classification and seismic category applicable to the safety-related portion of this system and supporting systems. Table 10.3.2-1 shows that the components meet the design and fabrication codes given in section 3.2. All the power supplies and controls necessary for safety-related functions of the main steam supply system are Class 1E, as described in chapters 7 and 8.

D. The safety-related portions of the main steam supply system are located in the containment and auxiliary buildings and the main steam isolation valve (MSIV) areas. These buildings and areas are designed to withstand the effects of earthquakes, tornadoes, hurricanes, floods, external missiles, and other appropriate natural phenomena. Sections 3.3, 3.4, 3.5, 3.7, and 3.8 provide the bases for the adequacy of the structural design of these buildings and areas.

   The safety-related portions of the main steam supply system are designed to remain functional after a safe shutdown earthquake. Sections 3.7 and 3.9 provide the design loading conditions that were considered. Sections 3.5, 3.6, and 9.5 provide the hazards analyses to ensure that a safe shutdown can be achieved and maintained.

E. As indicated by the failure mode and effects analysis in table 10.3.3-1, no single failure coincident with loss of offsite power compromises the system's safety functions. All vital power can be supplied from either onsite or offsite power systems, as described in chapter 8.

F. The main steam supply system is initially tested with the program given in chapter 14. Periodic inservice functional testing is done in accordance with subsection 10.3.4.

   Section 6.6 provides the American Society of Mechanical Engineers Boiler and Pressure Vessel Code, Section XI requirements that are appropriate for the main steam supply system.

G. The steam line to the auxiliary feedwater pump turbine is connected to a cross-connecting header upstream of the MSIVs for steam generators 1 and 2. This arrangement ensures a supply of steam to this turbine when the steam generators are isolated. Check valves are provided in each supply line from the main steam lines to preclude potential backflow during a postulated main steam line break. The auxiliary feedwater system is described in subsection 10.4.9.

H. The components of the main steam supply system are qualified to function in normal, test, and accident environmental conditions. The results of the environmental qualification program are provided in subsection 3.11.B.3.

I. A discussion of high energy pipe break locations and evaluation of effects is provided in section 3.6.

## 10.3.4 INSPECTION AND TESTING REQUIREMENTS

### 10.3.4.1 Preservice Valve Testing

The operability and relief setpoints of the main steam safety valves will be verified at temperature in a bench test using steam as the pressurization fluid. The advantages of this approach are: the testing at temperature will reduce the probability of having to adjust the valve

setpoints during hot functional testing heatup; the hot functional testing has, at other plants resulted in valves not reseating because of the combination of the booster device and limited steam flow which barely opens the valve; and the booster device test is not particularly accurate and will probably require retesting and averaging of the data to obtain acceptable results.

The lift-point of each power-operated atmospheric relief valve is checked against pressure gauges mounted in the main steam piping.

The main steam isolation valves and main steam bypass isolation valves are checked for closing time prior to initial startup.

### 10.3.4.2 Preservice System Testing

Preoperational testing is described in chapter 14.

The main steam supply system is designed to include the capability for testing through the full operational sequence that brings the system into operation for reactor shutdown and for main steam line break accidents. This includes operation of applicable portions of the protection system and the transfer between normal and standby power sources.

The safety-related components of the system, i.e., valves and piping, are designed and located to permit preservice and inservice inspections to the extent practicable.

### 10.3.4.3 Preservice Pipe Testing

Main steam piping located in the main steam tunnels adjacent to the diesel generator building is volumetrically inspected in the shop or in the field. This inspection includes a 100-percent inspection of circumferential and longitudinal welds.

The main steam lines from the five-way restraint forgings in the isolation valve areas to the main steam tunnel five-way restraint forgings are classified as non-Seismic Category 1 piping. However, these lines are designed to withstand a safe shutdown earthquake, and 100 percent volumetric inspection at installation is provided (i.e., 100 percent volumetric examination of shop and field longitudinal and circumferential welds).

### 10.3.4.4 Inservice Testing

The performance and structural leaktight integrity of all system components are demonstrated by continuous operation.

The redundant actuator power trains of each main steam isolation valve are subjected to the following tests:

A. Closure time; the valves are checked for closure time at each refueling.

B. While the nuclear steam supply system is in operation, the operability of the actuator may be checked periodically by exercising the valve to approximately 90 percent of full open.

The main steam isolation valve bypass valves may be tested for closure time at power as discussed in paragraph 7.1.2.5.C.

Additional discussion of inservice inspection of American Society of Mechanical Engineers Code Class 2 and 3 components is contained in section 6.6. Also, pump and valve testing is discussed in subsection 3.9.6.

## 10.3.5 WATER CHEMISTRY

The objectives of the secondary side water chemistry program are as follows:

A. Minimizing general corrosion in the steam generators, turbine, and feedwater system by maintaining proper pH control and by minimizing oxygen ingress (coupled with oxygen scavenging).

B. Minimizing localized corrosion in the steam generators, turbine, and feedwater system by minimizing chemical contaminant ingress and by controlling contaminant levels through condensate polishing and steam generator blowdown.

### 10.3.5.1 Chemistry Control Basis

Electric Power Research Institute (EPRI) Pressurized-Water Reactor (PWR) Secondary Water Chemistry Guidelines recommendations are considered in the secondary side water chemistry program.

Secondary side water chemistry at VEGP is optimized as shown below:

A. Design and Control Phase

1. Selection of secondary side materials to minimize corrosive species such as copper oxides.
2. Capability of deaeration in the condensate storage tank and hotwell.
3. Capability of continuous blowdown of the steam generator bulk water.
4. Post-construction cleaning of the feedwater system followed by wet layup of the feedwater system and steam generators.

B. Operation Phase

1. Early identification of contaminant ingress (salts, corrosion products, and oxygen).
2. Full flow condensate filtration and demineralization.
3. Chemical addition to establish and maintain an environment that minimizes system corrosion.
4. Identification of action levels based on chemistry conditions.

### 10.3.5.2 Contaminant Ingress

Contaminant may be introduced into the secondary side water system through three major mechanisms: make-up water; condenser tube leaks; atmospheric leaks at the condenser or pump seals. The following methods are used to detect the ingress of contaminants in the secondary water system.

A. Demineralized water is continuously monitored as it is being produced in the water treatment plant. Then the condenser makeup water from the condensate storage tanks and from the condensate storage tank degasifiers are monitored routinely for contaminants.

B. Ionic contaminants are detected by monitoring (either continuous process monitors or sample analysis) the condensate pump discharge.

C. Atmospheric contamination is detected by monitoring the condensate pump discharge for dissolved oxygen.

### 10.3.5.3 Condensate Polishing

Full-flow condensate filter/demineralizers are provided to filter suspended corrosion products from the condensate and also to remove ionic contaminants. This polishing system will not necessarily be employed during all phases of plant operation, but when the polisher is in service, provisions are made for monitoring the common inlet/outlet from each vessel, and the total outlet.

The secondary side water system has provisions for recirculating feedwater to the condenser prior to and during startup. The polisher may be used during this phase to remove corrosion products from the feedwater and thus prevent their ingress into the steam generators.

### 10.3.5.4 Chemical Addition

VEGP employs volatile treatment plus ammonium chloride addition to minimize general corrosion in the feedwater system, steam generators, and main steam piping. Hydrazine, alternate amine, and ammonium chloride are the chemicals to be injected into the condensate system.

Alkaline conditions reduce the general corrosion rate of ferrous alloys, so a combination of hydrazine and an alternate amine provide these conditions. The alternate amine allows for better protection of the wet steam areas of the secondary system.

Hydrazine is added to scavenge dissolved oxygen present in the feedwater system. Hydrazine also promotes the formation of a protective magnetite layer on ferrous surfaces and to keep this layer in a reduced state, further inhibiting general corrosion.

Ammonium chloride is added to the secondary system to control the steam generator sodium to chloride molar ratio. The objective in controlling molar ratio is to maintain a neutral crevice where soluble contaminants concentrate, such as the tube to tube sheet crevices. The relationship between the bulk water molar ratio and the hideout return molar ratio can then be determined to control crevice chemistry.

### 10.3.5.5 Action Levels for Abnormal Conditions

Prompt and appropriate responses to abnormal chemistry conditions are prudent to assure the long-term integrity of secondary cycle components. As such, three action levels have been defined for taking remedial action when control parameters are observed and confirmed to be outside of limits.

In general, these actions will be consistent with the EPRI "PWR Secondary Water Chemistry Guidelines."

### 10.3.5.6 Layup and Heatup

VEGP plans no long-term steam generator layup under dry conditions. When maintenance or inspection is required on the secondary side of the steam generators, the steam generators will be drained hot under nitrogen atmosphere. After cooling, the nitrogen purge will be lifted and the maintenance/inspection will begin.

Then, wet layup conditions will be established for corrosion protection during outages. Guidelines are given in EPRI, PWR Secondary Water Chemistry Guidelines.

Before heatup to full power, the bulk water in the steam generators is normally brought into power operation specifications by draining and refilling or by feeding and bleeding. Guidelines for heatup are provided in EPRI, PWR Secondary Water Chemistry Guidelines.

### 10.3.5.7 Chemical Analysis Basis

The chemical parameters listed below play a significant role in the corrosion of the feedwater system and the steam generators. Each parameter will be addressed as indicated below.

A. Oxygen in the presence of moisture rapidly corrodes carbon steel. These corrosion products may be carried through the feedwater system and form a sludge pile in the steam generator. This sludge pile forms an ideal environment for localized corrosion mechanisms on steam generator tubes. Thus, concentration of oxygen should be kept as low as practical in the feedwater system, and it is prudent to control dissolved oxygen at the condenser and not allow it to be transported to the feedwater system. The oxygen concentration is measured by process analyzers in which an electric current is generated at an electrode in proportion to oxygen dissolved in a flowing sample stream. Oxygen is also measured on a grab sample by color comparison after reaction with a reagent.

B. In the absence of significant impurities, the pH is controlled by the concentration of alternate amine and hydrazine. Maintaining the pH within the recommended band results in minimal corrosion rates of ferrous materials. The pH is measured in both process and bench instruments which measure the potential across an electrode sensitive to hydrogen ions and a reference electrode.

C. By passing the sample through a cation resin column, conductive anions (such as chloride, sulfate, and to a lesser extent carbonate) can be indicated in a conductivity cell. Cation conductivity is a sensitive method for indicating soluble species which have been indicated in many localized corrosion mechanisms.

D. Sodium is an effective continuous indicator of many forms of containment ingress. The ability to analyze for the species at the sub ppb level makes it a very useful chemical tracer. Increased sodium levels can be indicative of condenser leakage or makeup water contamination. Sodium is measured by process analyzers using specific ion electrodes. Grab samples are analyzed by furnace atomic absorption.

E. Silica is an effective indicator of many forms of contaminant ingress. The ability to analyze for this species at the low ppb level makes it a very useful chemical tracer.

F. Chloride is aggressive to ferrous materials at steam generator operating conditions, particularly in crevice regions. It also has been identified as an

aggravant relevant to inconel 600 pitting. Grab samples are analyzed for chloride by ion chromatography.

The chemical analysis methods listed above represent current technology applicable for power plant laboratories. These methods may be changed over the lifetime of the plant as new technology is proven to be more effective.

### 10.3.5.8 Sampling

Sampling points are identified in table 9.3.2-3 (grab sample points and process instrumentation) and table 9.3.2-4 (grab sample points). These sample points include hotwells, condensate storage tank, condensate, feedwater, auxiliary feedwater system, steam generator blowdown, reheat steam, and heater drains. Many of these points will be sampled and analyzed routinely, others only as needed for troubleshooting and problem diagnosis.

### 10.3.5.9 Condenser Inspection

The secondary side water chemistry program will include a comprehensive inspection program of the condenser and will be developed to ensure condenser integrity. This program includes a visual inspection of the condenser every refueling outage, waterbox inspection for tube leaks during plant operation, and component inspection for oxygen leaks during plant operation.

These water box inspections and component inspections will be performed as necessary to diagnose and troubleshoot abnormal chemistry levels.

### 10.3.5.10 Data Management and Corrective Actions for Out-of-Specification Condition

All analytical data will be recorded and stored in a retrievable fashion. The technician performing the analysis will compare the analyzed value to the specifications value, where specifications are established, and promptly report any values which are out of the established specification limits stated earlier to shift supervision. Analytical data for key parameters will be routinely trended.

When an analysis indicates an out-of-specification condition, laboratory personnel will promptly investigate the problem. Investigation may include confirmation of the reported value by resampling and reanalysis, checking on-line monitoring data where appropriate, and examining values of other parameters to obtain confirmation of the analyzed value.

Once confirmed, action to correct the out-of-specification condition will be initiated. The shift supervisor, utilizing technical help from the laboratory, will initiate corrective action based on the abnormal operating procedure.

Cases may exist, at all action levels, where prompt action by on-shift personnel can rapidly correct the out-of-specification condition. If the problem is promptly corrected, notification will not be necessary. Shift personnel will receive problem-solving assistance from site technical support personnel (chemists, plant engineers, maintenance personnel, etc.) and/or off-site personnel (vendors, consultants, and/or Corporate personnel) as appropriate to diagnose the root cause of out-of-specification occurrences and to apply corrective actions. VEGP management will assist both in technical resolutions and in assuring proper problem-solving support.

Corrective actions taken will depend on the specifics of the problem. Some typical corrective actions are listed below.

- Increase blowdown levels to lower chemical concentration in the steam generator water.
- Increase/decrease treatment chemicals addition to bring parameters back into specification.
- Locate and stop contaminant ingress; special samples and/or increased sampling frequency may be used to assist this effort.
- Decrease power or shutdown to limit potential damage while corrective action is being applied.
- Direct drains to the hotwell polishing.

#### 10.3.5.11 Conformance to Branch Technical Position MTEB 5-3

VEGP conformance to Branch Technical Position MTEB 5-3 is discussed in table 10.3.5-1.

### 10.3.6 STEAM AND FEEDWATER SYSTEM MATERIALS

#### 10.3.6.1 Fracture Toughness

Compliance with fracture toughness requirements of American Society of Mechanical Engineers (ASME) Code, Section III, Articles NC-2300 and ND-2300, is discussed in section 6.1.

#### 10.3.6.2 Material Selection and Fabrication

All pipe, flanges, fittings, valves, and other piping material conform to the referenced ASME, American Society of Testing Materials (ASTM), American National Standards Institute (ANSI), or Manufacturer Standardization Society-Standard Practice (MSS-SP) code.

The following code requirements apply.

| Component | Stainless Steel | Carbon Steel |
|---|---|---|
| Pipe | ANSI B36.19 | ANSI B36.10 |
| Fittings | ANSI B16.9, B16.11, or B16.28 | ANSI B16.9, B16.11, or B16.28 |
| Flanges | ANSI B16.5 | ANSI B16.5 |

The following ASME material specifications apply specifically:

- ASME SA-155, GR KCF 70 Class 1 (impact tested).
- ASME SA-155, GR KCF 70 Class 1.
- ASME SA-106, GR C (impact tested).
- ASME SA-106, GR B.
- ASME SA-106, GR B (normalized).
- ASME SA-234, GR WPB.
- ASME SA-234, GR WPBW (manufactured from grade 70 plate).
- ASME SA-516, GR70.
- ASME SA-216, GR WCC.
- ASME SA-515, GR70.
- ASME SA-106, GR B (impact tested).
- ASME SA-234, GR WPC or WPCW (impact tested).
- ASME SA-234, GR WPCW.
- ASME SA-234, GR WPC.
- ASME SA-105.
- ASME SA-193, GR B7.
- ASME SA-194, GR 2H, or ASME SA-194, GR 7
- ASME SA-216, GR WCB.
- ASME SA-333, GR 6 (impact tested).
- ASME SA-420, GR WPL6 (impact tested).
- ASME SA-508, Class 1 (impact tested).
- ASME SA-312, TP 304.
- ASME SA-403, WP-304.
- ASME SA-403, WP-304 W.
- ASME SA-182, F-304.
- ASME SA 672, GR C70.

Conformance with the following regulatory guides is discussed in section 1.9:

- Regulatory Guide 1.31, Control of Ferrite Content in Stainless Steel Weld Metal.
- Regulatory Guide 1.36, Nonmetallic Thermal Insulation of Austenitic Stainless Steel.
- Regulatory Guide 1.37, Quality Assurance Requirement for Cleaning of Fluid Systems and Associated Components of Water-Cooled Nuclear Power Plants. (Note: Regulatory Guide 1.37 has been replaced by NQA-1-1994 as described in the SNC Quality Assurance Topical Report (QATR)).
- Regulatory Guide 1.44, Control of the Use of Sensitized Stainless Steel.
- Regulatory Guide 1.50, Control of Preheat Temperatures for Welding of Low-Alloy Steels.
- Regulatory Guide 1.71, Welder Qualification for Areas of Limited Accessibility.
- Regulatory Guide 1.85, Materials Code Case Acceptability-ASME Section III, Division I.

TABLE 10.3.2-1 (SHEET 1 OF 2)

MAIN STEAM SUPPLY SYSTEM
DESIGN DATA

| Steam Flow (lb/h) | 100% Turbine Guarantee | Valves Wide Open | Power Uprate 100% Power |
|---|---|---|---|
| Per steam generator | 3,790,260 | 3,972,247 | 3,980,000 |
| Total | 15,161,041 | 15,888,986 | 15,920,000 |

Operating/Design Pressures/Temperatures

| | |
|---|---|
| 100% turbine guarantee | 1000 psia/545°F |
| No load (hot standby) | 1087 psia/547°F |
| Design conditions | 1200 psia/600°F |

Main Steam Piping: See table 10.3.2-3.

| | |
|---|---|
| Design Pressure Drop, Steam Generator Outlet to Turbine Stop Valves | 40.0 psi |

Steam Generator Flow Restrictor

| | |
|---|---|
| Number per steam generator outlet nozzle | 7 |
| Throat size | 6.03 in. |
| Total area | 1.388 $ft^2$ |

Design Codes

| | |
|---|---|
| From steam generators to restraint downstream of outboard MSIV plus associated branch connections up to and including first valve that is normally closed or capable of automatic closure | ASME B&PV Code, Section III, Class 2, Seismic Category 1 |
| Remainder | ANSI B31.1,Power Piping |

TABLE 10.3.2-1 (SHEET 2 OF 2)

Power-Operated Relief Valve

| | |
|---|---|
| Number per main steam line | 1 |
| Normal set pressure | 1120 psig |
| Design capacity | 64,000 lb/h at 100 psia inlet pressure, 596,000 lb/h at 1107 psia inlet pressure |
| Code | ASME B&PV Code, Section III, Class 2, Seismic Category 1 |
| Actuator | Electrohydraulic |

TABLE 10.3.2-2

DESIGN DATA FOR MAIN STEAM SAFETY VALVES

| | |
|---|---|
| Number per main steam line | 5 |
| Total number of valves required | 20 |
| Minimum relieving capacity at 110% of main steam design pressure | 16,852,500 lb/h |
| Available relieving capacity | 18,607,220 lb/h |
| Valve size | 6 x 10 in. |
| Orifice area (each) | 16.0 $in.^2$ |
| Design code | ASME B&PV Code, Section III, Class 2, Seismic Category 1 |

| Valve Number | Set Pressure (psig) | Relieving Capacity[(a)] (lb/h) |
|---|---|---|
| 1 | 1185 | 930,361 |
| 2 | 1200 | 930,361 |
| 3 | 1210 | 930,361 |
| 4 | 1220 | 930,361 |
| 5 | 1235 | 930,361 |
| Total capacity, per line | | 4,651,805 |
| Total capacity, 4 lines | | 18,607,220 |

a. Based on system accumulation pressure of 1272 psig per paragraph NC-7826 of ASME B&PV Code Section III Division 1 1974 Edition Subsection NC, Class 2 components including addenda through summer 1975.

TABLE 10.3.2-3

DESCRIPTION OF MAIN STEAM PIPING

| Segment | Applicable Steam Generators | Description | Dimensions (in.) Nominal OD | Nominal OD | Minimum ID |
|---|---|---|---|---|---|
| Steam generator outlet to penetration | All | 26-in. extruded pipe | 26.0 | 23.956 | - |
| Penetration to first safety valve | All | 28-in. forging | 28.0 $\binom{+0.125}{-0.031}$ | 23.875 | 23.75 |
| First safety valve to inboard MSIV | All | 29.5-in. forging | 29.5 $\binom{+0.125}{-0.031}$ | 23.875 | 23.625 |
| Between MSIVs | All | 29.5-in. forging | 29.5 $\binom{+0.125}{-0.031}$ | 23.875 | 23.625 |
| Outboard MSIV to forged section | All | 28-in. forging | 28.0 $\binom{+0.125}{-0.031}$ | 23.875 | 23.75 |
| Forged section to common run | All | 26-in. extruded pipe | 26.0 | 23.956 | - |
| Common run to branch point | 1 and 4 | 38-in. extruded pipe | 38.0 | 35.138 | - |
| Common run to branch point | 2 and 3 | 36-in. extruded pipe | 36.0 | 33.282 | - |
| Branch point to turbine stop valves | All | 28-in. extruded pipe | 28.0 | 25.808 | - |

TABLE 10.3.3-1 (SHEET 1 OF 13)

MAIN STEAM SYSTEM
FAILURE MODES AND EFFECTS ANALYSIS

| Item No. | Description of Component | Safety Function | Plant Operating Mode | Failure Mode(s) | Method of Failure Detection | Failure Effect on System Safety Function Capability | General Remarks |
|---|---|---|---|---|---|---|---|
| 1. | MSIVs HV-3006A and HV-3006B, normally open, fail closed, with self-contained hydraulic actuator | Isolate steam generator No. 1 in the event of a main steam line break (MSLB) to prevent blowdown of more than one steam generator | A. All but MSLB | A. Fails closed or fails to open upon command | A. Position indicator on QMCB and QPCP. If valve closes spuriously, a turbine followed by reactor trip will occur. | A. None; plant goes to or remains in a safe shutdown condition. | Two MSIVs are provided on each main steam line and operated from separate Class 1E power sources. Therefore, failure of one train, including loss of SLIS, will not impair isolation capability. |
| | | | B. MSLB (Cond. IV event) | B. Fails to close upon steam line isolation signal (SLIS) | B. Position indicator on QMCB and QPCP | B. None; closure of either MSIV will isolate steam generator No. 1 and thus prevent blowdown of more than one steam generator. | |
| 2. | MSIVs HV-3016A and HV-3016B | Same as item 1, except for steam generator No. 2 | Same as items 1A and 1B | Same as items 1A and 1B | Same as items 1A and 1B | Same as items 1A and 1B, except for steam generator No. 2 | Same as item 1. |
| 3. | MSIVs HV-3026A and HV-3026B | Same as items 1A and 1B, except for steam generator No. 3 | Same as items 1A and 1B | Same as items 1A and 1B | Same as items 1A and 1B | Same as items 1A and 1B, except for steam generator No. 3 | Same as item 1. |
| 4. | MSIVs HV-3036A and HV-3036B | Same as items 1A and 1B, except for steam generator No. 4 | Same as items 1A and 1B | Same as items 1A and 1B | Same as items 1A and 1B | Same as items 1A and 1B, except for steam generator No. 4 | Same as item 1. |

TABLE 10.3.3-1 (SHEET 2 OF 13)

| Item No. | Description of Component | Safety Function | Plant Operating Mode | Failure Mode(s) | Method of Failure Detection | Failure Effect on System Safety Function Capability | General Remarks |
|---|---|---|---|---|---|---|---|
| 5. | Main steam power-operated relief valve (PORV) PV-3000, normally closed, fail closed electro-hydraulic valve | Relieve steam generator No. 1 overpressure during transients to minimize lifting of safety valves and to regulate steam generator No. 1 pressure during reactor startup and cooldown | A. Normal startup and cooldown | A1. Fails closed or fails to open upon command including spurious operation | A1. Position indicator on QMCB and PSDA, plus higher pressure in steam generator No. 1 than in the other three steam generators | A1. None; manual actuation of turbine bypass valve provides steam generator pressure control. Overpressure protection is provided by main steam safety valves. | |
| | | | | A2. Fails open or fails to close upon command, including spurious operation | A2. Position indicator on QMCB and PSDA, low water level and/or pressure in steam generator No. 1, plus high temperature alarm (TE-3000) in exhaust stack | A2. None; maximum valve flow less than maximum allowed by Westinghouse to prevent excessive cooldown rate. PORV can be manually isolated. Plant shutdown can be effected with only one PORV out of four operable. | |
| | | | B. Plant shutdown with loss of offsite power (LOP) | B1. Fails closed or fails to open upon command | B1. Same as item 5A1 | B1. None; only one operable PORV required for safe shutdown. Overpressure provided by main steam safety valves | |
| | | | | B2. Fails open | B2. Same as item5A2 | B2. None; same as item 5A2 | |

TABLE 10.3.3-1 (SHEET 3 OF 13)

| Item No. | Description of Component | Safety Function | Plant Operating Mode | Failure Mode(s) | Method of Failure Detection | Failure Effect on System Safety Function Capability | General Remarks |
|---|---|---|---|---|---|---|---|
| | | | C. Power operation | C. Fails open | C. Same as 5A2 | C. None; plant shutdown not required. Valve can be isolated for repair. See also item 5B1. | |
| | | | D. All | D. Valve leaks | D. High temperature alarm TE-3000 in exhaust stack | D. None; same as item 5C | |
| 6. | PORV PV-3010 | Same as item 5, except for steam generator No. 2 | Same as items 5A, 5B, 5C, and 5D | Same as items 5A, 5B, 5C, and 5D, respectively | Same as items 5A, 5B, 5C, and 5D, except position indicators on QMCB and PSDB and temperature alarm TE-3010 | None; same as items 5A, 5B, 5C, and 5D, respectively | |
| 7. | PORV PV-3020 | Same as item 5, except for steam generator No. 3 | Same as items 5A, 5B, 5C, and 5D | Same as items 5A, 5B, 5C, and 5D, respectively | Same as items 5A, 5B, 5C, and 5D, except position indicators on QMCB and PSDB and temperature alarm TE-3020 | None; same as items 5A, 5B, 5C, and 5D, respectively | |
| 8. | PORV PV-3030 | Same as item 5, except for steam generator No. 4 | Same as items 5A, 5B, 5C, and 5D | Same as items 5A, 5B, 5C, and 5D, respectively | Same as items 5A, 5B, 5C, and 5D, except position indicators on QMCB and PSDB and temperature alarm TE-3030 | None; same as items 5A, 5B, 5C, and 5D, respectively | |

TABLE 10.3.3-1 (SHEET 4 OF 13)

| Item No. | Description of Component | Safety Function | Plant Operating Mode | Failure Mode(s) | Method of Failure Detection | Failure Effect on System Safety Function Capability | General Remarks |
|---|---|---|---|---|---|---|---|
| 9. | Steam generator blowdown (SGBD) isolation valve HV-7603A. Normally open, fail closed pneumatic valve | Isolates blowdown from steam generator No. 1 upon auxiliary feedwater start or be remote-manual operation | All | A. Fails closed or fails to open upon command | A. Position indicator on QMCB, plus zero flow indication, FIC-1171, on PSBP | A. None; blowdown is terminated, but blowdown has no safety function. | |
| | | | | B. Fails open or fails to close upon command | B. Position indicator on QMCB, plus flow indication, FIC-1171, on PSBP | B. None; isolation of 3 of 4 steam generators sufficient for safe shutdown. If coincident with SGBD line break, HV-15216A will close automatically to effect isolation and prevent room overpressurization. | |
| 10. | SGBD isolation valve HV-7603B | Same as item 9, except for steam generator No. 2 | All | Same as items 9A and 9B | Same as items 9A and 9B, except flow indication provided by FIC-1172 on PSBP | None; same as items 9A and 9B, except back up isolation provided by HV-15216B | |
| 11. | SGBD isolation valve HV-7603C | Same as item 9, except for steam generator No. 3 | All | Same as items 9A and 9B | Same as items 9A and 9B, except flow indication provided by FIC-1173 on PSBP | None; same as items 9A and 9B, except back up isolation provided by HV-15216C | |
| 12. | SGBD isolation valve HV-7603D | Same as item 9, except for steam generator No. 4 | All | Same as items 9A and 9B | Same as items 9A and 9B, except flow indication provided by FIC-1174 on PSBP | None; same as items 9A and 9B, except back up isolation provided by HV-15216D | |

TABLE 10.3.3-1 (SHEET 5 OF 13)

| Item No. | Description of Component | Safety Function | Plant Operating Mode | Failure Mode(s) | Method of Failure Detection | Failure Effect on System Safety Function Capability | General Remarks |
|---|---|---|---|---|---|---|---|
| 13. | SGBD isolation valves HV-15216A, HV-15212A normally open, fail closed pneumatic valve | Isolates blowdown from steam generator No. 1 upon SGBD line break in auxiliary building | All | A. Fails closed | A. Position indicator on QPCP, plus zero flow indication, FIC-1171, on PSBP | A. None; same as item 9A | |
| | | | | B. Fails open coincident with line break in auxiliary building | B. Position indicator on QPCP, plus high compartment pressure/temperature alarm | B. None; closure of either SGBD isolation valve will isolate SG no. 1 and prevent room high temperature and overpressurization. | |
| 14. | SGBD isolation valves HV-15216B and HV-15212B | Same as item 13, except for steam generator No. 2 | All | Same as items 13A and 13B | Same as items 13A and 13B, except flow indication provided by FIC-1172 on PSBP | None; same as items13A and 13B, except SG no. 2. | |
| 15. | SGBD isolation valves HV-15216C and HV-15212C | Same as item 13, except for steam generator No. 3 | All | Same as items 13A and 13B | Same as items 13A and 13B, except flow indication provided by FIC-1173 on PSBP | None; same as items13A and 13B, except SG no. 3 | |
| 16. | SGBD isolation valves HV-15216D and HV-15212D | Same as item 13, except for steam generator No. 4 | All | Same as items 13A and 13B | Same as items 13A and 13B, except flow indication provided by FIC-1174 on PSBP | None; same as items13A and 13B, except SG no. 4 | |
| 17. | SGBD sample isolation valve HV-9451, normally open, fail closed solenoid valve | Isolates SGBD sample from steam generator No. 1 upon auxiliary feedwater start or by remote-manual control | All | A. Fails closed | A. Position indicator on QPCP | A. None; sampling terminated, but not required for safety | |
| | | | | B. Fails open | B. Same as item 17A | B. None | |

TABLE 10.3.3-1 (SHEET 6 OF 13)

| Item No. | Description of Component | Safety Function | Plant Operating Mode | Failure Mode(s) | Method of Failure Detection | Failure Effect on System Safety Function Capability | General Remarks |
|---|---|---|---|---|---|---|---|
| 18. | SGBD sample isolation valve HV-9452 | Same as item 17, except for steam generator No. 2 | All | Same as items 17A and 17B | Same as item 17A | None; same as items 17A and 17B | |
| 19. | SGBD sample isolation valve HV-9453 | Same as item 17, except for steam generator No. 3 | All | Same as items 17A and 17B | Same as item 17A | None; same as items 17A and 17B | |
| 20. | SGBD sample isolation valve HV-9454 | Same as item 17, except for steam generator No. 4 | All | Same as items 17A and 17B | Same as item 17A | None; same as items 17A and 17B | |
| 21. | SGBD sample isolation HV-9553B, normally open, fail closed solenoid valve | Isolates SGBD sample from steam generator No. 1 upon remote-manual operation | All | A. Fails closed | A. Position indicator on PRP | A. None; same as item 17A | |
| | | | | B. Fails open | B. Same as item 21A | B. None | |
| 22. | SGBD sample isolation valve HV-9554B | Same as item 21, except for steam generator No. 2 | All | Same as items 21A and 21B | Same as item 21A | None; same as items 21A and 21B | |
| 23. | SGBD sample isolation valve HV-9555B | Same as item 21, except for steam generator No. 3 | All | Same as items 21A and 21B | Same as item 21A | None; same as items 21A and 21B | |
| 24. | SGBD sample isolation valve HV-9556B | Same as item 21, except for steam generator No. 4 | All | Same as items 21A and 21B | Same as item 21A | None; same as items 21A and 21B | |

TABLE 10.3.3-1 (SHEET 7 OF 13)

| Item No. | Description of Component | Safety Function | Plant Operating Mode | Failure Mode(s) | Method of Failure Detection | Failure Effect on System Safety Function Capability | General Remarks |
|---|---|---|---|---|---|---|---|
| 25. | SGBD sample isolation valve HV-9553A, normally closed, fail closed solenoid valve | Isolates SGDB (bottom) from steam generator No. 1 upon remote-manual actuation | All | A. Fails closed or fails to open upon command | A. Position indicator on PRP | A. None; same as item 17A | |
| | | | | B. Fails open | B. Same as item 25A | B. None | |
| 26. | SGBD sample isolation valve HV-9554A | Same as item 25, except for steam generator No. 2 | All | Same as items 25A and 25B | Same as item 25A | None; same as items 25A and 25B | |
| 27. | SGBD sample isolation valve HV-9555A | Same as item 25, except for steam generator No. 3 | All | Same as items 25A and 25B | Same as item 25A | None; same as items 25A and 25B | |
| 28. | SGBD sample isolation valve HV-9556A | Same as item 25, except for steam generator No. 4 | All | Same as items 25A and 25B | Same as item 25A | None; same as items 25A and 25B | |
| 29. | Steam generator chemical addition valve HV-5278, normally closed, fail closed pneumatic valve | Isolates chemical addition to steam generator No. 1 upon remote-manual actuation | All | A. Fails closed or fails to open upon command | A. Position indicator on QPCP | A. None; chemical addition is not possible, but this function is not required for safe shutdown | Chemical addition required only during cold shutdown. |
| | | | | B. Fails open | B. Same as item 29A | B. None | |
| 30. | Steam generator chemical addition valve HV-5279 | Same as item 29, except for steam generator No. 2 | All | Same as items 29A and 29B | Same as item 29A | None; same as items 29A and 29B | |

TABLE 10.3.3-1 (SHEET 8 OF 13)

| Item No. | Description of Component | Safety Function | Plant Operating Mode | Failure Mode(s) | Method of Failure Detection | Failure Effect on System Safety Function Capability | General Remarks |
|---|---|---|---|---|---|---|---|
| 31. | Steam generator chemical addition valve HV-5280 | Same as item 29, except for steam generator No. 3 | All | Same as items 29A and 29A | Same as item 29A | None; same as items 29A and 29B | |
| 32. | Steam generator chemical addition valve HV-5281 | Same as item 29, except for steam generator No. 4 | All | Same as items 29A and 29A | Same as item 29A | None; same as items 29A and 29B | |
| 33. | Steam supply valve HV-3009 from steam generator No. 1 to turbine-driven auxiliary feedwater pump (TDP) turbine, normally open, dc-powered MOV | Admits steam from steam generator No. 1 to TDP turbine and also provides for isolating the downstream piping in the event of a pipe break | A. Hot standby and cooldown | A. Fails closed or fails to open upon command (remote-manual) | A. Valve position indicators on QMCB and PAFP | A. None; 100% redundant steam supply to the TDP provided via HV-3019 from steam generator No. 2. Also, two motor-driven auxiliary feedwater pumps provide 100% of auxiliary feedwater requirements. | For discussion of valve failure impact on auxiliary feedwater system, see table 10.4.9-4. |
| | | | B. All | B. Fails open coincident with break in downstream piping | B. Same as item 33A | B. None; manual closure of HV-3019 will prevent blowdown of more than one steam generator. | |
| 34. | Steam supply valve HV-3019 from steam generator No. 2 to TDP turbine | Same as item 33, except for steam generator No. 2 | Same as items 33A and 33B | Same as items 33A and 33B, respectively | Same as item 33A | Same as item 33A and 33B, except that HV-3009 from steam generator No. 1 provides redundancy | See item 33. |

| Item No. | Description of Component | Safety Function | Plant Operating Mode | Failure Mode(s) | Method of Failure Detection | Failure Effect on System Safety Function Capability | General Remarks |
|---|---|---|---|---|---|---|---|
| 35. | Check valve 006 in steam supply to TDP turbine from steam generator No. 1 (Valve internals are removed. Valve is not active.) | None. Valve internals removed. | N/A | N/A | N/A | N/A | See item 33. |
| 36. | Check valve 008 in steam supply to TDP turbine from generator No. 2 | Same as item 35, except interchange steam generators No. 1 and No. 2 | All | A. Fails open coincident with MSLB from steam generator No. 2 | A. Same as item 35A, except for steam generator No. 1 | A. Same as item 33A, except HV-3019 provides protection of steam generator No. 1 | See items 35 and 33. |
| 37. | Main steam safety valves PSV-3001, PSV-3002, PSV-3003, PSV-3004, and PSV-3005 | Protect steam generator No. 1 from overpressurization | All | A. Fails to open when required | A. Higher pressure and/or water level in steam generator. No. 1 | A. None; 4 of 5 safety valves for No. 1 steam generator still available, with PORV PV-3000 available to supplement relieving capacity. Also, plant trip occurs upon high steam generator water level. | |
| | | | | B. Spurious opening or failure to reset after opening | B. High temperature alarm in exhaust stack | B. None; maximum flow from one valve less than maximum specified by Westinghouse to prevent excessive cooldown rate. | |

TABLE 10.3.3-1 (SHEET 10 OF 13)

| Item No. | Description of Component | Safety Function | Plant Operating Mode | Failure Mode(s) | Method of Failure Detection | Failure Effect on System Safety Function Capability | General Remarks |
|---|---|---|---|---|---|---|---|
| 38. | Main steam safety valves PSV-3011, PSV-3012, PSV-3013, PSV-3014, and PSV-3015 | Protect steam generator No. 2 from overpressurization | All | A. Same as item 37A | A. Same as item 37A except for steam generator No. 2 | A. None; same as item 37A except PORV is PV-3010 | |
| | | | | B. Same as item 37B | B. Same as item 37B | B. None; same as item 37B | |
| 39. | Main steam safety valves PSV-3021, PSV-3022, PSV-3023, PSV-3024, and PSV-3025 | Protect steam generator No. 3 from overpressurization | All | A. Same as item 37A | A. Same as item 37A except for steam generator No. 3 | A. None; same as item 37A except PORV is PV-3020 | |
| | | | | B. Same as item 37B | B. Same as item 37B | B. None; same as item 37B | |
| 40. | Main steam safety valves PSV-3031, PSV-3032, PSV-3033, PSV-3034, and PSV-3035 | Protect steam generator No. 4 from overpressurization | All | A. Same as item 37A | A. Same as item 37A except for steam generator No. 4 | A. None; same as item 37A except PORV is PV-3030 | |
| | | | | B. Same as item 37B | B. Same as item 37B | B. None; same as item 37B | |
| 41. | Manual valves 001 and 136, normally open | Isolate PV-3000 (item 5, steam generator No. 1) for repair | All | Inadvertent closure | None; PV-3000 (item 5) incapacitated | None; same as items 5A1 and 5B1 | |
| 42. | Manual valves 002 and 137, normally open | Isolate PV-3010 (item 6, steam generator No. 2) for repair | All | Inadvertent closure | None; PV-3010 (item 6) incapacitated | None; same as items 5A1 and 5B1 | |

TABLE 10.3.3-1 (SHEET 11 OF 13)

| Item No. | Description of Component | Safety Function | Plant Operating Mode | Failure Mode(s) | Method of Failure Detection | Failure Effect on System Safety Function Capability | General Remarks |
|---|---|---|---|---|---|---|---|
| 43. | Manual valves 003 and 138, normally open | Isolates PV-3020 (item 7, steam generator No. 3) for repair | All | Inadvertent closure | None; PV-3020 (item 7) incapacitated | None; same as items 5A1 and 5B1 | |
| 44. | Manual valves 004 and 139, normally open | Isolates PV-3030 (item 8, steam generator No. 4) for repair | All | Inadvertent closure | None; PV-3030 (item 8) incapacitated | None; same as items 5A1 and 5B1 | |
| 45. | Manual valve 005, normally open | Isolates HV-3009 (item 33) for repair | All | Inadvertent closure | None; HV-3009 (item 33) incapacitated | None; same as item 33A | See table 10.4.9-4 FMEA for impact on auxiliary feedwater system. See item 33. |
| 46. | Manual valve 007, normally open | Isolates HV-3019 (item 34) for reparation | All | Inadvertent closure | None; HV-3019 (item 33) incapacitated | None; same as item 33A | Same as item 45. See item 34. |
| 47. | MSIV bypass valves HV-13005A and HV-13005B, normally open (N.O.), failed closed (F.C.), pneumatic valves | Valves are used when MSIV are closed to permit warming the MSS of S/G no. 1 prior to startup and to drain the portion of the MS line located upstream of MSIV during startup and normal operation | A. All but MSLB | A. Fails closed or fails to open upon command | A. Position indicator and alarm on QMCB. If valve closes spuriously, a light is turned on and operator takes necessary action to open valve | A. Inability of the operator to correct a closed MSIV bypass valve during startup can cause water to accumulate in the bypass line. This water can be drained via the drain valve when the bypass valve is open | |
| | | | B. MSLB (Cond. IV event) | B. Fails to close on steam line isolation signal (SLIS) | B. Position indicator on QMCB. | B. None; closure of either MSIV bypass valve will isolate S/G no. 1 and thus prevent blowdown of more than one S/G | Two MSIV bypass valves are provided on each main steam bypass line and operated from separate Class 1E power source; therefore, failure of one train including loss of SLIS will not impair isolation capability |

TABLE 10.3.3-1 (SHEET 12 OF 13)

| Item No. | Description of Component | Safety Function | Plant Operating Mode | Failure Mode(s) | Method of Failure Detection | Failure Effect on System Safety Function Capability | General Remarks |
|---|---|---|---|---|---|---|---|
| 48. | MSIV bypass valves HV-13007A and HV-13007B | Same as item no. 47 except for S/G no. 2 | Same as 47A and 47B | Same as items 47A and 47B | Same as items 47A and 47B | Same as 47A and 47B except for S/G no. 2 | Same as 47B |
| 49. | MSIV bypass valves HV-13008A and HV-13008B | Same as item no. 47 except for S/G no. 3 | Same as 47A and 47B | Same as items 47A and 47B | Same as items 47A and 47B | Same as 47A and 47B except for S/G no. 3 | Same as 47B |
| 50. | MSIV bypass valves HV-13006A and HV-13006B | Same as item no. 47 except for S/G no. 4 | Same as 47A and 47B | Same as items 47A and 47B | Same as items 47A and 47B | Same as 47A and 47B except for S/G no. 4 | Same as 47B |
| 51. | Check valve no. 404 in steam supply to TDP turbine from steam generator no. 1 | Prevents cross flow between main steam lines to prevent blowdown of S/G no. 2 in event of a break in the steam line branch of AFW turbine from S/G no. 1 within the auxiliary FW pump building | All | A. Fails open coincident with branch line break within AFW pump building | A. Gradual decrease of water level in S/G no. 2 | A. None; HV-3009 can be manually closed to prevent blowdown of S/G no. 2 while ensuring a steam supply to the TDP turbine | See item no. 33 |
| | | | | B. Fails closed | B. None | B. None; same effect as item 33A | |
| | | | | C. Fails open coincident with MSLB from S/G no. 1 | C. None | C. None; HV-3009 can be manually closed in the event that level in S/G no. 2 is decreasing | |

TABLE 10.3.3-1 (SHEET 13 OF 13)

| Item No. | Description of Component | Safety Function | Plant Operating Mode | Failure Mode(s) | Method of Failure Detection | Failure Effect on System Safety Function Capability | General Remarks |
|---|---|---|---|---|---|---|---|
| 52. | Main steam isolation valve (MSIV) hydraulic reservoir desiccant type filler/ breather cap (Paul Munroe P/N 93334). This desiccant type filler/ breather cap screws into an adapter which is mounted on the MSIV's hydraulic reservoir cover plate. | None; the desiccant type filler/ breather cap retains moisture and other particles as the MSIVs stroke open and closed. The filler/ breather cap serves as an air exhaust path. | 1, 2, & 3 | A. The filler/ breather's polypropelene, PVC, and desiccant components will degrade during accident conditions, when maximum temperature and radiation levels are reached. During a seismic event, the breather may fall off or its standpipe may bend. | A. Periodic surveillance checks as well as post-accident walkdowns will ensure failure detection. | A. None; the filler/breather is assembled with a stainless steel standpipe such that degraded material will not enter the hydraulic reservoir. A bent standpipe during a seismic event will cause the reservoir's displaced air to exhaust through the hydraulic reservoir cover nonpressure tight gasket. | The desiccant type filler/ breather cap is classified as nonsafety related, nonseismic Category II. |

TABLE 10.3.5-1 (SHEET 1 of 4)

CONFORMANCE TO BRANCH TECHNICAL POSITION MTEB 5-3, MONITORING OF SECONDARY SIDE WATER CHEMISTRY IN PWR STEAM GENERATORS

| Branch Technical Position | |
|---|---|
| 1. Crevices between the tubing and the tube sheets or tubing supports should be minimized to prevent concentration of impurities or solids in these areas. To achieve this goal the tubes at the tube/tube sheet interface should be expanded for the full depth of the tube sheet.<br><br>To minimize the deposition of corrosion products and sludge between the tubes and the supporting structure, the tube/tube support interface should be designed to promote high velocity water flow at the interface. This would improve the washing of this area. | 1. Conform. Refer to 5.4.2.4. |
| 2. Regulatory Guide 1.37 endorses ANSI N45.2.1 and states in part, "The surface (of components) shall appear metal clear. Scattered areas of rust are permissible provided the aggregate area of rust does not exceed two square inches in any one square foot area." Experimental work has shown that a porous packing of oxide in the tube support annulus is one of the conditions resulting in the concentration of contaminants, which leads to runaway corrosion of the tube support plate. Nuclear plant operators should start up the steam generators with metal clean surfaces. A method of confirmation such as photographing the inside of the steam generator should be undertaken after hot functional testing to confirm the metal clean condition. | 2. Conform. Refer to section 1.9 and paragraph 5.4.2.4. (Note: Regulatory Guide 1.37 has been replaced by NQA-1-1994 as described in the SNC Quality Assurance Topical Report (QATR)). |
| 3. A. In the FSAR, the applicant should describe implementation of a secondary water chemistry and monitoring program (in accordance with reference nuclear steam system supplier's recommended procedure) to inhibit steam generator corrosion and tube degradation. This program should cover the following operational modes: power operation (normal), startup, hot standby, hot shutdown, and cold shutdown/ cold wet layup<br><br>Each of the above modes should be defined with regard to percent rated thermal power and approximate temperature range, °F. | 3.A. Conform. Refer to paragraphs 10.3.5.5 and 10.3.5.6. Table 1.1 in chapter 16 lists the percent rated thermal power and temperatures for the appropriate operational modes. |
| B. The secondary water chemistry monitoring and control program should include the following: | 3.B. Conform, except additives are injected into the condensate providing protection for the condensate/feedwater system. Concentrations in each steam generator can be controlled to a certain extent with blowdown. |

TABLE 10.3.5-1 (SHEET 2 of 4)

| | | | |
|---|---|---|---|
| | Identification of a sampling schedule for critical parameters during each mode of operation and of acceptance control criteria for these parameters. The program should include as a minimum the control of pH, cation conductivity, free sodium, and dissolved oxygen. However, other parameters such as specific conductivity, chlorine, fluorine, suspended solids, silica, total iron, copper, ammonia, and residual hydrazine merit consideration. In plants having more than one steam generator, additives to each steam generator should be controlled separately. | | |
| C. | (1) The Nuclear Regulatory Commission will review the secondary water chemistry control and monitoring program of each individual plant. The applicant should incorporate the technical recommendations of the steam generator supplier. Any significant deviation from the supplier's recommendations should be noted and justified technically. | 3.C.(1) | Conform. |
| | (2) Records should be made of the monitored item values, and in accordance with 10 CFR 50.71(a), they shall be made available for audit and inspection when deemed necessary. | 3.C.(2) | Conform. |
| | (3) Each licensee as part of his annual operating report should include an evaluation of the secondary side water chemistry program with an evaluation of the trends and a summary of the total time during the reporting period the various chemistry parameters were out of specification. | 3.C.(3) | Do not conform. Refer to section 1.9.16. |
| D. | For plants utilizing volatile chemistry | | |
| | (1) The composition, quantities, and addition rates of additives should be recorded. Routine changes in these items should be reported under the biannual FSAR update as required by 10 CFR 50.71. However, nonconservative changes, i.e., relaxation in sample frequency or changes in impurity limits, shall be submitted to the NRC for approval before the change is implemented | 3.D.(1) | Control of additives (types, amounts, impurities, sampling frequencies, etc.) is achieved in accordance with the VEGP procedure system described in section 13.5. |
| | (2) The electrical conductivity and the pH of the bulk steam generator water and feedwater should be measured continuously. Assurance should be provided that the sample taken at the blowdown is typical of the bulk steam generator water and that there is a minimum bypass between the feedwater inlet and the blowdown sampling point. | 3.D.(2) | Conform. Refer to subsection 9.3.2. |
| | (3) For once-through steam generators, the pH and electrical conductivity at the coolant inlet should be measured continuously. | 3.D.(3) | Not applicable to VEGP. |

TABLE 10.3.5-1 (SHEET 3 of 4)

| | | |
|---|---|---|
| (4) Free hydroxide concentration and impurities (particularly chloride, ammonia, and silica) in the steam generator water should be measured at least three times per week. | 3.D.(4) | Ammonia is not an impurity in the secondary side system as the alternate amine added will degrade to ammonium and hydroxide. Therefore, we do not plan to monitor free hydroxide in the steam generator water. Other parameters such as pH, pH/conductivity relationships, and sodium levels will be used to minimize corrosion. Chloride will be measured at least three times per week. Silica will be analyzed as a diagnostic parameter on an as-needed basis for troubleshooting |
| E. For plants utilizing phosphate treatment. | 3.E. | Not applicable to VEGP. |
| F. For all PWR plants: | | |
| (1) Condenser cooling water inleakage to the condensate has been identified as the major source of impurity ingress in the PWR secondary feedwater. The combination of impurity ingress with corrosion of copper containing alloys and corrosion product transport ($Fe_30_4$, $Ni0_2$, etc.) in the secondary water system produces a sludge that is difficult to remove and is reactive to steam generator materials. | | |
| In reporting the program the following guidelines should be observed: | | |
| a. Monitor the condensate water quality at the condensate pump discharge as a minimum. Supplement as necessary by samples from the condenser hot well and condenser discharge. | 3.F.(1)a. | Conform. Refer to subsection 9.3.2. |
| b. Measure the cation conductivity and oxygen. | 3.F.(1)b. | Conform. Refer to subsection 9.3.2. |
| c. Maintain condensate impurity level at 0.1 ppm $\pm$ 0.05 ppm, oxygen at $\leq$ 5 ppb. | 3.F.(1)c. | Condensate impurity level and oxygen are monitored and their control limits are outlined in the secondary water chemistry and monitoring program. |
| d. A cation conductivity increase of 0.05 to 0.10 ˜mh o/cm justifies online investigation of possible contamination | 3.F.(1)d. | An increase in control limit of cation conductivity which could initiate investigation of possible contamination is outlined in the secondary water chemistry and monitoring program. |

TABLE 10.3.5-1 (SHEET 4 of 4)

| | | |
|---|---|---|
| e. | An increase of 0.10 to 0.20 μmho/cm is considered an indication of condenser leakage. | 3.F.(1)e. An increase of 0.10 to 0.20 μmho/cm in cation conductivity will initiate investigation of possible condenser leakage.<br><br>The investigation procedure is a part of the secondary water chemistry and monitoring program. |
| f. | When a condenser leak is confirmed, the leak should be repaired or plugged within 96 h or before the total integrated conductivity increase reaches 20 μmho/cm h. The staff will consider other impurity-time limit proposals for limiting the quantity of impurities entering the steam generator. | 3.F.(1)f. Condenser leak identification and repair are outlined in the secondary water chemistry and monitoring program. |
| (2) | Identify the procedures used to measure the value of each of the critical parameters. Provide the procedure title, the applicant/licensee's procedure number, and the basis (i.e., ASTM No.). | 3.F.(2) Conform. |
| (3) | Identify sampling points. The program should consider sampling the steam generator blowdown, the hot well discharge, the feedwater, and the demineralizer effluent as a minimum of sampling points | 3.F.(3) Conform. Refer to subsection 9.3.2 |
| (4) | State the procedure for recording and management of data. | 3.F.(4) Conform. |
| (5) | State the procedures defining corrective action for various out-of-specification parameters. The procedures should define the allowable time for correction of out-of-specification chemistry. | 3.F.(5) Conform. |
| (6) | Identify the authority responsible for the interpretation of the data and the sequence and timing of administrative events required to initiate corrective action. | 3.F(6) Conform. |
| (7) | Identify major components of the secondary water system and materials in contact with secondary water coolant. | 3.F.(7) Conform. See section 10.3.6. |

## 10.4 OTHER FEATURES OF STEAM AND POWER CONVERSION SYSTEM

This section provides discussions of each of the principal design features of the steam and power conversion system.

### 10.4.1 MAIN CONDENSERS

The main condenser functions as the steam cycle heat sink, receiving and condensing exhaust steam from the main turbine, steam generator feed pump turbines, and the turbine bypass system.

#### 10.4.1.1 Design Bases

##### 10.4.1.1.1 Safety Design Bases

The main condenser serves no safety function and has no safety design bases.

##### 10.4.1.1.2 Power Generation Design Bases

A. The main condenser is designed to receive and condense the full-load main steamflow exhausted from the main turbine and the feedwater pump turbines.

B. The main condenser is designed to serve as a collection point for vents and drains from various components of the steam cycle system.

C. The main condenser is designed to receive and condense steam bypass flows up to 40 percent of guaranteed maximum steamflow at the same time the remaining flow is through the turbine without exceeding 5-in. Hg absolute pressure at the turbine exhaust flange.

D. The main condenser hotwells shall be sized to provide surge storage for the condensate system equal to that required for 5 min of operations at maximum guaranteed load.

E. The main condenser is designed to deaerate the condensate so that dissolved oxygen content of the condensate will not exceed 7 ppb over the entire load range.

#### 10.4.1.2 System Description

The main condenser is a three-shell, two-pass, divided water box, floor-supported unit. Each shell is located beneath its respective low-pressure turbine. The condenser is equipped with 1-in.-diameter titanium tubes. The outermost two rows of tubes in the entering steam sections are 18 Birmingham wire gauge (BWG), while the remainder are 22 BWG. The titanium material has good corrosion resisting properties. Design parameters of the condenser are listed in table 10.4.1-1.

Exhaust steam from the steam generator feedwater pump turbines is used to reheat the condensate in the condenser. Each hotwell is divided longitudinally by a vertical partition plate. The condensate pumps take suction from these hotwells.

The condenser shells are located below the turbine building operating floor and are supported above the turbine building foundation. Expansion joints are provided between each low-pressure turbine exhaust opening and the steam inlet connections of the condenser shell. The hotwells of the three shells are interconnected by steam-equalizing lines. Three low-pressure feedwater heaters are located in the steam dome of each shell. Piping is installed for hotwell level control and condensate sampling. The condensate system is shown in drawings 1X4DB168-1, 1X4DB168-2, and 1X4DB168-3.

#### 10.4.1.2.1 System Operation

During normal power operation, exhaust steam from the low-pressure turbines is directed into the main condenser shells. The condenser also receives auxiliary system flows such as feedwater heater vents and drains, feedwater pump turbine exhaust, and gland sealing steam spillover and drains.

The hotwell level controllers provide automatic makeup or rejection of condensate to maintain a normal level in the condenser hotwells. On low level, the makeup control valves open and admit condensate, by gravity flow, to the hotwell from the condensate storage tank. On high water level in a hotwell, the condensate reject control valves open to divert water from the condensate pump discharge to the condensate storage tank. This rejection automatically stops when the hotwell level falls to within normal operating range. Rejection to the storage tank can be manually overridden upon an indication of high hotwell conductivity to prevent transfer of contaminants into the condensate storage tank in the event of a condenser tube failure.

The main condenser, with the assistance of main steam at low loads, deaerates the condensate so that dissolved condensate oxygen does not exceed 7 ppb over the entire load range. Air leakages are estimated to be no greater than 60 $sf^3$/min, in accordance with Table ST-4 of the Heat Exchange Institute Standards for Steam Surface Condensers. Both the air inleakage and the noncondensable gases contained in the turbine exhaust steam are collected in the condenser and removed by the main condenser evacuation system. The main condenser evacuation system is discussed in subsection 10.4.2.

To protect the condenser shells and turbine exhaust hoods from overpressurization, steam relief blowout diaphragms are provided in the main low-pressure turbine exhaust hoods and the main feedwater pump steam turbine exhausts. The diaphragms for the main feedwater pump turbines are designed to rupture at 15 psig, and the main low-pressure turbine diaphragms are designed to rupture at 5 psig.

Upon activation of the turbine bypass system, the main condenser receives up to 40 percent of full-load main steamflow from that system. The main condenser is designed to accept this without either increasing the condenser backpressure to the turbine trip setpoint or exceeding the allowable exhaust temperature during normal operational load swings. If the main condenser, because of tube failure, cooling water failure, air leakage, or other reasons, is unavailable to receive this flow, the steam bypass system controls automatically block the path to the condenser and the steam is discharged to the atmosphere through the main steam relief or safety valves.

Perforated distribution headers and internal baffles are incorporated to protect the condenser tubes and components from turbine bypass or high-temperature drains into the condenser shell.

Leakage at the connections of the tubes to the tube sheets, should it occur, can be detected at either end of each tube bundle by the collection troughs and conductivity cells (24 monitoring points are provided for the 3 shells). These conductivity measurements are indicated and alarmed in the control room. This information permits determination of which tube bundle has sustained the leakage. Steps may be taken to repair or plug the leaking tubes. Leakage occurring in tube locations other than at the tube ends is detected and alarmed in the control room by monitoring the condensate leaving each hotwell (six monitoring points altogether). Detection, isolation, and repair are performed as above. Refer to table 10.3.5-1 Branch Technical Position F.(1) for permissible cooling water inleakage and time of operation which ensures that the condensate/feedwater quality can be maintained.

A condenser tube cleaning system is provided for each shell to perform mechanical cleaning of the circulating water side of the titanium tubes. This removes fouling in order to maintain optimum performance of the condenser.

During normal operation and shutdown the main condenser has a negligible inventory of radioactive contaminants. Radioactive contaminants may enter through a steam generator tube leak. A discussion of the radiological aspects of primary-to-secondary leakage, including anticipated operating concentrations of radioactive contaminants, is included in chapter 11. No hydrogen buildup in the main condenser is anticipated.

The failure of the main condenser and the resultant flooding will not preclude operation of any essential system since no essential safety-related equipment is located in the turbine building (however, see paragraph 7.2.1.1.2.F for trip instrumentation in the turbine building), and the water cannot reach safety-related equipment located in other Category 1 plant structures. Refer to subsection 10.4.5.

### 10.4.1.3 Safety Evaluation

Since there are no safety design bases for the main condenser, no safety evaluation is provided.

### 10.4.1.4 Tests and Inspections

The main condenser water boxes are hydrostatically tested after erection. Condenser shells are tested by completely filling them with water and then testing by the fluorescent tracer method, in accordance with the American Society of Mechanical Engineers Power Test Code 19.11. All tube joints are leak tested during construction and, again, prior to startup. The main condenser is not subject to inservice inspection testing.

### 10.4.1.5 Instrumentation Applications

The main condenser hotwell is equipped with level control devices for automatic makeup or rejection of condensate. Remote indicators are provided for monitoring pressures and water levels in the condenser shells. Local indicators are also provided for monitoring the water levels in the condenser shells. High and low hotwell level alarms are provided in the control room.

A sensor is provided to monitor condenser backpressure (Unit 2). Three pressure sensors are provided on each condenser zone to monitor condenser backpressure (Unit 1). An alarm for high pressure, set at 5 in. Hg absolute, and a turbine trip for high-high pressure, set at 7.5 in. Hg absolute, are provided.

Temperature indicators for monitoring condenser performance are provided.

## 10.4.2 MAIN CONDENSER EVACUATION SYSTEM

Main condenser evacuation is performed by the condenser air ejection system (CAES). The CAES removes noncondensable gases and air from the main condenser during plant startup, cooldown, and normal operation.

### 10.4.2.1 Design Bases

#### 10.4.2.1.1 Safety Design Bases

The CAES serves no safety function and has no safety design bases.

#### 10.4.2.1.2 Power Generation Design Bases

A. The CAES is designed to remove air and noncondensable gases from the condenser during plant startup, cooldown, and normal operation from the steam side of the three main condenser shells and to exhaust them to the atmosphere.

B. The CAES establishes and maintains a vacuum in the condenser during startup by the use of mechanical vacuum pumps. Steam jet air ejectors (SJAEs) are provided to remove noncondensables and hold vacuum during normal operation.

C. The CAES is designed to satisfy all applicable interface requirements and the design recommendations of both the turbine-generator and main condenser suppliers.

### 10.4.2.2 System Description

#### 10.4.2.2.1 General Description

The CAES, as shown in drawing 1X4DB182, consists of two mechanical vacuum pumps and two two-stage SJAEs, which remove air and noncondensable gases from the three condenser shells during normal operation and provide condenser hogging during startup. The noncondensable gases, together with a quantity of vapor, are drawn from the condenser shell, through the air cooler section, to the suction of the air removal equipment. These noncondensables consist mainly of air, nitrogen, and ammonia. Since no hydrogen buildup is anticipated in the main condenser (as discussed in paragraph 10.4.1.2.1), likewise, no hydrogen buildup is anticipated in the CAES. Dissolved oxygen will be present in the condensate and condenser hotwell inventory. Only trace amounts of this oxygen will be released in the condenser, and the amounts are considered negligible compared to the large amounts of air being evacuated by the CAES. Therefore, the potential for explosive mixtures within the CAES does not exist. The condensate and feedwater system provide the cooling medium for the SJAE condensers, while the turbine plant cooling water system provides the cooling for the vacuum pump seal water heat exchangers. The vacuum pump seal water cooler uses turbine

plant cooling water so that the seal water is kept cooler than the saturation temperature of the condenser at its operating pressure.

The noncondensable gases and vapor mixture discharged to the atmosphere from the system are not normally radioactive. However, it is possible for the mixture to become contaminated in the event of primary-to-secondary system leakage. Air inleakage and noncondensable gases are removed from the condenser during startup and normal operations by the mechanical vacuum pumps and steam jet air ejectors, respectively. The normal discharge flow path is directed to the unit vent through the charcoal adsorption train bypass. The discharge flow is monitored for radioactivity prior to discharge to the unit vent. Upon detection of radioactivity in the process stream, discharge flow will be automatically diverted through the charcoal adsorption train prior to discharge. (See subsection 9.4.4.) A discussion of the radiological aspects of primary-to-secondary leakage, including anticipated release from the system, is included in chapter 11.

As long as the CAES is functional, its operation does not affect the reactor coolant system. Should the air removal system fail completely, a gradual reduction in condenser vacuum would result from the buildup of noncondensable gases and air. This reduction in vacuum would cause a decrease in the turbine cycle efficiency necessitating an increase in reactor power to maintain a constant electrical power output. The reactor control system and the reactor protection system guarantee that reactor power is limited and maintained within safe operating limits. If the CAES remains inoperable, condenser vacuum decreases to the turbine trip setpoint and a turbine trip is initiated. A loss of condenser vacuum incident is discussed in subsection 15.2.5.

Loss of the main condenser vacuum causes a turbine trip but does not trip the main steam isolation valves. The causes of low condenser vacuum include air inleakage, poor heat transfer, and faulty air removal. Air inleakage is normally handled by the ejection system as described above. If significant air inleakage is suspected, the pump and valve seals vacuum gauge connections are checked for leaktightness. Faulty air removal by the air ejectors is checked by ensuring proper steamflow to the ejectors. In the unlikely event that air inleakage exceeds the removal capacity of the ejectors, the mechanical vacuum pumps, normally used only during startup, could be used to augment the air removal capability.

Poor heat transfer is caused by dirty tubes, air blanketing, inadequate circulating water, and high inlet circulating water temperature. As noted above, the interior of the tubes is kept clean by a tube cleaning system so dirty tubes are not likely to occur. Air blanketing is caused by insufficient venting of the circulating water system prior to operation. Normal startup procedures will eliminate this problem. The circulating water system is sized to provide adequate flow.

High inlet circulating water temperature may be caused by various atmospheric conditions. Under such conditions, power would have to be curtailed.

#### 10.4.2.2.2 Component Description

A. Mechanical Vacuum Pumps

The mechanical vacuum pump rated capacity is 25 $sf^3$/min each at 1.0 in. Hg absolute with 7.5°F each subcooling. Each pump is driven with a 150-hp motor. For startup hogging service, each vacuum pump capability is 800 $sf^3$/min at 10 in. Hg absolute.

B. Vacuum Pump Seal Water Coolers

The vacuum pump seal water coolers are shell and tube heat exchangers. Seal

water flows through the shell side of the coolers, and turbine plant cooling water flows through the tube side.

C. Steam Jet Air Ejectors

Two two-stage SJAEs are provided with each two-stage unit. They are capable of 60 $sf^3$/min at 1.0 in. Hg absolute with 7.5°F subcooling. The steam consumption for operating each two-stage SJAE is 4100 lb/h.

D. Steam Jet Air Ejector Condensers

One SJAE condenser, consisting of an inter- and after-condenser, services each two-stage SJAE. The steam condensed in the shell sides of the inter- and after-condensers is routed either to the main condenser hotwells or floor drain system, depending on chemistry requirements. The condensate from the condensate and feedwater system is pumped through the tube side of each SJAE inter- and after-condenser.

Piping and valves are carbon steel. All piping is designed to American National Standards Institute (ANSI) B31.1. In conformance with Regulatory Guide 1.26, the quality group and associated quality standards for the CAES are tabulated in table 3.2.2-1. The design parameters of the system are provided in table 10.4.2-1.

#### 10.4.2.2.3 System Operation

During startup operation, air is rapidly removed from the condenser by operating the two condenser mechanical vacuum pumps.

During normal plant operation, noncondensable gases are removed from the condenser by the operation of one of the two two-stage SJAEs. Should the SJAE trip during normal holding operation, automatic switchover is provided to the vacuum pumps.

### 10.4.2.3 Safety Evaluation

The condenser air ejection system has no safety-related function.

### 10.4.2.4 Tests and Inspections

Construction quality assurance was performed by Ecolaire Heat Transfer Company quality assurance personnel. This was supplemented by an extensive surveillance program by SCS/Bechtel inspectors. Welders and weld procedures were qualified in accordance with ASME B&PV, Section IX. Tensile testing, magnaflux testing, hydro tests, and material certificates of compliance were provided by Ecolaire. Inspection prior to shipment of components was required and performed. In addition, inspections were also performed by GPC quality control personnel upon receipt of material and components. During the erection/installation of CAES components, quality assurance was performed by qualified GPC personnel.

Testing and inspection of the system is performed prior to plant operation.

Components of the system are continuously monitored during operation to ensure satisfactory operation. Periodic inservice inspections of the evacuation system are performed in conjunction with the scheduled maintenance outages.

#### 10.4.2.5 Instrumentation Applications

Local indicating devices (e.g., pressure, temperature, and flow indicators) are provided as required for monitoring the system operation. Pressure switches are provided for automatic operation of the standby mechanical vacuum pump during normal operation.

Volumetric flow indication is provided locally to monitor the quantity of exhausted noncondensable gases.

A radiation detector is provided in the turbine building heating, ventilation, and air-conditioning system to monitor the discharge of the condenser mechanical vacuum pumps. The radiation detector is indicated and alarmed in the control room.

## 10.4.3 TURBINE STEAM SEALING SYSTEM

### 10.4.3.1 Design Bases

#### 10.4.3.1.1 Safety Design Bases

The turbine steam sealing system serves no safety function and has no safety design bases.

#### 10.4.3.1.2 Power Generation Design Bases

A. The turbine steam sealing system is designed to prevent air leakage into and steam leakage out of the casings of the turbine-generator and the steam generator feedwater pump turbines.

B. The turbine steam sealing system returns condensed steam to the condenser and exhausts noncondensable gases to the atmosphere.

C. The turbine steam sealing system is designed to detect the potential release of radioactive materials to the environment.

### 10.4.3.2 System Description

#### 10.4.3.2.1 General Description

The turbine steam sealing system is shown in drawing 1X4DB160-3 and includes the following items and assemblies.

- Steam supply header.

- Steam exhaust header.
- Steam control panel.
- Steam packing exhauster gauge panel.
- Two steam packing exhauster blowers, motor driven.
- Associated piping, valves, and controls.

In conformance with Regulatory Guide 1.26, the quality group and associated quality standards for the turbine steam sealing system are tabulated in table 3.2.2-1.

#### 10.4.3.2.2 System Operation

The annular space through which the turbine shaft penetrates the casing is sealed by steam supplied to shaft packings. Where the packing seals against positive pressure, the sealing steam connection acts as a leakoff. Where the packing seals against vacuum, the sealing steam either is drawn into the casing or leaks outward to a vent annulus that is maintained at a slight vacuum. The vent annulus also receives air leakage from the outside. The air-steam mixture is drawn to the steam packing exhauster.

Sealing steam is distributed to the turbine shaft seals through the steam-seal header. Steamflow to the header is controlled by the steam-seal feed valves which respond to maintain steam-seal header pressure. In case of low steam-seal header pressure, a pressure regulator signal opens the feed valves to admit steam from the main steam piping upstream of the turbine stop valves, from the auxiliary steam headers, or from ninth stage turbine extraction on Unit 2. On Unit 1, the DEHC provides the control signals for the steam seal feed valves. In case of high pressure, the steam packing unloading valve automatically opens to bypass excess steam directly to the main condenser.

During the startup phase of turbine-generator operation or at low turbine loads, steam is supplied to the turbine steam sealing system from the main steam piping or auxiliary steam header. During low-load operation, turbine-generator sealing steam is supplied from the main steam system through the steam-seal feed valve to maintain the necessary steamflow to the steam-seal header. As the turbine-generator load is increased, steam leakage from turbine high-pressure packings increases and enters the steam-seal header. When this leakage is sufficient to maintain steam-seal header pressure, sealing steam to all turbine seals, including the low-pressure turbine casings and the main feedwater pump turbine, is supplied entirely from these high-pressure packings. At full load, more steam leaks from the high-pressure packings than is required by vacuum packings, and excess steam is discharged directly to the main condenser. Steam leakoff from the turbine stop valves feeds into the high-pressure turbine exhaust.

The outer ends of all glands are provided with collection piping which routes the mixture of air and excess seal steam to the steam packing exhauster. The steam packing exhauster is a shell and tube heat exchanger; the steam-air mixture passes into the shell side, and service water flows through the tube side. The steam packing exhauster is maintained at a slight vacuum by a motor-operated blower, which discharges to the turbine building roof. There are two blowers mounted in parallel providing 100-percent redundancy. Condensate from the steam-air mixture drains to the main condensers, while noncondensables are exhausted to the atmosphere.

The mixture of noncondensable gases discharged from the steam packing exhauster blower is not normally radioactive; however, in the event of significant primary-to-secondary system leakage due to a steam generator tube leak, it is possible for the mixture discharged to be radioactively contaminated. The steam packing exhauster blower discharge bypasses an air filtration unit and is then headered with the condenser vacuum exhaust discharge. The headered discharge line contains a radiation monitor which diverts each discharge through its own filtration unit if radioactivity is detected in the headered discharge line. The filtered air then exhausts to the atmosphere. (See subsection 9.4.4.) A full discussion of the radiological aspects of primary-to-secondary system leakage is included in chapter 11.

Failure of the turbine steam seal system will result in no leakage of radioactivity to the atmosphere. A failure of this system would, however, result in a loss of condenser vacuum.

#### 10.4.3.3 Safety Evaluation

The turbine steam sealing system has no safety-related function.

#### 10.4.3.4 Tests and Inspections

Quality assurance for the design and construction of the turbine gland sealing system was provided utilizing the standard GE large steam turbine-generator quality assurance program (GEZ-4982A).

Construction phase quality assurance also included an extensive surveillance program by SCS/Bechtel inspectors. Weld procedures were reviewed by SCS stress personnel. Inspections are performed by GPC quality control personnel upon receipt of material and components and upon installation.

The system is tested, in accordance with written procedures, during the initial testing and operation program. Since the turbine steam sealing system is in constant use during normal plant operation, the satisfactory operation of the system components will be evident.

#### 10.4.3.5 Instrumentation Applications

Automatic pressure control is provided by the digital electro-hydraulic control system (DEHC) to maintain steam seal header pressure by providing signals to the steam seal feed valves (Unit 1). A pressure controller is provided to maintain steam-seal header pressure by providing signals to the steam-seal feed valves (Unit 2).

Local and remote indicators, as well as alarm devices, are provided for monitoring the operation of the system.

### 10.4.4 TURBINE BYPASS SYSTEM

The turbine bypass system has the capability to bypass main steam from the steam generators to the main condenser in a controlled manner to minimize transient effects on the reactor coolant system of startup, hot shutdown and cooldown, and the step-load reductions in generator load. The turbine bypass system is also called the steam dump system.

### 10.4.4.1 Design Bases

#### 10.4.4.1.1 Safety Design Bases

The turbine bypass system serves no safety function and has no safety design basis.

#### 10.4.4.1.2 Power Generation Design Bases

A. The turbine bypass system has the capacity to bypass 40 percent of the valves wide open (VWO) main steamflow to the main condenser.

B. The turbine bypass system is designed to bypass steam to the main condenser during plant startup and to permit a normal manual cooldown of the reactor coolant system from a hot shutdown condition to a point consistent with the initiation of residual heat removal system operation.

C. The turbine bypass system will permit a 50-percent electrical step-load reduction without reactor trip. The system under some upset conditions will also allow a turbine and reactor trip from full power without lifting the main steam relief and safety valves.

### 10.4.4.2 System Description

#### 10.4.4.2.1 General Description

The turbine bypass system is shown on drawing 1X4DB160-1. The system consists of a manifold connected to the main steam lines upstream of the turbine stop valves and of lines from the manifold with regulating valves to each condenser shell. The system is designed to directly bypass 40 percent of the VWO main steamflow to the condenser.

The capacity of the system, combined with the capacity of the reactor coolant system to accept a 10-percent step-load change, provides the capability to shed 50 percent of the turbine-generator rated load without reactor trip and without the operation of relief and safety valves. A load rejection in excess of 50 percent under certain transient conditions may result in a reactor trip and may result in the operation of the main steam relief and safety valves. The operation of the main steam power-operated relief valves and spring-loaded safety valves prevents overpressurization of the main steam system.

There are 12 turbine bypass valves. Four valves discharge into each condenser shell.

The steam bypassed to the main condenser is normally not radioactive. In the event of primary-to-secondary leakage, it is possible for the bypassed steam to become radioactively contaminated. A full discussion of the radiological aspects of primary-to-secondary leakage is contained in chapters 11 and 12.

#### 10.4.4.2.2 Component Description

The turbine bypass system contains 12 air-actuated globe valves. The valves are pilot operated, spring opposed, and fail closed upon loss of air or loss of power to the control system. Sparger piping distributes the steam within the condenser. Isolation valves permit maintenance of the bypass valves while the plant is in operation.

#### 10.4.4.2.3 System Operation

During normal operating transients, the turbine bypass system is automatically regulated by the reactor coolant temperature control system to maintain the programmed coolant temperature. The programmed coolant temperature is derived from the high-pressure turbine first-stage pressure, which is a load reference signal. The difference between programmed reactor coolant average temperature and measured reactor coolant average temperature is used to activate the steam dump system under automatic control. The system operates in two fundamental modes. In one mode, two groups of six valves each trip open sequentially within 3 s. This operational mode is activated during a large reactor-to-turbine power mismatch. In the other mode, four groups of three valves each modulate open sequentially. The valves will modulate with a maximum full-stroke time of 20 s over the steam pressure range of 100 to 1200 psig. A logic diagram is shown in drawing 1X6AA02-234.

When the plant is at no load and there is no turbine load reference, the system is operated in a pressure control mode. The measured main steam system pressure is compared against the pressure set by the operator in the control room. The pressure control mode is also used for plant cooldown. When the absolute pressure within the condenser shell reaches approximately 5 in. Hg, the valves are prevented from opening.

The turbine bypass control system can malfunction in either the open or closed mode. The effects of both these potential failure modes on the nuclear steam supply system and turbine system are addressed in chapter 15. If the bypass valves fail-open, additional heat load is placed on the condenser. If this load is great enough, the turbine is tripped on high-high condenser pressure. Ultimate overpressure protection for the condenser is provided by rupture discs. If the bypass valves fail-closed, the atmospheric relief valves permit controlled cool-down of the reactor.

### 10.4.4.3 Safety Evaluation

The turbine bypass system serves no safety function and has no safety design basis. There is no safety-related equipment in the vicinity of the turbine bypass system. All high-energy lines of the turbine bypass system are located in the turbine building.

The failure of a turbine bypass high-energy line will not disable the turbine speed control system. The turbine speed control system is designed in such a manner that its failure will cause a turbine trip. Additional information concerning speed control can be found in paragraphs 10.2.2.3.1.1 and 10.2.2.3.1.5.

### 10.4.4.4 Inspection and Testing Requirements

Before the system is placed in service, all turbine bypass valves are tested for operability. The steam lines are hydrostatically tested to confirm leaktightness. The bypass valves may be tested while the unit is in operation. All system piping and valves are accessible for inspection.

The turbine bypass system falls under American National Standards Institute B31.1 rules, which do not require inservice inspection and testing.

#### 10.4.4.5 Instrumentation Applications

The turbine bypass control system is described in section 7.7. Hand switches in the main control room are provided for selection of the system operating mode. Pressure controllers and valve position lights are also located in the main control room.

### 10.4.5 CIRCULATING WATER SYSTEM

#### 10.4.5.1 Design Bases

##### 10.4.5.1.1 Safety Design Bases

The circulating water system serves no safety function and has no safety design basis.

##### 10.4.5.1.2 Power Generation Design Bases

The circulating water system supplies cooling water to remove heat from the main condensers and from the components of the turbine plant cooling water system under all conditions of power plant loading and design weather conditions.

#### 10.4.5.2 System Description

##### 10.4.5.2.1 General Description

The circulating water system consists of two 50-percent capacity circulating water pumps, one hyperbolic natural draft cooling tower, and associated piping, valves, and instrumentation. The system is shown schematically in drawings 1X4DB150 and 1X4DB150-1. Design parameters for major components are provided in table 10.4.5-1.

##### 10.4.5.2.2 Component Description

A. Circulating Water Pumps

The two 50-percent capacity circulating water pumps are vertical, wet pit, single-stage pumps driven by solid-shaft electric motors. Each pump has a capacity of 242,300 gal/min. The pumps are mounted in an intake structure, which is connected to a cooling tower by a canal. The two-pump discharge lines connect to a common header, which connects to a three- section, six-flow-path condenser. Each pump discharge line has a motor-operated butterfly valve

located between the pump discharge and the main header. This permits isolation of one pump for maintenance and allows single-pump operation.

B. Cooling Tower

The cooling tower is a hyperbolic natural draft structure employing the counterflow principal of heat transfer. The design circulating water flowrate through the tower is 509,600 gal/min. The cooling tower basin has a storage volume of $6.0 \times 10^6$ gal of water. The cooling tower is designed to withstand winds of up to 110 mph and earthquake loads of 0.15 g horizontal and 0.10 g vertical ground acceleration.

C. River Water Makeup Pumps

The circulating water system makeup is accomplished by four 22,000-gal/min river water makeup pumps. These four pumps provide the circulating water makeup to both Units 1 and 2. They are vertical, turbine-type pumps driven by electric motors and are located in the river intake structure.

D. Piping and Valves

Most of the circulating water system piping is constructed of prestressed concrete. The remainder is carbon steel and has an internal coating of corrosion preventive compound. Motor-operated butterfly valves are provided in each of the circulating water lines at their inlet to and exit from the condenser shell, to allow isolation of faulty equipment. Control valves are provided for regulation of cooling tower blowdown.

The circulating water transport system is designed to withstand the maximum operating discharge pressure of the circulating water pumps. Piping is designed for 80 psig as are the expansion joints, butterfly valves, condenser water boxes, and tube bundles.

#### 10.4.5.2.3 System Operation

The two half-capacity circulating water pumps take suction from the circulating water intake structure and circulate the water through the tube side of the main condenser and back to the discharge network in the cooling tower. The natural draft cooling tower cools the circulating water by discharging the water over a network of baffles in the tower. The water then falls to the basin beneath the tower and, in the process, gives up some of its heat to the atmosphere. Provision is made during cold weather to direct all of the circulating waterflow to the periphery of the cooling tower. This directs the total heat load to the peripheral region. Air flowing through the peripheral spray is thus preheated which allows deicing in the central cooling tower spray region.

The flow to the cooling tower can be diverted directly to the basin, bypassing the tower internals. This is achieved by opening the motor-operated bypass valve while operating one of the two circulating pumps. The bypass is normally used only to achieve plant startup in cold weather or to maintain basic temperature above 40°F while operating at partial load during periods of cold weather.

The river water makeup pumps supply water to the circulating water system to replace water losses due to evaporation, wind drift, and blowdown. The makeup water is supplied to the

cooling tower basin. Normally, only one or two of the makeup pumps are operating, depending upon the makeup demand.

During normal plant operation, biocides are added to the circulating water to control biological growth as needed. Chemical treatment is added to the flume as required to maintain a Ryznar Index to prevent scaling and reduce corrosion. A condenser tube cleaning system is installed to clean the circulating water side of the main condenser.

Blowdown from the circulating water system is taken from the cooling tower basin and is dechlorinated, as required, and discharged. Water being discharged into the river meets appropriate regulatory requirements.

The circulating water system is normally used to supply cooling water to the main condenser in order to condense the steam exhausted from the main turbine. If the circulating water pumps, the cooling tower, or the circulating water piping malfunctions such that the flowrate of the circulating water falls below the minimum allowable value, the main condenser will no longer be able to adequately perform its functions. Cooldown of the reactor in such an event would be accomplished by using the power-operated atmospheric steam relief valves rather than the turbine bypass system.

Passage of secondary system condensate from the main condenser into the circulating water system through a condenser tube leak is not possible during power generation operation, since the circulating water system operates at a greater pressure than the condenser.

Small leaks around valves and fittings would drain into the turbine building drain sump via the floor drains. The condenser pit sump is provided with sump pumps and with high-level alarms. Large leaks due to pipe failures would be indicated in the control room by a gradual loss of vacuum in the condenser shell. The effects of turbine building flooding are discussed in appendix 3F.

In the event of collapse of the cooling tower, the potential for damage to other plant structures is slight. The structure would tend to collapse inwardly. To prevent damage, the structure is located a sufficient distance from any equipment or structure important to reactor safety.

### 10.4.5.3 Safety Evaluation

Because the circulating water system has no safety design basis, no safety evaluation is provided.

### 10.4.5.4 Tests and Inspections

All active components of the circulating water system are accessible for inspection during plant power generation. The circulating water pumps and river water makeup pumps are tested in accordance with standards of the Hydraulic Institute. Performance, hydrostatic, and leakage tests associated with preinstallation and preoperational testing are performed on the circulating water system in accordance with the standards of the Hydraulic Institute and the American Water Works Association Code 504-70. The performance along with structural and leaktight integrity of all system components are demonstrated by continuous operation.

#### 10.4.5.5 Instrumentation Applications

Indicating lights are provided in the control room to indicate open and closed positions of motor-operated butterfly valves in the circulating water piping. The motor-operated valve at each pump discharge is interlocked with the pump such that the pump will be shut down if the discharge valve fails to reach the full-open position shortly after starting the pump. The motor-operated valves are provided with position switches required for indicating lights and interlocking with the pumps.

The process sampling system discussed in subsection 9.3.2 periodically tests the circulating water quality to ensure that no harmful effects will result to the system piping and valves due to improper water chemistry.

Local pressure indicators are provided on the circulating water pump discharge lines. A local pressure indicator and a pressure transmitter are provided on one inlet branch to the condenser in the turbine building. The pressure transmitter also provides indication in the control room and provides a signal to the plant computer.

On two inlet branches to the condenser, local and remote indicators are provided for temperature. A signal for inlet temperature is also provided to the plant computer.

On each of the six outlet branches from the condenser, temperature instrumentation is provided for local indication and signal to the plant computer.

In each of the six condenser flow paths, differential pressure transmitters are provided to measure the inlet/outlet differential pressure and to feed this information to the plant computer.

Level instrumentation is provided in the circulating water pump intake structure to transmit level signals to the cooling tower makeup valves. Level instrumentation is also provided to annunciate a low water level in the pump structure and a high water level in the cooling tower basin. These annunciators are located in the main control room. Hand switches for the cooling tower makeup valves are also located in the main control room so that the valves may be aligned manually.

In order to maintain totally dissolved solids below a pre-established level, cooling tower blowdown is controlled as a function of plant load (condensate flow) and circulating water conductivity. Cooling tower blowdown also has a manual mode of operation.

### 10.4.6 CONDENSATE CLEANUP SYSTEM

The condensate cleanup function is performed by the condensate filter demineralizer system. The system maintains the required purity of feedwater for the steam generators by filtration to remove corrosion products and/or ion exchange to remove condenser leakage impurities.

#### 10.4.6.1 Design Bases

##### 10.4.6.1.1 Safety Design Bases

The condensate filter demineralizer system serves no safety function and has no safety design basis.

#### 10.4.6.1.2 Power Generation Design Bases

The condensate filter demineralizer system has the following power generation design bases.

A. Removes dissolved and suspended solids from the condensate prior to startups.

B. Removes impurities entering the secondary cycle from condenser leaks that would otherwise concentrate in the steam generators.

C. Removes corrosion products from the condensate and any drains returned to the hotwell, so as to limit accumulation of those products which are difficult to remove from the steam generator by blowdown.

D. Limits the entry of dissolved solids into the feedwater system in the event of large condenser leaks, such as a tube break, to permit a reasonable amount of time for plant shutdown.

E. Removes impurities from the condensate, which may occur in primary-to-secondary leaks in the steam generators until the leaks can be repaired.

F. Includes a backwash recovery subsystem, which is capable of recovering the backwash water to a quality suitable for return to the condensate system. The spent resins are separated as a pumpable slurry ready for disposal.

G. Includes a spent resin dewatering system. During normal operations, resins are dewatered. If a primary to secondary leak is suspected or has occurred, resins are sampled for radioactivity and either transferred to a high integrity container for radwaste disposal or dewatered. Dewatered resins are sampled for radioactivity and then either sent to low-level radwaste storage or disposed of as nonradioactive waste.

### 10.4.6.2 System Description

#### 10.4.6.2.1 General Description

The condensate cleanup system processes the full flow of condensate and has five condensate filter demineralizer vessels with 25-percent capacity each (one on standby), five holding pumps, piping, instrumentation, and controls. A backwash system cleans the filters. The filters are either operated without resin or are precoated with resins, and a backwash recovery system permits separation of spent demineralizer resins.

The condensate cleanup system filter demineralizer vessels are located at the discharge of the condensate pumps in the turbine building between el 220 and 245 ft. System arrangement is shown in drawings 1X4DB185-1, 1X4DB185-2, 1X4DB185-3 and 1X4DB185-5. Design data for major components are listed in table 10.4.6-1. Codes and standards applicable to the condensate cleanup system are listed in table 3.2.2-1. The system is designed and constructed in accordance with quality Group D specifications. The filter demineralizer system, including the backwash handling system, is designed to sustain power generating operations when condenser leaks are 5 gal/min or less, or primary-to-secondary leaks are 0.1 gal/min or less with a specified failed fuel of 0.25 percent.

Provisions are made in the building design for shielding against potential radioactivity, and space for 18-in. shielding walls around each vessel is allowed. Access is provided to the equipment for inspection and repair.

The spent powdered demineralizer resins, after backwashing from the filter demineralizer vessels, are normally dewatered, sampled for radioactivity, and then either sent to low-level radwaste disposal or disposed of as nonradioactive waste. If a primary to secondary leak is suspected or has occurred, resins are sampled for radioactivity, and either transferred to a high integrity container for radwaste disposal or dewatered.

### 10.4.6.2.2 Component Description

10.4.6.2.2.1 Condensate Filter Demineralizer. The condensate filter demineralizers are the precoatable type with the demineralizer vessels designed to operate at condensate pump discharge pressures and temperatures. Design specific flowrate does not exceed 4 gal/min/$ft^2$ of coated element surface. The demineralizer units are capable of operating with either mixed powdered resins or other approved precoat media as a precoat. In each filter vessel, 420 polypropylene-wound or nylon-wound elements, 70 in. long, are evenly coated with a total of 300 lb of mixed resins. The active surface area is 1281 $ft^2$ and is 1/4 to 3/8 in. thick.

The system is installed with a full-flow bypass which ensures full condensate flow and limits pressure drop across the cleanup system. The bypass can be used when water chemistry can be maintained below normal design limits by the steam generator blowdown system alone and/or during emergency conditions to maintain main feedwater pump suction pressure requirements.

For each of the condensate filter demineralizers, a holding pump is provided to recycle a minimum flow as required to retain the precoat material on the vessel element if the condensate flow should become too low or the vessel is in standby. A flow control system is provided to equalize the flows through the four vessels on line. Each vessel is also equipped with a resin trap to catch resin in the vessel effluent.

After the vessel is exhausted, the spent resin is purged out of the vessel sequentially by backwash water and service air. The backwash pump supplies the water required for backwashing from the condensate storage tank through the condensate makeup line. The backwashing air requirement is taken from the station oil free air supply system.

After all the spent resin is backwashed from the vessel, new resin is coated on the clean vessel elements. New resin is manually charged to the precoat tank, where it will flocculate and slurry with demineralized water by action of an agitator. The slurry is recirculated by a precoat pump through the filter demineralizer unit until the filter elements are completely coated. The freshly coated filter may then be returned to the standby mode.

10.4.6.2.2.2 Backwash Recovery Subsystem. The spent powdered resin or other approved precoat media is flushed to the backwash recovery tank by backwash water.

After an exhausted vessel is taken out of service and before the backwash operation of the spent resin is performed, the backwash water from the previous backwash operation stored in the backwash recovery tank can be pumped by the backwash recovery pump and processed through the exhausted vessel until the backwash water quality is good enough for recycling to the turbine cycle. If necessary, the exhausted vessel is overlaid with fresh resin for cleaning up the backwash water. An overlay tank with an agitator and an overlay pump is provided to overlay the spent vessel.

10.4.6.2.2.3 Spent Resin Dewatering Subsystem. The spent resin dewatering subsystem receives the resin slurry resulting from backwashing the spent powdered resins from the demineralizer vessels. The slurry is stored in spent resin holding tanks located within the turbine building, where it is allowed to clarify. Excess water is then removed by pumping the clarified water through cartridge-type filters to the turbine building drain system. This concentrates the resin slurry. During normal operations, resins are separated and dewatered by processing the concentrated slurry through belt-type pressure filters. The spent resin dewatering subsystem can transfer resin from the spent resin storage tanks to a high integrity container for radwaste disposal. Dewatered resins are sampled for radioactivity and then either sent to low-level radwaste storage or disposed of in the industrial waste landfill. The spent resin dewatering system is common to Units 1 and 2.

### 10.4.6.2.3 System Operation

The demineralizers are capable of operation in continuous or intermittent service. The system is operated to maintain the water quality requirements of the nuclear steam supply system supplier. In addition, the ion concentrations, particularly sodium and chloride, are maintained within the effluent requirements given in paragraph 10.4.6.2.4.4.

Service runs are terminated when the following occur.

A. The pressure drop from inlet to outlet reaches a preselected value between 25 and 35 psi due to buildup of suspended solids.

B. Conductivity measurements indicate that the resin bed has reached ion exchange exhaustion.

C. Sodium, silica, or other ion leakage is judged unacceptable.

The valving and controls are such that a ready standby vessel can be placed in service, or any or all of the operating vessels can be placed out of service from the local control panel.

The following functions are initiated by switch or push button at the local control panel:

A. Placing standby vessel online and exhausted vessel on hold.

B. Initiation of automatic backwash.

C. Initiation of automatic precoat operation.

D. Bypassing vessel system.

The backwash water from the previous backwashing operation can be polished through the exhausted vessel and subsequently sent to the condensate storage tank after the water quality is tested and proved suitable.

The spent resin is backwashed using condensate storage tank water and air. During normal operation, the backwash water and spent resin slurry are pumped to the spent resin dewatering subsystem for processing. After the elements are cleaned, fresh resin is coated on the elements.

Fresh resins are manually charged to the resin precoat tank, where they are slurried with demineralized water by an agitator. Flocculation of the resins occurs as the cation and anion forms agglomerate. The floc size is controlled to 4 to 8 mm. The slurry is recirculated by a slurry pump through the vessel until the filter elements are completely coated. (Leaking elements may be detected by taking filter samples from grab sample connections on each precoat system downstream of the elements.)

This vessel is then ready for service. These backwash recovery and backwash/precoat cycles are manually initiated but are carried out according to an automatic program sequence.

#### 10.4.6.2.4 Water Quality

10.4.6.2.4.1 Influent Water Quality. Table 10.4.6-2 lists the concentrations of corrosion products and other impurities in the influent condensate used as a design basis for two operating conditions, startup or design maximum and expected operation. The concentrations listed for the expected operation are those associated with a condenser leak of 1 gal/min. The concentrations listed for design maximum or startup have been calculated from a condenser leak of 5 gal/min, and the maximum radioactivity level has been set by a steam generator tube leak of 0.1 gal/min.

10.4.6.2.4.2 Primary Coolant. The primary coolant circulating through the steam generator haas the expected analysis shown in table 10.4.6-3.

10.4.6.2.4.3 Circulating Water. The circulating water has the expected analysis after six cycles as shown in table 10.4.6-4.

10.4.6.2.4.4 Effluent Requirements. The condensate filter demineralizer is operated to produce the effluent required to meet the secondary side water chemistry specifications during power operation provided in the EPRI PWR Secondary Water Chemistry Guidelines.

### 10.4.6.3 Safety Evaluation

There is no safety design basis for the condensate cleanup system and no safety evaluation is provided.

### 10.4.6.4 Tests and Inspections

Preoperational testing of the condensate cleanup system ensures proper functioning of the equipment and instrumentation. The system is checked functionally during power generation operation.

### 10.4.6.5 Instrumentation Applications

In-line sensors monitor equipment performance during demineralizer operation or their restoration cycle. Local and control room alarms annunciate trouble in components of the system. Systematic analysis of local samples monitors the accuracy of the automatic equipment. At each filter, demineralizer flow and differential pressure are continually monitored. Instruments of the condensate system continuously monitor the influent and effluent streams of

the filter demineralizer system. Ionic concentrations are monitored using sampling points and methods identified in table 9.3-2-3 (grab sample points and process instrumentation).

## 10.4.7 CONDENSATE AND FEEDWATER SYSTEM

The condensate and feedwater system provides for the condensing of high-pressure and low-pressure turbine extraction and exhaust steam and main feedwater pump turbine exhaust steam, collects the condensate in the condenser hotwell, and maintains steam generator water level by supplying preheated feedwater through all power operation modes of the plant.

### 10.4.7.1 Design Bases

#### 10.4.7.1.1 Safety Design Bases

The safety-related portions of the condensate and feedwater system are protected from wind and tornado effects, as discussed in section 3.3; flood protection is discussed in section 3.4; missile protection is discussed in section 3.5; protection against dynamic effects associated with the postulated rupture of piping is discussed in section 3.6; and environmental design is discussed in section 3.11.

The portion of the feedwater system from the steam generator main and bypass feedwater inlets to the first seismic restraint upstream of the containment feedwater isolation valves are designed in accordance with the requirements of Section III of the American Society of Mechanical Engineers (ASME) Boiler and Pressure Vessel Code for Class 2 components and are designed to Seismic Category 1 requirements. This same portion of the system is also required to function following a design basis accident (DBA) and to achieve and maintain the plant in a safe shutdown condition. The failure of the condensate and feedwater system upstream of the main feedwater isolation valves (MFIVs) and the bypass feedwater isolation valves (BFIVs) would not result in either exceeding containment design pressure or a departure from nucleate boiling ratio (DNBR) below the design basis limit. The system provides isolation valving on main and bypass feedwater lines into containment. The isolation valves close within 5 s after receipt of an isolation signal. In addition, the feedwater pumps are tripped simultaneously with the signal to close the feedwater control valves.

A. The safety-related portion of the condensate and feedwater system is designed to remain functional after a safe shutdown earthquake (SSE) or to perform its intended function following postulated hazards of fire, internal missiles, or pipe break.

B. The safety-related portion of the system is protected from the effects of natural phenomena such as earth- quakes, tornadoes, hurricanes, floods, and external missiles.

C. Safety functions can be performed, assuming a single active component failure coincident with the loss of offsite power.

D. The portion of the feedwater system to be constructed in accordance with ASME III, Class 2 requirements is provided with access to welds and removable insulation for inservice inspection, in accordance with ASME XI. The condensate and feedwater system is designed so that the active components are capable of limited testing during plant operation.

E. The condensate and feedwater system is designed and fabricated to codes consistent with the quality group classification assigned by Regulatory Guide 1.26 and the seismic category assigned by Regulatory Guide 1.29. The power supply and control functions are in accordance with Regulatory Guide 1.32.

F. For a main feedwater or main steam line break (MSLB) inside the containment, the condensate and feedwater system is designed to limit high-energy fluid to the broken loop. The feedwater bypass lines provide a path for the addition of auxiliary feedwater to the three intact loops.

G. For a main feedwater line break upstream of the MFIV (outside of the containment), the condensate and feedwater system is designed to prevent the blowdown of any steam generator. The feedwater bypass lines provide a path for the addition of auxiliary feedwater.

#### 10.4.7.1.2 Power Generation Design Bases

A. The condensate and feedwater system is designed to provide a continuous feedwater supply to the four steam generators at the required pressures and temperatures for steady-state and anticipated transient conditions.

B. The system is designed to maintain feedwater flow following a 50-percent step reduction in electrical load.

C. The system is designed to continue functioning in spite of a loss of one feedwater pump or one heater drain pump.

D. The system is designed to protect the steam generator main feedwater inlet nozzle and inlet feedwater piping during low-load and hot standby conditions.

### 10.4.7.2 System Description

#### 10.4.7.2.1 General Description

The condensate and feedwater system is shown schematically in drawings 1X4DB168-1, 1X4DB168-2, and 1X4DB168-3. The condensate and feedwater system supplies the steam generators with heated feedwater in a closed steam cycle using regenerative feedwater heating.

The main portion of the feedwater flow is deaerated condensate pumped from the main condenser hotwells by the condensate pumps. The main condenser hotwells receive makeup from the condensate tank. (Refer to subsection 9.2.6 for a discussion of the condensate storage system.) This stream passes, in sequence, through the condensate filter demineralizer system (described in subsection 10.4.6); the three trains of low-pressure heaters, each train consisting of a number 1, 2, and 3 low-pressure heater; two trains of low-pressure heaters, each train consisting of a number 4 and 5 low-pressure heater; the parallel steam generator feedwater pumps; the two trains of high-pressure heaters, each train consisting of a number 6 high-pressure heater; regulating and isolation valves; and on into the four steam generators. The balance of the feedwater flow is provided by the drains from the moisture separators, the reheaters, and number 6, 5, and 4 heaters and is collected into a drain tank and pumped into the feedwater pump suction stream by the heater drain pumps.

To allow for feedwater and condensate system flush, cleanup, and startup recirculation, a steam generator flush line is installed upstream of the main feedwater isolation valves. This flush system is designed for approximately 50 percent of the maximum condensate-rated flow and is used in flushing operations. The flush lines may be discharged to the condensers or to the waste water retention basins. Steam may be provided to the number 5 low-pressure heater from the auxiliary steam in order to preheat the feedwater to about 200°F during the initial cleanup and startup recirculation operations. This preheating action, along with added chemicals, prevents formation of iron oxides in the condensate system.

The condensate filter demineralizer system consists of five precoat-type vessels, as described in subsection 10.4.6, that may be in service or completely bypassed.

Each of the four main 16-in. feedwater lines to the four steam generators contains a feedwater flow element, a main feedwater regulating valve, a main feedwater bypass regulating valve, a power-operated MFIV, a tilting disc check valve, and a chemical injection connection.

A 6-in. bypass feedwater line tees off each main 16-in. feedwater line just upstream of the 16-in. MFIV and contains a BFIV, one tilting disc check valve located outside containment, an auxiliary feedwater connection, followed by one tilting disc check valve inside containment before it connects to the steam generator 6-in. auxiliary feedwater (bypass feedwater) nozzle.

The condensate and feedwater chemical injection system, as shown in drawing 1X4DB157, is provided to inject hydrazine and alternate amine into the condensate pump discharge downstream of the condensate demineralizers and additional hydrazine and alternate amine into the four 6-in. bypass feedwater lines connecting with the four steam generators. Injection points are shown in drawing 1X4DB157.

During normal power operation, the continuous addition of hydrazine to the condensate system downstream of the condensate demineralizers is under automatic control, with manual control optional. As discussed in subsection 10.3.5, the addition of alternate amine and hydrazine establishes the design pH according to the condensate and feedwater system chemistry requirements and establishes a constant initial hydrazine residual in the feedwater system, so that oxygen inleakage can be scavenged.

During hot standby conditions, the wet layup chemical feed pumps are available to inject a mixture of hydrazine and alternate amine from a batch chemical feed tank into the steam generators downstream of the connection to the 6-in. bypass feedwater line when the condensate and/or steam generator feedwater pumps are not operating.

Within the condensate and feedwater system component failures which affect the final feedwater temperature or flow have a direct effect on the reactor coolant system. These are listed in table 10.4.7-1. Occurrences which produce an increase in feedwater flow or decrease in feedwater temperature result in increased heat removal from the reactor coolant system (RCS), which is compensated for by control system action, as described in section 7.7. Events which produce the opposite effect (i.e., decreased feedwater flow or increased feedwater temperature) result in reduced heat transfer in the steam generators. Normally, automatic control system action is available to adjust feedwater flow and reactor power, which prevents excess energy accumulation in the RCS, and the increasing reactor coolant temperature provides a negative reactivity feedback which tends to reduce reactor power. In the absence of normal control action, either the high-outlet temperature or the high-pressure trips of the reactor by the reactor protection system are available to ensure reactor safety. Loss of all feedwater, the most severe transient of this type, is examined in chapter 15.

Refer to section 5.4 for a discussion of steam generator design features to preclude fluid flow instabilities, such as water hammer. The main 16-in. and bypass 6-in. feedwater connections on each of the steam generators are the highest point of each feedwater line downstream of the

MFIV and the BFIV. An elbow, with a short transition piece, is connected directly to the steam generator main 16-in. and bypass 6-in. feedwater nozzles, which tend to minimize the portion of feedwater piping that can drain into the steam generator and become filled with steam. The feedwater lines contain no high-point pockets that could trap steam and lead to water hammer. The horizontal pipe length from the main and bypass feedwater nozzles of each steam generator is minimized. This feedwater system configuration coupled with the steam generator design minimizes the potential for steam backleakage and bubble collapse water hammer in the main and auxiliary feedwater lines. To complement the system and steam generator design, Westinghouse recommended feedwater system maintenance and startup procedural guidelines to further preclude feedwater system water hammer conditions in the main and bypass feedwater lines. The routing of the main and bypass feedwater lines is shown in drawings 1K5-1305-062-01 and 1K5-1305-058-01.

#### 10.4.7.2.2 Component Description

Codes and standards applicable to the condensate and feedwater system are listed in table 3.2.2-1. The condensate and feedwater system is designed in accordance with the requirements of Section III of the ASME Boiler and Pressure Vessel Code for Class 2 components and Seismic Category 1 requirements from the steam generator out to the restraint upstream of the MFIVs and BFIVs. The remaining piping of the condensate and feedwater system meets American National Standards Institute (ANSI) B31.1 requirements. Refer to tables 10.1-1 and 10.4.7-2 for design data. Safety-related feedwater piping materials are discussed in subsection 10.3.6.

10.4.7.2.2.1 Feedwater Piping. Approximately 92 percent of the feedwater is supplied to the four main steam generators by four main 16-in. carbon steel lines during normal operation. Each of the lines is anchored at the containment wall and has sufficient flexibility to provide for relative movement of the steam generators resulting from thermal expansion. The main feedwater line and associated branch lines between the containment penetration and the restraint upstream of the MFIV are designed to meet the criteria, as described in Nuclear Regulatory Commission (NRC) Branch Technical Position (BTP) MEB 3-1. (Refer to section 3.6.)

The remaining 8 percent of the feedwater is supplied to the four steam generators by four bypass 6-in. carbon steel lines. Each of the lines is anchored at the containment wall and has sufficient flexibility to provide for relative movement of the steam generators resulting from thermal expansion. The bypass feedwater line and associated branch lines are designed and constructed in accordance with Quality Group B and Seismic Category 1 requirements.

The 6-in. bypass feedwater line is provided to a 6-in. auxiliary feedwater steam generator nozzle to add feedwater during low-load or hot standby conditions when the feedwater flow and temperatures are low. During these periods, no flow is added through the main 16-in. feedwater line and nozzle by closing the MFIVs in order to reduce the potential of main feedline and nozzle cracking.

The feedwater piping system is analyzed to eliminate or minimize the potential for water hammer and subsequent detrimental effects on the following:

- Steam generators with top feed ring design (BTP ASB 10-2).
- Main and bypass feedwater check valves due to line breaks (BTP MEB 3-1).

- Spurious isolation or feedwater regulating valve trips.
- Pump trips.
- Condensate filter-demineralizer flow balancing control valve trip.
- Local feedwater piping, anchors, supports, and snubbers as applicable.

10.4.7.2.2.2 Feedwater Isolation Valves. One MFIV is installed in each of the four main feedwater lines outside the containment and downstream of the feedwater regulating valve. The MFIVs are installed to prevent uncontrolled blowdown from more than one steam generator in the event of a feedwater pipe rupture. The main feedwater check valve provides backup isolation. The MFIVs isolate the nonsafety-related portions from the safety-related portions of the system. In the event of a secondary side pipe rupture inside the containment, the MFIVs limit the quantity of high-energy fluid that enters the containment through the broken loop and provides a pressure boundary for the controlled addition of auxiliary feedwater to the three intact loops.

The MFIVs are also provided to prevent feedwater from entering the main steam generator inlet nozzle to cause line cracking during low-load and/or hot standby conditions when the feedwater flow and temperatures are low. During these periods feedwater enters the steam generator through the 6-in. bypass line and 6-in. auxiliary feedwater nozzle. The valves are bidirectional, double disc, parallel slide gate valves. Stored energy for closing is supplied by accumulators which contain a fixed mass of high-pressure nitrogen and a variable mass of high-pressure hydraulic fluid. For emergency closure, a solenoid is deenergized, which causes the high-pressure hydraulic fluid to be admitted to the top of the valve stem driving piston and also causes the fluid stored below the piston to be dumped to the fluid reservoir. Two separate pneumatic/hydraulic power trains are provided for each MFIV. Electrical solenoids are energized from separate Class 1E sources.

One BFIV is installed in each of the four bypass feedwater lines outside containment. The BFIVs are installed to prevent uncontrolled blowdown from more than one steam generator in the event of a feedwater line rupture. Bypass feedwater check valves provide backup isolation. The BFIVs isolate the nonsafety-related portions from the safety-related portions of the system. The BFIVs provide the same function as do the MFIVs. The valve body configuration and type of BFIVs are similar to the MFIVs. Air-operated piston actuators are provided to close the valves. Electrical solenoids are energized from separate Class 1E sources.

10.4.7.2.2.3 Main Feedwater Regulating Valves and Regulating Bypass Valves. The 16-in. main feedwater regulating valves are air-operated angle valves which control feedwater flow between approximately 20 percent and full power. The bypass control valves are air-operated globe valves, which are used during startup to approximately 40-percent power.

The feedwater regulating and bypass valves automatically maintain the water level in the steam generators during all phases of operation, in accordance with the steam generator manufacturer's prescribed normal water level. Positioning of the main feedwater regulating valve during normal operation is the function of a three-element feedwater level control system. The bypass feedwater regulating valve is also positioned during normal operation by the three-element feedwater level control system. The three-element control system maintains feedwater flow equal to the steamflow, and steam generator water level is used as an input to trim

feedwater flow and maintain programmed water level. During startup operations from no-load to about 40-percent power, a 4-in. bypass valve is provided to automatically maintain the water level in each steam generator.

In the event of a secondary side pipe rupture inside the containment, the main feedwater regulating valve (and associated bypass valve) provides a secondary backup to the MFIV to limit the quantity of high-energy fluid that enters the containment through the broken loop. For emergency closure of the main feedwater regulating valves, both of the two separate solenoids, when deenergized, will result in valve closure in 5 s or less. For emergency closure of the regulating bypass valves, either of two separate solenoids, when deenergized, will result in valve closure in 5 s or less. Electrical solenoids are energized from separate Class 1E sources.

10.4.7.2.2.4 Feedwater Check Valves. The main feedwater tilting disc check valves are located immediately downstream of the MFIVs. The check valve provides a diverse backup to the MFIV for preventing blowdown of the steam generators during a main feedline break outside the containment.

Two tilting disc check valves are located downstream of the BFIVs. To prevent a possible backflow or spillage of auxiliary feedwater, the first check valve is located upstream of the auxiliary feedwater line connection to the bypass feedline. The check valve in the bypass line inside containment reduce the possibility of the steam generator blowing down through the 6-in. nozzles simultaneously following an auxiliary and bypass feedline rupture outside containment.

10.4.7.2.2.5 Chemical Addition Line Check Valves and Isolation Valves. The check valves are located downstream of the chemical feedline isolation valves in the chemical addition lines. The check valves provide a secondary backup to the isolation valves to ensure the pressure boundary. The normally closed isolation valves are air-operated valves which fail-closed.

10.4.7.2.2.6 Condensate Pumps. The three condensate pumps are motor-driven and operate in parallel. Valving is provided to allow individual pumps to be removed from service. Pump capacity is sufficient to meet full-power requirements with two of the three pumps in operation.

10.4.7.2.2.7 Low-Pressure Feedwater Heaters. Parallel strings of closed, low-pressure feedwater heaters number 1, 2, and 3 are located in each of three condenser necks. The number 1, 2, and 3 heaters have integral drain coolers, and their drains cascade to the next lower stage feedwater heater. The drains from the number 1 heaters are dumped to their respective condenser. Feedwater leaving the three trains of number 3 heaters is headered and enters the two trains of low-pressure feedwater heaters number 4 and 5. Heater number 4 has no integral drain cooler and drains directly into the heater drain tank. Heater number 5 has an integral drain cooler, and its drains cascade into the number 4 heater shell. The discharges from the two trains of number 5 heaters enter the suction of the two steam generator feedwater pumps.

10.4.7.2.2.8 High-Pressure Feedwater Heaters. The steam generator feedwater pumps discharge into two trains of number 6 high-pressure feedwater heaters which have integral drain

coolers. The number 6 heaters drain directly into the shell side of low-pressure heaters number 5.

Isolation valves and bypasses are provided which allow each string of feedwater heaters to be removed from service.

Provisions are made in all heater drain lines, except number 4, which drains via the heater drain tank to allow direct discharge to the condenser in the event the normal drain path is blocked or flooding occurs in the heater.

The low-pressure and high-pressure feedwater heater shells are carbon steel, and the tubes are stainless steel.

10.4.7.2.2.9 Heater Drain Tanks. Two heater drain tanks, each serving one train of heaters number 4, 5, and 6, receive the total drains from these heaters and provide a reservoir capacity for the two heater drain pumps. Each heater drain tank is installed below the number 4 heater and receives the total drain flow by gravity flow. The drain level is maintained within the tank by a level controller in conjunction with a heater drain pump.

Each heater drain tank is provided with an alternate drain line to the main condenser for automatic dumping upon high-liquid level. The alternate drain line is also used during startup and shutdown when it is desirable to bypass the drain flow to the condenser for feedwater quality purposes.

10.4.7.2.2.10 Heater Drain Pumps. Each motor-driven heater drain pump takes suction from its respective heater drain tank and discharges it into the condensate system between the no. 4 and no. 5 feedwater heaters.

10.4.7.2.2.11 Steam Generator Feedwater Pumps. The steam generator feedwater pumps operate in parallel and discharge to the number 6 high-pressure feedwater heaters. The pumps take suction from the number 5 low-pressure feedwater heaters and discharge through the high-pressure feedwater heaters. Each pump is turbine driven. The turbine drivers are regulated by a master speed controller for both pumps and one slave speed controller for each pump. Steam for the turbines is supplied from the main steam header at low loads and from the moisture separator/reheater steam outlet during normal operation.

Isolation valves are provided which allow each steam generator feedwater pump to be individually removed from service, while continuing operations at reduced capacity with the parallel pump.

10.4.7.2.2.12 Pump Recirculation Systems. Minimum flow control systems are provided to allow all pumps in the main condensate and feedwater trains to pump at the manufacturer's recommended minimum flowrate to prevent pump damage.

### 10.4.7.2.3 System Operation

10.4.7.2.3.1 Normal Operation. For normal operating conditions between approximately 3- to 100-percent load, system operation is automatic. Automatic level control systems control the water levels in all feedwater heaters, the heater drain tank, and the condenser hotwell. Feedwater heater water levels are controlled by modulating flow control valves. Control valves in the discharges of the heater drain pumps control heater drain pump flows in relation to the level in the heater drain tank. Level control valves in the makeup line to the condenser from the condensate storage tank and in the return line to the condensate storage tank control the level in the condenser hotwell.

At very low power levels, feedwater is available from the auxiliary feedwater system. (Refer to subsection 10.4.9.) Once sufficient steam pressure has been established, a steam generator feedwater pump turbine is started, and from this low power level to approximately 20-percent power, feedwater flow is controlled by the feedwater bypass regulating valves. During this phase of operation, the variable-speed steam generator feedwater pump turbine may be placed on manual or automatic control at a low pump speed. Open and shut valve position indicators are provided in the control room for the main and bypass feedwater regulating valves. Feedwater flow control is transitioned to include both the bypass regulating valves and the main feedwater regulating valves. This operation may be done entirely in automatic control or with the bypass and main regulating valves in manual control. The plant load may be increased and the plant placed on completely automatic control with both the main and bypass feedwater valves in operation.

Ten-percent step load and 5-percent/min ramp changes are accommodated without major effect to the condensate and feedwater system. The system is capable of accepting a 50-percent step-load rejection without tripping of reactor or turbine systems. Under this transient, heater drain pump flow may be lost and the feedwater heater drain flows may be subsequently dumped to the condenser via the heater drain tank. The condensate pumps furnish full feedwater flow until heater drain pump flow is restored.

There are basically two modes of operation involved in supplying feedwater into the steam generators: (1) zero and low powers and (2) moderate and high powers. For the first, the required feedwater is supplied through the feedwater bypass line; i.e., BFIV is open and MFIV is closed. At moderate and high powers, the steam generator is supplied simultaneously through the bypass and main lines; i.e., both BFIV and MFIV are open.

The MFIV should not be opened until the resulting flowrate through the main nozzle exceeds that where stratification and striping can occur. This is 500 gal/min per steam generator, which corresponds to a 6-percent power level.

With both BFIV and MFIV open, approximately 8 percent of the total flow will be through the bypass line and 92 percent through the main line. The simultaneous flow in the bypass line serves two purposes. It serves as tempering flow, maintaining the 6-in. nozzle at a reduced temperature, and it ensures that steam generator water or steam does not reverse flow in the bypass line into the auxiliary feedwater system.

During the plant loading operation, the MFIV will be opened when the power level is 12 to 20 percent, so that the required 500 gal/min per steam generator is significantly exceeded. To achieve the 1- to 2-percent tempering flow recommendation would require opening the MFIV at about 25-percent power. At this power level, sufficient differential pressure across the bypass line is available to provide the recommended flow.

During the operations of heatup, cooldown, and hot standby, the necessary steam generator makeup will be provided by the auxiliary feedwater system through the auxiliary feedwater nozzle.

10.4.7.2.3.2 Emergency Operation. In the event that the plant must be shut down and offsite power is lost or a DBA occurs which results in a feedwater isolation signal, the MFIV, BFIV, and the main feedwater control and bypass valves are automatically closed. Coordinated operation of the auxiliary feedwater system (refer to subsection 10.4.9) and the main steam supply system (refer to section 10.3) is employed to remove the primary loop sensible heat and reactor decay heat.

### 10.4.7.3 Safety Evaluation

A. The safety-related portions of the condensate and feedwater system are located in the control and auxiliary buildings. These buildings are designed to withstand the effects of earthquakes, tornadoes, hurricanes, floods, external missiles, and other appropriate natural phenomena. Sections 3.3, 3.4, 3.5, 3.7, and 3.8 provide the bases for the adequacy of the structural design of these buildings.

B. The safety-related portions of the condensate and feedwater system are designed to remain functional after an SSE. Subsection 3.7.2 and section 3.9 provide the design loading conditions that were considered. Sections 3.5 and 3.6 and subsection 9.5.1 provide the hazards analyses to ensure that a safe shutdown, as outlined in section 7.4, can be achieved and maintained.

C. The condensate and feedwater system safety functions are accomplished by redundant means. A single active component failure of the safety-related portion of the system will not compromise the safety function of the system. As indicated by table 10.4.7-1, no single failure of nonessential portions of the system will compromise the system's safety functions. All vital power can be supplied from either onsite or offsite power systems, as described in chapter 8.

D. Preoperational testing of the condensate and feedwater system is performed as described in chapter 14. Periodic inservice functional testing is done in accordance with paragraph 10.4.7.4.

   Section 6.6 provides the ASME Boiler and Pressure Vessel Code Section XI requirements that are appropriate for the condensate and feedwater system.

E. Section 3.2 delineates the quality group classification and seismic category applicable to the safety-related portion of this system and supporting systems. Table 10.4.7-2 shows that the components meet the design and fabrication codes given in section 3.2. All the power supplies and controls necessary for the safety-related functions of the condensate and feedwater system are Class 1E, as described in chapters 7 and 8.

F. For a main feedwater line break inside the containment or a MSLB, the MFIVs, BFIVs, and the main feedwater regulating valves (and associated bypass valves) located in the control and auxiliary buildings (main feedwater isolation valve areas) are automatically closed upon receipt of a feedwater isolation signal, which occurs on a safety injection initiation, steam generator high-high water level, or a reactor trip coincident with a low $T_{avg}$ condition. For each intact loop,

the main and bypass feedwater check valves, MFIV, BFIV, and associated redundant isolation of the chemical addition line will close, forming a pressure boundary to permit auxiliary feedwater addition. The auxiliary feedwater system is described in subsection 10.4.9.

G. For a main feedwater line break upstream of the MFIV, the MFIVs and BFIVs are supplied with redundant power supplies and power trains to ensure their closure to isolate safety- and nonsafety-related portions of the system. Branch lines downtream of the MFIVs and BFIVs contain normally-closed, fail closed with power locked out, or locked closed manual valves. These valves separate the safety-related and nonsafety-related portions of the system.

   Releases of radioactivity from the condensate and feedwater system resulting from the main feedwater line break are minimal because of the negligible amount of radioactivity in the system under normal operating conditions. Additionally, following a steam generator tube rupture, the main steam isolation system provides controls for reducing accidental releases, as discussed in section 10.3 and chapter 15. Detection of radioactive leakage into and out of the system is facilitated by area radiation monitoring (discussed in subsection 12.3.4), process radiation monitoring (discussed in section 11.5), and steam generator blowdown sampling (discussed in subsection 10.4.8).

H. In the event of loss of offsite power, loss of the steam generator feedwater pumps, or other situations that may result in a loss of main feedwater, the feedwater isolation signal will automatically isolate the feedwater system and permit the addition of auxiliary feedwater to allow a controlled reactor cooldown under emergency shutdown conditions. The auxiliary feedwater system is described in subsection 10.4.9.

I. Hydraulic instabilities are minimized from occurring during the various modes of operation. The feed ring, with J nozzles and thermal sleeve in each steam generator (model F), is designed so that the feed discharge nozzles are located on the top of the feed ring. The feedwater piping into the steam generators has been arranged so that it does not drain into the steam generators. This was accomplished by sloping the piping upward. These two features prevent the formation of a steam pocket in the feedwater piping which, when collapsed, creates the hydraulic instability.

### 10.4.7.4 Tests and Inspections

#### 10.4.7.4.1 Preservice Valve Testing

The MFIVs, BFIVs, and feedwater regulating valves are checked for closing time prior to initial startup.

#### 10.4.7.4.2 Preservice Pipe Testing

The main feedwater lines from the five-way restraint forgings in the isolation valve areas to the main steam tunnel five-way restraint forgings are classified as non-Seismic Category 1 piping. However, these lines are designed to withstand a safe shutdown earthquake, and 100 percent

volumetric inspection at installation is provided (i.e., 100 percent volumetric examination of shop and field longitudinal and circumferential welds).

#### 10.4.7.4.3 Preoperational System Testing

Preoperational testing of the condensate and feedwater system is performed as described in chapter 14.

#### 10.4.7.4.4 Inservice Inspections

The performance and structural and leaktight integrity of system components are demonstrated by continuous operation.

The redundant actuator power trains of each MFIV and BFIV are subjected to the following tests:

A. Closure Time

   The valves are checked for closure time at each refueling.

B. Online Operability

   While the condensate and feedwater system is in operation, the operability of the actuator may be checked periodically by exercising the MFIV to approximately 90 percent of full open.

Additional discussion of inservice inspection of ASME Code Class 2 and 3 components is presented in section 6.6.

### 10.4.7.5 Instrumentation Applications

The main feedwater instrumentation, as given in table 10.4.7-3, is designed to facilitate automatic operation, remote control, and continuous indication of system parameters. As described in chapter 7, certain devices are involved in the secondary cycle pipe rupture protection system.

The feedwater flow to each steam generator is controlled by a three-element feedwater flow control system to maintain a programmed water level in the steam generator. The three-element feedwater controllers regulate the feedwater regulating valves by continuously comparing the feedwater flow and steam generator water level with the programmed level and the pressure-compensated steamflow signal. (Refer to section 7.7.) The steam generator feedwater pump turbine speed is automatically varied to maintain sufficient discharge pressure to support feedwater flow demand. The pump speed is increased or decreased, in accordance with the speed signal by modulating the steam pressure at the inlet of the pump turbine drivers.

Both steam generator feedwater pump turbines are tripped upon either one of the following:

A. High-high level in any one steam generator.

B. Actuation of safety injection. (Refer to section 7.3.)

One steam generator feedwater pump turbine trips when any one of the following directly affects it:

A. Turbine low bearing oil pressure.

B. Pump low bearing oil pressure.

C. Turbine overspeed.

D. Low vacuum (two out of two).

E. Turbine exhaust valve not fully open.

F. Thrust bearing wear.

G. Manual.

Yarway automatic recirculation valves are provided in the discharge lines from each condensate and heater drain pump. The purpose of the valves is to protect the respective pumps from becoming vapor bound, with subsequent internal pump damage and/or failure to pumps resulting from pumping rates below the minimum pump flow conditions stipulated by the pump manufacturer.

A flow element is installed in the suction of each of the steam generator feedwater pumps to provide the control signal to open the minimum recirculation flow valves, in order to protect the steam generator feedwater pumps.

A flow element with two flow transmitters is located on the inlet to each of the four steam generators to provide signals for the three-element feedwater control system.

In the event that a feedwater heater experiences a sizable number of tube leaks or a feedwater heater water level control valve fails closed, the main turbine is protected from failure resulting from flooding on the shell side of a feedwater heater and subsequent water induction into the moving turbine blades. This is done by the steam extraction lines to that heater by closing the respective bleeder trip and isolation valves and opening the high-level dump control valves that dump the heater excess drains to the condenser.

The total water volume in the condensate and feedwater system is maintained through automatic makeup and rejection of condensate to the condensate storage tank. The system makeup and rejection are controlled by the condenser hotwell level controller.

The system water quality requirements are maintained through the injection of hydrazine and alternate amine into the condensate system. The alternate amine and hydrazine injection is controlled by pH and the hydrazine residual in the system, which is continuously monitored by the process sampling system.

A temperature sensor is installed just below each elbow in the main and bypass feedwater piping welded to the Westinghouse steam generator inlet feedwater nozzles. These sensors may be used to provide a backup source for plant calorimetric data. These RTDs may also be used as inputs in the normal plant calorimetric data.

Instrumentation, including pressure indicators, flow indicators, and temperature indicators, required for monitoring the system is provided in the control room.

### 10.4.8 STEAM GENERATOR BLOWDOWN PROCESSING SYSTEM

The steam generator blowdown processing system (SGBPS) accepts water from each steam generator blowdown line, processes the water as may be required, and delivers the processed water either to the condensate system or to the waste water retention basin. Process steps include cooling with heat recovery, pressure reduction, filtration, and ion exchange. The

purpose of the SGBPS is to remove impurities resulting from primary coolant or circulating water inleakage which are concentrated in the steam generator by the evaporative process.

### 10.4.8.1 Design Bases

The SGBPS is designed to maintain optimum secondary side water chemistry during normal operation and during anticipated operational occurrences of main condenser inleakage or primary to secondary leakage.

The ion removal capacity provided by the two series-operated mixed bed demineralizers is sufficient to allow the blowdown water to be recycled to the condensate system or discharged via the waste water retention basin. The demineralizers are designed for the full SGBPS design flow of 360 gal/min, although the blowdown rate normally is much less.

Design radioactivity in the untreated blowdown water assumes a primary-to-secondary side leakage of 1.0 gal/min (12,000 lb/day) coincident with operation with 1.0-percent failed fuel; i.e., clad leakage in the fuel rods providing 1 percent of full power. Additional assumptions are provided in table 10.4.8-1. The secondary side water isotopic concentrations under these assumptions are provided in table 10.4.8-2. This design basis radioactivity is used for in-plant design information such as shielding and radiation area evaluation.

Expected radioactivity levels in the secondary side water are determined based on a primary-to-secondary side leakage of 100 lb/day and reactor coolant activities based on American National Standards Institute (ANSI) Standard N237-1976. Specific activities for secondary side water using a realistic basis are provided in section 11.1.

The demineralizer resins are temperature sensitive, and if the process flow temperature exceeds 140°F, the resins will begin to degrade, releasing some of the stored ions. Process temperatures significantly greater than 140°F could potentially cause an unacceptable excursion in secondary side water chemistry and activity. To prevent this and to maintain the purification function of the demineralizers, the SGBPS is designed to reduce the blowdown liquid temperature to 130°F before it enters the demineralizers. The design includes instrumentation to actuate an alarm if the water going to the demineralizers exceeds 140°F and to halt blowdown if the water temperature reaches 147.5°F.

Because portions of the SGBPS are classified as high energy lines, design features are provided to rapidly isolate the blowdown path should a rupture occur in the system piping outside of the containment.

Other instrumentation is provided in the SGBPS design to ensure that the operation is maintained in established parameter ranges. Description of this instrumentation is provided in paragraph 10.4.8.2.

While the design flow capacity of the SGBPS is 360 gal/min, the flow from each steam generator can be varied from 5 to 90 gal/min at the discretion of the operator. Estimated average total blowdown flow is 240 gal/min. Continuous blowdown from each steam generator is necessary to counteract the evaporative concentration occurring in each steam generator. The SGBPS may be isolated for maintenance and repair provided that the secondary side water chemistry does not go beyond specifications (subsection 10.3.5). The SGBPS is capable of purging the steam generators during all modes of operation, including shutdown conditions.

Process design parameters and equipment design capacities are provided in tables 10.4.8-3 and 10.4.8-4, respectively.

The design of the SGBPS conforms with the recommendations of Regulatory Guide 1.143, Revision 1. Those portions of the SGBPS extending from the steam generators to the outermost containment isolation valves are Safety Class 2 and Seismic Category 1. The remainder of the system is classified Non-Nuclear Safety Class since it performs no function related to safety. Seismic and safety classifications and design codes of individual pieces of equipment are provided in table 3.2.2-1.

### 10.4.8.2 System Description

The SGBPS piping and instrumentation diagram is shown in drawings 1X4DB159-1, 2X4DB159-1, 1X4DB159-2, 1X4DB159-3, 1X4DB179-1, and 1X4DB179-2. Location of SGBPS major components is indicated on general arrangement drawings provided in section 1.2. The SGBPS process flow diagram in figure 10.4.8-1 includes process parameters for a selected number of points in the flow hydraulic path. These process parameters are based on a design basis maximum blowdown rate of 90 gal/min per steam generator, realistic secondary side water activity levels (see section 11.1 for values and assumptions), and demineralizer decontamination factors consistent with reference 1.

Blowdown is drawn from each of the four steam generators at 5 to 90 gal/min each. The blowdown flow is controlled administratively to permit blowdown to match the variable nature of water quality. The blowdown water is extracted from each steam generator through a sparger pipe located just above the tube sheet where impurities are expected to accumulate. A sample flow is continuously diverted from each steam generator blowdown flow. These sample flows are analyzed to ascertain water chemistry conditions for each of the steam generators.

The blowdown from each steam generator passes through a separate blowdown heat exchanger where it is cooled with condensate. The condensate outlet temperature from the heat exchanger is controlled to a value consistent with its disposition in the feedwater train and which optimizes heat recovery. After leaving the blowdown heat exchanger, each separate blowdown flow passes through a flow control valve. The four blowdown flows are then combined, and if the temperature of the blowdown was not sufficiently reduced by the blowdown heat exchangers due to high-condensate temperatures, the blowdown flow is directed to the trim heat exchanger where it is cooled to below the demineralizer resin temperature limits. Otherwise, the trim heat exchanger is bypassed. Downstream of the trim heat exchanger is a temperature element that measures the temperature of the flow entering the demineralizer. A high-temperature signal (140°F) actuates an alarm and a high-high signal (147.5° F) isolates blowdown. The blowdown next flows through a pressure control valve that, with varying blowdown rates, maintains a constant upstream pressure at a high level to limit flashing of blowdown upstream of the blowdown heat exchangers. This valve is the boundary between the high- and low-pressure portions of the SGBPS. The low-pressure portion of the SGBPS is protected from overpressurization by a relief valve and by automatic isolation of blowdown on high pressure.

The blowdown now normally passes through the steam generator blowdown backflushable prefilter, a cartridge type filter, and through a cation demineralizer and a mixed bed demineralizer, connected in series. An increase in conductivity, at the outlet of each demineralizer, signals resin bed exhaustion. When the upstream cation bed is exhausted, flow is directed around both beds to the waste water retention building. Spent resin is transferred to the spent resin storage tank with the aid of the spent resin sluice pump or demineralized water header pressure, and new resin is charged to the demineralizer.

From the demineralizers, the blowdown passes through the two steam generator blowdown filters in parallel and the discharge radiation monitor. It is then normally returned to the

condensate system or it can be discharged to the environment via the waste water retention basin. The discharge radiation monitor is provided downstream of the demineralizers to isolate the blowdown flow and to terminate the discharge to the waste water retention basin upon a high radiation signal.

Another radiation monitor is provided downstream of the pressure control valve and alarms on high radiation.

Although blowdown normally is processed through the demineralizers and is recycled, the prefilter, a cartridge type filter, and demineralizers may be bypassed, especially during plant startup and shutdown operations when the quality of the blowdown water is most likely to be poorest. This water is discharged to the environment.

The spent resin handling portion of the SGBPS serves to sluice spent resin out of the steam generator blowdown demineralizers and to a storage tank or directly to the solid waste disposal system. If the spent resin is stored in the spent resin storage tank, it is eventually transferred to the solid waste disposal system by pressurizing the tank with nitrogen.

#### 10.4.8.2.1 Component Descriptions

A summary of principal component data is given in table 10.4.8-4. Component design codes are given in section 3.2.

A. Steam Generator Blowdown Heat Exchangers

The blowdown heat exchangers are provided to subcool the blowdown fluid streams from steam generator exit conditions to approximately 130°F, 10° below the recommended maximum operating temperature for the resin in the demineralizers. Direct cooling of the blowdown with condensate recovers some of the heat loss due to blowdown.

B. Trim Heat Exchanger

A trim heat exchanger cooled by turbine plant cooling water is used when condensate temperature significantly exceeds the design temperature for the blowdown heat exchanger. When condensate temperatures as high as 130°F are expected for significant periods of time, the trim heat exchanger, located downstream of the blowdown heat exchangers, can be valved into operation.

C. Steam Generator Blowdown Prefilter

The prefilter is of the backflushable type. It is designed to prevent release of radioactive particulates and to prevent plugging of the resin beds.

D. Steam Generator Blowdown Cartridge Filter

The cartridge filter is of the cartridge type. It is designed to prevent release of radioactive particulates and to prevent plugging of the resin beds.

E. Steam Generator Blowdown Outlet Filter

The outlet filters are of the cartridge type. They operate in parallel to filter the demineralizer train effluent, eliminating the possibility of recycling or discharging radioactive resin particles.

F. Steam Generator Blowdown Demineralizers

A cation bed and a mixed bed demineralizer are provided to remove ionic contaminants, including radioisotopes if present, from the blowdown fluid. These beds prepare the blowdown fluid for discharge or recycling.

G. Spent Resin Storage Tank

The spent resin storage tank provides a collection point for spent ion exchange resins from the steam generator blowdown mixed bed demineralizers. The tank is a vertical, cylindrical type and is designed to withstand an internal pressure sufficient to push the resins to the resin disposal area.

H. Steam Generator Blowdown Spent Resin Sluice Pump

This centrifugal pump is designed to sluice resin in a 3-in. pipe at 110 gal/min.

I. Steam Generator Blowdown Spent Resin Sluice Filter

This disposable cartridge filter catches any resin fines contained in the sluice water.

J. Steam Generator Drain Pump

This centrifugal pump is designed to drain the steam generator. The drain pump is rated at 200 gal/min at 265 ft head.

### 10.4.8.2.2 Instrumentation Application

The instrumentation for the SGBPS is primarily provided on the steam generator blowdown process panel (PSBP), which is located on level B of the auxiliary building. Alarms and control functions are provided as noted. There is also a common alarm on the main control board, which indicates any alarms on the PSBP.

10.4.8.2.2.1 Temperature.

A. Blowdown Heat Exchanger Inlet Temperature

Instruments provide local indication of the temperature of the blowdown fluid entering each blowdown heat exchanger.

B. Blowdown Heat Exchanger Coolant Outlet Temperature

Instruments maintain the outlet temperature of the condensate used for cooling in the blowdown heat exchangers and control the condensate flowrate. Temperature indication, a variable control setpoint, and high and low alarms are provided on the PSBP for each heat exchanger.

C. Blowdown Heat Exchanger Outlet Temperature

An instrument monitors the temperature of the blowdown stream downstream of the trim heat exchanger. Indication and a high-temperature alarm are provided on the PSBP. A high-temperature interlock terminates blowdown flow to protect the demineralizer resin.

D. Trim Heat Exchanger Inlet Temperature

An instrument monitors the temperature of the blowdown entering the trim heat exchanger. Indication is provided on the PSBP.

10.4.8.2.2.2 Pressure.

A. Blowdown Heat Exchanger Outlet Pressure

An instrument maintains a constant backpressure on the high-pressure section of the SGBPS by modulating valve PV-1151. The instrument's setpoint can be manually set by the operator at the PSBP. The set pressure and the position of the individual flow control valves establish the flowrate from each steam generator.

Pressure indication, a variable control setpoint, and a variable high alarm are provided on the PSBP.

B. Prefilter Inlet Pressure

An instrument monitors the inlet pressure to the prefilter. Indication is provided locally and on the PSBP. A high-pressure signal sounds an alarm on the PSBP and a high-high-pressure signal terminates blowdown flow.

C. Prefilter Outlet Pressure

An instrument locally indicates the prefilter outlet pressure. This instrument, used in conjunction with the prefilter inlet pressure instrument, is used to determine the pressure drop across the prefilter.

D. Cartridge Filter Differential Pressure

Instrument locally indicates differential pressure across the filter.

E. Outlet Filter Differential Pressure

Instruments locally indicate differential pressure across each outlet filter.

F. Spent Resin Storage Tank Pressure

An instrument provides indication of the spent resin storage tank pressure on the PSBP. A high-pressure alarm is furnished on the PSBP. This alarm indicates that the setpoint on the sluice pump discharge relief valve may be reached if the sluice pump is started.

G. Spent Resin Sluice Filter Differential Pressure

An instrument locally indicates the differential pressure across the sluice filter.

10.4.8.2.2.3 Level.

A. Spent Resin Storage Tank Level

An instrument indicates the total level in the resin storage tank; i.e., combined resin slurry and sluice water level.

High- and low-level alarms are provided, as well as a low-level interlock which shuts off the resin sluice pump. Level indication and alarms are located on the PSBP. Level indication is also provided on the PSPP.

10.4.8.2.2.4 Flow.

A. Steam Generator Blowdown Flow

Instruments monitor the blowdown flowrate from each steam generator with varying steam generator pressure (load). The instruments modulate flow control valves to maintain the desired blowdown flowrate for each steam generator. The instruments' setpoints are set at the miscellaneous systems equipment panel in the main control room. Flowrate indication is provided on the miscellaneous systems equipment panel and on the PSBP. A high-flow alarm is provided on the PSBP.

B. Total Flow

An instrument monitors the total blowdown flowrate through the SGBPS. A high-flow alarm is provided on the PSBP. Indication is provided both on the miscellaneous systems equipment panel and on the PSBP.

C. Discharge Flow

An instrument provides indication on the PSBP of the flowrate of blowdown being discharged to the environment.

D. Spent Resin Sluice Pump Flow

An instrument provides indication of spent resin sluice pump flow. A low-flow or low-level interlock shuts off the spent resin sluice pump. Indication and a low-flow alarm are given at the PSBP.

10.4.8.2.2.5 Discharge Radiation. An instrument monitors the activity of the fluid leaving the SGBPS. A high-activity signal from this instrument automatically terminates blowdown flow and isolates the discharge line to the environment. Indication and alarm are given in the main control room. This instrument is discussed in more detail in section 11.5.

### 10.4.8.2.3 System Operation

10.4.8.2.3.1 Plant Startup. Plant startup encompasses the operations that bring the reactor plant from cold shutdown to no-load power operating temperature and pressure.

During startup operations the steam generators are brought into chemical specification as rapidly as possible. High-blowdown rates (90 gal/min per steam generator) are the most effective means the operator has to reduce the solids content of the steam generators. The filters and demineralizers are normally bypassed during this operation with the blowdown discharged to the environment.

10.4.8.2.3.2 Normal Operation. Normal operation includes operation at steady power level, load follow operation, and hot standby.

During normal operation blowdown rates range from 5 to 90 gal/min per steam generator, as required. The typical, average blowdown rate is 240 gal/min (60 gal/min per steam generator). The cooled, filtered, demineralized effluent is normally recycled to the condensate system although it may be discharged if desired.

If there is any significant primary-to-secondary side leakage or main condenser inleakage, the blowdown demineralizers provide continuous cleanup to counteract the inflow of contaminants.

10.4.8.2.3.3 Plant Shutdown. Plant shutdown is defined as the operations that bring the reactor plant from normal operating temperature and pressure to cold shutdown for maintenance or refueling.

High blowdown rates can again be used to clean up the steam generators. As during startup operations, if secondary side activity levels permit, the filters and demineralizers are bypassed and the blowdown is discharged to the environment.

10.4.8.2.3.4 Resin Handling.

A. Resin Fluffing

The demineralizer is valved out of service, and the flow path is aligned from the spent resin sluice pump or the demineralized water header, through the process line of the demineralizer, through the Johnson screen at the top of the demineralizer, and back to the spent resin storage tank. The resin bed is backflushed for about 10 min to loosen it for sluicing. This operation may also be used to recover pressure drop due to bed fouling by backwashing particulates from the top layer of the resin into the spent resin storage tank. Such a recovery is useful when the resin is not ionically depleted.

B. Resin Sluicing

After the resin is loosened by backflushing, the resin outlet isolation valve is opened and the resin flows to the spent resin storage tank.

When resin activity in a demineralizer is low, the spent resin can be sluiced to a bulk disposal unit for low-level radioactive waste instead of to the spent resin storage tank. Demineralized water is added to the spent resin storage tank to replace any sluice water lost to the bulk disposal facility.

C. Resin Fill

After sluicing is completed, fresh resin must be added. The path to the drain

header from the demineralizer is opened to allow overflow. The resin fill line is opened and resin added. After fresh resin is added, the fill line valve is closed and the flow path is realigned for normal demineralizer operation.

D. Resin Disposal

The resin in the spent resin storage tank is loosened before disposal by sending pressurized nitrogen or sluice water through the six sparging nozzles in the tank. The valves in the resin transfer line are opened to direct the spent resin to a container in the radwaste processing facility.

The tank is then pressurized with nitrogen to force the resin up through the resin transfer line to the disposal area. A single nozzle in the spent resin storage tank is provided to allow local fluidization with sluice water at the opening of the discharge pipe. During resin transfer, this nozzle is used to ease the flow of the resin slurry. After resin transfer is complete, the tank is vented to the plant vent and returned to atmospheric pressure. The resin transfer line is then backflushed to the spent resin storage tank to clear it of resin.

Since a certain amount of resin remains in the tank after a disposal operation, it may hinder the backflush operation. Therefore, the fluidizing nozzle is again utilized to facilitate the backwash operation.

### 10.4.8.3 Design Evaluation

#### 10.4.8.3.1 Discharge of Radioactivity to the Environment

Tritium is not treated by this system, and during discharge operation, it is released to the environment unaffected by the SGBPS. It is assumed that the noble gases are stripped out of the steam generator so that the blowdown water is free from isotopes of krypton and xenon.

A. Design Discharge

When operating without steam generator leakage or when the system effluent is being recycled back to the main condenser, there will be essentially zero release of radioactivity from this system. When the system effluent is discharged to the environment while operating with concurrent fuel defects and steam generator leakage, radioactivity discharge rates will depend on the combinations of parameters such as percent of fuel defects, steam generator leakage rates, blowdown rate, and quantity of dilution flow. The system is designed to limit average discharge concentrations under these various combinations of conditions to isotopic concentrations consistent with meeting the maximum permissible site boundary concentrations for intermittent or off-normal releases (Appendix B, Table II of 10 CFR 20.1 - 20.601) and the offsite dose criteria outlined in 10 CFR 50, Appendix I. The total decontamination factor provided by the two blowdown demineralizers is 100 for cesium and rubidium and 1000 for all other isotopes. These decontamination factors are based on reference 1.

B. Expected Discharge

A determination of expected annual releases from the SGBPS is included in table 11.2.3-1.

### 10.4.8.3.2 Failure Analysis

If a failure in the SGBPS blocks the blowdown flow or prevents necessary processing of the blowdown water, reactor operation can continue without blowdown until the required maintenance is performed or until plant shutdown is necessitated because of inability to maintain secondary side water chemistry specification or radioactivity level requirements. While the SGBPS is not a safety-related system, its interfaces with other plant systems can adversely affect safety-related portions of the plant following a postulated pipe rupture in the high-energy portion of the system outside of containment. For this reason, safety-related flow sensors are installed in each blowdown line inside containment and temperature sensors are installed in areas where SGBPS high-energy piping is routed. These sensors automatically terminate blowdown if the blowdown flow in a line or ambient temperature in the affected area exceeds predetermined setpoints. The single-failure adequacy of the safety-related portion of the SGBPS is addressed in table 10.3.3-1.

### 10.4.8.4 Tests and Inspections

The SGBPS is in continuous operation during normal reactor operation. Periodic visual inspection and preventive maintenance are conducted using normal industry practice. The Flow-Accelerated Corrosion Program is credited as a license renewal aging management program for carbon steel components which may be affected by acidic effluent from the SGBPS demineralizers (see subsection 19.2.10).

Periodic tests and recalibration are performed on the radiation monitor in the SGBPS.

Periodic tests of the SGBPS containment isolation valves are performed to check operability, as described in the Technical Specifications.

### 10.4.8.5 REFERENCE

1. U.S. Nuclear Regulatory Commission, "Calculation of Releases from Pressurized Water Reactors," NUREG-0017, April 1976.

## 10.4.9 AUXILIARY FEEDWATER SYSTEM

The auxiliary feedwater system (AFW) is designed to supply feedwater to the steam generators whenever the reactor coolant temperature is above 350°F and the main feedwater system is not in operation; i.e., during startup, cooldown, or emergency conditions resulting in a loss of main feedwater. The system is a safety grade, Seismic Category 1, redundant system with Class 1E electric components as indicated in table 3.2.2-1. The AFW meets the additional requirements of II.E.1.1 and II.E.1.2 of NUREG-0737 as described in the paragraphs below.

### 10.4.9.1 Design Bases

Protection of the AFW from wind and tornado effects is discussed in section 3.3; flood protection is discussed in section 3.4; missile protection is discussed in section 3.5; protection against dynamic effects associated with the postulated rupture of piping is discussed in section 3.6; and environmental design is discussed in section 3.11.

#### 10.4.9.1.1 Safety Design Bases

The AFW automatically provides feedwater for the removal of reactor core decay heat in order that there is no damage to the reactor core following a loss of main feedwater from a condition of full power to where the reactor coolant temperature is brought to the point at which the residual heat removal system (RHRS) may be placed in operation. The automatic initiating circuits are powered from the emergency buses. The motor-driven pumps are automatically sequenced on the emergency diesel generators.

The AFW is protected from the effects of natural phenomena. This system is designed to remain functional after a safe shutdown earthquake (SSE) or following a postulated hazard such as fire, internal missile, or high-energy line break (HELB).

The AFW and supporting systems ensure the required flow to the steam generators in the event of a single active failure. In the unlikely event that the control room must be evacuated, the AFW can be operated from the shutdown panels and the auxiliary feedwater panel as the primary means of feedwater supply. The AFW includes condensate storage tanks (CSTs) as described in subsection 9.2.6.

The AFW is designed with sufficient diversity to remain operable for a limited duration with neither offsite nor onsite ac power available. The system is designed to avoid the effects of hydraulic instability (water hammer). The AFW is also designed to permit testing of the pumps during normal plant operation.

The AFW is provided with flow limiting orifices to limit the auxiliary feedwater flow to a depressurized steam generator and to ensure that the required minimum flow is directed to the remaining three intact steam generators.

#### 10.4.9.1.2 Power Generation Design Basis

The AFW supplies feedwater to the steam generators at a sufficient flowrate to support normal low-power transients such as startup, cooldown, and hot standby. Flowrate is varied as necessary to maintain stable plant conditions by throttling the motor-operated discharge isolation/flow control valves.

#### 10.4.9.1.3 Codes and Standards

Codes and standards applicable to the AFW are listed in table 3.2.2-1. The AFW is designed and constructed in accordance with Safety Class 3 requirements up to the motor-operated valves in each branch line located upstream of the stop-check valves, which serve as the containment isolation valves; the remaining valves and piping to the main feedwater system are Safety Class 2.

### 10.4.9.2 System Description

#### 10.4.9.2.1 General Description

The system consists of two motor-driven pumps, one steam turbine-driven pump, and associated piping, valves, instruments, and controls as shown on drawings 1X4DB161-2, 1X4DB161-3, and 1X4DB168-3. Table 10.4.9-1 provides data for the various active components in the AFW. Table 10.4.9-2 presents minimum system flow requirements for various modes of operation. A plan view of the AFW pumphouse is provided in drawing 1X4DE316.

Each motor-driven auxiliary feedwater pump is sized to supply the feedwater flow required for removal of 100 percent of the decay heat from the reactor. The turbine-driven pump is sized to supply up to twice the capacity of a motor-driven pump. The system capacity is sufficient to remove decay heat and to provide adequate feedwater for cooldown of the reactor coolant system (RCS) at an average of 50°F per hour but not to exceed 100°F in any 1 hour. Following transients or accidents, cooldown may begin within 1 to 2 hours at zero load hot standby.

For a transient or accident condition, the minimum flow is delivered within 1 min of any automatic auxiliary feedwater actuation signal to at least two effective steam generators.

Normal flow is from the CST to the auxiliary feedwater pumps. The design of the condensate storage tank provides for cold shutdown capability for a period of 9 hours; 4 hours at hot standby, followed by a 5 hours cooldown of the primary system at an average rate of 50°F per hour but not to exceed 100°F in any 1 hour to a temperature of 350°F. For Unit 2, a combined safety-related capacity in two CSTs supports an additional 3 h hot standby/cooldown for a total of 12 h prior to placing RHR in service. Table 10.4.9-2 provides the nuclear steam supply system (NSSS) required makeup rates to the steam generators for various situations. Refer to subsection 9.2.6 for a description of the condensate storage system.

The design of the AFW provides that auxiliary feedwater can automatically be supplied to the steam generators in approximately 1 min in the event that the normal source of feedwater is lost.

Provisions are incorporated in the AFW design to allow for periodic operation to demonstrate performance and structural and leaktight integrity. Leak detection is provided by visual examination and in the floor drain system described in subsection 9.3.3.

Refer to section 5.4 for a discussion of steam generator design features to preclude fluid flow instabilities, such as water hammer. The main 16-in. and bypass 6-in. feedwater connections on each steam generator are the highest point of each feedwater line downstream of the MFIV and the BFIV. An elbow, with a short transition piece, is connected directly to the steam generator main 16-in. and bypass 6-in. feedwater nozzles, which tend to minimize the portion of feedwater piping that can drain into the steam generator and become filled with steam. The feedwater lines contain no high-point pockets that could trap steam and lead to water hammer. The horizontal pipe length from the main and bypass feedwater nozzles of each steam generator is minimized. This feedwater system configuration coupled with the steam generator design minimizes the potential for steam backleakage and bubble collapse water hammer in the main and auxiliary feedwater lines. To complement the system and steam generator design, Westinghouse recommended feedwater system maintenance and startup procedural guidelines further preclude feedwater system water hammer conditions in the main and bypass feedwater lines. The routing of the main and bypass feedwater lines is shown in drawings 1K5-1305-062-01 and 1K5-1305-058-01.

### 10.4.9.2.2 Component Description

10.4.9.2.2.1 Motor-Driven Pumps. Two of the three auxiliary feedwater pumps are horizontal, centrifugal pumps driven by electric motors which are supplied with power from independent, Class 1E essential switchgear buses. Each pump has a capacity of 630 gal/min. Each motor-driven auxiliary feedwater pump is normally aligned to feed two steam generators. The pumps can take suction from either of two Seismic Category 1 CSTs, and their discharge piping is directed to each of the four steam generators.

10.4.9.2.2.2 Turbine-Driven Pump. A turbine-driven auxiliary feedwater pump provides system redundancy of auxiliary feedwater supply and diversity of motive pumping power. The pump is a horizontal, centrifugal unit with a capacity of 1175 gal/min. The turbine-driven auxiliary pump is normally aligned to feed four steam generators. Steam supply piping to the turbine driver is taken from two of the four main steam lines between the containment penetrations and the main steam isolation valves. Each of the steam supply lines to the turbine driver is equipped with a check valve and a normally open, dc motor- operated gate valve. These steam supply lines join to form a header which leads to the turbine through a normally closed steam admission valve and normally open trip/throttle valve, both of which are dc motor-operated, and a normally open electro-hydraulically operated speed governing valve. The steam lines contain provisions to prevent the accumulation of condensate in the lines. The turbine driver can operate with steam inlet pressures ranging from 90 to 1130 psig. Exhaust steam from the turbine driver is vented to the atmosphere above the auxiliary feedwater pumphouse roof. In the unlikely event that neither offsite nor onsite ac power is available, the turbine-driven pump can function normally for up to 4 h, which is the time that the batteries can sustain dc power during a station blackout. Station blackout coping is discussed in section 8.4. Each auxiliary feedwater pump is located in a separate room in the auxiliary feedwater pumphouse.

10.4.9.2.2.3 Pump Discharge Isolation/Flow Control Valves. The motor-operated isolation/flow control valves in the branch lines from each motor-driven auxiliary feedwater pump that serves an individual steam generator are powered from the same Class 1E bus as the respective motor-driven auxiliary feedwater pump and associated controls. The motor-operated isolation/flow control valves in the branch lines from the turbine-driven auxiliary feedwater pump that serve an individual steam generator are powered from the train C Class 1E dc distribution system. The motor-operated isolation/flow control valves can be controlled from either the control room or local panels to modulate the auxiliary feedwater flow to maintain the required steam generator water level. The valves are normally open during power generation, but can be throttled or closed as necessary. The valves receive an open signal upon receipt of the same signal which starts the respective auxiliary feedwater pump. The valves can be manually closed to isolate the flow to a faulted steam generator and thus minimize spillage losses.

During normal operation, the isolation/flow control valve operability status is continuously indicated in the control room, thus informing the operators that a valve is closed or the automatic opening function is overridden. Also, as shown on the failure modes and effects analysis (table 10.4.9-4), inadvertent closure of any one valve terminates only the auxiliary feedwater flow from one pump to one steam generator, with all other flow paths from the three pumps to the four steam generators unaffected. Thus, adequate auxiliary feedwater flow is ensured during postulated accident conditions.

10.4.9.2.2.4 Piping and Valves. The piping from the CSTs to the pump suction and from the pump discharge to the pump discharge locked open manual gate valve is seamless stainless steel. The remainder of the discharge piping is seamless carbon steel. Welded joints are used throughout the system, except at flanged equipment connections. The auxiliary feedwater pump suction, recirculation, and discharge lines outside the auxiliary feedwater pumphouse are heat traced. The discharge line from each pump is equipped with a check valve, and manually operated isolation valve. Each branch auxiliary feedwater line from the pump discharge that serves an individual steam generator is equipped with a motor-operated isolation/flow control valve, a manually operated isolation gate valve, and two stop-check valves. The motor-operated steam supply valves for the turbine-driven pump and the turbine-driven pump discharge motor-operated isolation/flow control valves are supplied with Class 1E dc power. The motor-driven pump, motor-operated isolation/flow control valves are supplied with Class 1E ac power.

The lines that supply water to the auxiliary feedwater pumps from the No. 1 CST are equipped with two locked open, manual butterfly valves; the lines from the No. 2 CST have one locked open manual and one normally shut motor-operated butterfly valve. All motor-operated valve positions are monitored in the control room to provide the operators with the status of system alignment at all times.

Auxiliary feedwater is provided to each of the four steam generators by a 4-in. carbon steel line that ties into the 6-in. bypass feedwater line downstream of the 6-in. bypass feedwater check valve as shown in drawing 1X4DB168-3. The 6-in. bypass feedwater line connects to the steam generator 6-in. auxiliary feedwater nozzle, which is provided with an internal upward- inclined pipe extension that ends in a horizontal section so that the incoming feedwater enters below the steam generator normal water level in a stream horizontal to the steam generator deck plates. This flow path is provided at low load or hot standby conditions when auxiliary feedwater flowrate and temperatures are low and during normal power operation from 15-percent to 100-percent load.

10.4.9.2.2.5 Pump Miniflow. Auxiliary feedwater pumps are provided with miniflow protection. Each motor-driven pump has a miniflow return line with a flow control valve which isolates when flow to the generators exceeds a preset minimum. The turbine-driven pump miniflow return line has a manual valve which may be closed when flow to the generators exceeds a preset minimum. The motor-driven pump miniflow and turbine-driven pump miniflow may be routed to CST 1 or CST 2.

### 10.4.9.2.3 System Operation

10.4.9.2.3.1 Plant Startup. During startup, the auxiliary feedwater pumps are used under manual control to supply feedwater from the CST to the steam generators until sufficient steam is available to operate the turbine-driven main feedwater pumps.

10.4.9.2.3.2 Normal Plant Operation. The AFW is not required during normal power generation. The pumps are placed in the automatic mode, lined up with the CST, and are available if needed.

10.4.9.2.3.3 Normal Plant Cooldown. During cooldown, the auxiliary feedwater pumps are used under manual control to supply water from the CST to the steam generators. Auxiliary feedwater flow to each steam generator is regulated by the motor-operated isolation/flow control valves. The auxiliary feedwater pumps are used until reactor coolant temperature drops to 350°F, at which point the RHRS is placed in service and further cools the reactor.

10.4.9.2.3.4 Emergency Operation. In addition to manual actuation capabilities, the AFW is aligned to be placed in service automatically in the event of an emergency. Upon the occurrence of a safety injection signal, low-low steam generator water level in any one steam generator, or trip of both main feedwater pumps, or an Anticipated Transient Without Scram (ATWS) Mitigation System Actuation Circuitry (AMSAC) actuation, the auxiliary feedwater isolation/flow control valves go to the full-open position and the two motor-driven auxiliary feedwater pumps are started and begin to pump the contents of the CST into the steam generators to remove heat from the RCS. The turbine-driven pump is actuated automatically on low-low water level in any two steam generators or an AMSAC actuation. A blackout sequence starts the turbine-driven pump and both motor-driven pumps. To start the turbine-driven pump, a dc motor-operated turbine steam admission valve in the steam supply line to the turbine driver is opened. The turbine is designed to start immediately without warmup. The operator can remotely manipulate the position of the auxiliary feedwater motor-driven isolation/flow control valves in order to control steam generator water level. Provision for local manual operation of the isolation/flow control valves is also provided.

Two seismic Category 1 CSTs are provided for each unit. The CSTs are redundant because the motor-driven pump miniflow and turbine-driven pump miniflow may be routed to either CST 1 or CST 2. For Unit 1, a safety-related volume of 340,000 gal in either CST is sufficient to supply the auxiliary feedwater pumps. For Unit 2, a combined safety-related volume of 378,000 gal in two CSTs is sufficient to support an additional 3 h hot standby/cooldown for a total of 12 h prior to placing RHR in service. In addition to the 340,000 gal, the CSTs contain an additional capacity exceeding the equivalent heat generated by the remaining three reactor coolant pumps for 9 hours after reactor trip.

Decay heat is removed from the reactor by boiling the feedwater in the steam generators and venting steam to the atmosphere through the power-operated atmospheric relief valves or the main steam safety valves. If the main condenser is available, the steam may be discharged via the turbine bypass system to the main condensers. When reactor coolant temperature drops to 350°F, cooldown is shifted to the RHRS.

A summary of system performance for various accident conditions is provided in table 10.4.9-3. As noted, the table includes flows to both the faulted and intact steam generators. Comparing these data with those in table 10.4.9-2, it is seen that minimum flow requirements for the intact steam generators are satisfied under all failure modes. In accordance with latest Westinghouse criteria, the total flow to the intact steam generators is less than 2500 gal/min for all cases.

### 10.4.9.3 Safety Evaluation

The AFW, in conjunction with two Seismic Category 1 CSTs, provides a means of pumping sufficient feedwater to prevent damage to the reactor following a loss-of-main-feedwater incident as well as to cool down the RCS at a rate not to exceed 100°F in any 1 hour to a temperature of 350°F, at which point the RHRS can operate. At a minimum, the 340,000 gal of water in the CSTs for the exclusive use of the AFW is sufficient to supply water as follows:

- 226,500 gal of water to hold the plant for 4 hours at hot standby conditions followed by an additional 5 hours during plant cooldown to 350°F.
- 19,800 gal of water to dissipate the heat generated by one reactor coolant pump for 9 hours after the reactor trip.
- 72,000 gal of water to compensate the turbine-driven auxiliary feedwater pump runout flow for 30 min (single failure until operator action isolates failure).
- 17,790 gal of water to compensate the auxiliary feedwater flow spilled through a pipe break for 30 min (initiating event).

For Unit 2, an additional 38,000 gal of water is required to compensate for an additional 3 h of hot standby/cooldown prior to placing RHR in service. Therefore, two CSTs with a combined safety-related volume of 378,000 gal are sufficient to meet the safety-related AFW requirements for Unit 2.

Pump capacities and start times are calculated by the NSSS vendor such that these objectives can be realized. At minimum water levels in the CSTs, there is sufficient net positive suction head (NPSH) to the motor- and turbine-driven auxiliary feedwater pumps.

Pump discharge head is sufficient to establish the minimum necessary flowrate against a steam generator pressure corresponding to the accumulation pressure of the lowest main steam safety valve. The maximum times to start the electric motors and the steam turbine which drive the auxiliary feedwater pumps are chosen so that sufficient flowrates are established before the loss of feedwater can result in damage to the reactor core. Minimum flow requirements are satisfied by one motor-driven pump assuming a HELB in the turbine-driven pump circuit and a single active failure.

In accordance with Branch Technical Position (BTP) ASB 10-1, the AFW provides redundant and diverse means of supplying feedwater to the steam generators for cooling the RCS under emergency conditions. Either of the motor-driven pumps has the capability of supplying a minimum of 100 percent of the feedwater requirements for safe cooldown of the RCS. The turbine-driven auxiliary feedwater train is capable of providing 200 percent of the minimum required feedwater flow for residual heat removal.

The three-train configuration can thus perform its safety-related function, assuming a postulated failure in the discharge piping of one train concurrent with a single active component failure in another train.

In the event the steam generators become contaminated, manual action is required to secure steamflow from the affected steam generators in order to prevent spread of contamination to the turbine-driven pump.

The motor-driven pump discharges are cross-connected through two normally closed, manually-actuated isolation valves.

Complete physical and electrical separation is maintained throughout the pump controls, control signals, electrical power supplies, and instrumentation for each auxiliary feedwater pump. Each motor-driven pump and associated motor-operated isolation/flow control valves are powered from the corresponding diesel generator, while the auxiliary feedwater turbine-driven pump components are powered from the train C, Class 1E dc distribution system.

Instrumentation and controls for the operation of the AFW are provided at the shutdown panels and the auxiliary feedwater panel as well as in the control room. Instrumentation controls are discussed in paragraph 10.4.9.5.

The AFW, including the CSTs, is designed to Seismic Category 1 requirements. The components (and supporting structures) of any system, equipment, or structure which are not Seismic Category 1 and whose collapse could result in loss of a required function of the AFW through either impact or flooding are analytically checked to determine that they will not collapse when subjected to seismic loading resulting from the SSE or are upgraded to Seismic Category 1.

Diversity is provided in the type and number of pumps, power supplies, piping arrangements, and pump and valve controls so that any single failure will not negate the AFW's ability to perform its safety function. The features which provide this capability are shown in the system schematic, drawings 1X4DB161-2 and 1X4DB161-3. In the event of a total loss of all ac power, the AFW can fulfill its safety function with any coincident, single active mechanical failure except the failure of the turbine-driven pump itself and its steam inlet valve.

An AFW analysis was performed in accordance with Action Items II.E.1.1 and II.E.1.2 of NUREG-0737 using the methodology defined by Appendix III and Annex 1 of Appendix X of NUREG-0611 and NUREG-0635 to determine the system reliability and major contributions to AFW failure under various loss-of-main-feedwater transients. This reliability analysis is presented in appendix 10A.

Auxiliary feedwater capabilities to cope during a station blackout (SBO) are discussed in section 8.4. An SBO coping analysis was performed as required by 10 CFR 50, Section 50.63.

A failure modes and effects analysis is provided in table 10.4.9-4.

### 10.4.9.4 Inspection and Testing Requirements

Each of the auxiliary feedwater pumps is hydrostatically tested in accordance with American Society of Mechanical Engineers (ASME) Section III, Class 3. The turbine driver is given a hydrostatic test, a mechanical running test, and an overspeed trip test prior to shipment. Inservice inspection and testing is performed as discussed in section 3.9.

The wall thickness of pressure boundary castings of the pumps, turbines, trip valve, throttle valve, and governor valve are checked and recorded by the manufacturer. The entire AFW is hydrostatically tested after assembly is completed. The system, its initiating signals, and its circuits are capable of being tested periodically while the plant is at power, in accordance with the frequency specified in the Technical Specifications. In addition, there will be double verification of proper valve alignment after system maintenance.

### 10.4.9.5 Instrumentation Requirements

Safety-related display instrumentation related to the AFW is discussed in section 7.5. Information indicative of the readiness of the AFW prior to operation and the status of active components during system operation is displayed for the operator in the main control room, at the shutdown panels, and at the auxiliary feedwater panel.

Equipment normally relied upon to be operable is monitored by the bypass/inoperable status monitoring system. Should the component be bypassed (e.g., by operation of a control switch, loss of control power, etc.), it is indicated at the system level on the bypass/inoperable status board. This indication satisfies the intent of the bypass indication requirement of Institute of Electrical and Electronics Engineers (IEEE) 279-1971 and Regulatory Guide 1.47 and the control power surveillance requirement of IEEE 308-1973.

The indication and controls provided for the AFW are summarized in table 10.4.9-5.

The logic for auxiliary feedwater initiation upon loss of both main feedwater pumps is shown in drawing 1X6AA02-239. Hydraulic fluid pressure for the feedwater pump turbine controls is sensed by instrumentation in each turbine. Upon loss of hydraulic fluid pressure (indicative of loss of main feedwater pumps) in both turbines, the pressure switches actuate relays which transmit the auxiliary feedwater pump start signal. The pressure switches are nonsafety grade; the actuating relays serve as isolation devices, with the remainder of the circuitry being safety grade.

The AFW can be manually initiated from the control room and the remote shutdown panels. A single failure in the manual circuits will not result in the loss of system function as described in section 7.3. The motor-driven auxiliary feedwater pumps and valves are automatically actuated and sequenced to the emergency bus as described in drawings 1X3D-AA-K02A, 2X3D-AA-K02A, 1X3D-AA-K02B, and 2X3D-AA-K02B. The automatic initiation signals and circuits are designed so that their failure will not result in the loss of manual initiation from the control room in accordance with Regulatory Guide 1.62.

Two channels of safety grade auxiliary feedwater flow instrumentation are provided per steam generator. Diversity is provided by steam generator water level for each steam generator.

## 10.4.10 CONDENSATE AND FEEDWATER CHEMICAL INJECTION SYSTEM

### 10.4.10.1 Design Bases

#### 10.4.10.1.1 Safety Design Bases

The condensate and feedwater chemical injection system serves no safety function and has no safety design basis.

#### 10.4.10.1.2 Power Generation Design Basis

The system is designed to control the dissolved oxygen content and pH in the turbine cycle and the steam generators.

### 10.4.10.2 System Description

#### 10.4.10.2.1 General Description

The condensate and feedwater chemical injection system is shown in drawing 1X4DB157. The system is designed and constructed in accordance with Quality Group D specifications. Design parameters for the major components are provided in table 10.4.10-1.

#### 10.4.10.2.2 Component Description

A. Methoxypropylamine (MPA) Storage Tank

Stainless steel, 12,789-gal-capacity tank used for bulk storage of MPA.

B. MPA Storage Transfer Pumps

Vertical, centrifugal type pump mounted atop the MPA storage tank (single pump per unit).

C. MPA and Hydrazine Day Tanks

Stainless steel, 250-gal-capacity tanks furnished with gauge glass level indicator, low-level switch, and 10-liter metering cylinder for Hydrazine day tanks (meter cylinder removed from Unit 1 and Unit 2 MPA tanks).

D. Batch Day Tank

Stainless steel, 250-gal-capacity tank furnished with gauge glass level indicator, and 10-liter metering cylinder.

E. Condensate Chemical Feed Pump

Simplex unit, positive displacement, disc diaphragm type pump, with autoelectric stroke adjustment control. The pump is driven by an electric motor.

F. Steam Generator Layup Chemical Pump

Duplex unit, positive displacement, disc diaphragm type pump, with manual stroke adjustment control. The pump is driven by an electric motor.

G. Auxiliary Feedwater Layup Chemical Pump

Electric motor-driven, simplex unit, positive displacement, disc diaphragm type pump with manual stroke adjustment control.

H. Hydrazine Dispensing Pump/High Volume

Electric motor-driven, magnetic drive, portable drum pump.

I. Tank Mixing Pump

Electric motor-driven, horizontal centrifugal type pump.

J. Hydrazine Storage Tank

Stainless steel, 6,000-gal-capacity tank used for bulk storage of hydrazine.

K. Hydrazine Storage Transfer Pump

Vertical, centrifugal type pump mounted atop the hydrazine storage tank.

L. Deleted

M. Ammonium Chloride Pump (Unit 1 only)

Positive displacement piston-type pump – high head – for small quantity chemical injection.

N. Ammonium Chloride Tank (Unit 1 only)

Polypropylene plastic, 7.5-gal tank with a polypropylene plastic lid.

#### 10.4.10.2.3 System Operation

The methoxypropylamine (MPA) consumed by the VEGP is brought to the plant by truck or rail and is stored in the 12,789-gal storage tank in the yard. Two MPA transfer pumps, one for each unit, send the MPA to the unit ammonia day tank and the unit batch tank. The hydrazine is brought to the plant in two ways: (1) bulk delivery by truck or rail and stored in the 6644-gal storage tank in the yard and (2) purchased and stored in 55-gal drums. Hydrazine is transferred to the hydrazine day tank and the batch day tank of each unit by using the hydrazine transfer pump or hydrazine dispensing pump.

The chemical injection systems for both units are located in the turbine building at el 195 ft 0 in. Each injection system consists of two subsystems. The first subsystem (for normal operation) has two day tanks, one for MPA and one for hydrazine. The tanks are connected to two autoelectric-controlled simplex metering pumps. One pump is capable of supplying the MPA and hydrazine required for one unit; the second pump is a 100-percent spare. Each hydrazine simplex pump has a capacity of 11.7 gal/h at a rated head of 650 psig. Each MPA simplex pump has a capacity of 0.5 gal/h at a rated head of 650 psig. Two separate lines, one for hydrazine and one for MPA, and individual automatic controls are provided for injecting chemicals to the condensate and feedwater system during normal operation and the hot standby condition.

The second subsystem is principally for wet layup operation. It consists of one batch tank for storing mixed chemicals in predetermined concentrations and one duplex metering pump for sending the mixed chemicals to the four steam generators during wet layup conditions. Two chemical lines run from the chemical injection area in the turbine building to the steam generators, one line to steam generators 1 and 4 and the other to steam generators 2 and 3. One simplex metering pump is also provided to send the mixed chemicals to lay up the condensate and feedwater system during cold shutdown, as well as the auxiliary feedwater system whenever necessary. The duplex metering pump and the simplex metering pump are rated at 100 gal/h and a head of 1200 psig and are interconnected to serve as spare for each other.

For the hydrazine day tank, MPA day tank, and batch tank a condensate supply, chemical metering cylinder, and mixing pump are provided. The chemical is measured in the metering cylinder before being drained to the day tank or batch day. The condensate supply provides water from the condensate pump discharge to the day tanks and the batch tank to dilute the concentrated chemicals to predetermined concentrations. The mixing pump provides the agitation required for the thorough mixing of the chemicals and the condensate.

### 10.4.10.3 Safety Evaluation

The condensate and feedwater chemical injection system is not a safety-related system.

### 10.4.10.4 Tests and Inspections

The condensate and feedwater chemical injection system is operationally checked before initial plant startup to ensure proper functioning of the feed systems and chemical makeup sensors.

### 10.4.10.5 Instrumentation Applications

Instrumentation is provided for manual and automatic control of the chemical injection system; a condensate flow signal and hydrazine and specific conductivity analyzers provide indication of water quality and provide inputs to the automatic mode chemical injectors. Local and remote hydrazine and conductivity alarms are provided to monitor the performance and protect components of the system. Wet layup operations are manually controlled, relying on the results of grab samples and local indications.

TABLE 10.4.1-1 (SHEET 1 OF 2)

MAIN CONDENSER DESIGN DATA

Condenser Data

| | |
|---|---|
| Condenser type | Ecolaire Model 165-TSMC2-45, two-pass, single-pressure, vertically divided surface condenser unit in three shells |
| Condenser overall dimensions per shell | 73 ft 11 in. x 27 ft 0 in. x 61 ft 8 in. high |
| Hotwell storage capacity | 8600 $ft^3$/3 shells |
| Heat transfer capability | 7.9514 x $10^9$ Btu/h |
| Surface area | 825,000 $ft^2$ |
| Steam load | 8,575,935 lb/h |
| Shell pressure (design) | Full vacuum to 27 psig (hydrostatic head) |
| Circulating water | 509,600 gal/min |
| Water box pressure (design) | 80 psig |
| Tube-side inlet temperature | 89°F |
| Tube-side temperature rise | 33.01°F |
| Condenser friction (ft of $H_2O$) | 18 ft at 89°F |
| Condenser outlet temperature | 120.6°F |
| Water box material | Steel (ASTM A283, Grade C) |

Condenser Tube Data

| | |
|---|---|
| Condenser tubes (main section) | 1 in. OD 22 BWG |
| Tube material | Titanium (ASTM B338-76, Grade B) |
| Quantity | 22,264/condenser shell |

TABLE 10.4.1-1 (SHEET 2 OF 2)

| | |
|---|---|
| Condenser tubes (periphery) | 1 in. OD 18 BWG |
| Tube material | Titanium (ASTM B338-76, Grade B) |
| Quantity | 1168/condenser shell |
| Total number of tubes | 23,432/condenser shell |
| Effective tube length | 44 ft 9 1/4 in. |
| Tube sheet material | Aluminum bronze (ASTM B171, alloy 614) |
| Support plates | 15 |

TABLE 10.4.2-1

CONDENSER AIR EJECTION SYSTEM
DESIGN DATA

| | |
|---|---|
| **Steam Jet Air Ejectors** | |
| Quantity | Two per unit |
| Holding requirements | 3.5 in. Hg absolute at full load (VWO) |
| Available steam | 930 psig and 1191.7 Btu/lb (normal operation) |
| Auxiliary steam | 170 psig and 1238 Btu/lb |
| Applicable standards | ANSI B31.1<br>HEI Standards |
| **Condenser Exhausters** | |
| Quantity | Two per unit |
| Hogging requirements | Evacuate total condenser volume from atmospheric pressure to less than 5 in. Hg absolute in 2 h or less. |
| Holding requirements | Maintain 36 $sf^3$/min exhaust flow at 1.0 in. Hg absolute based on an initial temperature difference of 25°F. |

TABLE 10.4.5-1

DESIGN PARAMETERS FOR MAJOR COMPONENTS
OF CIRCULATING WATER SYSTEM

Circulating Water Pump

| | |
|---|---|
| Quantity | Two per unit |
| Flowrate (gal/min) | 242,300 |
| Head (ft) | 95 |

Natural Draft Cooling Tower

| | |
|---|---|
| Quantity | One per unit |
| Design wet bulb temperature (°F) | 78 |
| Design relative humidity (%) | 51 |
| Design inlet temperature (°F) | 122 |
| Design outlet temperature (°F) | 89 |
| Design temperature rise (°F) | 33 |
| Design flowrate (gal/min) | 509,600 |
| Design bypass flowrate (gal/min) | 300,000 |

TABLE 10.4.6-1 (SHEET 1 OF 3)

CONDENSATE FILTER/DEMINERALIZER SYSTEM
MAJOR COMPONENTS

1. Powdex Filter/Demineralizer Vessel

   Qty.: 5 per unit
   Dia.: 72 in. (x 96 in. straight side)
   Code: ASME Boiler and Pressure Vessel Code, Section VIII
   Design press.: 700 psig
   Internal filter elements: qty. 420 per vessel, polypropylene-wound or nylon-wound elements with stainless steel support hardware
   Design flow: 5340 gal/min per vessel

2. Backwash Recovery Pump

   Qty.: 1 per unit
   Size: 6 x 8-15, single-stage, vertically split casing
   Type: horizontal centrifugal pump
   Rating: 1280 gal/min at 200 ft TDH, 150°F
   Motor: 125 hp, 1800 rpm

3. Spent Resin Pump

   Qty.: 1 per unit
   Size: 1 x 1 1/2-8, single-stage, vertically split casing
   Type: horizontal centrifugal pump
   Rating: 108 gal/min at 83 ft TDH, 150°F
   Motor: 10 hp at 3600 rpm

4. Hold Pump

   Qty.: 5 per unit
   Size: 1 1/2 x 2-9, single-stage, vertically split casing
   Type: horizontal centrifugal
   Rating: 70 gal/min at 60 ft TDH, 150°F
   Motor: 5 hp at 1800 rpm

TABLE 10.4.6-1 (SHEET 2 OF 3)

5. Precoat Pump

   Qty.: 1 per unit
   Size: 6 x 8-13
   Type: horizontal centrifugal pump
   Rating: 1680 gal/min of 5% slurry at 150°F and 100 ft TDH
   Motor: 60 hp at 1800 rpm

6. Backwash Pump

   Qty.: 1 per unit
   Size: 3 x 4-10
   Type: horizontal centrifugal
   Rating: 500 gal/min at 100 ft TDH
   Motor: 25 hp at 3550 rpm

7. Precoat Tank

   Dia.: 60 in. (x 5 ft high), lined with Plasite No. 7155 HHB

8. Overlay Tank

   Dia.: 36 in. (x 4 ft high), lined with Plasite No. 7155 HHB

9. Backwash Recovery Tank

   Dia.: 12 ft (x 10 ft high), lined with Plasite No. 7155 HHB

10. Overlay Pump

    Qty.: 1 per unit
    Size: 1 x 1 1/2-6
    Type: horizontal centrifugal
    Rating: 25 gal/min at 100 ft TDH, 10% resin slurry
    Motor: 5 hp at 3600 rpm

11. Resin Trap

    Qty.: 5 per unit
    Type: Basket, stainless steel

TABLE 10.4.6-1 (SHEET 3 OF 3)

12. Dirty Spent Resin Holding Tank

    Dia: 18 ft (x 10.5-ft high) lined with Plasite No. 7122

13. Clean Spent Resin Holding Tank

    Dia.: 18 ft (x 10.5-ft high) lined with Plasite No. 7122

14. Recirculation Pump

    Qty.: 2 shared
    Size: 3 x 4-13
    Type: horizontal centrifugal
    Rating: 360 gal/min at 118-ft TDH
    Motor: 20 hp at 1780 rpm

15. Dewatering Pump

    Qty.: 2 shared
    Size: 2 x 2
    Type: double diaphram, air powered
    Rating: 120 gal/min at 70-ft TDH for 100 psi air

16. Dewatering Filter

    Qty.: 4, shared
    Dia.: 14 1/8 in. (x 45 7/8-in. high)
    Design press.: 150 psi
    Internal filter elements: 10-micron cotton matrix
    Design flow: 120 gal/min

17. Pressure Filter Skid

    Qty.: 2 shared
    Pump rating: 120 gal/min at 70-ft TDH for 100 psi air

TABLE 10.4.6-2 (SHEET 1 OF 2)

IMPURITY CONCENTRATIONS IN INFLUENT CONDENSATE

| Impurity | | Concentration: Design Max. or Startup | Concentration: Expected Full-Load Operation at 1-gal/min Condenser Leak[a] |
|---|---|---|---|
| Iron (as Fe) | Soluble | 40 ppb | 4 ppb |
| | Insoluble | 1000 ppb[b] | 6 ppb |
| Copper (as Cu) | (Total) | Trace | Trace[c] |
| Chloride (as Cl) | Soluble | 10 ppb[d] | 1.7 ppb |
| Sodium (as Na) | Soluble | 10.6 ppb[d] | 2.5 ppb |
| Silica (as $SO_2$) | Soluble | 10 ppb | 0.4 ppb |
| Suspended solids | Insoluble | 1400 ppb[b] | Equivalent to the insoluble iron |
| Influent pH at 25°C | | 7 to 9.6 | 8.8 to 9.2 during leak (9.2 to 9.6 if operated full time) |
| Influent conductivity at 25°C | | ~10 μmho | 10 μmho at pH 9.6 |
| Influent cation conductivity at 25°C | | 0.5 μmho | 0.2 μmho |
| Radioactivity level (μ/Ci per ml condensate) | | <5.7 ($10^{-5}$) | Negligible |

a. The maximum condenser leak for allowable water quality maintenance that can be accommodated continuously at full-load operation with continuous recoating (no 2-h-lag allowance) and with a 0.1-gal/min steam generator leak of primary coolant is 61 gal/min.

b. This might be as much as 4000 ppb for several hours at initial plant startup. This value is used for equipment specification purposes, since without condenser tube leaks, sodium and chloride concentrations are much lower as a result of cleanup by the steam generator blowdown demineralizer. The maximum sodium and chloride input is that which occurs during a condenser tube leak. The maximum allowable concentration is determined by the rate at which the demineralizer units can be kept continuously in service without breakthrough.

TABLE 10.4.6-2 (SHEET 2 OF 2)

c. The secondary system does not use copper or copper nickel.

d. The design maximum influent impurity level is operationally accommodated by the powdered resin units at normal coating coverage. The time for exhaustion is not shorter than the time required to backwash, precoat, and apply resins to a unit, allowing 2 additional hours for these operations to start. The allowable influent impurity level depends on the pounds of resin, exchange capacity of the resin, backwash time, backwash handling time (1 h), and the circulating water analysis. Paragraph 10.4.6.2.4.3 gives a typical analysis of the condenser leakage circulating water. The maximum condenser leak under these conditions is 27 gal/min.

TABLE 10.4.6-3

WATER CHEMISTRY FOR REACTOR COOLANT SYSTEM

| Item | Value |
|---|---|
| Electrical conductivity | Determined by the concentration of boric acid and alkali present. Expected range is < 1 to 40 μmho/cm at 25°C. |
| Solution pH | Determined by the concentration of boric acid and alkali present. Expected values range between 4.2 (high boric acid concentration) to 10.5 (low boric acid concentration) at 25°C. |
| Oxygen | 0.1 ppm max. |
| Chloride | 0.15 ppm max. |
| Fluoride | 0.15 ppm max. |
| Hydrogen | 25 to 50 $cm^3$ (STP)/kg $H_2O$ |
| Total suspended solids | 1.0 ppm max. |
| pH control agent ($Li^7$/OH) | $0.3 \times 10^{-4}$ to $3.2 \times 10^{-4}$ molal strong-base alkali (equivalent to 0.22 to 2.2 ppm $Li^7$) |
| Boric acid | Variable from 0 to 4000 ppm boron |

TABLE 10.4.6-4 (SHEET 1 OF 2)

WATER CHEMISTRY FOR CIRCULATING WATER SYSTEM

| Constituent | Concentration |
|---|---|
| Hardness (ppm as $CaCO_3$) | 184 |
| Total dissolved solids (ppm) | 407 |
| Suspended matter (ppm) | Estimated to vary from 20-100 |
| Ammonia (ppm as N) | 0.22 |
| Total sulfides (ppm as S) | 0.0 |
| Silica (ppm as $SiO_2$) | 45.0 |
| Iron (ppm as Fe) | 1.8 |
| Manganese (ppm as Mn) | 0.0 |
| Calcium (ppm as Ca) | 39.0 |
| Magnesium (ppm as Mg) | 21.0 |
| Sodium (ppm as Na) | 43.8 |
| Potassium (ppm as K) | 11.4 |
| Bicarbonate (ppm as $HCO_3$) | 76.9 |
| Sulfate (ppm as $SO_4$) | 137.01[(a)] |
| Chloride (ppm as Cl) | 28.8 |
| Nitrate (ppm as $NO_3$) | 1.7 |
| Fluoride (ppm as F) | 0.2 |
| Free chlorine (intermittent ppm) | 0.2 |
| Phosphate (ppm as $PO_4$) | 0.5 |
| pH value | 7.0-7.5[(b)] |
| Free $CO_2$ (ppm) | 3 |

TABLE 10.4.6-4 (SHEET 2 OF 2)

a. This sulfate concentration results from the sulfuric acid treatment for scale control when adjusting the pH to the 7.0 to 7.5 range.

b. The VEGP will optimize the water stability, adjusting acid feed toward a Ryzner index in the range of 7.0 to 7.5. The pH will be higher (maximum 8.5), the sulfate lower, and the bicarbonates higher than shown.

TABLE 10.4.7-1

CONDENSATE AND FEEDWATER SYSTEM COMPONENT FAILURE ANALYSIS

| Component | Failure Effect on Train | Failure Effect on System | Failure Effect on RCS |
|---|---|---|---|
| Condensate pump | None. Condenser hotwells are interconnected into a common condensate pump suction header. | Operation continues at full capacity, using standby pumps. | None. |
| No. 1, 2, or 3 feedwater heater | One train of No. 1, 2, and 3 feedwater heaters is isolated. Remaining trains continue to operate. | Operation continues at reduced capacity, using parallel feedwater heater train and bypass line. Load must not exceed 91 percent of VWO rating to protect the low-pressure turbines from excessive exhaust flow. | None. No. 4 feedwater heater will maintain normal outlet feedwater temperature under this condition. |
| Heater drain tank | Extraction steam to corresponding No. 4 feedwater heater must be isolated. Drains from respective No. 6 and 5 feedwater heaters are dumped to condenser. | Operation continues at reduced capacity. | RCS reduces reactor power to compensate for reduced feedwater temperature. |
| Heater drain pump | None. Parallel pump with condensate pumps have sufficient capacity to handle full load. | No. 6, 5, and 4 heater drains of affected train are dumped to condenser. | RCS reduces reactor power to compensate for reduced feedwater temperature. |
| No. 4 or 5 feedwater heater | One train is isolated. | Condensate and feedwater system operation continues at full capacity, using parallel train and bypass line. | RCS reduces reactor and generator output power to compensate for reduced feedwater temperature. |
| Steam generator feedwater pump | None. Two parallel trains are interconnected. | Operations may continue at reduced capacity, using parallel pump if the reactor does not trip. | RCS reduces reactor power to compensate for reduced feedwater flow. |
| No. 6 feedwater heater | One train is isolated. | Condensate and feedwater system operation continues at full capacity, using parallel train and bypass line. | RCS reduces reactor and generator output power to compensate for reduced feedwater temperature. |

TABLE 10.4.7-2 (SHEET 1 OF 2)

CONDENSATE AND FEEDWATER SYSTEM
DESIGN DATA FOR MAJOR COMPONENTS

| Component | Data |
|---|---|
| **Condensate Pump** | |
| Type | Vertical, 34APKD - 8-stage |
| Driver | 4500 hp, 1180 rpm motor |
| Quantity | Three per unit |
| Capacity | 10,640 gal/min at 1250 ft head |
| **Condenser** | |
| Type | Two-pass, single-pressure, vertically-divided surface condenser |
| Size | 825,000 $ft^2$ condensing surface in three shells |
| Tube Sheet | Aluminum bronze |
| Tubes | 1-in. outside diameter - 22 BWG and 18 BWG titanium |
| Design Pressure | Water box - 80 psig; shell - 30 psig and vacuum |
| **Feedwater Heaters (Low Pressure)** | |
| Type | Horizontal, U-tube, shell and tube |
| Quantity | First through third stage - three each mounted in condenser neck; fourth and fifth stage - two each |
| Tubes | 20 BWG - AW stainless steel |
| Shell | Carbon steel |

| Heater No. | Tube Design Pressure/Temperature | Shell Design Pressure/temperature |
|---|---|---|
| 1 | 700 psig/300°F | 50 psig and vacuum/300°F |
| 2 | 700 psig/335°F | 50 psig and vacuum/370°F |
| 3 | 700 psig/375°F | 100 psig and vacuum/460°F |
| 4 | 700 psig/410°F | 225 psig and vacuum/410°F |
| 5 | 700 psig/450°F | 350 psig and vacuum/450°F |

TABLE 10.4.7-2 (SHEET 2 OF 2)

| Component | Data |
|---|---|
| Feedwater Heater (High Pressure) | |
| Type | Vertical head down, U-tube, shell and tube |
| Quantity | Two - sixth stage |
| Tubes | 18 BWG - AW stainless steel |
| Shell | Carbon steel |

| Heater No. | Tube Design Pressure/Temperature | Shell Design Pressure/temperature |
|---|---|---|
| 6 | 2000 psig/490°F | 525 psig/490°F |

| | |
|---|---|
| Feedwater Pump | |
| Type | Single-stage, centrifugal 20 x 17 HVF, variable speed |
| Driver | 15170 BHP steam drive turbine |
| Quantity | Two per unit |
| Capacity | 20,400 gal/min at 3000 ft NTDH at 5950 rpm |

TABLE 10.4.7-3

CONDENSATE AND FEEDWATER SYSTEM CONTROLS AND INDICATION

| Control/Indication | Control Room | Local |
|---|---|---|
| Steam generator feedwater control valves | X | |
| Steam generator startup control valves | X | |
| Steam generator feedwater isolation valves | X | |
| Hotwell temperature | X (computer) | X |
| Hotwell level | X | X |
| Condensate pump discharge temperature | X | X |
| Condensate pump discharge pressure | X | X |
| Condensate pump recirculation pressure | | X |
| Condensate to steam jet air ejector (SJAE) condenser temperature | X (computer) | X |
| SJAE condenser outlet temperature | | X |
| Steam packing exhauster condenser outlet temperature | | X |
| Condensate to heater 1 flowrate | X | |
| Heater 1 condensate inlet temperature | X (computer) | X |
| Heater 2 condensate inlet temperature | X (computer) | X |
| Heater 3 condensate inlet temperature | X (computer) | X |
| Heater 3 condensate outlet temperature | X (computer) | X |
| Heater 4 condensate inlet temperature | X (computer) | X |
| Heater 4 condensate outlet temperature | X (computer) | X |
| Heater 5 condensate outlet temperature | X (computer) | X |
| Steam generator feedwater pump condensate inlet flowrate | X | |
| Steam generator feedwater pump discharge pressure | X | X |
| Heater 6 discharge header temperature | X | |
| Steam generator feedwater flow | X | |
| Feedwater bypass flow | | X |

TABLE 10.4.8-1

PARAMETERS USED TO CALCULATE DESIGN BASIS SGBPS SOURCES

| | |
|---|---|
| Steam generator type | Model F |
| Total secondary side water mass (lb) | $4.7 \times 10^5$ |
| Steam generator steam fraction | 0.0548 |
| Total steam flowrate (lb/h) | $1.6 \times 10^7$ |
| Moisture carryover (percent) | 0.25 |
| Total makeup water feedrate (lb/h) | $9.9 \times 10^2$ |
| Total blowdown rate (gal/min) | 108 |
| Total primary-to-secondary leakrate (gal/min) | 1.0 |
| Iodine partition factor (mass basis) | 100 |

TABLE 10.4.8-2

STEAM GENERATOR SECONDARY SIDE WATER SPECIFIC ACTIVITY

| Nuclide | Activity (μCi/g) | Nuclide | Activity (μCi/g) |
|---|---|---|---|
| Br-83 | $2.5 \times 10^{-4}$ | Y-90 | $5.3 \times 10^{-7}$ |
| Br-84 | $3.5 \times 10^{-5}$ | Y-91m | $2.9 \times 10^{-5}$ |
| Br-85 | $4.3 \times 10^{-7}$ | Y-91 | $6.1 \times 10^{-6}$ |
| I-129 | $4.2 \times 10^{-10}$ | Y-92 | $4.0 \times 10^{-6}$ |
| I-130 | $1.3 \times 10^{-4}$ | Y-93 | $2.3 \times 10^{-6}$ |
| I-131 | $2.7 \times 10^{-2}$ | Zr-95 | $6.6 \times 10^{-6}$ |
| I-132 | $9.3 \times 10^{-3}$ | Nb-95 | $6.6 \times 10^{-6}$ |
| I-133 | $3.1 \times 10^{-2}$ | Mo-99 | $6.9 \times 10^{-3}$ |
| I-134 | $7.0 \times 10^{-4}$ | Tc-99m | $6.5 \times 10^{-3}$ |
| I-135 | $1.1 \times 10^{-2}$ | Ru-103 | $5.8 \times 10^{-6}$ |
| Rb-86 | $3.9 \times 10^{-4}$ | Ru-106 | $1.4 \times 10^{-6}$ |
| Rb-88 | $2.1 \times 10^{-3}$ | Rh-103m | $5.8 \times 10^{-6}$ |
| Rb-89 | $8.0 \times 10^{-5}$ | Rh-106 | $1.4 \times 10^{-6}$ |
| Cs-134 | $4.1 \times 10^{-2}$ | Ag-110m | $1.4 \times 10^{-5}$ |
| Cs-136 | $5.2 \times 10^{-2}$ | Te-125m | $2.8 \times 10^{-6}$ |
| Cs-137 | $2.7 \times 10^{-2}$ | Te-127m | $3.0 \times 10^{-5}$ |
| Ba-137m | $2.5 \times 10^{-2}$ | Te-127 | $8.1 \times 10^{-5}$ |
| Cs-138 | $7.5 \times 10^{-4}$ | Te-129m | $1.9 \times 10^{-4}$ |
| H-3 | 1.8 | Te-129 | $1.3 \times 10^{-4}$ |
| Cr-51 | $5.6 \times 10^{-5}$ | Te-131m | $2.2 \times 10^{-4}$ |
| Mn-54 | $4.5 \times 10^{-6}$ | Te-131 | $4.4 \times 10^{-5}$ |
| Mn-56 | $5.7 \times 10^{-5}$ | Te-132 | $2.8 \times 10^{-3}$ |
| Fe-55 | $2.0 \times 10^{-5}$ | Te-134 | $2.9 \times 10^{-5}$ |
| Fe-59 | $5.3 \times 10^{-6}$ | Ba-140 | $4.2 \times 10^{-5}$ |
| Co-58 | $1.5 \times 10^{-4}$ | La-140 | $1.8 \times 10^{-5}$ |
| Co-60 | $1.9 \times 10^{-5}$ | Ce-141 | $6.4 \times 10^{-6}$ |
| Sr-89 | $7.8 \times 10^{-5}$ | Ce-143 | $4.3 \times 10^{-6}$ |
| Sr-90 | $2.2 \times 10^{-6}$ | Ce-144 | $3.9 \times 10^{-6}$ |
| Sr-91 | $5.0 \times 10^{-5}$ | Pr-143 | $6.4 \times 10^{-6}$ |
| Sr-92 | $4.3 \times 10^{-6}$ | Pr-144 | $3.9 \times 10^{-6}$ |

TABLE 10.4.8-3

STEAM GENERATOR BLOWDOWN PROCESSING SYSTEM PARAMETERS

| | |
|---|---|
| Design blowdown flowrate, per steam generator (gal/min) | 90 |
| Normal temperature of blowdown entering system (°F) | 539-567 |
| Normal temperature of blowdown leaving system (°F) | 130 |
| Normal temperature of condensate water entering system (°F) | 100 |
| Normal temperature of condensate water leaving system (°F) | 350 |
| Design pressure of high-pressure section (psig) | 1185 |
| Design temperature of high-pressure section (°F) | 600 |
| Design pressure of low-pressure section (psig) | 300 |
| Design temperature of low-pressure section (°F) | 200 |
| Design pressure of spent resin sluice section (psig) | 150/220 |
| Design temperature of spent resin sluice section (°F) | 200 |
| Design pressure of condensate section (psig) | 600 |
| Design temperature of condensate section (°F) | 400 |

TABLE 10.4.8-4 (SHEET 1 OF 3)

STEAM GENERATOR BLOWDOWN PROCESSING SYSTEM
MAJOR COMPONENT PARAMETERS

Steam Generator Blowdown Heat Exchanger

| | |
|---|---|
| Number | 4 |
| Design heat transfer rate (Btu/h) | $2.22 \times 10^7$ |

| | Process Side | Coolant Side |
|---|---|---|
| Design pressure (psig) | 1300 | 600 |
| Design temperature (°F) | 600 | 400 |
| Design flow (lb/h) | 4000 (steam)/ 41,000 (liquid) | 88,000 |
| Inlet temperature (°F) | 539 | 100 |
| Outlet temperature (°F) | 130 | 350 |
| Fluid | Blowdown | Condensate |
| Material | | |
| Tubes | Stainless steel | |
| Shell | Carbon steel | |

Trim Heat Exchanger

| | |
|---|---|
| Number | 1 |
| Design heat transfer rate (Btu/h) | $5.4 \times 10^6$ |

| | Process Side | Coolant Side |
|---|---|---|
| Design pressure (psig) | 1300 | 150 |
| Design temperature (°F) | 250 | 250 |
| Design flow (gal/min) | 360 | 540 |
| Inlet temperature (°F) | 160 | 95 |
| Outlet temperature (°F) | 130 | 115 |
| Fluid | Blowdown | Turbine plant cooling water |
| Material | | |
| Tubes | Stainless steel | |
| Shell | Stainless steel | |

TABLE 10.4.8-4 (SHEET 2 OF 3)

Steam Generator Blowdown Prefilter

| | |
|---|---|
| Number | 1 |
| Type | Backflusable |
| Design pressure (psig) | 375 |
| Design temperature (°F) | 200 |
| Design flow (gal/min) | 360 |
| Particle retention | 98 percent of 2-mm size |
| Material (vessel) | Stainless steel |

Steam Generator Blowdown Cartridge Filter

| | |
|---|---|
| Number | 1 |
| Type | Cartridge |
| Design pressure (psig) | 300 |
| Design temperature (°F) | 150 |
| Design flow (gal/min) | 360 |
| Particle retention | 99 percent of 2-mm size |
| Material (vessel) | Carbon steel |

Steam Generator Blowdown Outlet Filters

| | |
|---|---|
| Number | 2 |
| Type | Cartridge |
| Design pressure (psig) | 300 |
| Design temperature (°F) | 250 |
| Design flow (gal/min) | 250 |
| Particle retention | 98 percent of 25-mm size |
| Material (vessel) | Stainless steel |

Steam Generator Blowdown Demineralizers

| | |
|---|---|
| Number | 2 |
| Design pressure (psig) | 300 |
| Design temperature (°F) | 250 |
| Design flow (gal/min) | 360 |
| Resin volume ($ft^3$) | 75 |
| Material (vessel) | Stainless steel |

TABLE 10.4.8-4 (SHEET 3 OF 3)

Steam Generator Blowdown Spent Resin Storage Tank

| | |
|---|---|
| Number | 1 |
| Capacity ($ft^3$) | 550 |
| Design pressure (psig) | 150 |
| Design temperature (°F) | 200 |
| Material | Stainless steel |

Steam Generator Blowdown Spent Resin Sluice Pump

| | |
|---|---|
| Number | 1 |
| Design pressure (psig) | 220 |
| Design temperature (°F) | 200 |
| Design flow (gal/min) | 110 |
| Design head (ft) | 165 |
| Material | Stainless steel |
| Motor horsepower | 15 |

Steam Generator Blowdown Spent Resin Sluice Filter

| | |
|---|---|
| Number | 1 |
| Design pressure (psig) | 300 |
| Design temperature (°F) | 250 |
| Design flow (gal/min) | 250 |
| Particle retention | 98 percent of 25-mm size |
| Material (vessel) | Stainless steel |

Steam Generator Drain Pump

| | |
|---|---|
| Number | 1 |
| Design pressure (psig) | 255 |
| Design temperature (°F) | 140 |
| Design flow (gal/min) | 200 |
| Design head (ft) | 265 |
| Material | Stainless steel |
| Rotational speed (rev/min) | 3600 |
| Motor horsepower | 40 |

TABLE 10.4.9-1 (SHEET 1 OF 2)

AUXILIARY FEEDWATER SYSTEM COMPONENT DATA

Motor-Driven Auxiliary Feedwater Pump (per pump)

| | |
|---|---|
| Quantity | 2 |
| Type | Horizontal, centrifugal, multistage, split case with packing or mechanical seals |
| Capacity, each (gal/min) | 630 |
| Minimum flow - design | 175 |
| - allowable | 150 |
| TDH (ft) - | 3500 |
| NPSH required, maximum (ft) | 19 |
| NPSH available, minimum (ft) | 30 |

Material

| | |
|---|---|
| Case | Cast Steel |
| Impellers | Cast steel |
| Shaft | Forged steel |
| Design code | ASME Section III, Class 3 |
| Seismic design | Category 1 |

Driver

| | |
|---|---|
| Type | Electric motor |
| Horsepower (hp) | 900 |
| Rpm | 3600 |
| Power supply | 4160 V, 60 Hz, 3-phase, Class 1E |
| Source of power | Train A 1AA02<br>Train B 1BA03 |
| Design code | National Electrical Manufacturers Association (NEMA) |
| Seismic design | Category 1 |

Turbine-Driven Auxiliary Feedwater Pump

| | |
|---|---|
| Quantity | 1 |
| Type | Horizontal, centrifugal, multistage, split case with packing or mechanical seals |
| Capacity (gal/min) | 1175 |
| Minimum flow (gal/min) | 145 |
| TDH (ft) | 3500 |
| NPSH required, maximum (ft) | 19 |
| NPSH available, minimum (ft) | 30 |

TABLE 10.4.9-1 (SHEET 2 OF 2)

| | |
|---|---|
| **Material** | |
| Case | Cast Steel |
| Impellers | Cast steel |
| Shaft | Forged steel |
| Design code | ASME Section III, Class 3 |
| Seismic Design | Category 1 |
| **Driver** | |
| Type | Noncondensing, single stage, mechanical-drive steam turbine |
| Rpm | 4200 |
| Horsepower (hp) | 1590 |
| Design code | NEMA |
| Seismic design | Category 1 |
| **Motor-Driven Pump Isolation/Flow Control Valves** | |
| Quantity | 4 (2 per pump) |
| Type | ac-powered, motor-operated globe valve |
| Size (in.) | 4 |
| Cv | 145 |
| Design pressure (psig) | 1800 |
| Design temperature (°F) | 120 |
| Material | Carbon steel |
| Design code | ASME Section III, Class 2 |
| Seismic design | Category 1 |
| **Turbine-Driven Pump Isolation/Flow Control Valves** | |
| Quantity | 4 |
| Type | dc-powered, motor-operated globe valve |
| Size (in.) | 4 |
| Cv | 145 |
| Design pressure (psig) | 1975 |
| Design temperature (°F) | 120 |
| Material | Carbon steel |
| Design code | ASME Section III, Class 2 |
| Seismic design | Category 1 |

TABLE 10.4.9-2

NSSS STEAM GENERATOR MAKEUP REQUIREMENTS

| | Flow Requirements (gal/min) | | | |
|---|---|---|---|---|
| Power Level (MWt) | Main Feedline Rupture | Loss of Normal Feedwater | Plant Cooldown | HELB in Turbine-Driven Pump System |
| 3579 | 510 | 510 | 470 | 510 |

TABLE 10.4.9-3 (SHEET 1 OF 4)

UNIT 1 AUXILIARY FEEDWATER FLOW INFORMATION FOR VARIOUS POSTULATED EVENTS

| | Accident Description | | | | Flow to the Intact Steam Generator (gal/min) | | | | | AFW Flow Information | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | Pressure (psia) | | | | | | | | | Pump Discharge Flow (gal/min) | | | Pump Recirculation Flow (gal/min) | | |
| Case | Faulted SG | Intact SG | Temp. Water (°F) | Number of Pumps in Operation | SG1 | SG2 | SG3 | Total | Flow to Faulted SG (gal/min) | Turbine-Driven Pump | Motor-Driven Pump A | Motor-Driven Pump B | Turbine-Driven Pump | Motor-Driven Pump A | Motor-Driven Pump B |
| Section 1.1[a] | | | | | | | | | | | | | | | |
| Main steam line rupture (hot shutdown) | 14.7 | 1200 | 32 | All three pumps are running | 347 | 484 | 482 | 1313 | 966 | 1095 | 750 | 586 | 151.7 | 0[b] | 0[b] |
| Section 1.2 | | | | | | | | | | | | | | | |
| Main steam system failure (depressurization of secondary side) | 400 | 1107 | 32 | Same as above | 443 | 531 | 528 | 1502 | 825 | 1111 | 734 | 634 | 151.6 | 0[b] | 0[b] |
| Section 1.3-1 | | | | | | | | | | | | | | | |
| Main feedwater line rupture[c] | 14.7 | 1250 | 120 | Turbine-driven pump and the motor-driven pump A are running | 317 | 183 | 183 | 683 | 972 | 1071 | 736 | 0 | 152 | 0[b] | 0 |
| Section 1.3-2 | | | | | | | | | | | | | | | |
| Main feedwater line rupture[c] | 14.7 | 1250 | 120 | Only the two motor-driven pumps are running | 148 | 284 | 282 | 714 | 596 | 0 | 745 | 566 | 0 | 0[b] | 0[b] |
| Section 1.3-3[a] | | | | | | | | | | | | | | | |
| Main feedwater line rupture[c] | 14.7 | 1250 | 120 | Only the turbine-driven pump is running | 167 | 182 | 182 | 531 | 398 | 1081 | 0 | 0 | 151.9 | 0 | 0 |

TABLE 10.4.9-3 (SHEET 2 OF 4)

| Case | Accident Description: Pressure (psia): Faulted SG | Accident Description: Pressure (psia): Intact SG | Accident Description: Temp. Water (°F) | Accident Description: Number of Pumps in Operation | Flow to the Steam Generator (gal/min): Intact SG1 | Flow to the Steam Generator (gal/min): Intact SG2 | Flow to the Steam Generator (gal/min): Intact SG3 | Flow to the Steam Generator (gal/min): Intact SG4 | Total Flow to SG (ga//min) | AFW Flow Information: Pump Discharge Flow (gal/min): Turbine-Driven Pump | AFW Flow Information: Pump Discharge Flow (gal/min): Motor-Driven Pump A | AFW Flow Information: Pump Discharge Flow (gal/min): Motor-Driven Pump B | AFW Flow Information: Pump Recirculation Flow (gal/min): Turbine-Driven Pump | AFW Flow Information: Pump Recirculation Flow (gal/min): Motor-Driven Pump A | AFW Flow Information: Pump Recirculation Flow (gal/min): Motor-Driven Pump B |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Section 1.4 A[d] | NA | 1250 | 120 | Turbine-driven pump and the motor-driven pump A are running | 456 | 193 | 192 | 457 | 1298 | 904 | 547 | 0 | 153.6 | 0 | 0 |
| Section 1.4 B[d] | NA | 1250 | 120 | Only motor-driven pump A running | 277 | 0 | 0 | 277 | 554 | 0 | 554 | 0 | 0 | 0 | 0 |
| Section 1.4 C[d] | NA | 1250 | 120 | Only motor-driven pump B running | 0 | 284 | 282 | 0 | 566 | 0 | 0 | 566 | 0 | 0 | 0 |
| Section 1.5 C[e] | NA | 1250 | 120 | Only turbine-driven pump running | 187 | 192 | 192 | 187 | 758 | 912 | 0 | 0 | 153.5 | 0 | 0 |

a. These bounding cases were recalculated based on orifice field test data taken during startup.

b. As shown in the description columns (see fifth column), three different cases were considered.

c. The motor-driven valves in the motor-driven pump recirculation lines are intended to close within a minute when discharging to the main discharge line and flow is above the minimum required. Thus, the motor-driven pump circulation flow was not considered.

d. Sections 1.4A, 1.4B, and 1.4C are for accident case involving loss of normal feedwater with loss of offsite power and with different pumps operating.

e. Section 1.5 is accident case involving station blackout with loss of normal feedwater and with only the turbine-driven pump operating.

TABLE 10.4.9-3 (SHEET 3 OF 4)

UNIT 2 AUXILIARY FEEDWATER FLOW INFORMATION FOR VARIOUS POSTULATED EVENTS

| | Accident Description | | | | Flow to the Intact Steam Generator (gal/min) | | | | | AFW Flow Information | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | Pressure (psia) | | | | | | | | | Pump Discharge Flow (gal/min) | | | Pump Recirculation Flow (gal/min) | | |
| Case | Faulted SG | Intact SG | Temp. Water (°F) | Number of Pumps in Operation | SG1 | SG2 | SG3 | Total | Flow to Faulted SG (gal/min) | Turbine-Driven Pump | Motor-Driven Pump A | Motor-Driven Pump B | Turbine-Driven Pump | Motor-Driven Pump A | Motor-Driven Pump B |
| Section 1.1[a] | | | | | | | | | | | | | | | |
| Main steam line rupture (hot shutdown) | 14.7 | 1200 | 32 | All three pumps are running | 349 | 486 | 484 | 1319 | 967 | 1106 | 749 | 585 | 152.3 | 0[b] | 0[b] |
| Section 1.2 | | | | | | | | | | | | | | | |
| Main steam system failure (depressurization of secondary side) | 400 | 1107 | 32 | Same as above | 446 | 535 | 532 | 1513 | 827 | 1122 | 733 | 637 | 152.2 | 0[b] | 0[b] |
| Section 1.3-1 | | | | | | | | | | | | | | | |
| Main feedwater line rupture[c] | 14.7 | 1250 | 120 | Turbine-driven pump and the motor-driven pump A are running | 318 | 186 | 186 | 690 | 973 | 1081 | 734 | 0 | 152.5 | 0[b] | 0 |
| Section 1.3-2 | | | | | | | | | | | | | | | |
| Main feedwater line rupture[c] | 14.7 | 1250 | 120 | Only the two motor-driven pumps are running | 147 | 282 | 281 | 710 | 596 | 0 | 743 | 563 | 0 | 0[b] | 0[b] |
| Section 1.3-3[a] | | | | | | | | | | | | | | | |
| Main feedwater line rupture[c] | 14.7 | 1250 | 120 | Only the turbine-driven pump is running | 170 | 185 | 185 | 540 | 399 | 1092 | 0 | 0 | 152.4 | 0 | 0 |

TABLE 10.4.9-3 (SHEET 4 OF 4)

| Case | Accident Description: Pressure (psia): Faulted SG | Accident Description: Pressure (psia): Intact SG | Accident Description: Temp. Water (°F) | Number of Pumps in Operation | Flow to the Steam Generator (gal/min): Intact SG1 | Flow to the Steam Generator (gal/min): Intact SG2 | Flow to the Steam Generator (gal/min): Intact SG3 | Flow to the Steam Generator (gal/min): Intact SG4 | Total Flow to SG (ga//min) | AFW Flow Information: Pump Discharge Flow (gal/min): Turbine-Driven Pump | AFW Flow Information: Pump Discharge Flow (gal/min): Motor-Driven Pump A | AFW Flow Information: Pump Discharge Flow (gal/min): Motor-Driven Pump B | AFW Flow Information: Pump Recirculation Flow (gal/min): Turbine-Driven Pump | AFW Flow Information: Pump Recirculation Flow (gal/min): Motor-Driven Pump A | AFW Flow Information: Pump Recirculation Flow (gal/min): Motor-Driven Pump B |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Section 1.4 A[d] | NA | 1250 | 120 | Turbine-driven pump and the motor-driven pump A are running | 448 | 193 | 193 | 448 | 1282 | 907 | 529 | 0 | 153.7 | 0 | 0 |
| Section 1.4 B[d] | NA | 1250 | 120 | Only motor-driven pump A running | 268 | 0 | 0 | 268 | 536 | 0 | 536 | 0 | 0 | 0 | 0 |
| Section 1.4 C[d] | NA | 1250 | 120 | Only motor-driven pump B running | 0 | 282 | 281 | 0 | 563 | 0 | 0 | 563 | 0 | 0 | 0 |
| Section 1.5 C[e] | NA | 1250 | 120 | Only turbine-driven pump running | 188 | 193 | 192 | 188 | 761 | 914 | 0 | 0 | 153.6 | 0 | 0 |

d. These bounding cases were recalculated based on orifice field test data taken during startup.

e. As shown in the description columns (see fifth column), three different cases were considered.

f. The motor-driven valves in the motor-driven pump recirculation lines are intended to close within a minute when discharging to the main discharge line and flow is above the minimum required. Thus, the motor-driven pump circulation flow was not considered.

d. Sections 1.4A, 1.4B, and 1.4C are for accident case involving loss of normal feedwater with loss of offsite power and with different pumps operating.

e. Section 1.5 is accident case involving station blackout with loss of normal feedwater and with only the turbine-driven pump operating.

TABLE 10.4.9-4 (SHEET 1 OF 31)

AUXILIARY FEEDWATER SYSTEM FAILURE MODES AND EFFECTS ANALYSIS

| Item No. | Description of Component[a] | Safety Function | Plant Operating Mode[b] | Failure Mode(s) | Method of Failure Detection[c] | Failure Effect on System Safety Function Capability | General Remarks |
|---|---|---|---|---|---|---|---|
| 1 | Condensate tank makeup valve LV-5158, normally closed, fail closed air-operated valve | Maintains inventory or condensate storage tank V4-001 at nominal level of 480,000 ± 22,750 gal; also prevents inventory loss post-SSE | All | 1A. Fails closed or fails to open upon command | 1A. Position lights on QPCP coincident with low water level alarm LSL-5146 on QMCB | 1A. None - One tank contains sufficient reserve without makeup to satisfy Regulatory Guide 1.139 with regard to 4 h at hot standby followed by a 5-h cooldown to RHR cut-in at RCS temperature of 350°F. | Total capacity of both tanks is 960,000 ± 45,500 gal. |
| | | | | 1B. Fails open or fails to close upon command | 1B. Position lights on QPCP coincident with high water level alarm LSH-5146 on QMCB | 1B. None - Tank may overflow, but surrounding dike holds more than 30-min makeup at maximum flow thus permitting operator corrective action. | |
| 2 | Condensate tank makeup valve LV-5162, normally closed, fail closed air-operated valve | Same as item 1, except tank is V4-002 | All | Same as item 1 | Same as item 1, except water level alarms are LSL-5147 and LSH-5147 on QMCB | None - Same as item 1 | See item 1. |
| 3 | Degasifier feed pump suction valve HV-5087, normally open, fail closed, air-operated valve | Isolates condensate storage tank V4-001 from degasifier manually or upon low pump suction pressure | All | 3A. Fails closed or fails to open upon command | 3A. Position lights on QPCP | 3A. None - Degasification not required for safe shutdown. Also, tank diaphragm limits oxygen buildup. | The degasifier is used periodically to control concentration of dissolved oxygen in the tanks to reduce potential for corrosion of the steam generators and the AFW piping (see also item 1). |

TABLE 10.4.9-4 (SHEET 2 OF 31)

| Item No. | Description of Component[a] | Safety Function | Plant Operating Mode[b] | Failure Mode(s) | Method of Failure Detection[c] | Failure Effect on System Safety Function Capability | General Remarks |
|---|---|---|---|---|---|---|---|
| | | | | 3B. Fails open or fails to close upon command | 3B. Position lights on QPCP coincident with low pump suction pressure alarm PSL-5087 on QPCP | 3B. None - Possible loss of tank inventory, but connection located above minimum level required for AFW reserve. | |
| 4 | Degasifier feed pump suction valve HV-5088, normally open, fail closed, air-operated valve | Same as item 3, except for condensate storage tank V4-002 | All | Same as item 3 | Same as item 3, except low suction pressure alarm is PSL-5088 on QPCP | None - Same as item 3 | See item 3. |
| 5 | Not used | | | | | | |
| 6 | Not used | | | | | | |
| 7 | AFW pump suction valve HV-5119, normally closed MOV, train A | Admits water from condensate storage tank V4-002 to suction of motor-driven AFW pump P4-003 upon depletion of condensate storage tank V4-001 (valve actuated by remote-manual operation from either the control room or shutdown panel PSDA) | 7A. Modes 1 and 2 | 7A1. Fails closed or fails to open upon command with condensate storage tank V4-001 depleted | 7A1. Position lights on PSDA or QMCB, plus low suction pressure (PI-5129A on QMCB or PI-5129B on PSDA), plus open indication for pump miniflow valve FV-5155 on QMCB and PSDA due to low pump flow | 7A1. None - AFW pumps P4-001 and P4-002 available to satisfy AFW requirements taking suction from tank V4-002 via HV-5113 and HV-5118, respectively. (Tank V4-001 presumed depleted. See item 21.) | For discussion of the ability of the AFW pumps to satisfy system requirements during various failure modes, see items 10, 11, and 12. Irrespective of source, pump miniflow returned to tank V4-002 (see also item 2). For the effects of loss of all ac power, see item 67. |
| | | | | 7A2. Spurious opening while tank V4-001 not depleted and pump running | 7A2. Position lights on PSDA or QMCB and reduced tank V4-002 inventory (LI-5116A and LI-5104 on QMCB, LI-5116B on PSDA, and LI-5104A on PSDB) | 7A2. None - Both tanks supplying AFW simultaneously but total consumption not affected. Also, makeup (see item 1) presumed available during modes 1 and 2. | |

TABLE 10.4.9-4 (SHEET 3 OF 31)

| Item No. | Description of Component[a] | Safety Function | Plant Operating Mode[b] | Failure Mode(s) | Method of Failure Detection[c] | Failure Effect on System Safety Function Capability | General Remarks |
|---|---|---|---|---|---|---|---|
| | | | 7B. Modes 3 and 4 | 7B1. Same as item 7A1 | 7B1. Same as item 7A1 7 | 7B1. None - Same as 7A1 | |
| | | | | 7B2. Same as item 7A2 | 7B2. Same as item 7A2 | 7B2. None - Same as item 7A2, except makeup not operable, but not required (see item 1A) | |
| | | | 7C. Mode 6 | 7C1. Same as item 7A1 | 7C1. Same as item 7A1 | 7C1. None - Same as item 7A1 | |
| | | | | 7C2. Same as item 7A2 | 7C2. Same as item 7A1 | 7C2. None - Same as item 7B2 | |
| | | | 7D. Mode 7 | 7D. Spurious opening | 7D. Position indicator on QMCB or PSDA | 7D. None - No loss of tank V4-002 inventory, since pump P4-003 is not operating. | |
| | | | 7E. Mode 5 | 7E1. Same as item 7A1 | 7E1. Same as item 7A1 | 7E1. None - Same as item 7A1, except only pump P4-002 available, but this is sufficient for safe shutdown. | |
| | | | | 7E2. Same as item 7A2 | 7E2. Same as item 7A2 | 7E2. None - Same as item 7B2 | |
| 8 | AFW pump suction valve HV-5118, normally closed MOV, train B | Same as item 7, except to suction of motor-driven AFW pump P4-002 and shutdown panel is PSDB | 8A. Modes 1 and 2 | 8A1. Same as item 7A1 | 8A1. Same as item 7A1, except pressure indicators are PI-5128A on QMCB or PI-5128B on PSDB, and miniflow valve FV-5154 position indicated on QMCB and PSDB. | 8A1. None - AFW pumps P4-001 and P4-003 available to satisfy AFW requirements taking suction from tank V4-002 via HV-5113 and HV-5119, respectively. | Same as item 7 |
| | | | | 8A2. Same as item 7A2 | 8A2. Same as item 7A2 | 8A2. None - Same as item 7A2 | |

TABLE 10.4.9-4 (SHEET 4 OF 31)

| Item No. | Description of Component[a] | Safety Function | Plant Operating Mode[b] | Failure Mode(s) | Method of Failure Detection[c] | Failure Effect on System Safety Function Capability | General Remarks |
|---|---|---|---|---|---|---|---|
| | | | 8B. Modes 3 and 4 | 8B. Same as items 7A1 and 7A2 | 8B. Same as items 8A1 and 8A2, respectively | 8B. None - Same as items 7B1 and 7B2, respectively | |
| | | | 8C. Mode 6 | 8C. Same as items 7A1 and 7A2 | 8C. Same as items 8A1 and 8A2, respectively | 8C. None - Same as items 7C1 and 7B2, respectively | |
| | | | 8D. Mode 7 | 8D. Spurious opening | 8D. Position lights on QMCB or PSDB | 8D. None - No loss of tank V4-002 inventory, since pump P4-002 is not operating. | |
| | | | 8E. Mode 5 | 8E1. Same as item 7A1 | 8E1. Same as item 8A1 | 8E1. None - Same as item 8A1, except only pump P4-003 available, but this is sufficient for safe shutdown. | |
| | | | | 8E2. Same as item 7A2 | 8E2. Same as item 7A2 | 8E2. None - Same as item 7B2 | |
| 9 | AFW suction valve HV-5113, normally closed, dc-operated MOV, train C | Same as item 7, except to suction of turbine-driven AFW pump P4-001, and controlled from the control room and AFW panel PAFP | 9A. Modes 3 and 4 | 9A1. Fails closed or fails to open upon command with condensate storage tank V4-001 depleted | 9A1. Position lights on QMCB or PAFP, plus low pump suction pressure PI-5110A or PI-5110B on QMCB or PAFP, respectively | 9A1. None - AFW pumps P4-002 and P4-003 available to satisfy AFW requirements taking suction from tank V4-002 via HV-5118 and HV-5119, respectively. | Turbine-driven AFW pump not ordinarily used during modes 1 and 2. (For discussion of the ability of the AFW pumps to satisfy system requirements during various failure modes, see items 10, 11, and 12.) The condensate storage facility is sized to accommodate 30-min spillage at turbine-driven pump (TDP) runout to allow for operator action to isolate the turbine and/or the TDP (for mode 5, |
| | | | | 9A2. Spurious opening while tank V4-001 not depleted and TDP pump running | 9A2. Same as item 7A2, except position indicator on QMCB or PAFP | 9A2. None - Both tanks supplying AFW simultaneously, but total consumption not effected. Makeup not operating but not required. | |

TABLE 10.4.9-4 (SHEET 5 OF 31)

| Item No. | Description of Component[a] | Safety Function | Plant Operating Mode[b] | Failure Mode(s) | Method of Failure Detection[c] | Failure Effect on System Safety Function Capability | General Remarks |
|---|---|---|---|---|---|---|---|
| | | | | | | | see item 9B). For the TDP, irrespective of source, the miniflow is returned to tank V4-001 via fixed flow orifice FO-5109. |
| | | | 9B. Mode 5 | 9B. Fails open or fails to close upon command | 9B. Same as item 9A2 | 9B. None - Manually operated valve HV-5097 can be closed to isolate pump, and HV-3009 (item 14) and HV-3019 (item 15) closed to stop turbine. | |
| | | | 9C. Mode 6 | 9C. Same as items 9A1 and 9A2 | 9C. Same as items 9A1 and 9A2, respectively | 9C. None - Same as items 9A1 and 9A2, respectively | |
| | | | 9D. Mode 7 | 9D. Spurious opening | 9D. Position indicator on QMCB or PAFP | 9D. None - No loss of tank V4-002 inventory, since pump P4-001 is not operating. | |
| 10 | Motor-driven AFW pump P4-003 (train A) | Provides AFW to SGs 1 and 4 automatically as required or by remote manual actuation from the control room or shutdown panel | 10A. Modes 1, 2, and 3 | 10A. Fails to start or stops running when required | 10A. Pump status light shows green on QMCB or PSDA, plus low discharge pressure indicators PI-5141A and PI-5141B on QMCB or PSDA, respectively. Also miniflow valve FV-5155 (item 27) is open. | 10A. None - Motordriven AFW pump (MDP) P4-002 available to satisfy 100% of AFW requirements. TDP P4-001 also available if required. | For HELB in TDP discharge (mode 5, items 10D and 11D), only one MDP is required to satisfy AFW requirements. (For the effects of loss of all ac power, see item 67.) |
| | | | 10B. Mode 4, SG 1 or 4 | 10B. Spurious start or failure to stop upon command | 10B. Pump status light shows red on QMCB or PSDA with same low pressure indicators | 10B. None - MDP P4-002 and TDP P4-001 operate automatically to satisfy AFW | |

TABLE 10.4.9-4 (SHEET 6 OF 31)

| Item No. | Description of Component[a] | Safety Function | Plant Operating Mode[b] | Failure Mode(s) | Method of Failure Detection[c] | Failure Effect on System Safety Function Capability | General Remarks |
|---|---|---|---|---|---|---|---|
| | | | | | as for item 10A, plus low discharge pressure alarm PSL-5149 on QMCB if break upstream of flow limiting orifices. | requirements. Thirty-minute spillage through break accounted for in condensate tank sizing. | |
| | | | 10C. Mode 4, SG 2 or 3 | 10C. Fails to start or stops running | 10C. Same as item 10A | 10C. None - TDP P4-001 operates automatically to satisfy AFW requirements. | |
| | | | 10D. Mode 5 | 10D. Fails to start or stops running | 10D. Same as item 10A | 10D. None - MDP P4-002 operates automatically to satisfy AFW requirements. | |
| | | | 10E. Mode 6 | 10E. Fails to start or stops running | 10E. Same as item 10A | 10E. None - MDP P4-002 and TDP P4-001 available. | |
| 11 | Motor-driven AFW pump P4-002 (train B) | Same as item 10, except provides AFW to SGs 2 and 3 and shutdown panel is PSDB | 11A. Modes 1, 2, and 3 | 11A. Fails to start or stops running when required | 11A. Pump status light shows green on QMCB or PSDB, plus low discharge pressure indicators PI-5140A or PI-1541B on QMCB and PSDB, respectively. Also, miniflow valve FV-5154 (item 28) is open. | 11A. None - Same as item 10A, except MDP P4-003 provides backup AFW capability along with TDP P4-001. | See item 10. |
| | | | 11B. Mode 4, SG 2 or 3 | 11B. Spurious start or failure to stop upon command | 11B. Pump status light shows red on QMCB or PSDB, plus same low pressure indicators as item 11A, plus low discharge pressure alarm PSL-5148 on QMCB | 11B. None - Same as item 10B, except backup MDP is P4-003. | |

TABLE 10.4.9-4 (SHEET 7 OF 31)

| Item No. | Description of Component[a] | Safety Function | Plant Operating Mode[b] | Failure Mode(s) | Method of Failure Detection[c] | Failure Effect on System Safety Function Capability | General Remarks |
|---|---|---|---|---|---|---|---|
| | | | 11C. Mode 4, SG 1 or 4 | 11C. Fails to start or stops running when required | 11C. Same as item 11A | 11C. None - Same as item 10C | |
| | | | 11D. Mode 5 | 11D. Fails to start or stops running | 11D. Same as item 11A | 11D. None - MDP P4-003 operates automatically to satisfy AFW requirements. | |
| | | | 11E. Mode 6 | 11E. Fails to start or stops running | 11E. Same as item 11A | 11E. None - Same as item 10E, except backup MDP is P4-003. | |
| 12 | Turbine-driven AFW pump P4-001 (train C, with dc-powered controls) | Same as item 10, except provides AFW to all four SGs | 12A. Mode 3 | 12A. Fails to start or stops running with steam admission valve HV-5106 (item 13) open | 12A. Low pump discharge pressure alarm PSL-5108, plus low speed indication SI-15109A on QMCB, SI-15109B on PAFP, SC-15109 on PAFT, and SI-15109. | 12A. None - MDPs P4-002 and P4-003 provide 100% of AFW requirements. | The turbine-driven AFW pump not ordinarily used during modes 1 and 2. (For FMEA of turbine steam admission valve, see item 13; and for main steam supply valves, see items 14 and 15.) |
| | | | 12B. Mode 4 any SG | 12B. Same as item 12A | 12B. Same as item 12A | 12B. None - Same as item 12A. Also, the MDPs satisfy minimum AFW requirements with allowance for 30-min spillage from faulted feedline. | |
| | | | 12C. Mode 5 | 12C. Continues to run or fails to stop upon command | 12C. Same as item 12A, except high pump speed indication SI-15109A on QMCB, SI-15109B on PAFP, SC-15109 on PAFT, and SI-15109; HV-5106 position | 12C. None - MDPs P4-002 and P4-003 provide 100% of AFW requirements. The condensate storage tank capacity accounts for 30-min spillage at pump runout conditions. Main | |

TABLE 10.4.9-4 (SHEET 8 OF 31)

| Item No. | Description of Component[a] | Safety Function | Plant Operating Mode[b] | Failure Mode(s) | Method of Failure Detection[c] | Failure Effect on System Safety Function Capability | General Remarks |
|---|---|---|---|---|---|---|---|
| | | | | | lights on PAFP or QMCB show valve open | steam turbine supply valves HV-3009 (item 14) and HV-3019 (item 15) can be closed to terminate turbine steam supply. | |
| | | | 12D. Mode 6 | 12D. Fails to start or stops running (HV-5106 open) | 12D. Same as item 12A | 12D. None - MDPs P4-002 and P6-003 available. | |
| 13 | AFW TDP turbine steam admission valve HV-5106, normally closed dc-powered MOV, train C | Controls steam admission to TDP turbine driver | 13A. Modes 3, 4, and 6 | 13A. Fails closed or fails to open upon command | 13A. Position lights on QMCB or PAFP coincident with low pump discharge pressure alarm PSL-5108 on QMCB | 13A. None - MDPs P4-002 and P4-003 provide 100% of AFW requirements. (See also item 12B, mode 4.) | The turbine-driven AFW pump not ordinarily used for modes, 1, 2, or 7. (For FMEA of TDP, see item 12; and for main steam supply valves, see items 14 and 15.) |
| | | | 13B. Mode 5 | 13B. Fails open or fails to close upon command | 13B. Same as item 13A | 13B. None - Same as item 12C | |
| 14 | Steam supply valve HV-3009 from SG 1 to TDP P4-001 turbine, normally open dc-powered MOV, train B | Admits steam from SG 1 to TDP turbine and also provides for isolation of the downstream piping in the event of a line break | 14A. Modes 3, 4, and 6 | 14A. Fails closed or fails to open upon command | 14A. Valve position lights on QMCB or PAFP | 14A. None - 100% redundant steam supply to the TDP provided via HV-3019 from SG 2. Also, MDPs P4-002 and P4-003 provide 100% of AFW requirements. | See also table 10.3.3-1, item 33. |
| | | | 14B. Mode 5 | 14B. Fails open or fails to close upon command | 14B. Same as item 14A | 14B. None - TDP steam admission valve can be closed to terminate steam supply. (See also item 12 for pump redundancy and design of CST capacity.) | |

TABLE 10.4.9-4 (SHEET 9 OF 31)

| Item No. | Description of Component[a] | Safety Function | Plant Operating Mode[b] | Failure Mode(s) | Method of Failure Detection[c] | Failure Effect on System Safety Function Capability | General Remarks |
|---|---|---|---|---|---|---|---|
| 15 | Steam supply valve HV-3019 from SG 2 to TDP P4-001 turbine, normally open dc-powered MOV, train A | Same as item 14A, but from SG 2 | 15A. Modes 3, 4, and 6 | 15A. Fails closed or fails to open upon command | 15A. Valve position lights on QMCB or PAFP | 15A. None - Same as item 14A, except redundant steam supply from SG 1 via HV-3009. | See also table 10.3.3-1, item 34. For containment isolation function, see table 6.2.4-1, item 19. |
| | | | 15B. Mode 5 | 15B. Fails open or fails to close upon command | 15B. Same as item 15A | 15B. None - Same as item 14B | |
| 16 | Check valve 1301-U4-404 in steam supply line to TDP turbine from SG 1 | Prevents crossflow between main steam lines to prevent blowdown of SG 2 in the event of MSLB in the No. 1 main steam line | 16A. Mode 6 (MSLB in SG 1) | 16A. Fails open | 16A. Gradual decrease of water level in SG 2 | 16A. None - TDP performance degraded but MDPs available. Degrading upon break location, HV-3009 can be closed to terminate blowdown from SG 2 and permit continued TDP operation. | See also table 10.3.3-1, item 35. |
| | | | 16B. Modes 3, 4, and 5 | 16B. Fails closed | 16B. None | 16B. None - Same as item 14A | |
| 17 | Check valve 1301-U4-008 in steam supply line to TDP turbine from SG 2 | Same as item 16, except SG designations reversed | 17A. Mode 6 (MSLB in SG 2) | 17A. Fails open | 17A. Gradual decrease of water level in SG 1 | 17A. None - Same as item 16A, except HV-3019 can be closed to terminate blowdown from SG 1 and permit continuous TDP operation. | See also table 10.3.3-1, item 36. |
| | | | 17B. Modes 3, 4, and 5 | 17B. Fails closed | 17B. None | 17B. None - Same as item 15A | |
| 18 | Check valve 013 in supply line from condensate storage tank V4-001 to TDP P4-001 suction | Prevents backflow to tank V4-001 from V4-002 when former is depleted | Modes 1 through 6 | 18A. Fails open with HV-5113 open | 18A. Water level equalized in tanks V4-001 and V4-002 | 18A. None - No loss of inventory also, HV-5090 and HV-5093 can be closed manually to prevent backflow. | This valve is required to function only if leaks are present between this valve and V4-001. |

TABLE 10.4.9-4 (SHEET 10 OF 31)

| Item No. | Description of Component[a] | Safety Function | Plant Operating Mode[b] | Failure Mode(s) | Method of Failure Detection[c] | Failure Effect on System Safety Function Capability | General Remarks |
|---|---|---|---|---|---|---|---|
| | | | | 18B. Fails closed with HV-5113 closed (HV-5113 normally closed - see item 9). | 18B. Decreased TDP P4-001 suction pressure indication PI-5110A on QMCB and PI-5110 on PAFP | 18B. None - Potentially damage to TDP P4-001 but MDP's P4-002 and P4-003 available. Also, HV-5113 can be opened to restore flow to MDP P4-001. | |
| 19 | Check valve 058 in supply line from condensate storage tank V4-001 to MDP P4-002 suction | Same as item 18 | Modes 1 through 6 | 19A. Fails open with HV-5118 open | 19A. Same as item 18A | 19A. None - Same as item 18A | See item 18. |
| | | | | 19B. Fails closed with HV-5118 closed (HV-5118 normally closed - see item 8) | 19B. Decreased MDP P4-002 suction pressure indication PI-5128A on QMCB and PI-5128 on PSDB | 19B. None - Potential damage to MDP P4-002 but, TDP P4-001 and MDP P4-003 are available. Also, HV-5118 can be opened to restore flow to MDP P4-002. | |
| 20 | Check valve 033 in supply line from condensate storage tank V4-001 to MDP P4-003 suction | Same as item 18 | Modes 1 through 6 | 20A. Fails open with HV-5119 open | 20A. Same as item 18A. | 20A. None - same as item 18A except manual valves are HV-5092 and HV-5095 | See item 18. |
| | | | | 20B. Fails closed with HV-5119 closed (HV-5119 normally closed - see item 7). | 20B. Decreased MDP P4-003 suction pressure indication PI-5129A on QMCB and PI-5129B on PSDA | 20B. None - Potential damage to MDP P4-003, but TDP P4-001 and MDP P4-003 available. Also, HV-5119 can be opened to restore flow to MDP P4-003. | See item 18. |
| 21 | Check valve 051 in supply line from condensate storage tank V4-002 to TDP P4-001 suction | Prevents backflow from tank V4-001 to tank V4-002 when latter depleted. | Modes 1 through 6 | 21A. Fails open with HV-5113 closed | 21A. None | 21A. None - HV-5113 prevents backflow | During normal auxiliary feedwater operation, tank V4-001 is depleted before pump aligned |

TABLE 10.4.9-4 (SHEET 11 OF 31)

| Item No. | Description of Component[a] | Safety Function | Plant Operating Mode[b] | Failure Mode(s) | Method of Failure Detection[c] | Failure Effect on System Safety Function Capability | General Remarks |
|---|---|---|---|---|---|---|---|
| | | | | | | | to tank V4-002. (See also item 18A except tank is V4-002). |
| | | | | 21B.Fails open with HV-5113 open | 21B.Same as item 18A | 21B. None - no loss of inventory MDV HV-5113 and manual valve HV-5097 can be closed to prevent backflow. | |
| | | | | 21C. Fails closed | 21C. Same as item 18B | 21C. None - same as item 9A1 | |
| 22 | Check valve 061 in supply line from condensate storage tank V4-002 to MDP P4-002 suction | Same as item 21. | Modes 1 through 6 | 22A. Fails open with HV-5118 closed | 22A. None | 22A. None - HV-5118 prevents backflow. | See items 18 and 21. |
| | | | | 22B. Fails open with HV-5118 open | 22B. Same as item 18A | 22B. None - Same as item 21B except valves are HV-5118 and HV-5098 | |
| | | | | 22C. Fails closed | 22C. Same as item 19B | 22C. None - Same as item 8A1 | |
| 23 | Check valve 052 in supply line from condensate storage tank V4-002 to MDP P4-003 suction. | Same as item 21 | Modes 1 through 6 | 23A. Fails open with HV-5119 closed | 23A. None | 23A. None - HV-5119 prevents backflow | See item 10. |
| | | | | 23B. Fails open with HV-5119 open | 23B. Same as item 18A | 23B. None - Same as 21B except valves are HV-5119 and HV-5099 | |
| | | | | 23C. Fails closed | 23C. Same as item 20B | 23C. None - Same as item 10A | |
| 24 | Check valve 001 in MDP P4-003 discharge | Prevents reduction in AFW flow to SG via backflow through P4-003 when pump is not running | 24A. Modes 1, 2, and 3 | 24A. Fails closed with MDP P4-003 running | 24A. Position indicators for miniflow valve FV-5155 on QMCB | 24A. None - Same as item 10A | See item 10. |

TABLE 10.4.9-4 (SHEET 12 OF 31)

| Item No. | Description of Component[a] | Safety Function | Plant Operating Mode[b] | Failure Mode(s) | Method of Failure Detection[c] | Failure Effect on System Safety Function Capability | General Remarks |
|---|---|---|---|---|---|---|---|
| | | | | | or PSDA show FV-5155 open | | |
| | | | 24B. Mode 4, SG 2 | 24B. Same as item 24A | 24B. Same as item 24A | 24B. None - Same as item 10C | |
| | | | 24C. Mode 5 | 24C. Same as item 24A | 24C. Same as item 24A | 24C. None - Same as item 10D | |
| | | | 24D. Mode 6 | 24D. Same as item 24A | 24D. Same as item 24A | 24D. None - Same as item 10E | |
| | | | 24E. All | 24E. Fails open with MDP P4-003 not running | 24E. None | 24E. None - Stop checks 043 and 046 prevent backflow, thus ensuring adequate AFW flow. | |
| 25 | Check valve 002 in MDP P4-002 discharge | Same as item 24, except pump is MDP P4-002 | 25A. Modes 1, 2, and 3 | 25A. Fails closed with MDP P4-002 running | 25A. Position lights for miniflow valve FV-5154 on QMCB or PSDB show FV-5154 open | 25A. None - Same as item 11A | See item 10. |
| | | | 25B. Mode 4, SG 1 or 4 | 25B. Same as item 25A | 25B. Same as item 25A | 25B. None - Same as item 11C | |
| | | | 25C. Mode 5 | 25C. Same as item 25A | 25C. Same as item 25A | 25C. None - Same as item 11D | |
| | | | 25D. Mode 6 | 25D. Same as item 25A | 25D. Same as item 25A | 25D. Same as item 11E | |
| | | | 25E. All | 25E. Fails open with MDP P4-002 not running | 25E. None | 25E. None - Stop checks 037 and 040 prevent backflow, thus ensuring adequate AFW flow. | |
| 26 | Check valve 014 in TDP P4-001 discharge | Same as item 24, except pump is TDP P4-001 | 26A. Mode 3 | 26A. Fails closed with TDP P4-001 running | 26A. None | 26A. None - Same as item 12A | See item 12. |
| | | | 26B. Mode 4, any SG | 26B. Same as item 26A | 26B. None | 26B. None - Same as item 12B | |

TABLE 10.4.9-4 (SHEET 13 OF 31)

| Item No. | Description of Component[a] | Safety Function | Plant Operating Mode[b] | Failure Mode(s) | Method of Failure Detection[c] | Failure Effect on System Safety Function Capability | General Remarks |
|---|---|---|---|---|---|---|---|
| | | | 26C. Mode 6 | 26C. Same as item 26A | 26C. None | 26C. None - Same as item 12D | |
| | | | 26D. All | 26D. Fails open with TDP P4-001 not running | 26D. None | 26D. None - Stop checks 017, 020, 023, and 026 prevent backflow, thus ensuring adequate AFW flow. | |
| 27 | AFW pump P4-003 miniflow valve FV-5155, normally open MOV, train A | Opens automatically or manually from PSDA to ensure MDP P4-003 miniflow requirements are satisfied at all times | 27A. Modes 1, 2, 3, 4, and 6 | 27A1. Fails open with MDP P4-003 running | 27A1. Position lights on QMCB or PSDA | 27A1. None - Reduced flow available from MDP P4-003, but MDP P4-002 and TDP P4-001 are available. | See item 10 for effect of loss of one MDP for HELB at TDP discharge (mode 5). |
| | | | | 27A2. Fails closed with MDP P4-003 running | 27A2. Same as item 27A1 | 27A2. None - Under some conditions MDP P4-003 may cavitate, but MDP P4-002 and TDP P4-001 are available. | |
| | | | 27B. Mode 5 | 27B1. Same as item 27A1 | 27B1. Same as item 27A1 | 27B1. None - Reduced flow available from MDP P4-003, but MDP P4-002 satisfies 100% AFW requirements. | |
| | | | | 27B2. Same as item 27A2 | 27B2. Same as item 27A1 | 27B2. None - Same as item 27A2, except only MDP P4-002 is available; this is sufficient for this case. | |
| 28 | AFW P4-002 miniflow valve FV-5154, normally open MOV, train B | Same as item 27, except MDP is P4-002 and manual control from PSDB | 28A. Modes 1, 2, 3, 4, and 6 | 28A1. Fails open with MDP P4-002 running | 28A1. Position indicator on QMCB or PSDB | 28A1. None - Reduced flow available from MDP P4-002, but MDP P4-003 and TDP P4-001 are available. | See items 27 and 10. |

TABLE 10.4.9-4 (SHEET 14 OF 31)

| Item No. | Description of Component[a] | Safety Function | Plant Operating Mode[b] | Failure Mode(s) | Method of Failure Detection[c] | Failure Effect on System Safety Function Capability | General Remarks |
|---|---|---|---|---|---|---|---|
| | | | | 28A2. Fails closed with MDP P4-002 running | 28A2. Same as item 28A1 | 28A2. None - Under same conditions, MDP P4-002 may cavitate, but MDP P4-003 and TDP P4-001 are available. | |
| | | | 28B. Mode 5 | 28B1. Same as item 28A1 | 28B1. Same as item 28A1 | 28B1. None - Reduced flow available from MDP P4-002, but MDP P4-003 satisfies 100% AFW requirements. | |
| | | | | 28B2. Same as item 28A2 | 28B2. Same as item 28A1 | 28B2. None - Same as item 28B1, except only MDP-003 available; this is sufficient for this case. | |
| 29 | AFW control valve HV-5139 from MDP P4-003 to SG 1, normally open MOV, train A | Regulates AFW flow from MDP P4-003 to SG 1 via remote manual control from QMCB and PSDA; opens automatically upon MDP automatic start signal | 29A. Modes 1, 2, and 3 | 29A. Fails closed or fails to open upon command | 29A. Position light on QMCB or position light on PSDA plus reduced flow indicators to SG 1; FI-5152A on QMCB, FI-5152B on PSDA, and FI-5152C on PAFP | 29A. None - No flow to SG 1 from MDP P4-003, but MDP P4-002 and TDP P4-001 are available. Also, MDP P4-003 still supplies AFW SG 4. | For FMEA of MDP P4-003, see item 10. For system capability to withstand 30-min spillage. See item 10. HV-5122 can be closed to isolate TDP P4-001 from SG 1 for mode 4 (item 29B). |
| | | | 29B. Mode 4, SG 1 | 29B. Fails open or fails to close upon command | 29B. Same as item 29A | 29B. None - MDP P4-003 can be turned off to terminate spillage. MDP P4-002 and TDP P4-001 supply 100% of flow. | |
| | | | 29C. Mode 4, SG 4 | 29C. Fails closed or fails to open upon command | 29C. Same as item 29A | 29C. None - MDP P4-003 not effective, but MDP P4-002 available to feed SGs 2 and 3. Also, | |

TABLE 10.4.9-4 (SHEET 15 OF 31)

| Item No. | Description of Component[a] | Safety Function | Plant Operating Mode[b] | Failure Mode(s) | Method of Failure Detection[c] | Failure Effect on System Safety Function Capability | General Remarks |
|---|---|---|---|---|---|---|---|
| | | | | | | TDP P4-001 available. MDP P4-003 can be turned off and HV-5120 can be closed to terminate AFW spillage from No. 4 feedline. | |
| | | | 29D. Mode 5 | 29D. Fails closed or fails to open upon command | 29D. Same as item 29A, plus zero flow indication to SG 1; FI-5152A on QMCB, FI-5152B on PSDA, and FI-5152C on PAFP | 29D. None - MDP P4-003 still supplies SG 4 while MDP P4-002 supplies SGs 2 and 3. | |
| | | | 29E. Mode 6 (LOCA) | 29E. Fails closed or fails to open upon command | 29E. Same as item 29A | 29E. None - Same as item 29A | |
| | | | 29F. Mode 6, MSLB in SG 1 | 29F. Fails open or fails to close upon command | 29F. Same as item 29A | 29F. None - Same as item 29B | |
| 30 | AFW control valve HV-5137 from MDP P4-003 to SG 4, normally open MOV, train A | Same as item 29, except SG 4 | 30A. Modes 1, 2, and 3 | 30A. Same as item 29A | 30A. Same as item 29A, except flow indicators to SG 4 are FI-5150A on QMCB, FI-5150B on PSDA, and FI-5150C on PAFP | 30A. None - Same as item 29A, except no flow to SG 4, but with flow to SG 1 from P4-003. | See items 29 and 10. HV-5120 can be closed to isolate TDP P4-001 from SG 4 for mode 4 (item 30B). |
| | | | 30B. Mode 4, SG 4 | 30B. Same as item 29B | 30B. Same as item 30A | 30B. None - Same as item 29B | |
| | | | 30C. Mode 4, SG 1 | 30C. Same as item 29C | 30C. Same as item 30A | 30C. None - Same as item 29C, except HV-5122 can be closed to terminate AFW spillage from No. 1 feedline. | |
| | | | 30D. Mode 5 | 30D. Same as item 29D | 30D. Same as item 30A, plus zero | 30D. None - Same as item 29D, except | |

TABLE 10.4.9-4 (SHEET 16 OF 31)

| Item No. | Description of Component[a] | Safety Function | Plant Operating Mode[b] | Failure Mode(s) | Method of Failure Detection[c] | Failure Effect on System Safety Function Capability | General Remarks |
|---|---|---|---|---|---|---|---|
| | | | | | flow indication to SG 4; FI-5150A on QMCB, FI-5150B on PSDA, and 5150C on PAFP | MDP P4-003 still supplies SG 1. | |
| | | | 30E. Mode 6 (LOCA) | 30E. Same as item 29E | 30E. Same as item 30A | 30E. None - Same as item 30A | |
| | | | 30F. Mode 6, MSLB in SG 4 | 30F. Same as item 29F | 30F. Same as item 30A | 30F. None - Same as item 29B | |
| 31 | AFW control valve HV-5134 from MDP P4-002 to SG 3, normally open MOV, train B | Same as item 29, except from MDP P4-002 to SG 3 and control from QMCB and PSDB | 31A. Modes 1, 2, and 3 | 31A. Same as item 29A | 31A. Same as item 29A, except flow indicators to SG 3 are FI-5153A on QMCB, FI-5153V on PSDB, and FI-5153C on PAFP | 31A. None - No flow to SG 3 from MDP P4-002, but MDP P4-003 and TDP P4-001 are available. Also, MDP P4-002 still supplies AFW to SG 2. | See items 10 and 29. HV-5127 can be closed to isolate TDP P4-001 from SG 3 for mode 4 (item 31B). |
| | | | 31B. Mode 4, SG 3 | 31B. Same as item 29B | 31B. Same as item 31A | 31B. None - MDP P4-002 can be turned off to terminate spillage. MDP P4-002 and TDP P4-001 supply 100% AFW flow. | |
| | | | 31C. Mode 4, SG 2 | 31C. Same as item 29C | 31C. Same as item 31A | 31C. None - MDP P4-002 not effective, but MDP P4-003 is available to feed SGs 1 and 4. Also, TDP P4-001 is available. MDP P4-002 can be turned off and HV-5125 closed to terminate spillage. | |
| | | | 31D. Mode 5 | 31D. Same as item 29D | 31D. Same as item 31A, plus zero flow indication to SG 3; FI-5153A on QMCB, FI-5153B | 31D. None - MDP P4-002 still supplies SG 2 while MDP P4-003 supplies SGs 1 and 4. | |

TABLE 10.4.9-4 (SHEET 17 OF 31)

| Item No. | Description of Component[a] | Safety Function | Plant Operating Mode[b] | Failure Mode(s) | Method of Failure Detection[c] | Failure Effect on System Safety Function Capability | General Remarks |
|---|---|---|---|---|---|---|---|
| | | | | | on PSDB, and FI-5153C on PAFP | | |
| | | | 31E. Mode 6 (LOCA) | 31E. Same as item 29E | 31E. Same as item 31A | 31E. None - Same as item 31A | |
| | | | 31F. Mode 6 (MSLB in SG 3) | 31F. Same as item 29F | 31F. Same as item 31A | 31F. None - Same as item 31B | |
| 32 | AFW control valve HV-5132 from MDP P4-002 to SG 2, normally open MOV, train B | Same as item 31, except from MDP P4-002 to SG 2 | 32A. Modes 1, 2, and 3 | 32A. Same as item 29A | 32A. Same as item 29A, except flow indicators to SG 2 are FI-5151A on QMCB, FI-5151B on PSDB, and FI-5151C on PAFP | 32A. None - Same as item 31A, except no flow to SG 2, but with flow from MDP P4-002 to SG 3. | See items 10 and 29. HV-5125 can be closed to isolate TDP P4-001 from SG 2 for mode 4 (item 32B). |
| | | | 32B. Mode 4, SG 2 | 32B. Same as item 29B | 32B. Same as item 32A | 32B. None - Same as item 31B | |
| | | | 32C. Mode 4, SG 3 | 32C. Same as item 29C | 32C. Same as item 32A | 32C. None - Same as item 31C, except HV-5127 can be closed to terminate AFW spillage from No. 3 feedline. | |
| | | | 32D. Mode 5 | 32D. Same as item 29D | 32D. Same as item 32A, plus zero flow indication to SG 2; FI-5151A on QMCB, FI-5151B on PSDB, and FI-5151C on PAFP | 32D. None - Same as item 31E, except MDP P4-002 still supplies SG 3. | |
| | | | 32E. Mode 6 (LOCA) | 32E. Same as item 29E | 32E. Same as item 32A | 32E. None - Same as item 32A | |
| | | | 32F. Mode 6 (MSLB in SG 2) | 32F. Same as item 29F | 32F. Same as item 32A | 32F. None - Same as item 31B | |
| 33 | AFW control valve HV-5127 from TDP P4-001 to SG 3, normally open dc- | Regulates AFW flow from TDP P4-001 to SG 3 via remote manual control from QMCB and PAFP; opens | 33A. Mode 3 | 33A. Same as item 29A | 33A. Same as item 29A, except flow indicators to SG 3 are FI-5153A | 33A. None - No flow to SG 3 from TDP P4-001, but MDPs P4-002 and 003 are | See items 12 and 19. The faulted SG (mode 4, item 33C) can be |

TABLE 10.4.9-4 (SHEET 18 OF 31)

| Item No. | Description of Component[a] | Safety Function | Plant Operating Mode[b] | Failure Mode(s) | Method of Failure Detection[c] | Failure Effect on System Safety Function Capability | General Remarks |
|---|---|---|---|---|---|---|---|
| | powered MOV, train C | automatically upon TDP auto start signal | | | on QMCB, FI-5153B on PSDB, and FI-5153C on PAFP | available. Also, TDP P4-001 still supplies AFW to SGs 1, 2, and 4. | isolated from its respective MDP by closing the appropriate AFW control valve. Mode 5 is not applicable since break is assumed to be upstream of flow control valves. |
| | | | 33B. Mode 4, SG 3 | 33B. Same as item 29B | 33B. Same as item 33A | 33B. None - TDP P4001 can be turned off and HV-5134 closed to terminate spillage. MPDs P4-002 and P4-003 provide 100% AFW flow. | |
| | | | 33C. Mode 4, SG 1, 2, or 4 | 33C. Same as item 29C | 33C. Same as item 33A | 33C. None - TDP P4-001 still supplies two effective steam generators. Also, MDPs P4-002 and P4-003 available. (See remarks.) | |
| | | | 33D. Mode 6 (LOCA) | 33D. Same as item 29E | 33D. Same as item 33A | 33D. None - Same as item 33A | |
| | | | 33E. Mode 6 (MSLB in SG 3) | 33E. Same as item 29F | 33E. Same as item 33A | 33E. None - Same as item 33B | |
| 34 | AFW control valve HV-5125 from TDP P4-001 to SG 2, normally open, dc-powered MOV, train C | Same as item 33, except to SG 2 | 34A. Mode 3 | 34A. Same as item 29A | 34A. Same as item 29A, except flow indicators to SG 2 are FI-5151A on QMCB, FI-5151B on PSDB, and FI-5151C on PAFP | 34A. None - Same as item 33A, except no flow from TDP P4-001 to SG 2, but with flow to SGs 1, 3, and 4. | See items 12, 29, and 33. |
| | | | 34B. Mode 4, SG 2 | 34B. Same as item 29B | 34B. Same as item 34A | 34B. None - Same as item 33B, except HV-5132 can be closed to isolate SG 2. | |

TABLE 10.4.9-4 (SHEET 19 OF 31)

| Item No. | Description of Component[a] | Safety Function | Plant Operating Mode[b] | Failure Mode(s) | Method of Failure Detection[c] | Failure Effect on System Safety Function Capability | General Remarks |
|---|---|---|---|---|---|---|---|
| | | | 34C. Mode 4, SG 1, 3, or 4 | 34C. Same as item 29C | 34C. Same as item 34A | 34C. None - Same as item 33C (see remarks item 33) | |
| | | | 34D. Mode 6 (LOCA) | 34D. Same as item 29E | 34D. Same as item 34A | 34D. None - Same as item 34A | |
| | | | 34E. Mode 6 (MSLB in SG 2) | 34E. Same as item 29F | 34E. Same as item 34A | 34E. None - Same as item 34B | |
| 35 | AFW control valve HV-5122 from TDP P4-001 to SG 1, normally open dc-powered MOV, train C | Same as item 33, except to SG 1 | 35A. Mode 3 | 35A. Same as item 29A | 35A. Same as item 29A, except flow indicators to SG 1 are FI-5152A on QMCB, FI-5152B on PSDA, and FI-5152C on PAFP | 35A. None - Same as item 33A, except no flow from TDP P4-001 to SG 1, but flow to SGs 2, 3, and 4. | See items 12, 29, and 33. |
| | | | 35B. Mode 4, SG 1 | 35B. Same as item 29B | 35B. Same as item 35A | 35B. None - Same as item 33B, except HV-5139 can be closed to isolate SG 1. | |
| | | | 35C. Mode 4, SG 2, 3, or 4 | 35C. Same as item 29C | 35C. Same as item 35A | 35C. None - Same as item 33C (see remarks item 33) | |
| | | | 35D. Mode 6 (LOCA) | 35D. Same as item 29E | 35D. Same as item 35A | 35D. None - Same as item 35A | |
| | | | 35E. Mode 6 (MSLB in SG 2) | 35E. Same as item 29E | 35E. Same as item 35A | 35E. None - Same as item 35B | |
| 36 | AFW control valve HV-5120 from TDP P4-001 to SG 4, normally open dc-powered MOV, train C | Same as item 33, except to SG 4 | 36A. Mode 3 | 36A. Same as item 29F | 36A. Same as item 29A, except flow indicators to SG 4 are FI-5150A on QMCB, FI-5150B on PSDA, and FI-5150C on PAFP | 36A. None - Same as item 33A, except no flow from TDP P4-001 to SG 4, but with flow to SGs 1, 2, and 3 | See items 12, 29, and 33. |
| | | | 36B. Mode 4, SG 4 | 36B. Same as item 29B | 36B. Same as item 36A | 36B. None - Same as item 33B, except | |

TABLE 10.4.9-4 (SHEET 20 OF 31)

| Item No. | Description of Component[a] | Safety Function | Plant Operating Mode[b] | Failure Mode(s) | Method of Failure Detection[c] | Failure Effect on System Safety Function Capability | General Remarks |
|---|---|---|---|---|---|---|---|
| | | | | | | HV-5137 can be closed to isolate SG 4. | |
| | | | 36C. Mode 4, SG 2, 3, or 4 | 36C. Same as item 29C | 36C. Same as item 36A | 36C. None - Same as item 33C (see remarks item 33) | |
| | | | 36D. Mode 6 (LOCA) | 36D. Same as item 29E | 36D Same as item 36A | 36D. None - Same as item 36A | |
| | | | 36E. Mode 6 (MSLB in SG 4) | 36E. Same as item 29F | 36E. Same as item 36A | 36E. None - Same as item 36B | |
| 37 | Stop-check valve 046 in AFW supply from MDP P4-003 to SG 1 | Prevents AFW backflow when MDP P4-003 not running and HV-5139 open | 37A. Modes 1, 2, and 3 | 37A. Fails closed with MDP P4-003 running and HV-5139 open | 37A. None | 37A. None - Same as item 29A | |
| | | | 37B. Mode 4, SG 4 | 37B. Same as item 37A | 37B. None | 37B. None - Same as item 29C | |
| | | | 37C. Mode 5 | 37C. Same as item 37A | 37C. Zero flow indication to SG 1 (see item 29D) | 37C. None - Same as item 29D | |
| | | | 37D. Mode 6 (LOCA) | 37D. Same as item 37A | 37D. None | 37D. None - Same as item 29E | |
| | | | 37E. All | 37E. Fails open with MDP P4-003 not running and HV-5139 open | 37E. None | 37E. None - Other check valves in flow path prevent backflow. Also, HV-1539 can be closed. | |
| 38 | Stop-check valve 043 in AFW supply from MDP P4-003 to SG 4 | Same as item 37, except HV is HV-5137 | 38A. Modes 1, 2, and 3 | 38A. Same as item 37A, except HV-5137 open | 38A. None | 38A. None - Same as item 30A | |
| | | | 38B. Mode 4, SG 1 | 38B. Same as item 38A | 38B. None | 38B. None - Same as item 30C | |
| | | | 38C. Mode 5 | 38C. Same as item 38A | 38C. Zero flow indication to SG 4 (see item 30D) | 38C. None - Same as item 30D | |

TABLE 10.4.9-4 (SHEET 21 OF 31)

| Item No. | Description of Component[a] | Safety Function | Plant Operating Mode[b] | Failure Mode(s) | Method of Failure Detection[c] | Failure Effect on System Safety Function Capability | General Remarks |
|---|---|---|---|---|---|---|---|
| | | | 38D. Mode 6 (LOCA) | 38D. Same as item 38A | 38D. None | 38D. None - Same as item 30E | |
| | | | 38E. All | 38E. Same as item 37E, except HV-5137 open | 38E. None | 38E. None - Same as item 37E, except HV-5137 can be closed. | |
| 39 | Stop-check valve 040 in AFW supply from MDP P4-002 to SG 3 | Prevents AFW backflow when MDP P4-002 not running and HV-5134 open | 39A. Modes 1, 2, and 3 | 39A. Fails closed with MDP P4-002 running and HV-5134 open | 39A. None | 39A. None - Same as item 31A | |
| | | | 39B. Mode 4, SG 2 | 39B. Same as item 39A | 39B. None | 39B. None - Same as item 31C | |
| | | | 39C. Mode 5 | 39C. Same as item 39A | 39C. Zero flow indication to SG 3 (see item 31D) | 39C. None - Same as item 31D | |
| | | | 39D. Mode 6 (LOCA) | 39D. Same as item 39A | 39D. None | 39D. None - Same as item 31E | |
| | | | 39E. All | 39E. Fails open with MDP P4-002 not running and HV-5134 open | 39E. None | 39E. None - Other check valves in flow path prevent backflow. Also, HV-5134 can be closed. | |
| 40 | Stop-check valve 037 in AFW supply from MDP P4-002 to SG 2 | Same as item 39, except HV is HV-5132 | 40A. Modes 1, 2, and 3 | 40A. Same as item 39A, except HV-5132 open | 40A. None | 40A. None - Same as item 32A | |
| | | | 40B. Mode 4, SG 3 | 40B. Same as item 40A | 40B. None | 40B. None - Same as item 32C | |
| | | | 40C. Mode 5 | 40C. Same as item 40A | 40C. Zero flow indication to SG 3 (see item 32D) | 40C. None - Same as item 32D | |
| | | | 40D. Mode 6 (LOCA) | 40D. Same as item 40A | 40D. None | 40D. None - Same as item 32E | |
| | | | 40E. All | 40E. Same as item 39E, except HV-1532 open | 40E. None | 40E. None - Same as item 39E, except HV-5132 can be closed. | |

TABLE 10.4.9-4 (SHEET 22 OF 31)

| Item No. | Description of Component[a] | Safety Function | Plant Operating Mode[b] | Failure Mode(s) | Method of Failure Detection[c] | Failure Effect on System Safety Function Capability | General Remarks |
|---|---|---|---|---|---|---|---|
| 41 | Stop-check valve 026 in AFW supply from TDP P4-001 to SG 3 | Prevents AFW backflow when TDP P4-001 not running and HV-5127 open | 41A. Mode 3 | 41A. Fails closed with TDP P4-001 running and HV-5127 open | 41A. None | 41A. None - Same as item 33A | See items 12 and 33. |
| | | | 41B. Mode 4, SG 1, 2, or 4 | 41B. Same as item 41A | 41B. None | 41B. None - Same as item 33C | |
| | | | 41C. Mode 5 | 41C. Fails open with HV-5127 open | 41C. None | 41C. None - Same as item 12C. Also, HV-5127 can be closed. | |
| | | | 41D. Mode 6 (LOCA) | 41D. Same as item 41C | 41D. None | 41D. None - Same as item 33D | |
| | | | 41E. All except 5 | 41E. Fails open with TDP P4-001 not running and HV-5127 open | 41E. None | 41E. None - Other check valves in flow path prevent backflow. Also, HV-5127 can be closed. | |
| 42 | Stop-check valve 023 in AFW supply from TDP P4-001 to SG 2 | Same as item 41, except HV is HV-5125 | 42A. Mode 3 | 42A. Same as item 41A, except HV-5125 open | 42A. None | 42A. None - Same as item 34A | See items 12, 33, and 34. |
| | | | 42B. Mode 4, SG 1, 3, or 4 | 42B. Same as item 42A | 42B. None | 42B. None - Same as item 34C | |
| | | | 42C. Mode 5 | 42C. Fails open with HV-5125 open | 42C. None | 42C. None - Same as item 12C. Also, HV-5127 can be closed. | |
| | | | 42D. Mode 6 (LOCA) | 42D. Same as item 42A | 42D. None | 42D. None - Same as item 34D | |
| | | | 42E. All except 5 | 42E. Same as item 41E, except HV-5125 open | 42E. None | 42E. None - Same as item 41E, except HV-5125 can be closed. | |
| 43 | Stop-check valve 020 in AFW supply from TDP P4-001 to SG 1 | Same as item 41, except HV is HV-5122 | 43A. Mode 3 | 43A. Same as item 41A, except HV-5122 open | 43A. None | 43A. None - Same as item 35A | See items 12, 33, and 35 |

TABLE 10.4.9-4 (SHEET 23 OF 31)

| Item No. | Description of Component[a] | Safety Function | Plant Operating Mode[b] | Failure Mode(s) | Method of Failure Detection[c] | Failure Effect on System Safety Function Capability | General Remarks |
|---|---|---|---|---|---|---|---|
| | | | 43B. Mode 4, SG 2, 3, or 4 | 43B. Same as item 43A | 43B. None | 43B. None - Same as item 35C | |
| | | | 43C. Mode 5 | 43C. Fails open with HV-5122 open | 43C. None | 43C. None - Same as item 12C. Also, HV-5122 can be closed. | |
| | | | 43D. Mode 6 (LOCA) | 43D. Same as item 43A | 43D. None | 43D. None - Same as item 35D | |
| | | | 43E. All except 5 | 43E. Same as item 41E, except HV-5122 open | 43E. None | 43E. None - Same as item 41E, except HV-5122 can be closed. | |
| 44 | Stop-check valve 017 in AFW supply from TDP to SG 4 | Same as item 41, except HV is HV-5120 | 44A. Mode 3 | 44A. Same as item 41A, except HV-5120 open | 44A. None | 44A. None - Same as item 36A | See items 12, 33, and 36. |
| | | | 44B. Mode 4, SG 2, 3, or 4 | 44B. Same as item 41A | 44B. None | 44B. None - Same as item 36C | |
| | | | 44C. Mode 5 | 44C. Fails open with HV-5120 open | 44C. None | 44C. None - Same as item 12C. Also, HV-5120 can be closed. | |
| | | | 44D. Mode 6 (LOCA) | 44D. Same as item 44A | 44D. None | 44D. None - Same as item 36D | |
| | | | 44E. All except 5 | 44E. Same as item 41E except HV-5120 open | 44E. None | 44E. None - Same as item 41E, except HV-5120 can be closed. | |
| 45 | Stop-check valve 113 in AFW supply lines to SG 1 | Prevents backflow from No. 1 feedwater bypass line whenever the AFW system not in operation | 45A. Mode 7 | 45A. Fails open | 45A. None | 45A. None - Other check valves in flow path prevent backflow. | |
| | | | 45B. Modes 1 through 6 | 45B. Fails closed | 45B. Zero flow indication to SG 1; FI-5152A on QMCB, FI-5152B on PSDA, and FI-5152C on PAFP | 45B. None - Same as item 29D, plus TDP P4-001 available to supply AFW to SGs 2, 3, and 4. | |

TABLE 10.4.9-4 (SHEET 24 OF 31)

| Item No. | Description of Component[a] | Safety Function | Plant Operating Mode[b] | Failure Mode(s) | Method of Failure Detection[c] | Failure Effect on System Safety Function Capability | General Remarks |
|---|---|---|---|---|---|---|---|
| 46 | Stop-check valve 114 in AFW supply line to SG 2 | Same as item 45, except for No. 2 feedwater bypass line | 46A. Mode 7 | 46A. Fails open | 46A. None | 46A. None - Same as item 45A | |
| | | | 46B. Modes 1 through 6 | 46B. Fails closed | 46B. Zero flow indication to SG 2; FI-5151A on QMCB, FI-5151B on PSDB, and FI-5151C on PAFP | 46B. None - Same as item 32D, plus TDP P4-001 is available to supply AFW to SGs 1, 3, and 4. | |
| 47 | Stop-check valve 115 in AFW supply line to SG 3 | Same as item 45, except for No. 3 feedwater bypass line | 47A. Mode 7 | 47A. Fails open | 47A. None | 47A. None - Same as item 45A | |
| | | | 47B. Modes 1 through 6 | 47B. Fails closed | 47B. Zero flow indication to SG 3; FI-5153A on QMCB, FI-5153B on PSDB, and FI-5153C on PAFP | 47B. None - Same as item 31D, plus TDP P4-001 is available to supply AFW to SGs 1, 2, and 4. | |
| 48 | Stop-check valve 116 in AFW supply line to SG 4 | Same as item 45, except for No. 4 feedwater bypass line | 48A. Mode 7 | 48A. Fails open | 48A. None | 48A. None - Same as item 45A | |
| | | | 48B. Modes 1 through 6 | 48B. Fails closed | 48B. Zero flow indication to SG 4; FI-5150A on QMCB, FI-5150B on PSDA, and FI-5150C on PAFP | 48B. None - Same as item 30D, plus TDP P4-001 is available to supply AFW to SGs 1, 2, and 3. | |
| 49 | Feedwater bypass isolation valve HV-15196, normally open, fail closed air-operated valve | Isolates feedwater bypass line to SG 1 upon feedwater isolation signal; admits main feedwater to AFW nozzle during power generation | 49A. Modes 1 through 6 | 49A. Fails open or fails to close upon command | 49A. Position lights on QMCB | 49A. None - Check valve 053 prevents backflow or injection of AFW into main feedwater nozzle. | This valve is normally open only during power generation. |
| | | | 49B. Mode 7 | 49B. Fails closed or fails to open upon command | 49B. Position lights on QMCB | 49B. None - AFW path not affected. All main feedwater enters via main feedwater nozzle. | |

TABLE 10.4.9-4 (SHEET 25 OF 31)

| Item No. | Description of Component[a] | Safety Function | Plant Operating Mode[b] | Failure Mode(s) | Method of Failure Detection[c] | Failure Effect on System Safety Function Capability | General Remarks |
|---|---|---|---|---|---|---|---|
| 50 | Feedwater bypass isolation valve HV-15197, normally open, fail closed air-operated valve | Same as item 49, except for SG 2 | 50A. Modes 1 through 6 | 50A. Same as item 49A | 50A. Same as item 49A | 50A. None - Same as item 49A, except check valve is 118. | See item 49. |
| | | | 50B. Mode 7 | 50B. Same as item 49B | 50B. Same as item 49B | 50B. None - Same as item 49B | |
| 51 | Feedwater bypass isolation valve HV-15198, normally open, fail closed air-operated valve | Same as item 49, except for SG 3 | 51A. Modes 1 through 6 | 51A. Same as item 49A | 51A. Same as item 49A | 51A. None - Same as item 49A, except check valve is 120. | See item 49. |
| | | | 51B. Mode 7 | 51B. Same as item 49B | 51B. Same as item 49B | 51B. None - Same as item 49B | |
| 52 | Feedwater bypass isolation valve HV-15199, normally open, fail closed air-operated valve | Same as item 49, except for SG 4 | 52A. Modes 1 through 6 | 52A. Same as item 49A | 52A. Same as item 49A | 52A. None - Same as item 49A, except check valve is 119. | See item 49. |
| | | | 52B. Mode 7 | 52B. Same as item 49B | 52B. Same as item 49B | 52B. None - Same as item 49B | |
| 53 | Feedwater bypass check valve 117 | Prevents backflow of SG 1 AFW flow into main feedwater system whenever latter not operating | 53A. Modes 1 through 6 | 53A. Fails open | 53A. None | 53A. None - Condensate storage facility sized to accommodate loss due to 30-min backflow. Also, HV-15196 can be closed. | See item 10. |
| | | | 53B. Mode 7 | 53B. Fails closed | 53B. None | 53B. None - Same as item 49B | |
| 54 | Feedwater bypass check valve 118 | Same as item 53, except for SG 2 | 54A. Modes 1 through 6 | 54A. Fails open | 54A. None | 54A. None - Same as item 53A, except HV is HV-15197. | See item 10. |
| | | | 54B. Mode 7 | 54B. Fails closed | 54B. None | 54B. None - Same as item 49B | |

TABLE 10.4.9-4 (SHEET 26 OF 31)

| Item No. | Description of Component[a] | Safety Function | Plant Operating Mode[b] | Failure Mode(s) | Method of Failure Detection[c] | Failure Effect on System Safety Function Capability | General Remarks |
|---|---|---|---|---|---|---|---|
| 55 | Feedwater bypass check valve 120 | Same as item 53, except for SG 3 | 55A. Modes 1 through 6 | 55A. Fails open | 55A. None | 55A. None - Same as item 53A, except HV is HV-15198. | See item 10. |
| | | | 55B. Mode 7 | 55B. Fails closed | 55B. None | 55B. None - Same as item 49B | |
| 56 | Feedwater bypass check valve 119 | Same as item 53, except for SG 4 | 56A. Modes 1 through 6 | 56A. Fails open | 56A. None | 56A. None - Same as item 53A, except HV is HV-15199. | See item 10. |
| | | | 56B. Mode 7 | 56B. Fails closed | 56B. None | 56B. None - Same as item 49B | |
| 57 | Not used | | | | | | |
| 58 | AFW line check valve 125 | Prevents blowdown at No. 1 steam generator or auxiliary building overpressurization in the event of a HELBA in the No. 1 steam generator auxiliary feedwater line outside containment. | 58A. All but AFW HELB | 58A. Fails closed | 58A. Same as for item 45B except "None" for Mode 7 | 58A. None - Same as item 45B | |
| | | | 58B. AFW HELB | 58B. Fails open | 58B. None | 58B. None - Check valve 117, 113, and 133 provides redundancy. | |
| 59 | Not Used | | | | | | |
| 60 | AFW line check valve 126 | Same as item 58 except for steam generator No. 2 and for control building overpressure protection. | 60A. All but AFW HELB | 60A. Fails closed | 60A. Same as for item 46B except "None" for Mode 7 | 60A. None - Same as item 46B | |
| | | | 60B. AFW HELBA | 60B. Fails open | 60B. None | 60B. None - Check valve 118, 114, 134 provides redundancy. | |
| 61 | Not used | | | | | | |
| 62 | AFW line check valve 128 | Same as item 58 except for steam generator No. 3 and for control building overpressure protection. | 62A. All but AFW HELB | 62A. Fails closed | 62A. Same as for item 47B except "None" for Mode 7 | 62A. None - Same as item 47B | |

TABLE 10.4.9-4 (SHEET 27 OF 31)

| Item No. | Description of Component[a] | Safety Function | Plant Operating Mode[b] | Failure Mode(s) | Method of Failure Detection[c] | Failure Effect on System Safety Function Capability | General Remarks |
|---|---|---|---|---|---|---|---|
| | | | 62B. AFW HELB | 62B. Fails open | 62B. None | 62B. None - Check valve 120, 116, 136 provides redundancy. | |
| 63 | Not used | | | | | | |
| 64 | AFW line check valve 127 | Same as item 58 except for steam generator No. 4. | 64A. All but AFW HELB | 64A. Fails closed | 64A. Same as for item 48B except "None" for Mode 7 | 64A. None - Same as item 48B | |
| | | | 64B. AFW HELB | 64B. Fails open | 64B. None | 64B. None - Check valve 119, 116, 135 provides redundancy. | |
| 65 | TDP steam line drain valve HV-5178, normally closed MOV, not train aligned | Opens manually from QMCB or automatically to drain excessive amounts of condensate from TDP steam supply line | Modes 3, 4, 5, and 6 | 65A. Fails open | 65A. Position light on QMCB | 65A. None - Possible degraded TDP P4-001 performance, but MDPs are available. | |
| | | | | 65B. Fails closed | 65B. Position light on QMCB | 65B. None - Condensate will drain via FO-15133. Also, turbine accepts water slugs. | |
| 66 | Not used | | | | | | |
| 67 | System components powered from essential ac bus: HV-5119 (item 7), HV-5118 (item 8), MDP P4-003 (item 10), and MDP P4-002 (item 11) | Refer to items 7, 8, 9, and 10 as indicated | Mode 8 | Both MDPs fail to operate and the HVs remain in position (normally closed) | Status lights indicate loss of all ac power, plus all pump valve indicators remain dark | None - TDP P4-001 (item 12) provides 100% of AFW requirements. All valves, controls, etc., associated with the TDPs are dc powered from emergency battery supply. | See also items 9, 12, 13, 14, and 15 for discussion of valves associated with operation of TDP P4-100. |
| 68 | Chemical injection system isolation valve HV-5194 (normally closed, fail closed air-operated valve, non-1E) | Provides isolation of SG 1 from chemical injection system by remote-manual control from QPCP | All | 68A. Fails closed or fails to open upon command | 68A. Position light on QPCP | 68A. None – SG chemistry control not required for safe shutdown | |

TABLE 10.4.9-4 (SHEET 28 OF 31)

| Item No. | Description of Component[a] | Safety Function | Plant Operating Mode[b] | Failure Mode(s) | Method of Failure Detection[c] | Failure Effect on System Safety Function Capability | General Remarks |
|---|---|---|---|---|---|---|---|
| | | | | 68B. Fails open or fails to close upon command | 68B. Same as item 68A | 68B. None - Check valve 133 provides isolation. | |
| 69 | Chemical injection system isolation valve HV-5195 (normally closed, fail closed air-operated valve, non-1E) | Same as item 68 except for SG 2 | All | 69A. Fails closed or fails to open upon command | 69A. Position light on QPCP | 69A. None - Same as item 68A | |
| | | | | 69B. Fails open or fails to close upon command | 69B. Same as item 69A | 69B. None - Check valve 134 provides isolation | |
| 70 | Chemical injection system isolation valve HV-5196 (normally closed, fail closed air-operated valve, non-1E) | Same as item 68 except for SG 3 | All | 70A. Fails closed or fails to open upon command | 70A. Position light on QPCP | 70A. None - Same as item 68A | |
| | | | | 70B. Fails open or fails to close upon command | 70B. Same as item 70A | 70B. None - Check valve 136 provides isolation. | |
| 71 | Chemical injection system isolation valve HV-5197 (normally closed, fail closed air-operated valve, non-1E) | Same as item 68 except for SG 4 | All | 71A. Fails closed or fails to open upon command | 71A. Position light on QPCP | 71A. None - Same as item 68A | |
| | | | | 71B. Fails open or fails to close upon command | 71B. Same as item 71A | 71B. None - Check valve 135 provides isolation. | |
| 72 | Check valve 133 in chemical injection line to SG 1 | Prevents loss of AFW through failure of the chemical injection system | All | 72A. Fails closed | 72A. None | 72A. None - Same as item 68A | |
| | | | | 72B. Fails open | 72B. None | 72B. None - HV-5122 and HV-5139 can be closed to prevent loss of AFW inventory. | |

TABLE 10.4.9-4 (SHEET 29 OF 31)

| Item No. | Description of Component[a] | Safety Function | Plant Operating Mode[b] | Failure Mode(s) | Method of Failure Detection[c] | Failure Effect on System Safety Function Capability | General Remarks |
|---|---|---|---|---|---|---|---|
| 73 | Check valve 134 in chemical injection line to SG 2 | Same as item 72 | All | 73A. Fails closed | 73A. None | 73A. None - Same as item 68A | |
| | | | | 73B. Fails open | 73B. None | 73B. None - HV-5125 and HV-5132 can be closed to prevent loss of AFW inventory. | |
| 74 | Check valve 136 in chemical injection line to SG 3 | Same as item 72 | All | 74A. Fails closed | 74A. None | 74A. None - Same as item 68A | |
| | | | | 74B. Fails open | 74B. None | 74B. None - HV-5127 and HV-5134 can be closed to prevent loss of AFW inventory. | |
| 75 | Check valve 135 in chemical injection line to SG 4 | Same as item 72 | All | 75A. Fails closed | 75A. None | 75A. None - Same as item 68A | |
| | | | | 75B. Fails open | 75B. None | 75B. None - HV-5120 and HV-5137 can be closed to prevent loss of AFW inventory. | |
| 76 | TDP steam line drain valve HV-5179 (normally closed MOV not train aligned) | Opens manually from QMCB or automatically drain excessive amounts of condensate from top steam supply line | Modes 3, 4, 5, and 6 | 76A. Fails open | 76A. Position light on QMCB | 76A. None - Possible degraded TDP P4-001 performance, but MDP available. | |
| | | | | 76B. Fails closed | 76B. Position light on QMCB | 76B. None - Condensate will drain via FO-15141. Also, turbine accepts water slugs. | |
| 77 | Relief valve PSV-5110 in TDP/P-4-001 | Prevents over pressure and damage to pump suction line and line instrumentation if | 77A. All but mode 5 | 77A. PSV fails to open with pump out of service | 77A. None | 77A. None - Potential suction line/instrumentation | |

TABLE 10.4.9-4 (SHEET 30 OF 31)

| Item No. | Description of Component[a] | Safety Function | Plant Operating Mode[b] | Failure Mode(s) | Method of Failure Detection[c] | Failure Effect on System Safety Function Capability | General Remarks |
|---|---|---|---|---|---|---|---|
| | train "C" suction line 010-10 | pump suction check valves 013 & 051 hold tight & pump discharge check valve 014 leaks even TDP is not in service | | | | damage: remaining two MDP's, train "B" & "A" each available, either satisfies 100% flow requirement. | |
| | | | 77B. All but mode 5 | 77B. Spurious PSV opening with TDP out of service | 77B. None | 77B. None | |
| | | | 77C. All but mode 7 | 77C. Spurious PSV opening with TDP in service | 77C. None | 77C. None - TDP still delivers flow requirement, since flow thru PSV is insignificant. | |
| 78 | Relief valve PSV-5128 in MDP/P-4-002 train "B" suction line 017-8 | Prevents over pressure and damage to pump suction line & instrumentation if pump suction check valves 050 & 061 hold tight & pump discharge check valve 002 leaks when MDP is not in service | 78A. All modes | 78A. Same as item 77A above with pump-002 | 78A. Same as item 77A above | 78A. None - Potential suction line/ instrumentation damage; remaining MDP & TDP available; MDP provides 100%, TDP provides 200% flow requirement | |
| | | | 78B. All modes | 78B. Same as item 77B above with pump-002 | 78B. Same as item 77B above | 78B. Same as item 77B above | |
| | | | 78C. All but mode 7 | 78C. Same as item 77C above with pump-002 | 78C. Same as item 77C above | 78C. Same as item 77C above | |
| 79. | Relief valve PSV-5129 in MDP/P-4-002 train "A" suction line 016-8 | Prevents over pressure and damage to pump suction line & line instrumentation if pump suction check valve 033 & 052 hold tight & pump discharge check valve 001 leaks when MDP is not in service | 79A. All modes | 79A. Same as item 77A above with pump-003 | 79A. Same as item 77A above | 79A. None - potential suction line/instrumentation damage remaining MDP & TDP available. MDP provides 100%, TDP pump provides 200% flow requirement | |
| | | | 79B. All modes | 79B. Same as item 77B above with pump-003 | 79B. Same as item 77B above | 79B. Same as item 77B above | |

TABLE 10.4.9-4 (SHEET 31 OF 31)

| Item No. | Description of Component[a] | Safety Function | Plant Operating Mode[b] | Failure Mode(s) | Method of Failure Detection[c] | Failure Effect on System Safety Function Capability | General Remarks |
|---|---|---|---|---|---|---|---|
| | | | 79C. All but mode 7 | 79C. Same as item 77C above with pump-003 | 79C. Same as item 77C above | 79C. Same as item 77C above | |

a. P&IDs IX4DB159-2, rev 15; 1X4DB161-1, rev 15; 1X4DB161-2, rev 15; 1X4DB161-3, rev 13; 1X4DB168-3, rev 15. Logic Diagrams 1X5DN108-1, rev 4, 1X5DN109-1, rev 2; 1X5DN109-2, rev 2; 1X5DN117-1, rev 4; 1X5DN120-1, rev 2; 1X5DN120-2, rev 3; 1X5DN120-3, rev 1; 1X5DN120-4, rev 3; 1X5DN120-5, rev 3; 1X5DN120-6, rev 2; 1X5DN121-1, rev 3; 1X5DN121-2, rev 3; 1X5DN122-1, rev 3; 1X5DN122-2, rev 3.

b. Mode 1, plant startup; mode 2, plant cooldown; mode 3, loss of normal feedwater coincident with loss of offsite power (LOP); mode 4, auxiliary feedline rupture coincident with LOP; mode 5, HELB in TDP discharge coincident with LOP; mode 6, safety injection coincident with LOP; mode 7, power generation; and mode 8, loss of all ac power.

c. In general, valve and other indicators will illuminate either on the main control board or on the shutdown panel, but not on both simultaneously, depending on the position of the transfer switch.

TABLE 10.4.9-5 (SHEET 1 OF 2)

AUXILIARY FEEDWATER SYSTEM SUMMARY OF INDICATION AND CONTROLS

INDICATION

| Parameter | Main Control Board | Shutdown Panel | Turbine-Driven Pump AFW Panel |
|---|---|---|---|
| Steam generator water level | X | X | X |
| Steam generator pressure | X | X | |
| AFW motor-driven pump suction pressure | X | X | |
| AFW motor-driven pump discharge pressure | X | X | |
| Turbine-driven pump steam inlet pressure | X | | X |
| Turbine-driven pump differential steam pressure | X | | X |
| Auxiliary feedwater flowrate | X | X | X |
| AFW isolation/control valve position | X | X | X |
| Turbine-driven pump steam inlet valve position | X | | X |
| CST level | X | X | X |
| Motor-driven pump miniflow valve position | X | X | |
| CST motor operated outlet valve position | X | X | X |
| AFW pump motor running | X | X | |
| CST level alarms | X | | X |
| AFW pump discharge pressure low alarm | X | | |
| Control transferred to local panel alarm | X | | |

TABLE 10.4.9-5 (SHEET 2 OF 2)

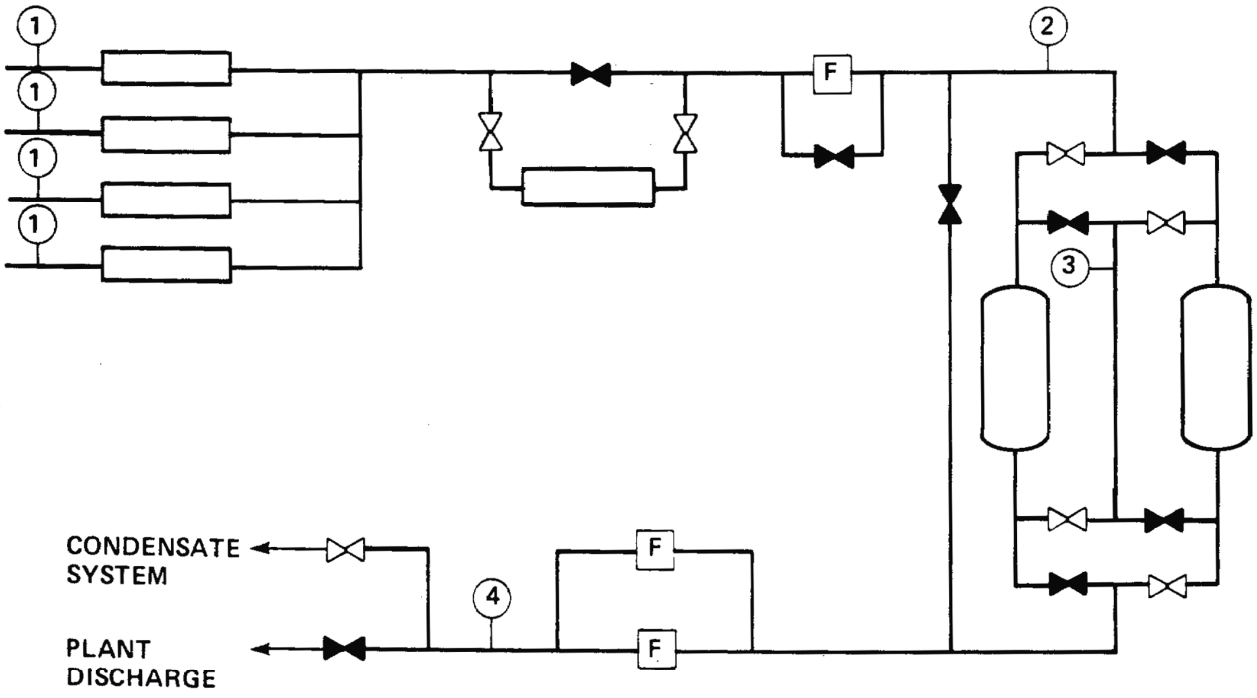
CONTROLS

| Parameter | Main Control Board | Shutdown Panel | Turbine-Driven Pump AFW Panel |
|---|---|---|---|
| Motor-driven pumps | X | X | |
| Turbine-driven pump | X | | X |
| AFW isolation/control valves | X | X | X |
| Turbine-driven pump speed control | X | | X |

TABLE 10.4.10-1

CONDENSATE AND FEEDWATER CHEMICAL
INJECTION SYSTEM

| Major Component | Design Parameters |
|---|---|
| Methoxypropylamine (MPA) storage tank | 12,789 gal |
| Hydrazine storage tank | 6644 gal |
| Day tanks | 250 gal |
| Metering cylinders | 10 liter |
| MPA storage transfer pump | 43 gal/min at 178 ft head |
| Hydrazine storage transfer pump | 43 gal/min at 110 ft head |
| Condensate chemical pump | Hydrazine - 11.7 gal/h at 650 psig<br>MPA - 0.5 gal/h at 650 psig |
| Steam generator layup chemical pump | 100 gal/h at 1200 psig |
| Auxiliary feedwater layup chemical pump | 100 gal/h at 1200 psig |
| Hydrazine dispensing pump/high volume | 3 gal/min at 24 ft head |
| Day tank mixing pump | 10 gal/min at 20 ft head |
| Ammonium chloride pump (Unit 1 only) | 0.2 gal/day |
| Ammonium chloride tank (Unit 1 only) | 7.5 gal |




| LOCATION | | TEMP. (°F) | PRESSURE (PSIG) | FLOW. (GPM)(1) | ISOTOPIC CONC. |
|---|---|---|---|---|---|
| 1 | INPUT FROM STEAM GENERATOR | 545 | 927-1075(5) | 90 | (2) |
| 2 | INPUT TO DEMINERALIZERS | 130 | <200 | 360 | (2) |
| 3 | OUTLET FROM 1st DEMINERALIZER | 130 | <200 | 360 | (3) |
| 4 | DISCHARGE FROM SGBPS | 130 | <200 | 360 | (4) |

NOTES:
1. DESIGN MAXIMUM FLOW AT 60°F
2. ISOTOPIC CONCENTRATIONS AS LISTED ON TABLE 11.1-7
3. CONCENTRATIONS OF CESIUM AND RUBIDIUM ISOTOPES ARE 0.1 OF THE VALUES LISTED ON TABLE 11.1-7. CONCENTRATIONS OF OTHER ISOTOPES ARE 0.01 OF THE VALUES LISTED ON TABLE 11.1-7.
4. CONCENTRATIONS OF CESIUM AND RUBIDIUM ISOTOPES ARE 0.01 OF THE VALUES LISTED ON TABLE 11.1-7. CONCENTRATIONS OF OTHER ISOTOPES ARE 0.001 OF THE VALUES LISTED ON TABLE 11.1-7.
5. ZERO-TO 100-PERCENT STEAM GENERATOR LOAD.



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VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

STEAM GENERATOR BLOWDOWN PROCESSING SYSTEM PROCESS FLOW DIAGRAM

FIGURE 10.4.8–1

# APPENDIX 10A

# VEGP AUXILIARY FEEDWATER SYSTEM RELIABILITY ANALYSIS

## 10A.1 INTRODUCTION

This appendix documents the results of the reliability analysis discussed below. This analysis was performed in accordance with Item II.E.1.1 of NUREG-0737 at the time of initial licensing. Since that time, the analysis has not been updated to reflect current plant configuration. Therefore, this appendix is retained for historical information only.

### 10A.1.1 STATEMENT OF PURPOSE

The purpose of this appendix is to address the conformance of the VEGP auxiliary feedwater system (AFWS) with the Standard Review Plan's recommendation that a reliability analysis of the AFWS be performed.

### 10A.1.2 BACKGROUND AND OBJECTIVES

The Three Mile Island Unit 2 (TMI-2) accident occurred on March 28, 1979. The TMI-2 accident and subsequent investigations and studies highlighted the importance of the AFWS in the mitigation of transients and accidents. Prompted by the TMI-2 accident, the staff of the Nuclear Regulatory Commission (NRC) evaluated the AFWS for all operating plants having nuclear steam supply systems designed by Westinghouse (NUREG-0611, January 1980[1]). The NRC subsequently mandated in its March 10, 1980, letter to all pending operating license applicants of Westinghouse and Combustion Engineering nuclear steam supply systems that a comparable reliability evaluation of the AFWS be performed for each licensee. The requirements outlined in that letter have since been integrated into the Standard Review Plan.[2] Consistent with the recommendations of the Standard Review Plan, this analysis has the following objectives:

A. Outline the details of the AFWS and present the information required to perform a reliability evaluation.

B. Perform a reliability evaluation of the AFWS which corresponds with the outline recommended by the NRC in Appendix III and Annex 1 of Appendix X in NUREG-0611 and Item II.E.1.1-1 in NUREG-0737[3] (section 10A.3).

C. Identify and address the important contributors to the AFWS unreliability (subsection 10A.4.2).

D. Develop a comparative basis for the AFWS reliability study versus the generic AFWS reliability study which has been performed by the NRC staff (subsection 10A.4.3).

## 10A.2 AFWS DESCRIPTION

### 10A.2.1 SAFETY-RELATED FUNCTION

The safety-related function of the AFWS is to maintain water inventory in the steam generators for reactor residual heat removal when the main feedwater system is unavailable. The AFWS in conjunction with the steam generators and the main steam line atmospheric relief and/or safety valves, is used to cool the reactor coolant system to 350°F, at which point the residual heat removal system is used to further cool the reactor coolant system. The AFWS may also be used to temporarily hold the plant in a hot standby condition while main feedwater flow is restored, with the option of cooling the reactor coolant system to the residual heat removal initiation temperature.

### 10A.2.2 SYSTEM CONFIGURATION

The AFWS for a given unit, shown schematically in figures 10A-1 and 10A-2, provides auxiliary feedwater from the condensate storage tanks (CSTs) (two per unit) to the four steam generators by means of mechanical trains A, B, and C.

Each train initially takes suction from a CST which is maintained above the minimum level of 330,000 gal, with the capability of manual switchover to the other CST. The minimum water level of a CST is designed to maintain the reactor in a hot standby condition for 4 h followed by a 5-h cooldown period, at which time the residual heat removal system can be used to further cool the reactor coolant system. The combined minimum operating capacity of the CSTs (660,000 gal) is designed to allow a hot standby condition for 31 h followed by a 5-h cooldown period until operation of the residual heat removal system is initiated.

From the CSTs, the auxiliary feedwater passes through the respective pump of each mechanical train and exits through a discharge header into the steam generators. The motor-driven pumps of mechanical trains A and B each provide more than 100 percent of the required auxiliary feedwater flow, with mechanical train A capable of providing feedwater to steam generators 1 and 4 and mechanical train B capable of providing feedwater to steam generators 2 and 3. The steam-driven turbine pump of mechanical train C is capable of providing 200 percent of the minimum required auxiliary feedwater flow for cooldown of the reactor. Therefore, any one of the AFWS mechanical trains operating will provide sufficient feedwater to two effective steam generators to allow cooldown of the reactor.

With a few exceptions, all valves in the AFWS flowpath from the CSTs to the steam generators are normally open. The exceptions to this are the motor-operated valves located between the normal standby CST and the suction of each auxiliary feedwater pump. These valves are opened from the control room when the water from the normal standby CST is required.

Though normally open, the motor-operated valves in the discharge of the AFWS receive an actuation signal to open automatically (5120, 5122, 5125, 5127, 5132, 5134, 5137, and 5139). Also, these valves are used by the operator to control auxiliary feedwater flow to the steam generators.

The turbine-driven pump of mechanical train C is powered by steam tapped from the main steam lines of steam generators 1 and 2. The motor-operated valves in each of these lines (3009 and 3019, respectively) are normally open. The steam from either steam generator is sufficient to drive the turbine-driven pump.

All three mechanical trains of the AFWS have miniflow lines on the discharge side of the pumps which allow a minimum flow through the pumps during their start. Additionally, these miniflow lines are used for inservice testing of the pumps. The flow, once passing through the pumps, is returned to the CSTs. The miniflow lines of mechanical trains A and B each have a motor-operated valve (5155 and 5154, respectively) which closes automatically after flow has been established through each respective pump. The miniflow lines of mechanical trains A and B also have flow-limiting orifices in each line which limit the flow to 175 gal/min/line. The miniflow line of the turbine-driven pump has flow through it whenever the turbine-driven pump is operating. A flow-limiting orifice in the turbine-driven pump miniflow line limits the flow to 160 gal/min.

On the discharge side of the motor-driven pumps there is a crossover tieline which has been added to allow versatility in the operation of the AFWS. The two valves in this line are normally locked closed (055 and 056); therefore, this crossover tieline is exempted from normal use. However, for emergency use, the operator may use the crossover tieline to maintain the water level in all four steam generators with only one motor-driven pump if an extended use of the AFWS is required.

For flushing of the AFWS, an auxiliary feedwater recirculation line draws water from the AFWS header of each steam generator and returns it to the condensate system.

Each recirculation line is fitted with an orifice that limits the amount of flow diverted away from the steam generators.

When the AFWS is not being flushed, the recirculation valves (81, 82, 83, and 84) must remain closed.

### 10A.2.3 INSTRUMENTATION AND CONTROLS

#### 10A.2.3.1 Automatic Controls

The AFWS is aligned to be placed in service automatically in the event of a demand. Following the receipt of a safety injection signal, a two-out-of-four low-low steam generator water level signal from any one steam generator, a trip signal from both main feedwater pumps, or a loss of offsite power signal, the auxiliary feedwater discharge valves go to the full-open position if not already open and the two motor-driven auxiliary feedwater pumps are actuated and begin to deliver flow from the online CST to the steam generators. Once flow has been established, the motor-operated valves in the miniflow lines close automatically. The turbine-driven pump is actuated automatically on two-out-of-four low-low water level in any two steam generators or on a loss of offsite power signal. To actuate the turbine-driven pump, a normally closed dc motor-operated valve (5106) in the steam supply line to the turbine is opened automatically. The speed governing valve and the trip/throttle valve, which are in the same line as the steam inlet valve, are automatically controlled by the speed governor on the turbine-driven pump. Following a transient or accident, the minimum flow is delivered to at least two effective steam generators within 1 min of an automatic auxiliary feedwater actuation signal. Once actuated, the operator can remotely manipulate the position of the auxiliary feedwater control valves in order to control steam generator water level.

### 10A.2.3.2 Manual Controls

For normal operation, the AFWS is used to fill and/or maintain the water level in the steam generators during startup, shutdown, and hot standby conditions. The AFWS may be actuated and controlled manually during normal operation or abnormal conditions. The components designated in table 10A-1 can be remotely controlled from the control room.

### 10A.2.3.3 Information Available to Operator

The parameters which the control room operators can use to judge the status of the AFWS are designated in table 10A-1.

## 10A.2.4 SUPPORT SYSTEMS

### 10A.2.4.1 Electric Power

The components dependent on electric power in mechanical trains A and B receive their electric power from the Class 1E switchgear which have diesel generators in reserve, should offsite power be lost. The diesel generators are designed to actuate on a safety injection signal or a loss of ac power signal. The loss of ac power signal is generated by an undervoltage condition on the 4.16-kV bus of train A or B. The safety injection signal is generated by low steam pressure, low pressurizer pressure, high containment pressure, or manual actuation. The components of mechanical trains A and B are powered from the corresponding emergency ac electric power trains A and B, respectively. Motor-operated valves 3019 and 3009 of mechanical train C are powered from electrical trains A and B, respectively. The electrically powered components of mechanical train C, with the exception of motor-operated valves 3009 and 3019, are all powered from dc electrical train C, which is backed up by the station batteries. The batteries for each electrical train are sized for a LOCA/LOSP duration of 2 3/4 h and a station blackout (SBO) duration of 4 h to provide continuous dc power without recharging. The electrical power sources for the components of the AFWS are listed in table 10A-2. As indicated in this table, the power to the controls of each component is the same as the component itself.

### 10A.2.4.2 Heating and Cooling

The heating and cooling systems are designed to maintain the temperature in the pump rooms between 40°F and 120°F. The fans that regulate the temperature in each of the motor-driven pump rooms are powered from the Class 1E onsite electrical system. In the pump rooms there are heaters powered from the non-Class 1E onsite electrical system. The turbine-driven pump room is provided with a nonsafety grade fan that is powered by a nonemergency electrical bus. The heaters and TDAFW pump room fan are intended for nonemergency use of the AFWS. For emergency use of the turbine-driven pump, the turbine-driven pump room is designed to be cooled by natural ventilation.

Independent cooling is provided for each of the AFWS pumps. Each motor-driven pump has air-cooled motors and water-cooled pumps. The turbine-driven pump uses water to cool the pump and the lube oil. The three pumps are located in separate rooms in the AFWS pumphouse, with each room cooled by its own ventilation system.

The stuffing boxes of each of the pumps are cooled by water tapped from the respective pump so that cooling water is provided if a pump is running. The lube oil of the turbine-driven pump is cooled by water drawn from the discharge of the turbine-driven pump.

### 10A.2.4.3 Water Sources

There are two water sources to the AFWS. They are the two CSTs. Each tank is a Seismic Category 1 structure and has a capacity of 480,000 gal. A total of 330,000 gal from one tank is sufficient to maintain the reactor in the hot standby mode for 4 h followed by a 5-h cooldown period. The minimum safety capacity is ensured by all nozzles of nonsafety systems being located on the storage tanks above the corresponding elevation. The condensate level in each tank is automatically maintained by a level control valve in the line (to the tank) from the demineralized water system, which actuates when the volume in the tank drops to 472,250 gal.

As the water in the online CST is depleted, the operator may manually realign the system so that the standby CST serves all three pumps. A separate line connects each pump to each CST.

### 10A.2.4.4 Main Steam System

The main steam system provides the steam supply to the AFWS turbine-driven pump. The steam to the turbine is provided from steam generators 1 and 2, either of which are sufficient for operation of the AFWS turbine-driven pump. Each of the steam supply lines to the turbine is equipped with a check valve, a locked open gate valve, and a normally open dc motor-operated gate valve. These steam supply lines join to form a header which leads to the turbine through a normally closed supply valve and normally open trip/throttle valve, both of which are dc motor-operated, and a normally open electro-hydraulically operated speed governing valve.

The main steam system is not the only supply of steam to the AFWS turbine-driven pump. A backup source is provided by a connection on the turbine steam supply header from an auxiliary steam system boiler inter-tie between units. This connection is normally isolated from the turbine by a manually operated normally closed gate valve.

## 10A.2.5 TESTING, MAINTENANCE, AND PLANT PROCEDURES

As of the date of this evaluation, the Technical Specifications, operating procedures, maintenance procedures, and testing procedures applicable to the VEGP AFWS have not been written.

Thus, in order to model and analyze the contribution of human error, testing and maintenance to the reliability of the VEGP AFWS relevant generic documents were used.

The Technical Specifications used were extracted from the Westinghouse Standard Technical Specifications.[4] The most notable factors of these preliminary Technical Specifications are:

A. With one AFWS pump inoperable, the limiting condition of operation action time to hot standby is 78 h.

B. With two AFWS pumps inoperable, the limiting condition of operation action time to hot standby is 6 h.

C. The testing frequency for AFWS pumps is once per 31 days.

D. The verification frequency of valves in the flowpath is once per 31 days.

E. The testing frequency of pumps and valves with automatic actuation is performed once per 18 months.

F. With one or more steam generators inoperable, the limiting condition of operation action time is 1 h.

G. With less than 330,000 gal in the CSTs, the limiting condition for operation action time to hot shutdown is 16 h.

H. The verification frequency of the CSTs water volume is once per 12 h.

I. With one 125-V dc train inoperable, the limiting condition for operation action time to hot standby is 2 h.

J. The testing frequency of each dc train is once per 7 days.

The generic plant operating procedures were assumed in the VEGP AFWS reliability evaluation. The most notable factors of these generic operating procedures are:

A. As part of the reactor trip procedures, the operator is to verify the proper automatic actuation of the AFWS (with at least one AFWS pump feeding to two steam generators).

B. As part of the station blackout procedures, the operator is to verify the proper automatic actuation of the AFWS (with at least one AFWS pump feeding to two steam generators). Station blackout coping is discussed in section 8.4.

C. As part of the station blackout procedures, the operator is to immediately align AFWS flowpaths to all steam generators to minimize the possibility of emptying any steam generator. Station blackout coping is discussed in section 8.4.

D. As part of the station blackout procedures, the operator is to maintain the steam generator level by manual control of the AFWS. Station blackout coping is discussed in section 8.4.

E. As part of the loss of secondary coolant procedures, the operator is to verify the proper automatic actuation of the AFWS.

F. As part of the loss of secondary coolant procedures, the operator is to isolate the AFWS flow to the affected steam generator to protect it from undesirable thermal transient.

G. As part of any procedure, manual actuation of the AFWS by the operator will be a situation the operator is readily familiar with.

H. As part of any procedure, manual actuation of the AFWS by the operator can be accomplished from the control room.

The generic plant testing and maintenance procedures used in the AFWS reliability evaluation were a synthesis of generic procedures. These generic procedures are based on current industry practice, lessons learned from previous human reliability analysis, and the VEGP AFWS design capabilities. The design capabilities of the AFWS allow testing while the plant is operating without affecting main feedwater flow. The alignment of any train of the AFWS for testing is such that suction is taken from a CST and the flow passes through the pump through the miniflow line, and back to the CST. Closure of the motor-operated valves in the discharge lines prevents flow to the steam generators during AFWS testing.

A. Human events for redundant trains as modeled are noncoupled events.

B. The motor-operated valves from CST 002 (5113, 5118, and 5119) are manually controlled with no automatic signals to close (if CST 002 is being used for testing or flushing of an AFWS train).

C. The motor-operated valves in the miniflow lines of trains A and B (5154 and 5155) are subject to maintenance for calibration of the flow element actuation device in these valves.

D. The motor-operated valves in the discharge lines (5120, 5122, 5125, 5127, 5132, 5134, and 5137) are used to manually throttle AFWS flow and pressure during testing to keep AFWS flow from entering a steam generator.

E. The performance of maintenance on a component requires the component be manually isolated on both the upstream and downstream sides.

F. The motor-operated valves in the discharge lines receive an automatic actuation signal to go to their full-open position even if they are being used for testing.

### 10A.2.6 PHYSICAL SEPARATION

Physical separation between the trains of the AFWS is maintained in regard to the prevention of common cause failures created by fire, flooding, and missiles. A simplified piping layout schematic of the AFWS is provided in figure 10A-3. Excluding the containment building, there are only two locations where a portion of all three trains lie in a common area. The first is in the building that houses the CSTs and the second is in a pipe chase in the auxiliary feedwater pumphouse. Both of these locations:

A. Are protected from external missiles and have no internal source for missiles.

B. Have no components subject to disabling damage due to flooding.

C. Have minimal sources of fire.

Physical separation between electrical components of the AFWS is provided in accordance with Regulatory Guide 1.75 and Institute of Electrical and Electronics Engineers (IEEE) Standard 384.

## 10A.3 RELIABILITY EVALUATION

### 10A.3.1 METHODOLOGY

As recommended in NUREG-0800, the reliability evaluation of the AFWS was performed utilizing the methodology in Appendix III and Annex I of Appendix X in NUREG-0611. This subsection provides the specific VEGP AFWS methodology utilized in the development and analysis of the reliability model. The outline of the methodology is presented as a flow chart in figure 10A-4. A discussion of the flow chart follows.

#### 10A.3.1.1 Development of System Bounds

To perform the reliability analysis, it was necessary to define the system boundaries. The AFWS was defined to be bounded by:

A. The CSTs (i.e., begins at).

B. The inlet to the steam generators (i.e., ends at).

C. The Class 1E electric power system immediate to the components of the AFWS (i.e., begins at).

D. The steam lines that lead from the main steam system to the turbine-driven pump (i.e., begins at).

E. The cooling and ventilation system immediate to the AFWS (i.e., begins at).

### 10A.3.1.2 Development of the Reliability Block Diagram

In accordance with NUREG-0611, fault tree analysis was utilized to model and quantify the reliability of the AFWS. (The methodology of fault tree analysis is not explained herein; references 5, 6, and 7 are recommended for the unfamiliar reader.) To develop the appropriate fault tree logic, a reliability block diagram (block diagram for short) was developed. The block diagram developed is shown in figure 10A-5. A block diagram was developed, as an intermediate step because it essentially depicts the relationship between groups of components (for a fault tree analyst, the blocks may be thought of as representing supercomponents) and because it represents the logical relationship of components in a format similar to the system diagram. The components within each block of figure 10A-5 are listed in table 10A-3. From the original block diagram, an expanded block diagram was developed. This second block diagram, shown in figure 10A-6, was developed to emphasize the component groupings which can be unavailable due to maintenance or which can fail randomly. As is evident, the steam generator blocks are allowed to only fail randomly because maintenance upon the components represented by these blocks is not allowed while the plant is operating.

### 10A.3.1.3 Review of the Generic AFWS Analysis Performed by the NRC

A review of the generic AFWS analysis (NUREG-0611) was performed to ensure the assumptions and boundary conditions used in the AFWS reliability analysis are consistent with similar studies so as not to influence the results disproportionably.

The review of the generic AFWS reliability analysis revealed the generic analysis consisted of:

A. Quantifying a point estimate of the probability of failure on demand of the AFWS.

B. Internal plant events (not including fire).

C. A qualitative evaluation of common cause.

D. A time frame equivalent to the time it takes a steam generator to boil dry with no makeup available.

E. An evaluation of the AFWS as it pertains to three accident sequences. The accident sequences are:

1. Case 1 - Loss of main feedwater with a reactor trip occurring and offsite ac power available (referred to as LMFW).

2. Case 2 - Loss of offsite ac power causing main feedwater to be lost and the reactor to trip (referred to as LMFW/LOOP).

3. Case 3 - Loss of main feedwater and only dc electrical power available (referred to as LMFW/LOAC).

The aforementioned potential accident sequences and time frame designations are actually boundary conditions of the reliability evaluation. But because they have a major impact on the quantification of the AFWS reliability, they are discussed herein. The major impact these two designations have is that they change the evaluation from the misnomered system reliability to a system conditional unavailability. The reasons that the two designations change the evaluation from a reliability evaluation to a conditional unavailability are:

A. The three accident sequences place the plant in an unconventional state, and as such there is a probability associated with the plant being placed in any of the three states. Also, there is an associated probability of the plant recovering from any of the three states; thus, the result is conditional to these probabilities.

B. The designated time frame does not encompass the amount of time the AFWS would be expected to operate, i.e., the mission time. Because of this, the calculations (quantification of the fault tree) are toward calculating the system availability or unavailability. (System unavailability is defined as the probability of the system being unavailable due to testing or maintenance or the probability of the system failing on demand.)

The effect of three accident sequences places the unavailabilities into a conditional state. This means that the unavailabilities are important only so far as the likelihood (or probability) of the plant being placed in any of these states. The effect of the quantification of a system unavailability (versus reliability) is that the result is an indication of system performance and is not complete. To emphasize this point, the quantification of the fault tree will be referred to as a conditional unavailability.

### 10A.3.1.4 Development of the AFWS Fault Tree Model to Component Failure Causes

One of the objectives of the AFWS reliability analysis was to develop a logic model which could be quantified. Toward this end, the basic tool utilized was a fault tree model. The fault tree model was developed as a logical progression from the reliability block diagram. Also, the fault tree was synthesized from the defined boundary conditions and necessary assumptions including those from the NRC's generic analysis.

The initial fault tree was developed to the component failure mode level and then expanded to the component failure cause level. The component failure causes considered were:

A. Random failure on demand.

B. Unavailability due to maintenance.

The fault tree developed for the analysis is shown in figure 10A-7.

### 10A.3.1.5 Minimal Cut Set Analysis

Once developed, the fault tree was analyzed for determining the minimal cut sets using the WAM-CUT computer code.[8] The list of minimal cut sets was then analyzed for their viability in reference to plant Technical Specifications. Also, the possibility of operator action (within the constricting time frame) was evaluated.

### 10A.3.1.6 Quantification

The final version of the fault tree was quantified for each of the three cases. The quantification process included applying a statistically independent probability to each basic event in the fault tree for each of the three cases. The three cases were evaluated to determine a point estimate of the conditional unavailability of the VEGP AFWS given:

A. Case 1 - Loss of main feedwater with a reactor trip and electric power available (LMFW).

B. Case 2 - Loss of offsite ac power causing a loss of main feedwater with a reactor trip and onsite electric power available (LMFW/LOOP).

C. Case 3 - Loss of main feedwater with a reactor trip and station blackout so that only dc electric power is available (LMFW/LOAC).

### 10A.3.1.7 Deterministic Common Cause Analysis

To assess the reliability of the VEGP AFWS against common cause failures, a common cause analysis was performed. This analysis was performed deterministically and in two parts. The first part was performed explicitly for common cause hardware failure by location. The analysis is discussed in subsection 10A.2.6. The second part of the common cause analysis was performed implicitly throughout the evaluation. That is, any potential failure of a support system (such as electrical power) resulting in the failure of supposedly redundant components, was accounted for in the logic fault tree. The results of the entire common cause analysis revealed no significant common cause potential within the VEGP AFWS.

### 10A.3.1.8 Derivation of Analysis Conclusions

Following the completion of the evaluation, conclusions were derived drawn on:.

A. The quantified conditional unavailability.

B. The dominant contributors to the conditional unavailability.

C. A comparison of the VEGP AFWS reliability with those AFWS evaluations already performed as reported in NUREG-0611.

## 10A.3.2 SYSTEM SUCCESS CRITERIA

The AFWS is composed of three mechanical trains which serve the four steam generators at a given unit. The steam generators have been analyzed to require 510 gal/min of flow under the most severe accident conditions. Each motor-driven pump of mechanical trains A and B has a capacity of 630 gal/min and provides more than 100 percent of the required auxiliary feedwater flow. Mechanical train A provides feedwater to steam generators 1 and 4, and mechanical train B provides feedwater to steam generators 2 and 3. The (steam) turbine-driven pump of mechanical train C has a capacity of 1300 gal/min and provides more than 200 percent of the required auxiliary feedwater flow. The turbine-driven pump provides feedwater to all four steam generators. The success criterion for the AFWS is flow to any two steam generators. Furthermore, as outlined by the NRC evaluation of generic AFWSs (NUREG-0611), the AFWS must actuate within the time it takes for the steam generators to boil dry when no flow is

provided to the steam generators. At VEGP, the boiloff time (and therefore the limit on the AFWS actuation time) is approximately 30 min.

### 10A.3.3 COMPONENT FAILURE DATA

The component failure data (data for short) used in quantifying the fault trees were taken, when possible, from NUREG-0611, which was derived in part from WASH-1400. When the appropriate data were not available from NUREG-0611, WASH-1400 data were used. Table 10A-4 is a listing of the data used in the VEGP AFWS reliability evaluation. All data were used to quantify point estimates of unavailability on demand, and uncertainty is not accounted for in the analysis. It should be noted that the data utilized in the reliability analysis is generic, and as such the results are an evaluation of the AFWS design. The implication of the data is that they do not account for the actual characteristics of how the plant is to be operated and maintained.

It should be noted that the formula used to calculate maintenance unavailability in NUREG-0611 was believed to be far too conservative for estimating valve maintenance unavailability. Instead a formula from NUREG-0492 was adapted. The NUREG-0611 formula appears to assume maintenance is performed every 4.5 months. Possibly this is routine maintenance. However, it was believed that maintenance on valves would be performed only when the valve fails. Therefore the formula used to estimate valve unavailability due to maintenance was:

$$\lambda_s T_R$$

where:

$\lambda_s$ = valve standby failure rate.

$T_R$ = average repair or maintenance time for valve.

### 10A.3.4 ANALYSIS BOUNDARY CONDITIONS AND ASSUMPTIONS

In order to properly model the AFWS, certain assumptions had to be made; in order to be consistent with the NRC's generic evaluation of AFWSs, the scope of the analysis had to be altered accordingly. The following is a list of the boundary conditions and assumptions which were used to model the conditional unavailability of the AFWS.

A. The AFWS is modeled for its safety function and not for its use during normal operation.

B. The AFWS can only be configured in modes consistent with the preliminary Technical Specifications and the preliminary plant procedures.

C. It is assumed that steam is available to the main steam lines that lead to the turbine of the turbine-driven pump.

D. The crossover tieline between trains A and B is not modeled because the use of this line falls outside of the 30-min limit on operator action.

E. The operator is allowed 30 min to actuate the AFWS from within the control room. This is based on the conservative assumption that it will take 30 min for the steam generators to boil dry after loss of main feedwater.

F. The loss of actuation of the AFWS is modeled as a basic event.

G. Sample lines are not modeled.

H. The loss of dc electric power is modeled as a basic event.

I. The loss of electric power to components of different voltages but powered from the same electrical train is modeled as the same basic event.

J. Common cause failures such as missiles, fires, floods, etc., are not modeled.

K. Pipe ruptures are not modeled (because of their negligible probability).

L. A single unit of VEGP is modeled and quantified.

## 10A.4 QUANTITATIVE FINDINGS

### 10A.4.1 CONDITIONAL UNAVAILABILITIES

The quantitative results of the conditional unavailabilities for the three cases designated by the NRC for the AFWS are:

A. Case 1 - LMFW

For the case where there is an assumed loss of main feedwater with a reactor trip occurring and offsite ac power available, the conditional unavailability of the AFWS was calculated to be $6.3 \times 10^{-6}$.

B. Case 2 - LMFW/LOOP

For the case where there is an assumed loss of main feedwater with a reactor trip occurring and offsite ac power not available, the conditional unavailability of the AFWS was calculated to be 2.6 x 10-5.

C. Case 3 - LMFW/LOAC

For the case where there is an assumed loss of main feedwater with a reactor trip occurring and no ac power available, the conditional unavailability of the AFWS was calculated to be 1.0 x 10-2.

### 10A.4.2 DOMINANT CONTRIBUTORS

The quantitative measure of importance was used as an indication of the dominant contributors to the AFWS conditional unavailability. The value of importance was taken as the sum of all cut set probabilities containing a component failure divided by the top event probability. The failure categories analyzed for each case are: random failure of valves on demand; unavailability of valves due to maintenance; operator error; and pump unavailabilities (random or maintenance).

A. Case 1 - LMFW

The most significant contributor to system failure was pump unavailabilities. The importance value to pump unavailabilities was calculated to be 86 percent. An examination of the category of pump unavailabilities revealed that pump failures were occurring in combination with electric power system failure.

Furthermore, it was determined that the unavailability of the turbine-driven pump was the most significant single component of the AFWS, but this pump did not dominate system unavailability. This finding is then an indication that due to redundancy of the AFWS, no single component of the AFWS can be thought of as dominating (or controlling) system unavailability.

B. Case 2 - LMFW/LOOP

The findings for case 2 revealed pump unavailabilities contribute 80 percent to system unavailability. An examination of this category revealed, as did case 1, no single component of the AFWS can be thought of as dominating (or controlling) system unavailability. The reduction of the system conditional availability for this case was found to be directly attributable to the assumed loss of redundancy in ac power sources.

C. Case 3 - LMFW/LOAC

The findings for case 3 revealed (under assumed conditions) that the AFWS is reduced to only the turbine-driven pump. Thus, any single failure along this pump train would be sufficient to fail the AFWS. The dominant contributors to system unavailability were as follows:

1. The turbine-driven pump package (pump, trip throttle valve, and speed governing valve).
2. The steam inlet valve (motor-operated valve 5106).

### 10A.4.3 COMPARATIVE STUDIES RESULTS

The availability of the AFWS for various plants, including VEGP, is compared in table 10A-5. This table is derived from NUREG-0611. For the case of LMFW, the AFWS compares very favorably, ranking in the very high category. For the case of LMFW/LOOP, the AFWS compares favorably, ranking in the high category. For the case of LMFW/LOAC, the AFWS ranks in the medium category.

### 10A.4.4 REFERENCES

1. U.S. Nuclear Regulatory Commission, "Generic Evaluation of Feedwater Transients and Small-Break Loss-of-Coolant Accidents in Westinghouse-Designed Operating Plants," NUREG-0611, Bulletins and Orders Task Force, Office of Nuclear Reactor Regulation, January 1980.
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4. U.S. Nuclear Regulatory Commission, "Standard Technical Specifications for Westinghouse Pressurized Water Reactors," NUREG-0452, Revision 4, Office of Nuclear Reactor Regulation, Fall 1981.
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7. Vesely, W. E., et al., "Fault Tree Handbook," NUREG-0492, January 1981.

8. Erdmann, R. C., Leverenz, F. L., and Kirch, H., "WAM-CUT: A Computer Code for Fault Tree Evaluation," EPRI-NP-803, Science Applications, Inc., June 1978.

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TABLE 10A-1 (SHEET 1 OF 2)

AFWS CONTROL CAPABILITIES[a]

| Component | Automatic Actuation | Indication | | Remote/Manual Controls | | Alarms/Light | |
|---|---|---|---|---|---|---|---|
| | | CR | Local | CR | Local | CR | Local |
| Motor-operated valves from CST 002 (5113, 5118, and 5119) | N | N | N | Y | Y | L(P) | L(P) |
| Motor-operated valves on miniflow line (5154 and 5155) | Y(F)[b] | N | Y(F) | N | Y | L(P) | L(P) |
| Motor-operated valves on discharge line from motor-driven pumps (5132, 5134, 5137, and 5139) | Y(AFS) | Y(P) | Y(P) | Y | Y | L(P) | L(P) |
| Motor-driven pumps (002 and 003) | Y(AFS) | Y(DP)<br>Y(SP) | Y(DP)<br>Y(SP) | Y | Y | L(P)<br>A(DP) | L(P)<br>A(DP) |
| Pump room drain sump | N | N | N | N | N | L(P)<br>A(WL) | L(P)<br>N |
| CSTs (001 and 002) | Y(WL) | Y(WL) | Y(WL) | Y | Y | L(P)[d]<br>A(WL) | L(P)[d]<br>A(WL) |
| Steam generators (1, 2, 3, and 4) | NA | Y(WL)<br>Y(T) | Y(WL)<br>Y(PR)<br>Y(T) | NA<br>Y(PR) | NA | A(T) | A(T)<br>A(WL)<br>A(PR) |
| Steam generators intake from AFWS (each of four lines) | NA | Y(F) | Y(F) | NA | NA | N | N |
| Motor-operated valves on turbine-driven pump discharge lines (5120, 5122, 5125, and 5127) | Y(AFS) | Y(P) | Y(P) | Y | Y | L(P) | L(P) |
| Turbine-driven pump (001) | Y(AFS)<br>Y(SV)[f] | Y(DP)<br>Y(SP)<br>Y( P)<br>Y(S) | Y(DP)<br>Y(SP)<br>Y( P)<br>Y(MF)<br>Y(ST) | Y | N | A(DP) | A(DP) |
| Turbine-steam intake motor-operated valve (5106) | Y(AFS) | N | N | Y | Y | L(P) | L(P) |
| Turbine-steam generator intake motor-operated valves (3009 and 3019) | N | N | N | Y | Y | L(P) | L(P) |
| Turbine steam intake | NA | Y(PR) | Y(PR)<br>Y(T) | NA | NA | N | N |
| Turbine speed governing valve | Y(SG) | Y(S) | Y(S) | Y[c] | Y[c] | A(OS) | N |
| Turbine trip and throttle valve | Y(SG) | Y(S) | Y(S) | Y[c] | Y[c] | L(P) | L(P) |

TABLE 10A-1 (SHEET 2 OF 2)

| Component | Automatic Actuation | Indication | | Remote/Manual Controls | | Alarms/Light | |
|---|---|---|---|---|---|---|---|
| | | CR | Local | CR | Local | CR | Local |
| Turbine lube oil cooler | | | | | | | |
| Lube oil inlet | NA | N | Y(T) | NA | NA | N | N |
| Lube oil return | NA | N | Y(PR) | NA | NA | N | N |

a. Table explanation:

CR = Location inside the main control room.
Local = Location anywhere in the plant outside of the main control room.
Alarm/Light = A bipositional annunciator.
Indication = A continuous variable display.

N = No.
Y( ) = Yes (variable).
L( ) = Annunciating light (variable).
A( ) = Annunciating alarm (variable).
NA = Not applicable.

Variables:

F = Flow.
P = Position.
MF = Miniflow.
ST = Shaft temperature.
DP = Discharge pressure.
SP = Suction pressure.
$\Delta$P = Discharge pressure between pump discharge pressure and turbine inlet pressure.
AFS = AFWS actuation signal.
PR = Pressure.
T = Temperature.
S = Speed.
WL = Water level.
OS = Overspeed.
SG = Speed governor.
SV = Turbine speed governing valve.
TV = Turbine trip and throttle valve.

b. Flow on pump discharge line.

c. Controls are not on valve per se, but are on the speed governor.

d. Position of motor-operated valve on tank fill line.

e. Low-low discharge pressure

f. See component 14.

TABLE 10A-2

ELECTRIC POWER SOURCE LIST FOR THE AFWS[a]

| Component Description | Component Number | 125-V dc Voltage Level/ Electrical Train | Normal Position[b] | AFWS Actuation Signal Train |
|---|---|---|---|---|
| Motor-driven pump controls | 002 | 4.16 kV/B | S | - |
| | | 125-V dc/B | S | B |
| Motor-driven pump controls | 003 | 4.16 kV/A | S | - |
| | | 125-V dc/A | S | A |
| Motor-driven pump miniflow motor-operated valves | 5154 | 480 V/B | NO | - |
| | 5155 | 480 V/A | NO | - |
| Motor-driven pump intake motor-operated valves | 5118 | 480 V/B | NC | - |
| | 5119 | 480 V/A | NC | - |
| Motor-driven pump discharge motor-operated valves | 5132 | 480 V/B | NO | B |
| | 5134 | 480 V/B | NO | B |
| | 5137 | 480 V/A | NO | A |
| | 5139 | 480 V/A | NO | A |
| Turbine-driven pump intake motor-operated valve | 5113 | 125-V dc/C | NC | - |
| Turbine-driven pump discharge motor-operated valves | 5120 | 125-V dc/C | NO | A, B[c] |
| | 5122 | 125-V dc/C | NO | A, B[c] |
| | 5125 | 125-V dc/C | NO | A, B[c] |
| | 5127 | 125-V dc/C | NO | A, B[c] |
| Turbine-steam inlet motor-operated valve | 5106 | 125-V dc/C | NC | A, B[c] |
| Turbine-steam generators inlet motor-operated valves | 3009 | 125-V dc/B | NO | - |
| | 3019 | 125-V dc/A | NO | - |

a. Abbreviations:
S = Standby.
NO = Normally open.
NC = Normally closed.

NOTE: Component control power is from the same voltage level as component unless designated otherwise; 125-V dc/C derives power from ac electric power train A with backup power provided by batteries.

b. Normal position is during plant operating mode of power generation.
c. Actuation signal from train A or B.

TABLE 10A-3

COMPONENT GROUPINGS IN AFWS BLOCK DIAGRAM
(Corresponds to Figure 10A-5)

| Block | Components in Block |
|---|---|
| SG1 | Check valve 121, check valve 125, check valve 113 |
| SG2 | Check valve 122, check valve 126, check valve 114 |
| SG3 | Check valve 124, check valve 128, check valve 115 |
| SG4 | Check valve 123, check valve 127, check valve 116 |
| PMPA | Gate valve 035, check valve 001, motor-driven pump 003, butterfly valve 5095, butterfly valve 5092, motor-operated valve 5119, butterfly valve 5099 |
| PMPB | Gate valve 060, check valve 002, motor-driven pump 002, butterfly valve 5094, butterfly valve 5091, motor-operated valve 5118, butterfly valve 5098 |
| PMPC | Check valve 014, turbine-driven pump 001, butterfly valve 5093, butterfly valve 5090, motor-operated valve 5113, butterfly valve 5097, motor-operated valve 5106, trip and throttle valve, speed governing valve, motor-operated valve 3009, motor-operated valve 3019, check valve 008, check valve 006, gate valve 007, gate valve 005 |
| A1 | Check valve 046, motor-operated valve 5139, gate valve 045 |
| A4 | Check valve 043, motor-operated valve 5137, gate valve 042 |
| B2 | Check valve 037, motor-operated valve 5132, gate valve 036 |
| B3 | Check valve 040, motor-operated valve 5134, gate valve 039 |
| C1 | Check valve 020, motor-operated valve 5122, gate valve 019 |
| C2 | Check valve 023, motor-operated valve 5125, gate valve 022 |
| C3 | Check valve 026, motor-operated valve 5127, gate valve 025 |
| C4 | Check valve 017, motor-operated valve 5120, gate valve 016 |

TABLE 10A-4 (SHEET 1 OF 3)

AFWS COMPONENT FAILURE DATA

| Fault Event/Tree Description | Component | Failure on Demand | Reference[a] | Repair Time (h) | Unavailability Due to Maintenance[b] | Reference[a] |
|---|---|---|---|---|---|---|
| Check valve (at steam generator intake) fails to open on demand | 121, 122, 123, 124, 125, 126, 127, 128 | $1 \times 10^{-4}$ | 1 | NA | NA | NA |
| Stop check valve (at steam generator intake) fails to open on demand | 113, 114, 115, 116 | $1 \times 10^{-4}$ | 1 | NA | NA | NA |
| Stop check valve (on AFWS discharge) fails to open on demand | 017, 020, 023, 026, 037, 040, 043, 046 | $1 \times 10^{-4}$ | 1 | 7 | $2.17 \times 10^{-6}$ | 1, 3 |
| Motor-operated valve (on discharge line) transfers closed | 5120, 5122, 5125, 5127, 5132, 5134 5137, 5139 | $1 \times 10^{-4}$ | 1 | 7 | $2.17 \times 10^{-6}$ | 1, 3 |
| Gate valve (on discharge line) transfers closed | 015, 016, 019, 022, 025, 035, 036, 039, 042, 045, 060 | $1 \times 10^{-4}$ | 1 | 7 | $7 \times 10^{-8}$ | 1, 3 |
| Check valve (on discharge line) fails to open on demand | 001, 002, 014 | $1 \times 10^{-4}$ | 1 | 7 | $2.17 \times 10^{-8}$ | 1, 3 |
| Motor-driven pump fails (includes controls) | 003, 002 | $5 \times 10^{-3}$ | 1 | 19 | $5.81 \times 10^{-3}$ | 1 |
| Turbine-driven pump fails (includes controls) | 001 | $5 \times 10^{-3}$ | 1 | 19 | $5.81 \times 10^{-3}$ | 1 |

TABLE 10A-4 (SHEET 2 OF 3)

| Fault Event/Tree Description | Component | Failure on Demand | Reference[a] | Repair Time (h) | Unavailability Due to Maintenance[b] | Reference[a] |
|---|---|---|---|---|---|---|
| Motor-operated valve (on turbine intake) fails on demand | 5106 | $3.1 \times 10^{-3}$ | 1 | 7 | $2.17 \times 10^{-6}$ | 1 |
| Check valves (on turbine steam intake) fail to open on demand | 006, 008 | $1 \times 10^{-4}$ | 1 | 7 | $2.17 \times 10^{-6}$ | 1, 3 |
| Motor-operated valves (on turbine steam intake) transfer closed on demand | 3009, 3019 | $1 \times 10^{-4}$ | 1 | 7 | $2.17 \times 10^{-6}$ | 1, 3 |
| Gate valve (on turbine steam intake) transfers closed on demand | 005, 007 | $1 \times 10^{-4}$ | 1 | NA | NA | NA |
| Butterfly valve (on suction line) transfers closed | 093, 094, 095 | $1 \times 10^{-4}$ | 1 | 7 | $7.0 \times 10^{-8}$ | 1, 3 |
| Motor-operated valve (pump suction line) fails on demand | 5113, 5118, 5119 | $3.1 \times 10^{-3}$ | 1 | 7 | $2.17 \times 10^{-6}$ | 1, 3 |
| Butterfly valve (on CST discharge line) transfers closed | 090, 091 092, 097, 098, 099 | $1 \times 10^{-4}$ | 1 | 40 | $4 \times 10^{-7}$ | 2, 3 |
| CST fails | 001, 002 | $1 \times 10^{-8}$ | 3 | NA | NA | NA |
| Failure of actuation signal | Train A, train B, speed governor | $7 \times 10^{-3}$ | 1 | NA | NA | NA |
| Loss of offsite power | Case 1 | 0.2 | 3 | NA | NA | NA |
| Failure of 125-V dc electric power | Train A, train B, train C | $2.4 \times 10^{-6}$ | NA | 2 | NA | 3 |

TABLE 10A-4 (SHEET 3 OF 3)

| Fault Event/Tree Description | Component | Failure on Demand | Reference[a] | Repair Time (h) | Unavailability Due to Maintenance[b] | Reference[a] |
|---|---|---|---|---|---|---|
| Failure of ac electric power (onsite - case 1 and 2) | Train A, train B | $3 \times 10^{-2}$ | 3 | NA | NA | NA |
| Motor-operated valve closed by error | 3009, 3019, 5120, 5122, 5125, 5127, 5132, 5134, 5137, 5139 | $5 \times 10^{-4}$ | 1 | NA | NA | NA |
| No manual open signal to motor-operated valve | 3009, 3019, 5106, 5113, 5118, 5119, 5120, 5122, 5125, 5132, 5134, 5137, 5139 | $5 \times 10^{-3}$ | 1 | NA | NA | NA |
| No manual start signal to pump | 001, 002, 003, speed governor | $5 \times 10^{-3}$ | 1 | NA | NA | NA |
| Trip and throttle valve or speed governing valve fails to open on demand | Trip and throttle valve, speed governing valve | $1.1 \times 10^{-3}$ | 1 | 7 | $2.17 \times 10^{-6}$ | 3 |

a. References

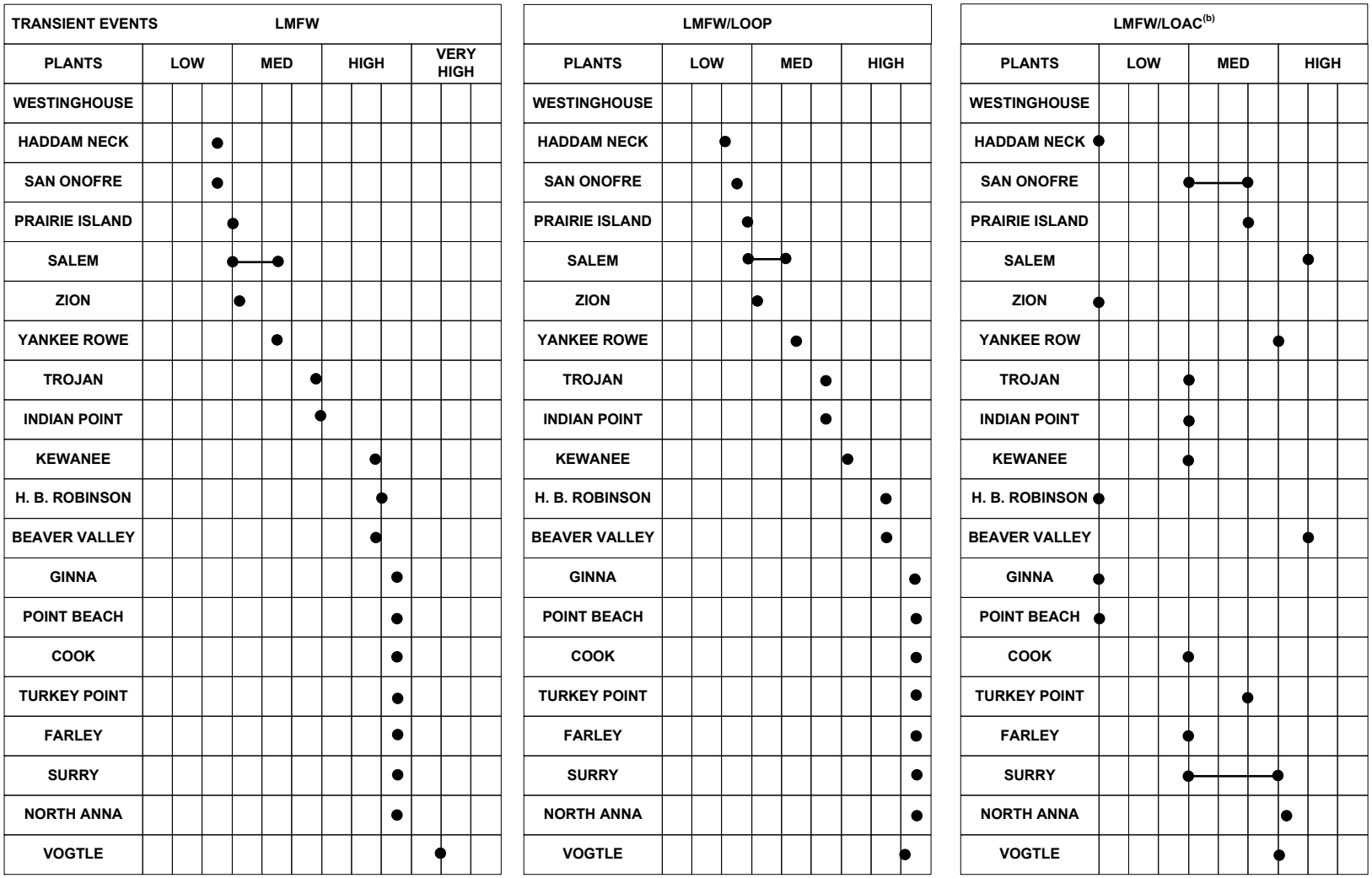

1. U.S. Nuclear Regulatory Commission, "Generic Evaluation of Feedwater Transients and Small-Break Loss-of-Coolant Accidents in Westinghouse-Designed Operating Plants," NUREG-0611, Bulletins and Orders Task Force, Office of Nuclear Reactor Regulation, January 1980.

2. Engineering judgment.

3. Rasmussen, N. C., et al., "Reactor Safety Study - An Assessment of Accident Risks in U.S. Commercial Nuclear Power Plants," U.S. Nuclear Regulatory Commission, WASH-1400 (NUREG-75/014), October 1975.


b. Maintenance is defined to be maintenance whereby the component is unable to perform its function. Also, unavailability due to maintenance is calculated as the frequency of failure times the repair time.

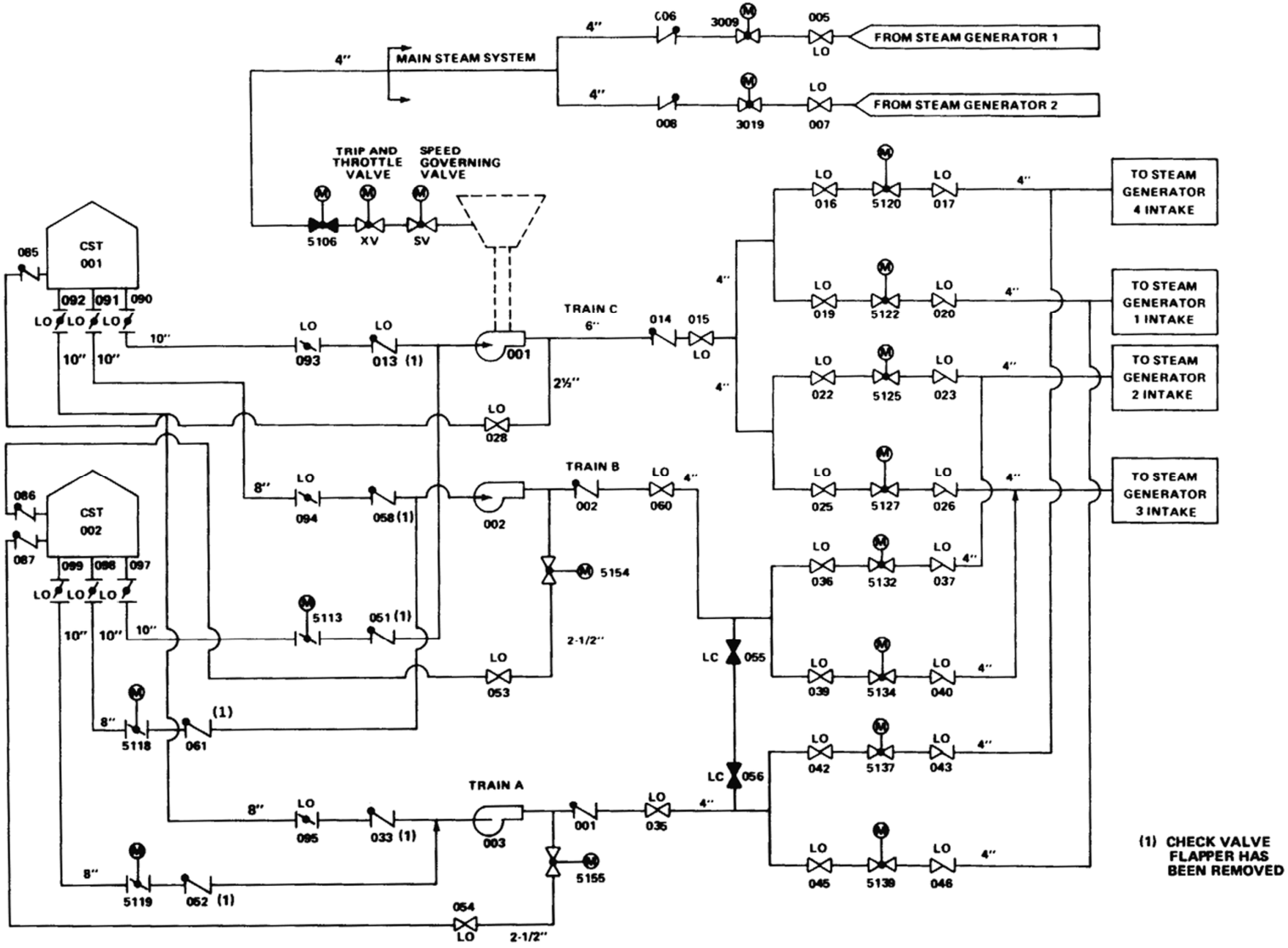
TABLE 10A-5
AFWS CONDITIONAL AVAILABILITY COMPARISON




A. →| |← ORDER OF MAGNITUDE IN UNAVAILABILITY REPRESENTED.

→ INCREASING AVAILABILITY.

B. THE SCALE FOR THIS EVENT IS NOT THE SAME AS THAT FOR THE LMFW AND THE LMFW/LOOP.



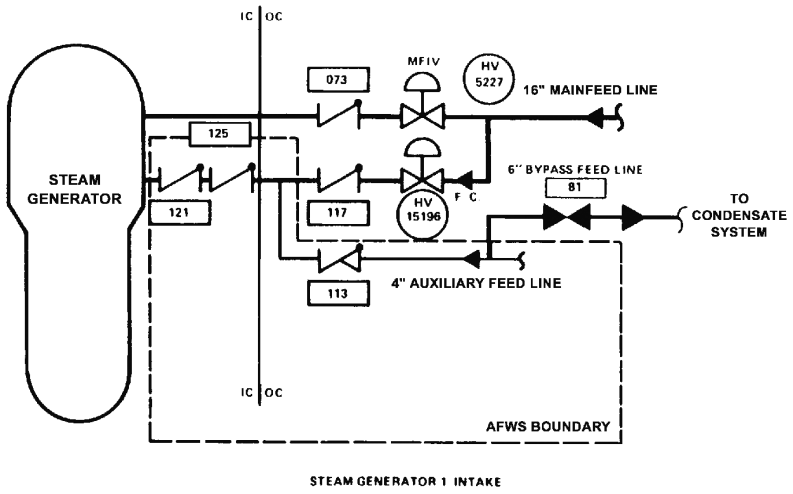

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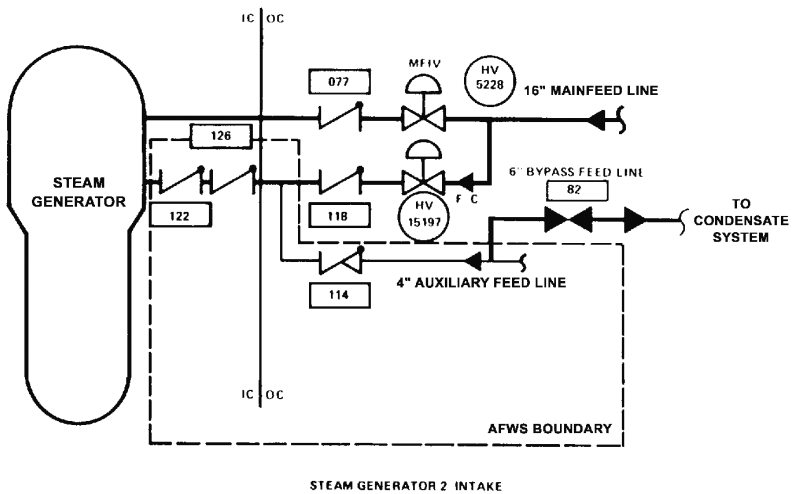
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AFWS

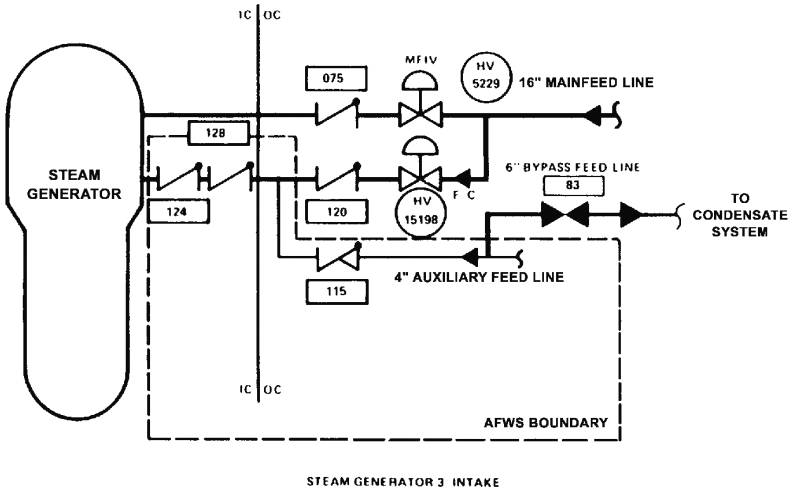
FIGURE 10A-1



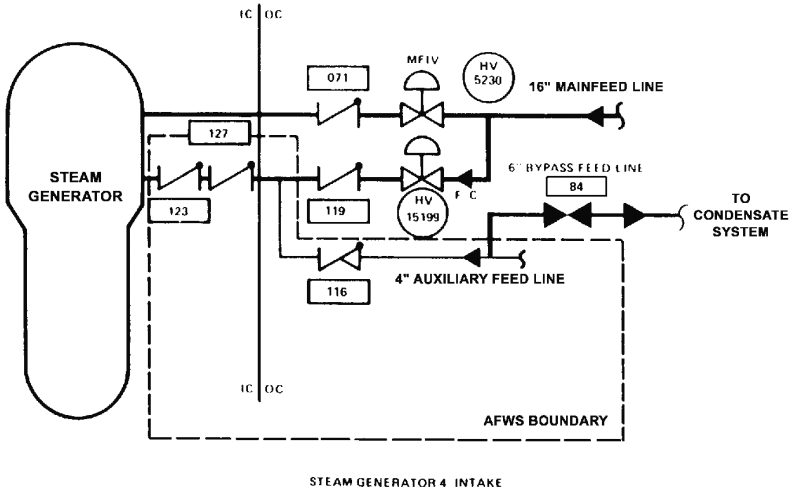

STEAM GENERATOR 1 INTAKE




STEAM GENERATOR 2 INTAKE




STEAM GENERATOR 3 INTAKE




STEAM GENERATOR 4 INTAKE

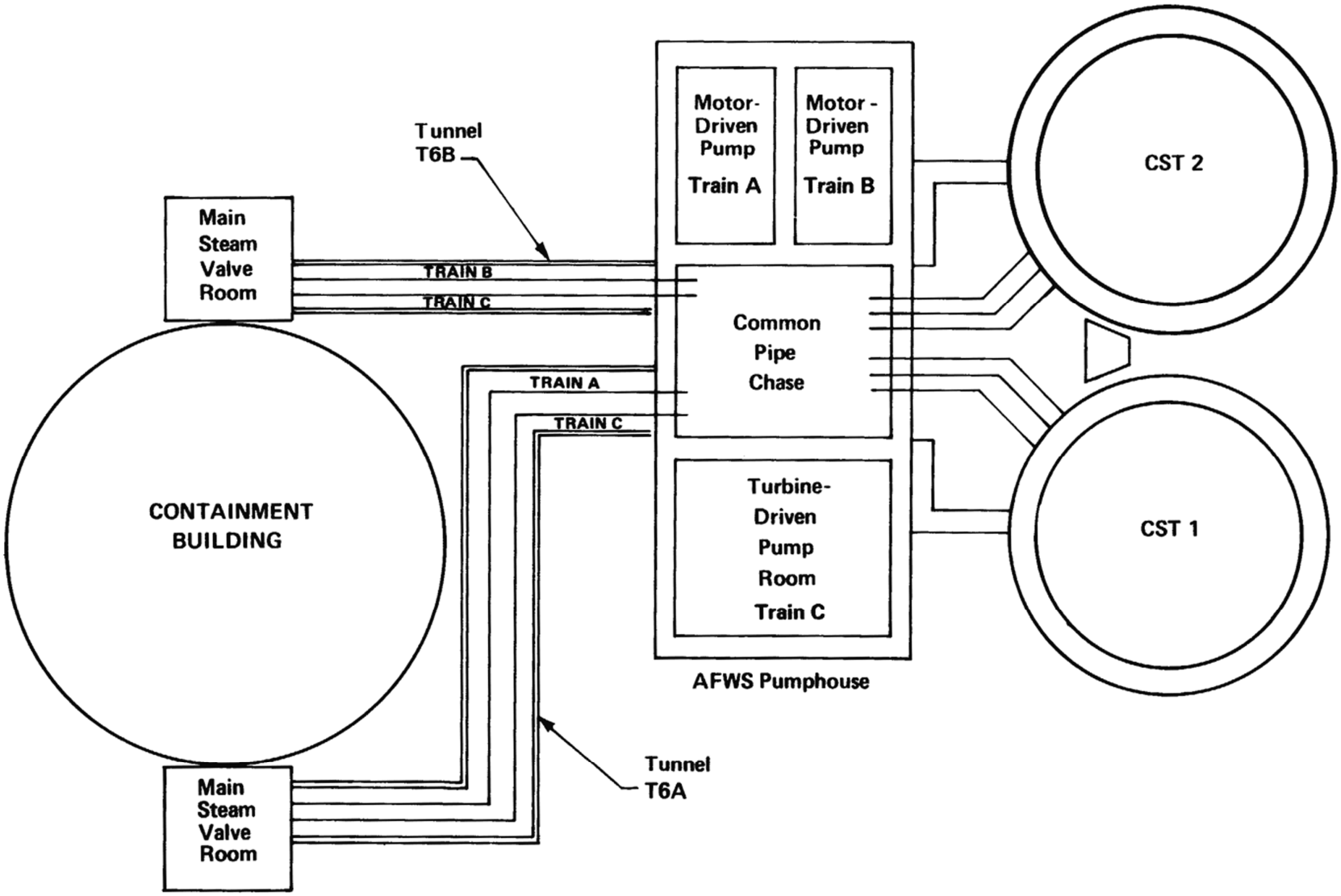
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UNIT 1 AUXILIARY FEEDWATER/ STEAM GENERATORS INTAKE

FIGURE 10A-2



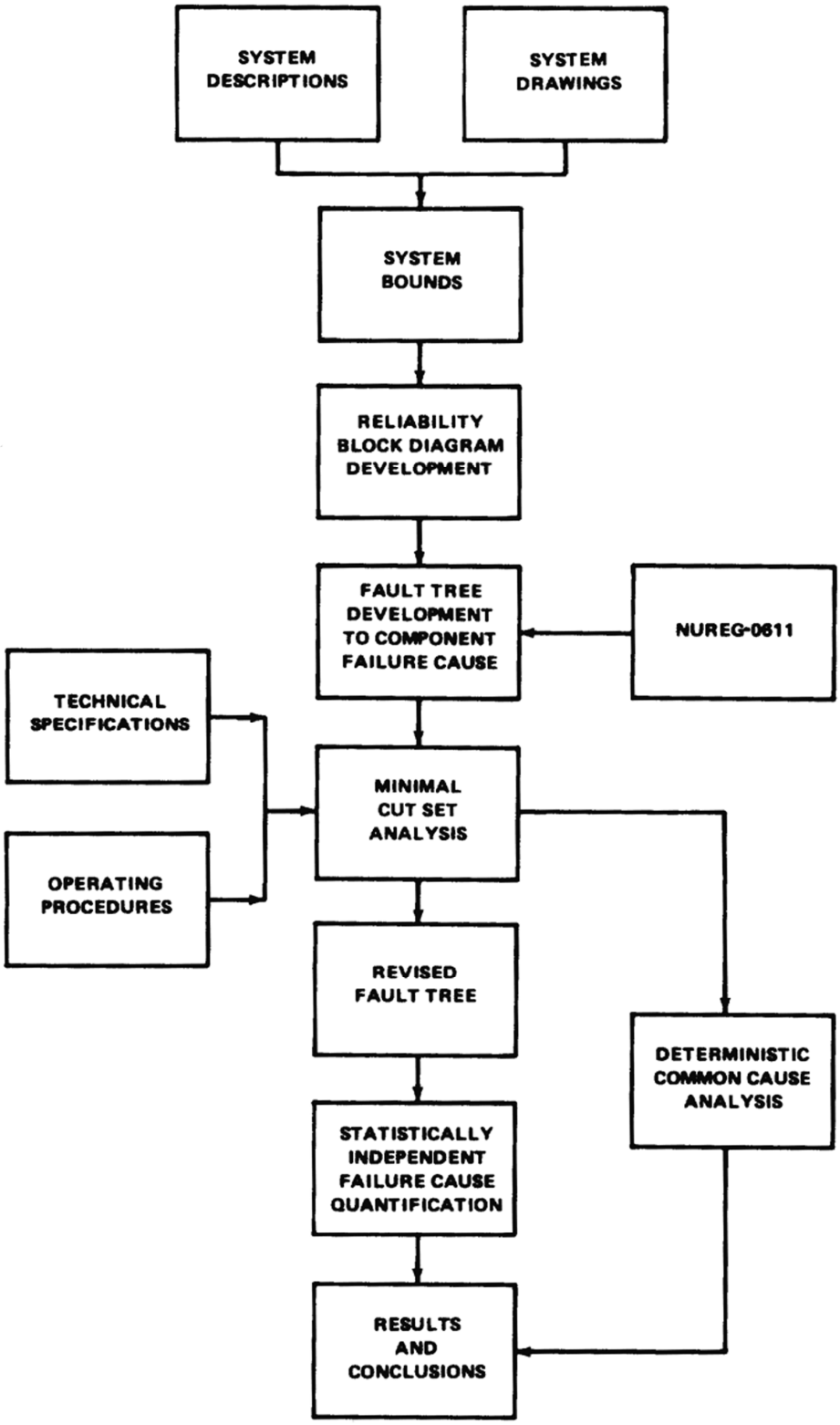

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AFWS SIMPLIFIED PIPING LAYOUT

FIGURE 10A-3



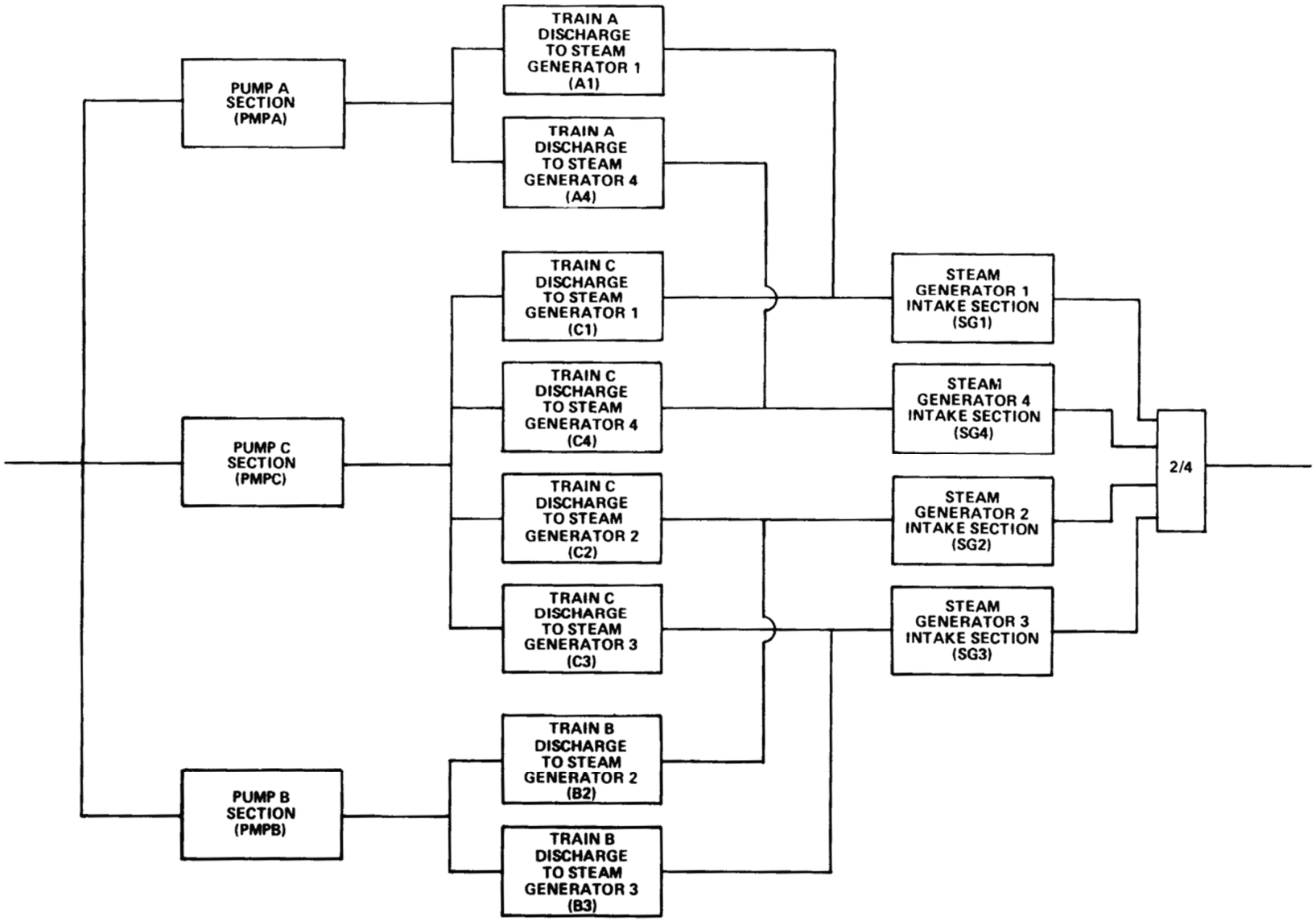

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AFWS RELIABILITY EVALUATION METHODOLOGY FLOW CHART

FIGURE 10A-4



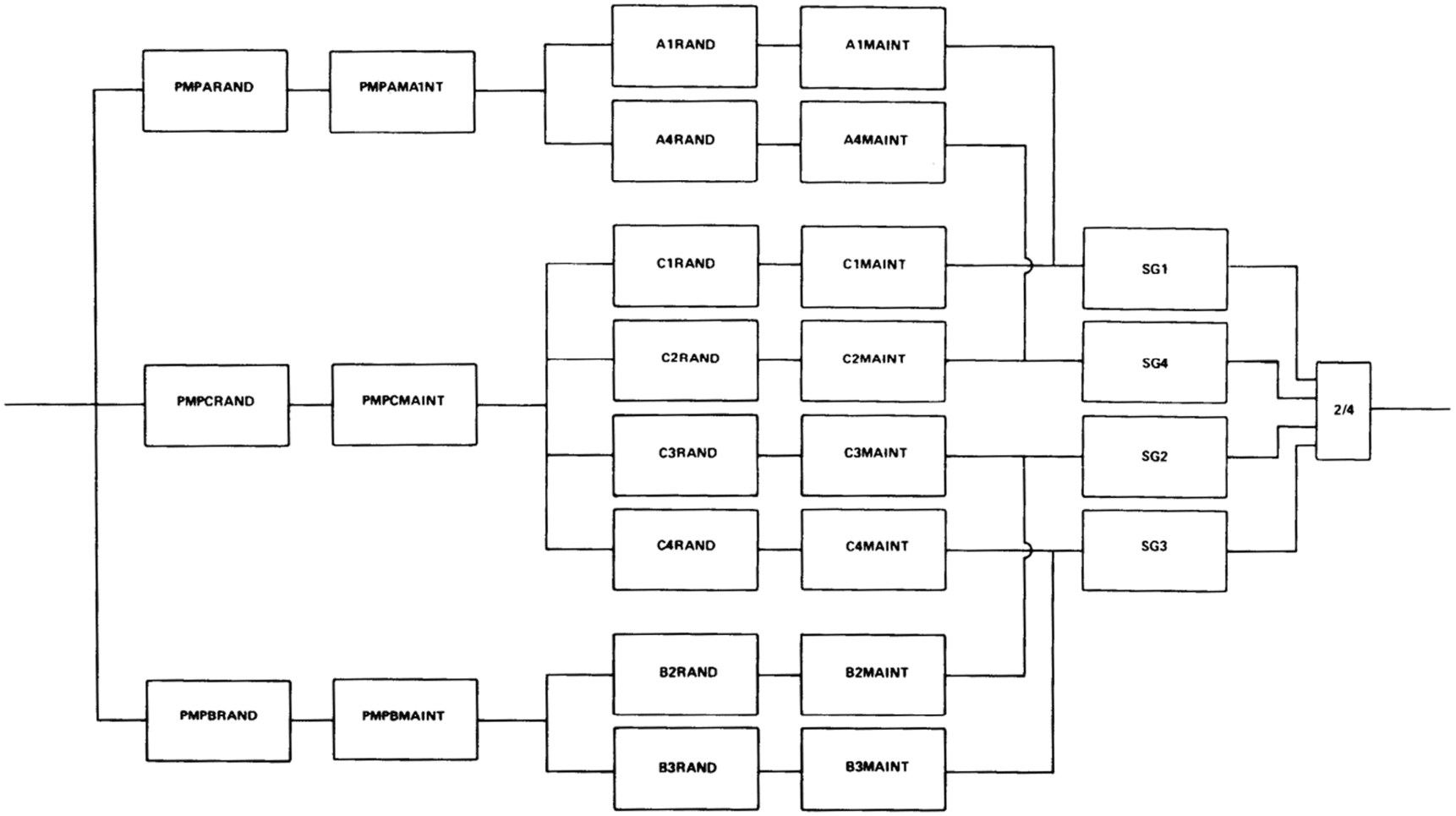

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UNIT 1 AFWS BLOCK DIAGRAM

FIGURE 10A-5




AFWS BLOCK DIAGRAM

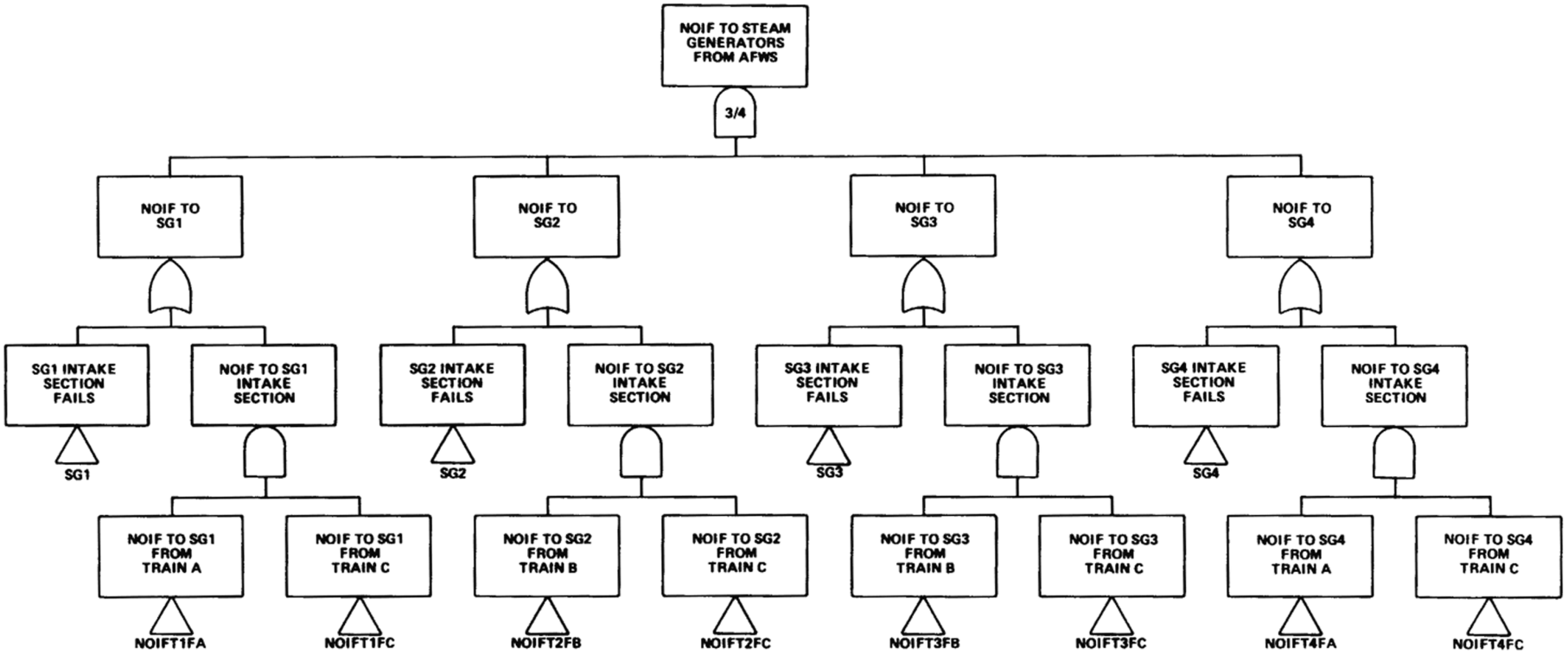
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AFWS EXPANDED BLOCK DIAGRAM

FIGURE 10A-6




NOIF = NO OR INSUFFICIENT FLOW

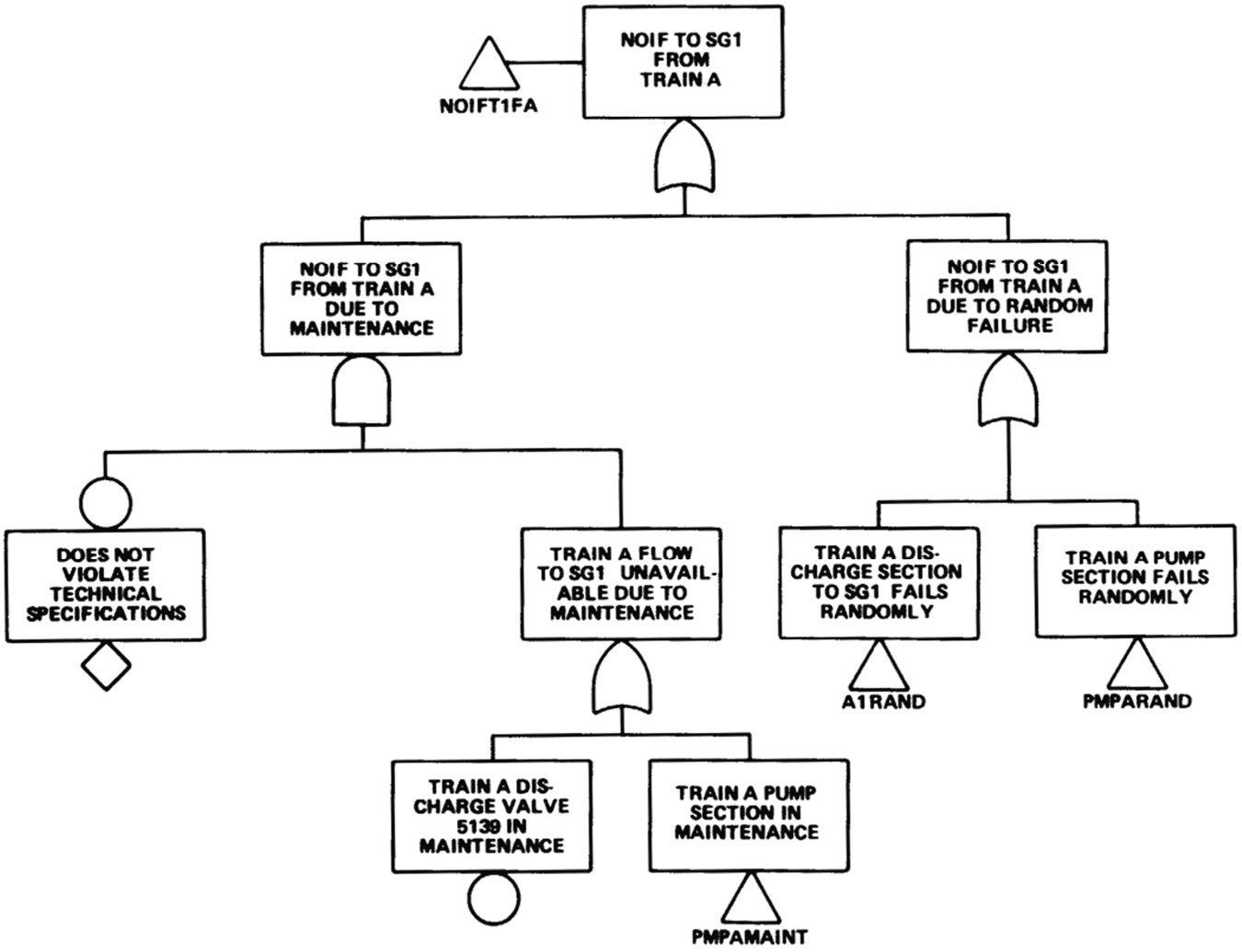
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UNIT 1 AFWS FAULT TREE MODEL

FIGURE 10A-7 (SHEET 1 OF 30)



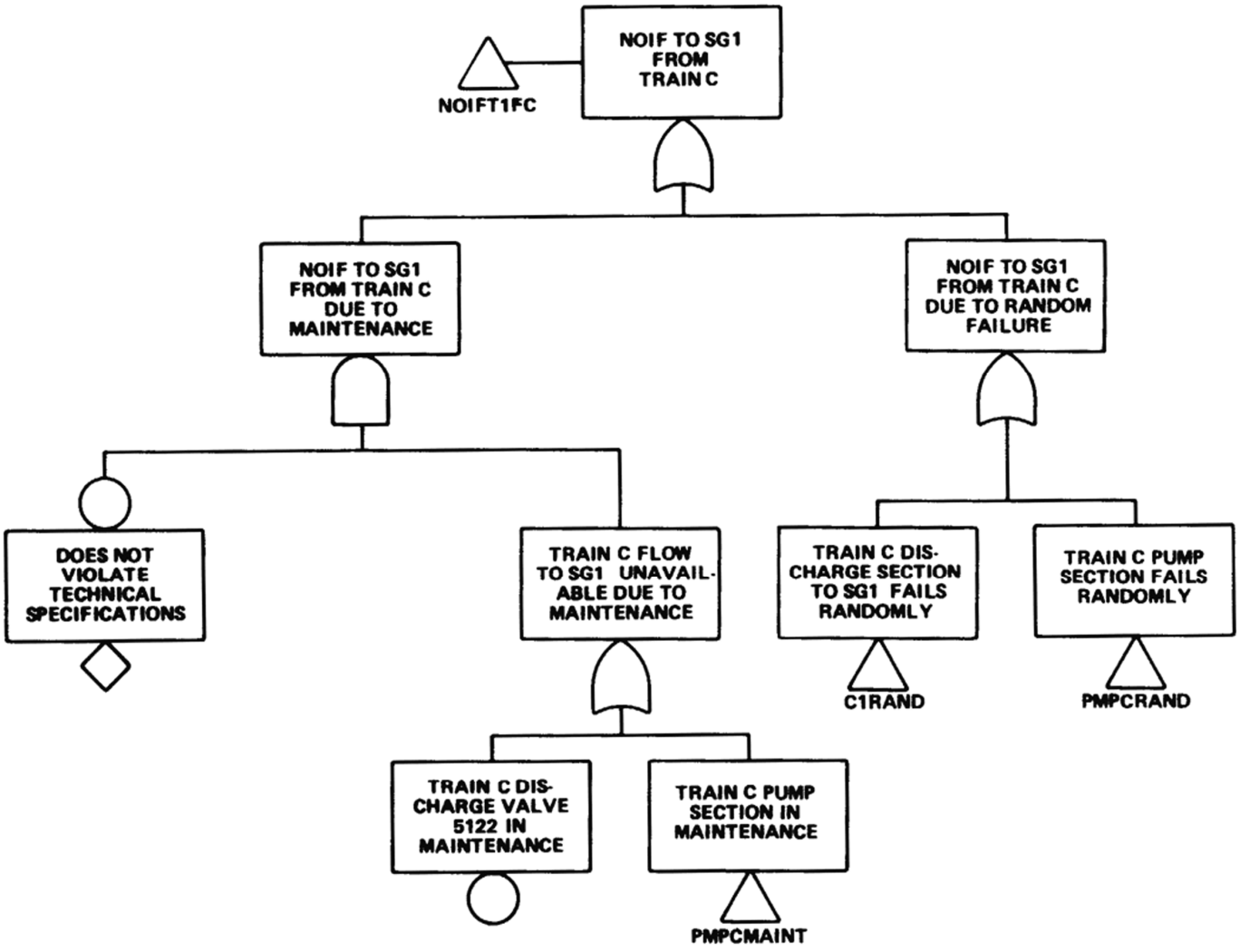

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UNIT 1 AFWS FAULT TREE MODEL

FIGURE 10A-7 (SHEET 2 OF 30)



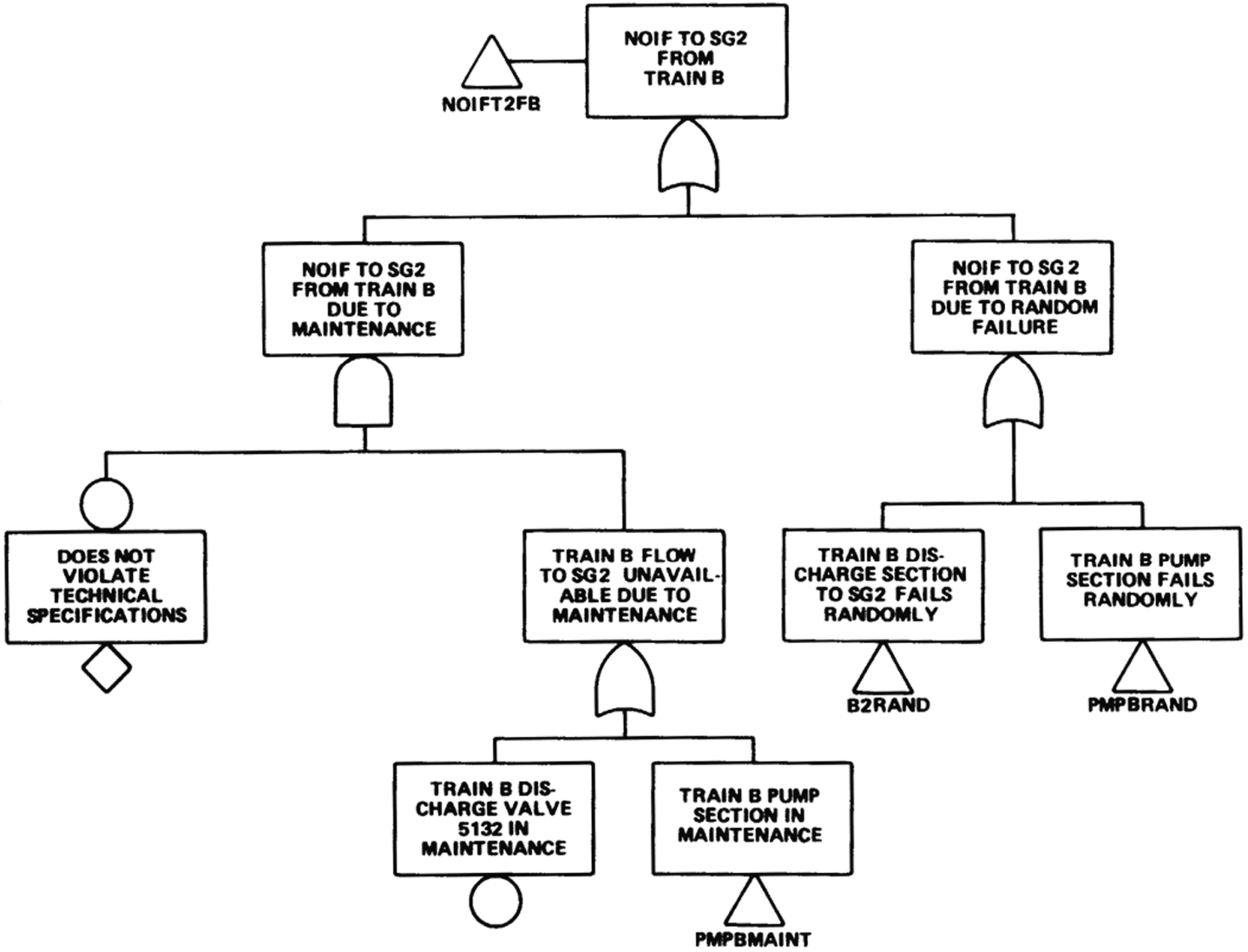

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**UNIT 1 AND UNIT 2**

UNIT 1 AFWS FAULT TREE MODEL

FIGURE 10A-7 (SHEET 3 OF 30)



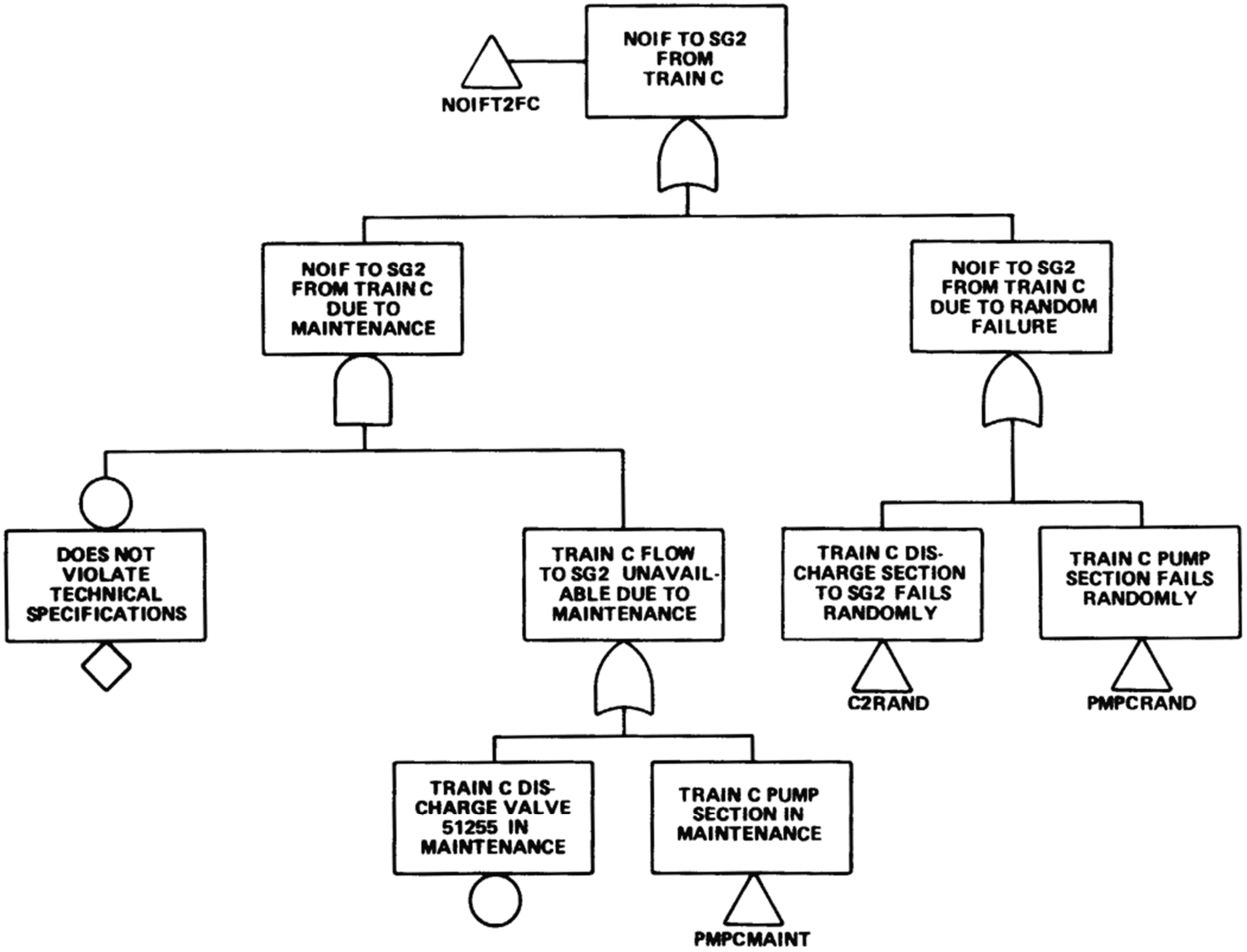

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**UNIT 1 AND UNIT 2**

UNIT 1 AFWS FAULT TREE MODEL

FIGURE 10A-7 (SHEET 4 OF 30)



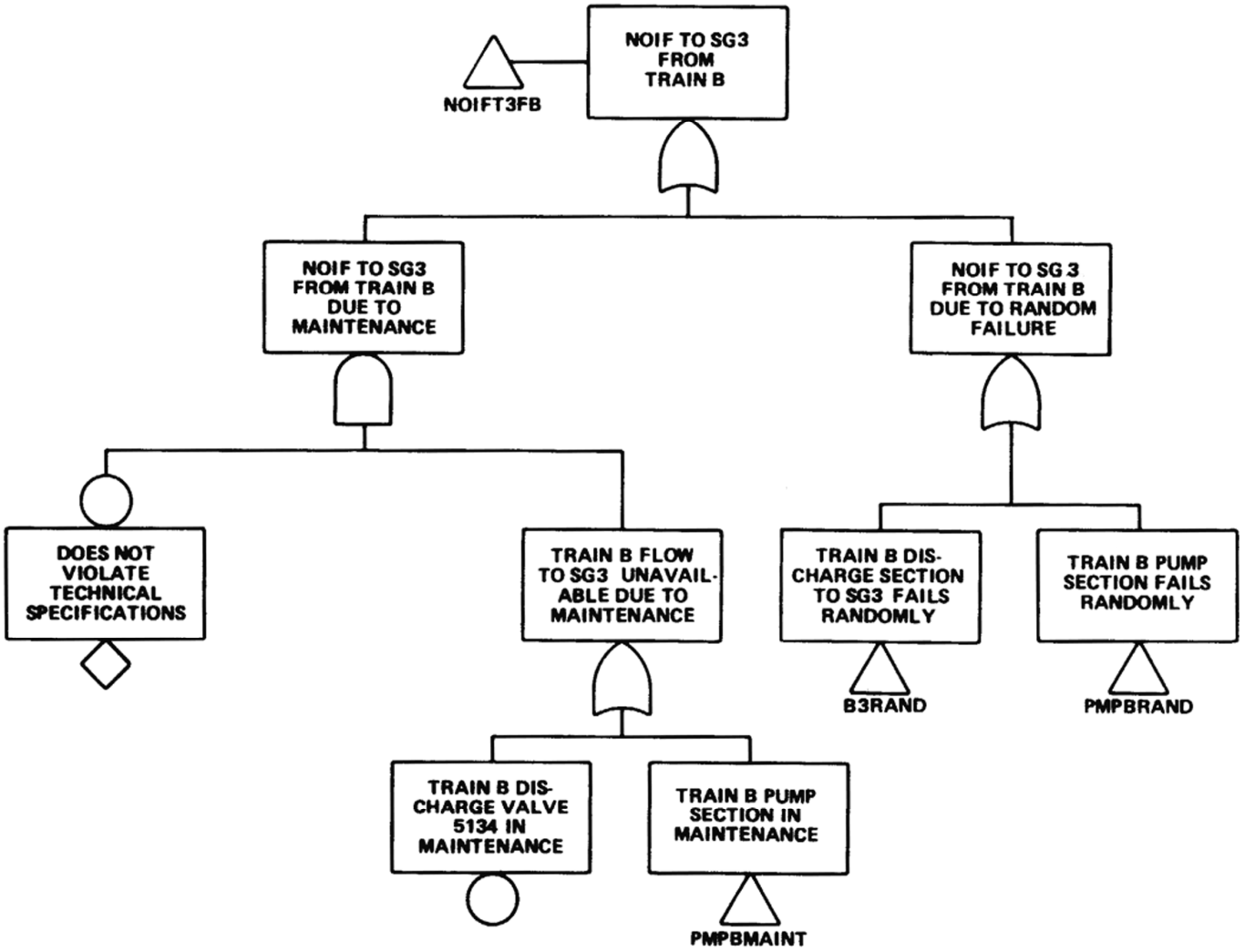

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**UNIT 1 AND UNIT 2**

UNIT 1 AFWS FAULT TREE MODEL

FIGURE 10A-7 (SHEET 5 OF 30)



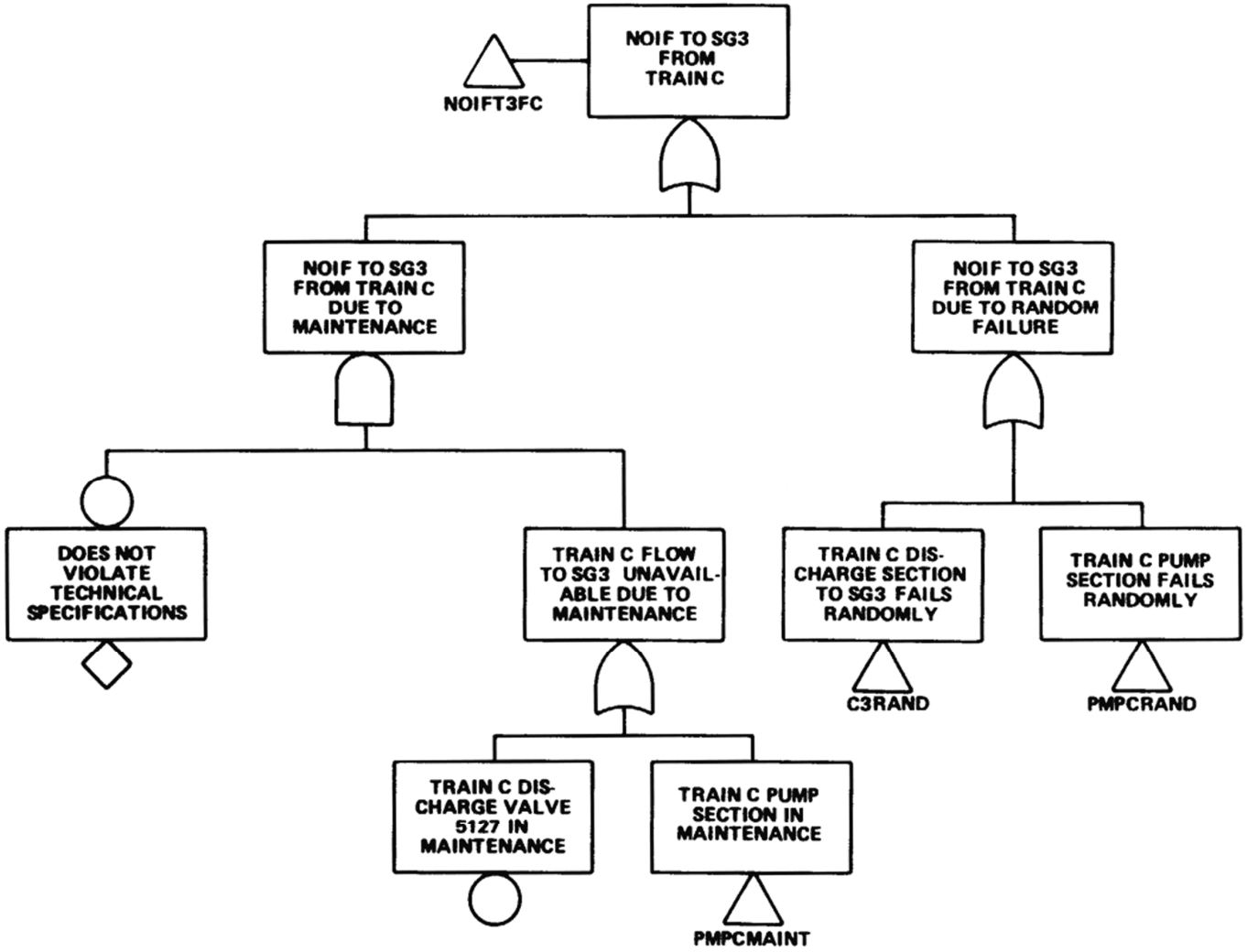

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**UNIT 1 AND UNIT 2**

UNIT 1 AFWS FAULT TREE MODEL

FIGURE 10A-7 (SHEET 6 OF 30)



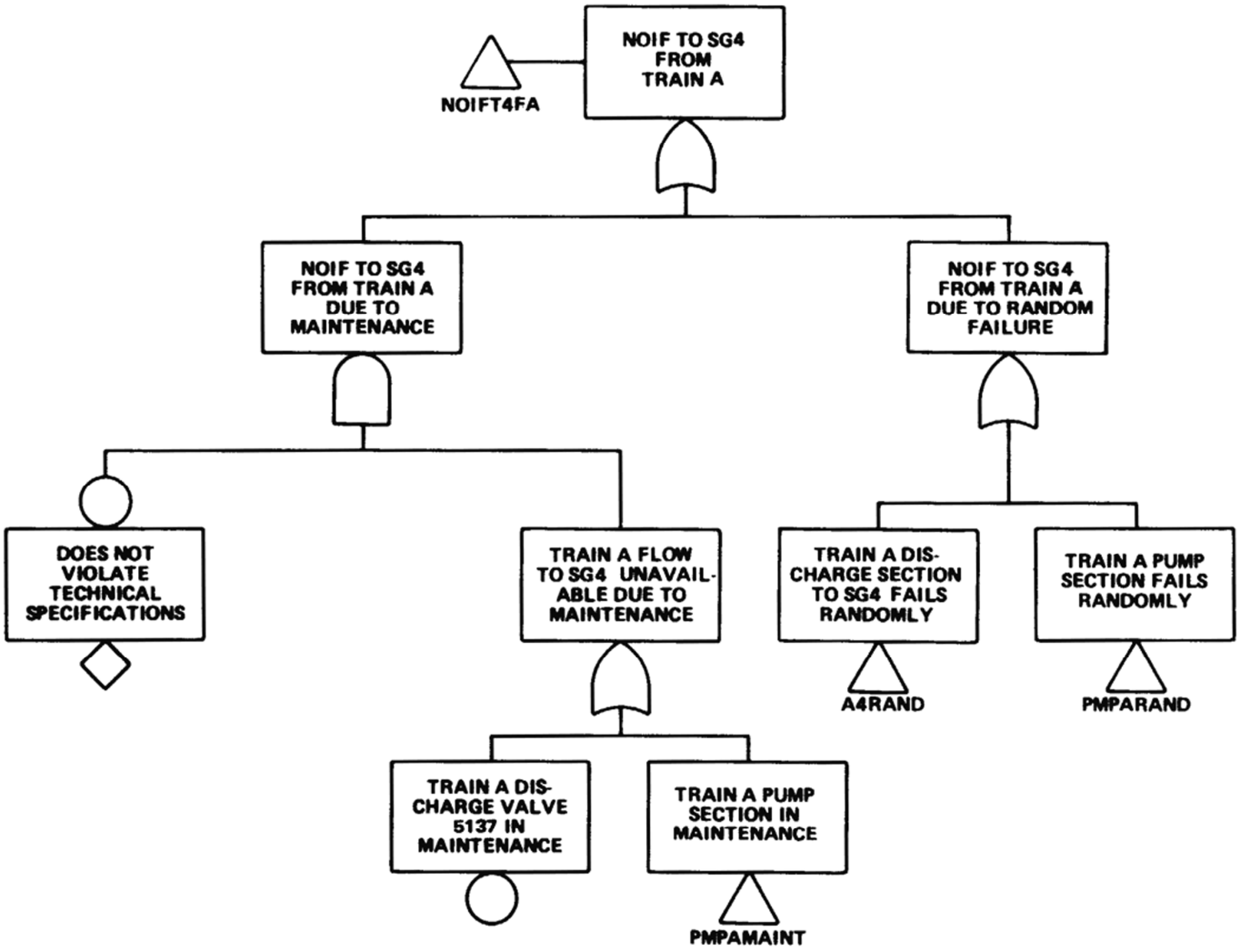

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**UNIT 1 AND UNIT 2**

UNIT 1 AFWS FAULT TREE MODEL

FIGURE 10A-7 (SHEET 7 OF 30)



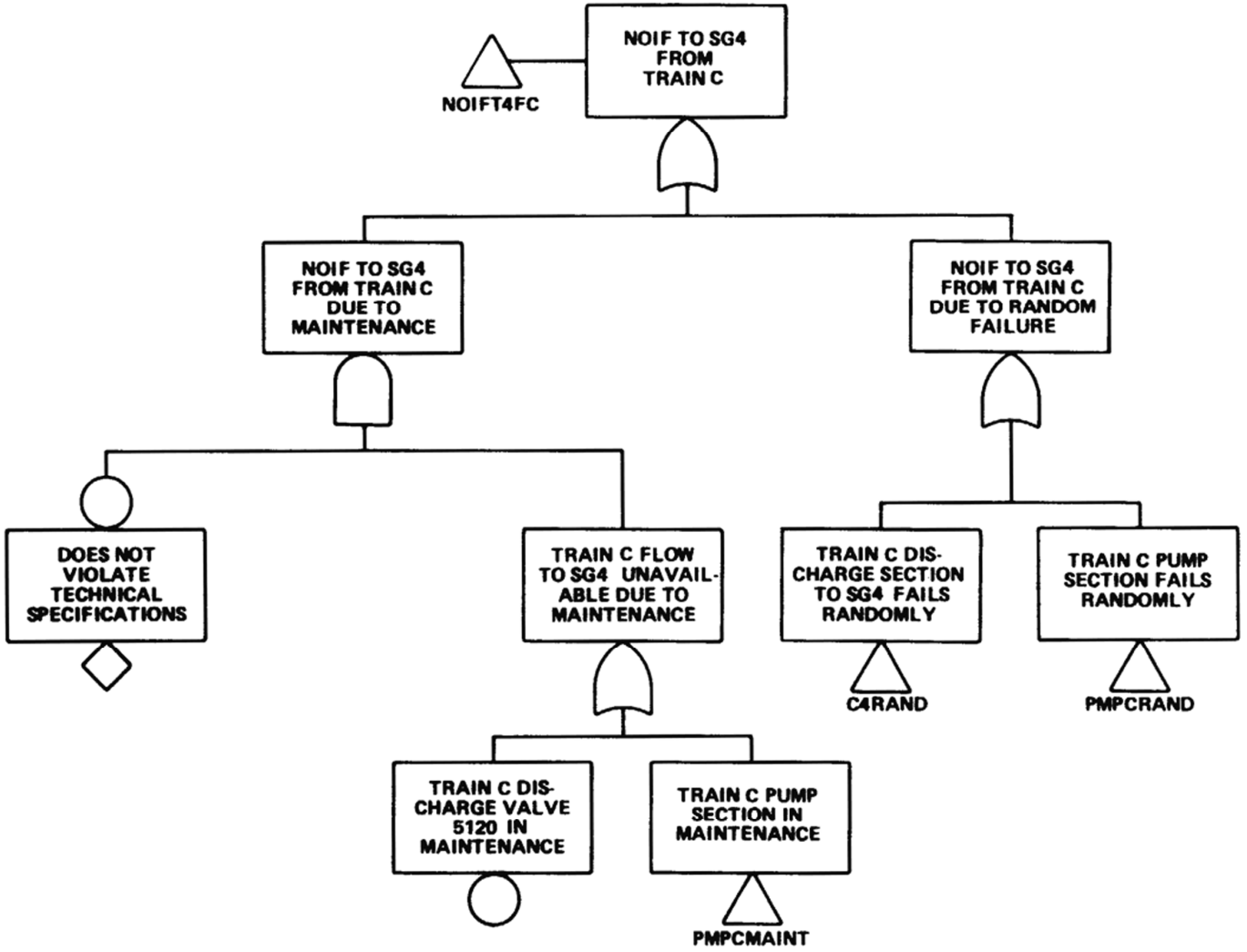

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**UNIT 1 AND UNIT 2**

UNIT 1 AFWS FAULT TREE MODEL

FIGURE 10A-7 (SHEET 8 OF 30)



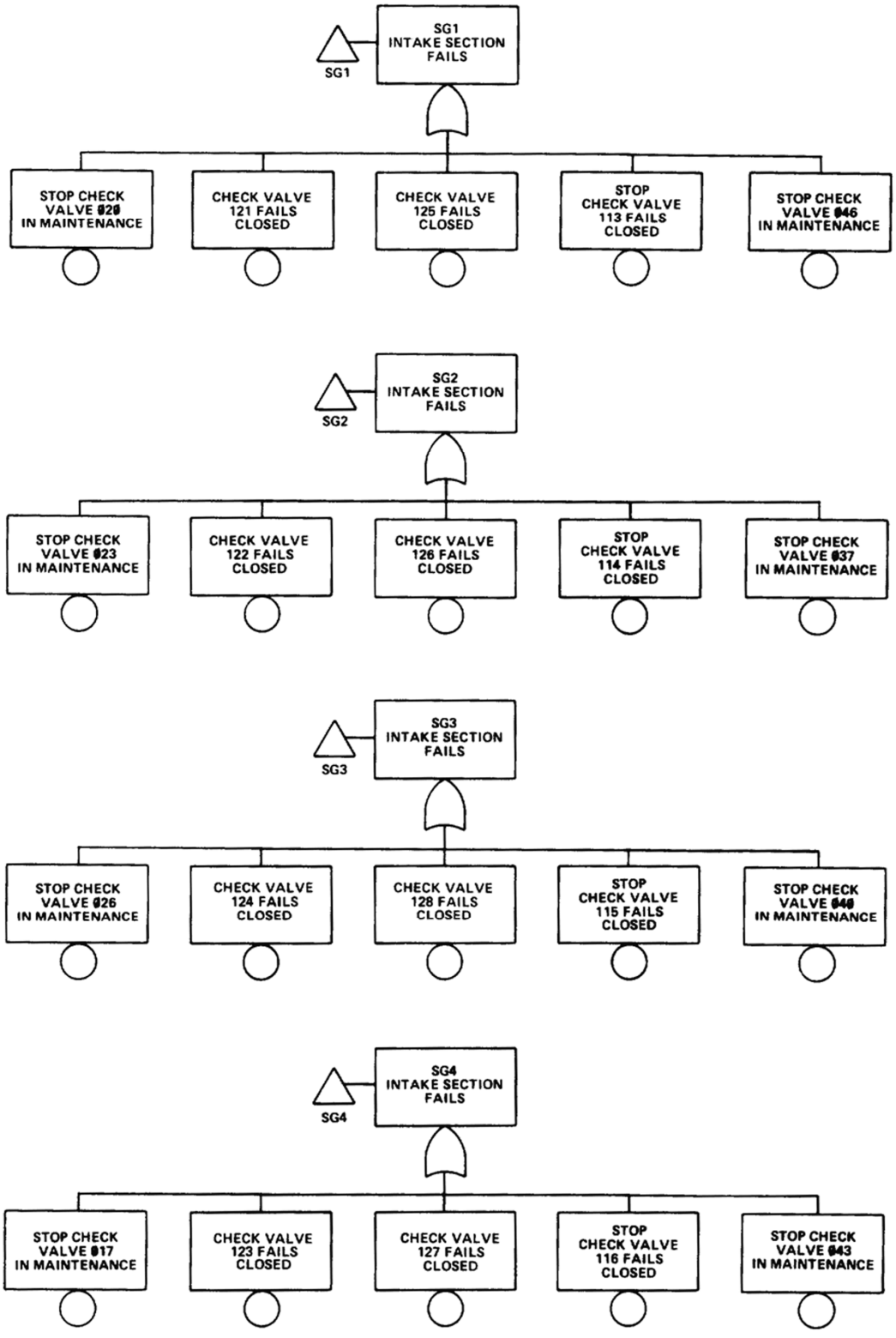

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**UNIT 1 AND UNIT 2**

UNIT 1 AFWS FAULT TREE MODEL

FIGURE 10A-7 (SHEET 9 OF 30)



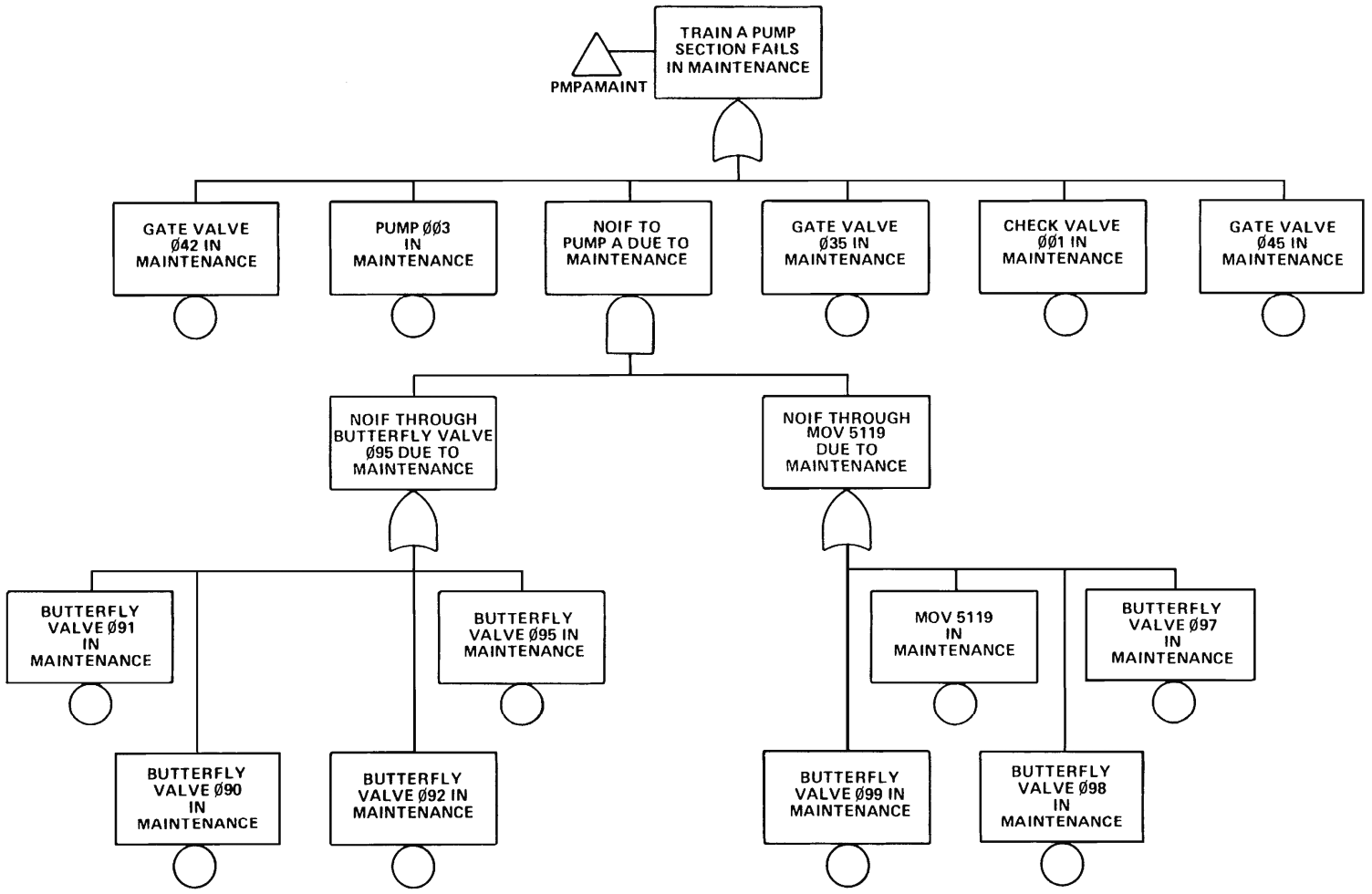

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**UNIT 1 AND UNIT 2**

UNIT 1 AFWS FAULT TREE MODEL

FIGURE 10A-7 (SHEET 10 OF 30)




NOIF = NO OR INSUFFICIENT FLOW

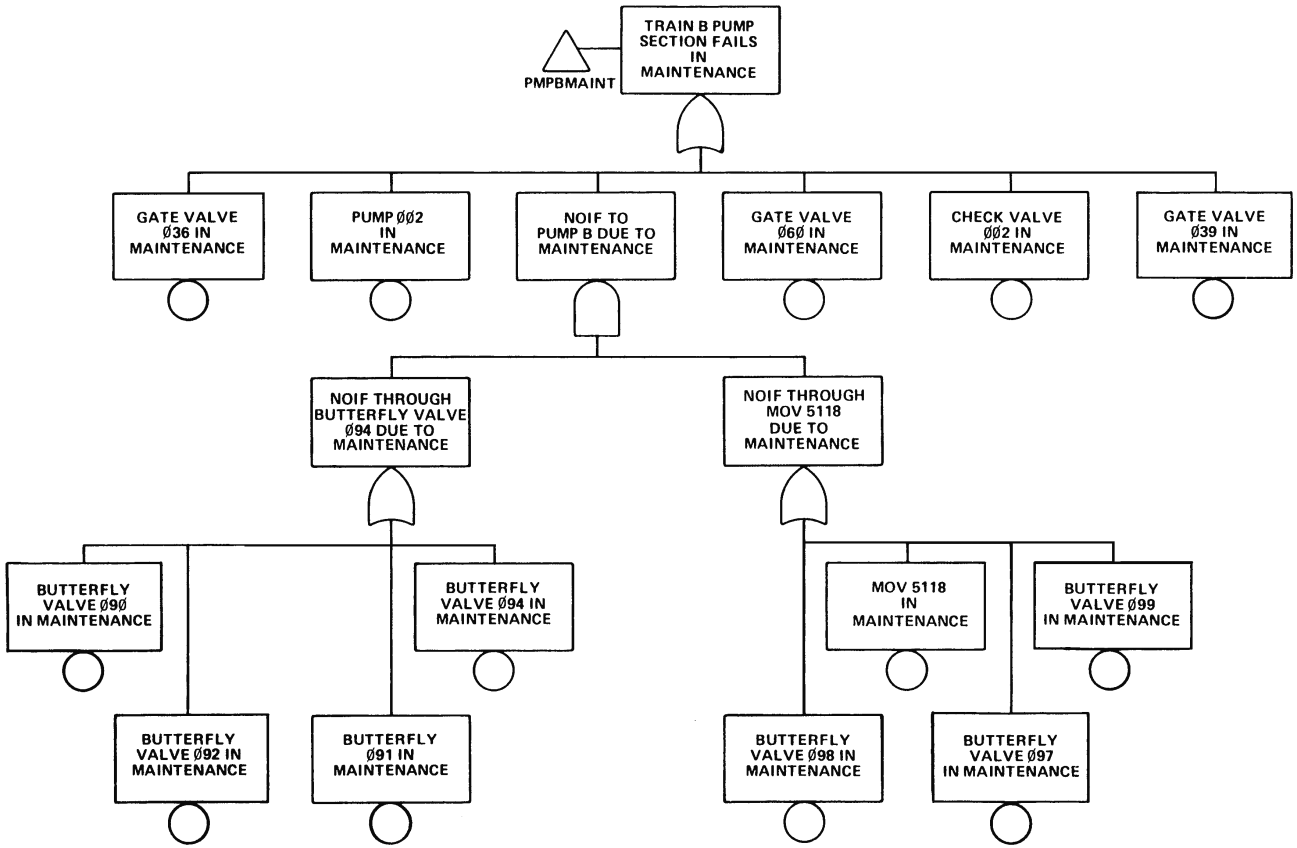
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**UNIT 1 AND UNIT 2**

UNIT 1 AFWS FAULT TREE MODEL

FIGURE 10A-7 (SHEET 11 OF 30)




NOIF = NO OR INSUFFICIENT FLOW

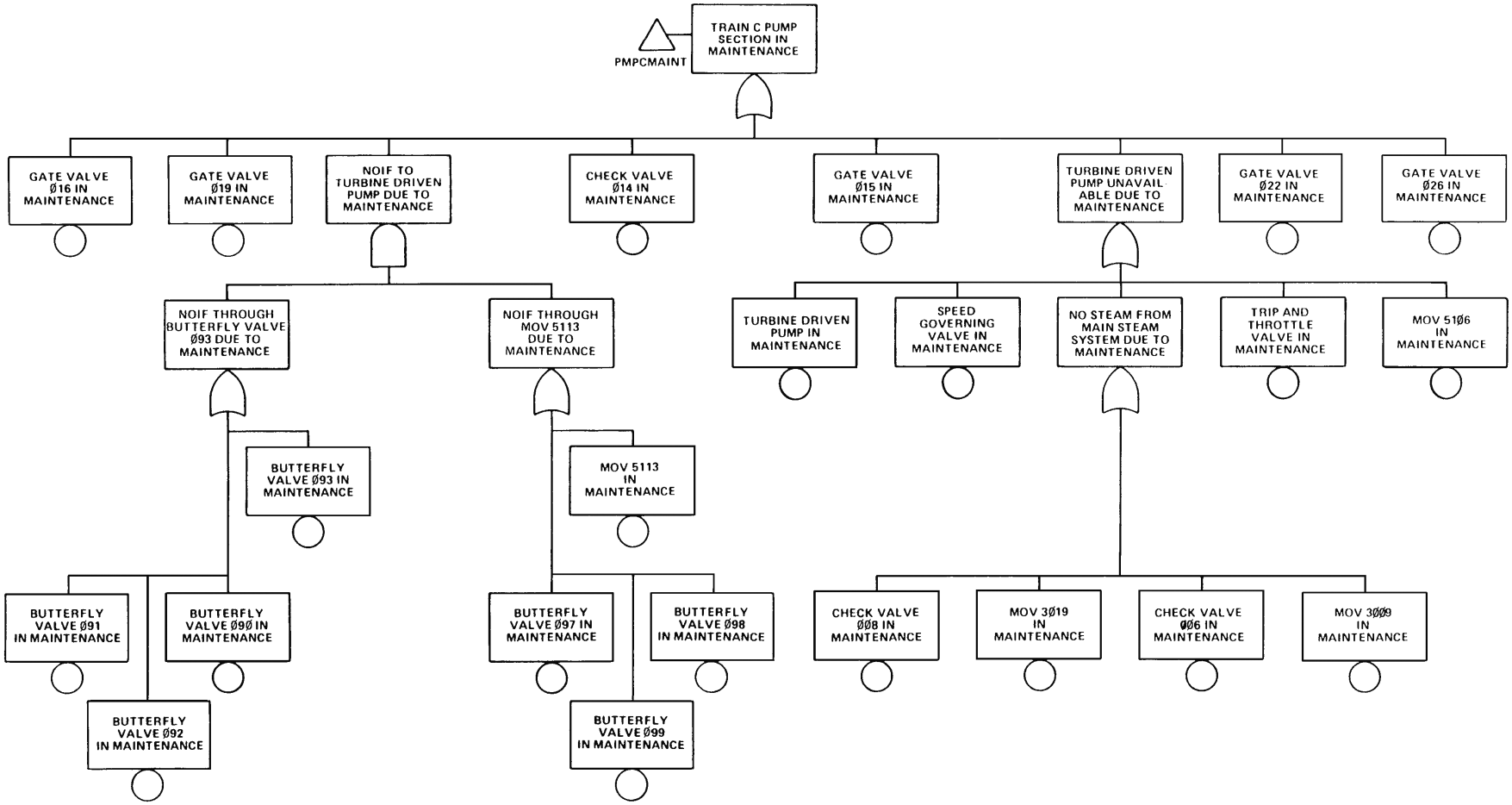
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**UNIT 1 AND UNIT 2**

UNIT 1 AFWS FAULT TREE MODEL

FIGURE 10A-7 (SHEET 12 OF 30)



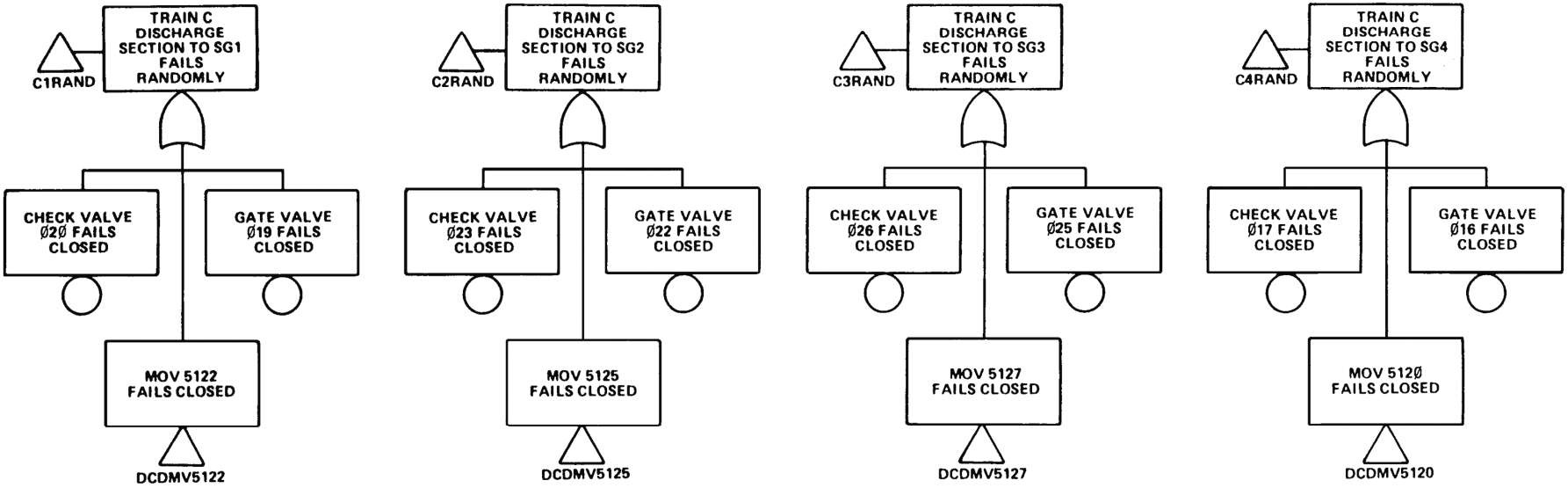

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**UNIT 1 AND UNIT 2**

UNIT 1 AFWS FAULT TREE MODEL

FIGURE 10A-7 (SHEET 13 OF 30)



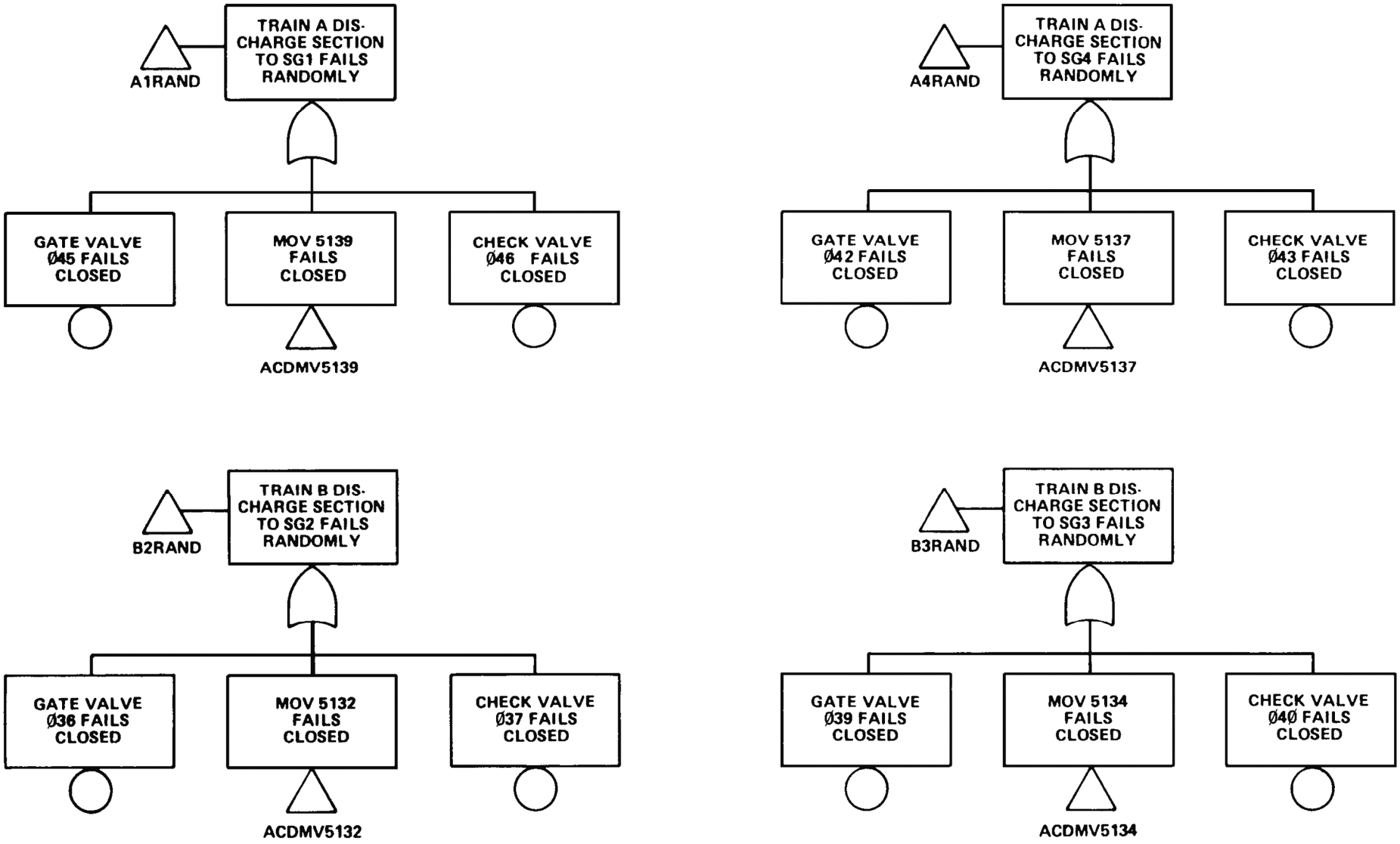

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**UNIT 1 AND UNIT 2**

UNIT 1 AFWS FAULT TREE MODEL

FIGURE 10A-7 (SHEET 14 OF 30)



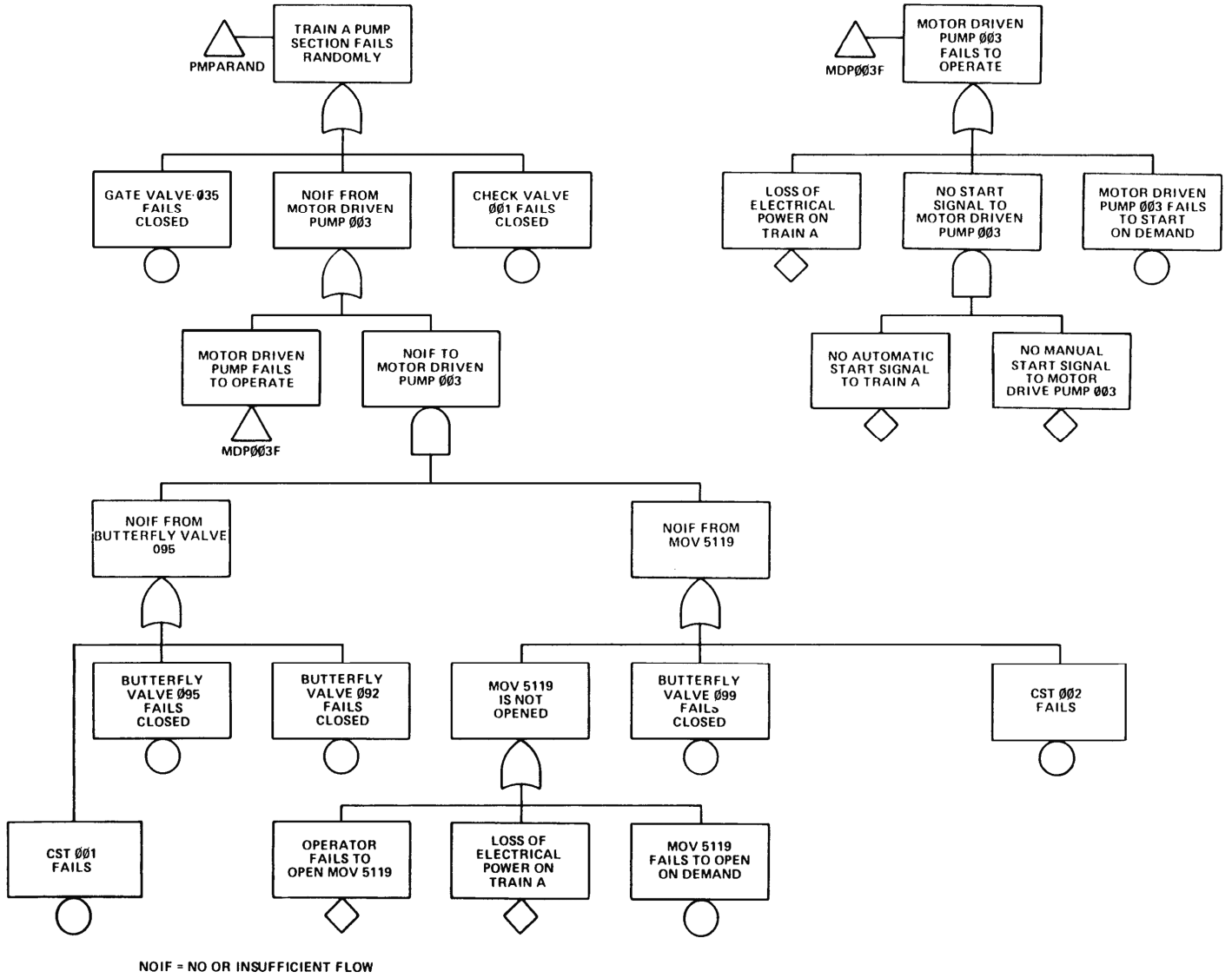

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**UNIT 1 AND UNIT 2**

UNIT 1 AFWS FAULT TREE MODEL

FIGURE 10A-7 (SHEET 15 OF 30)




NOIF = NO OR INSUFFICIENT FLOW

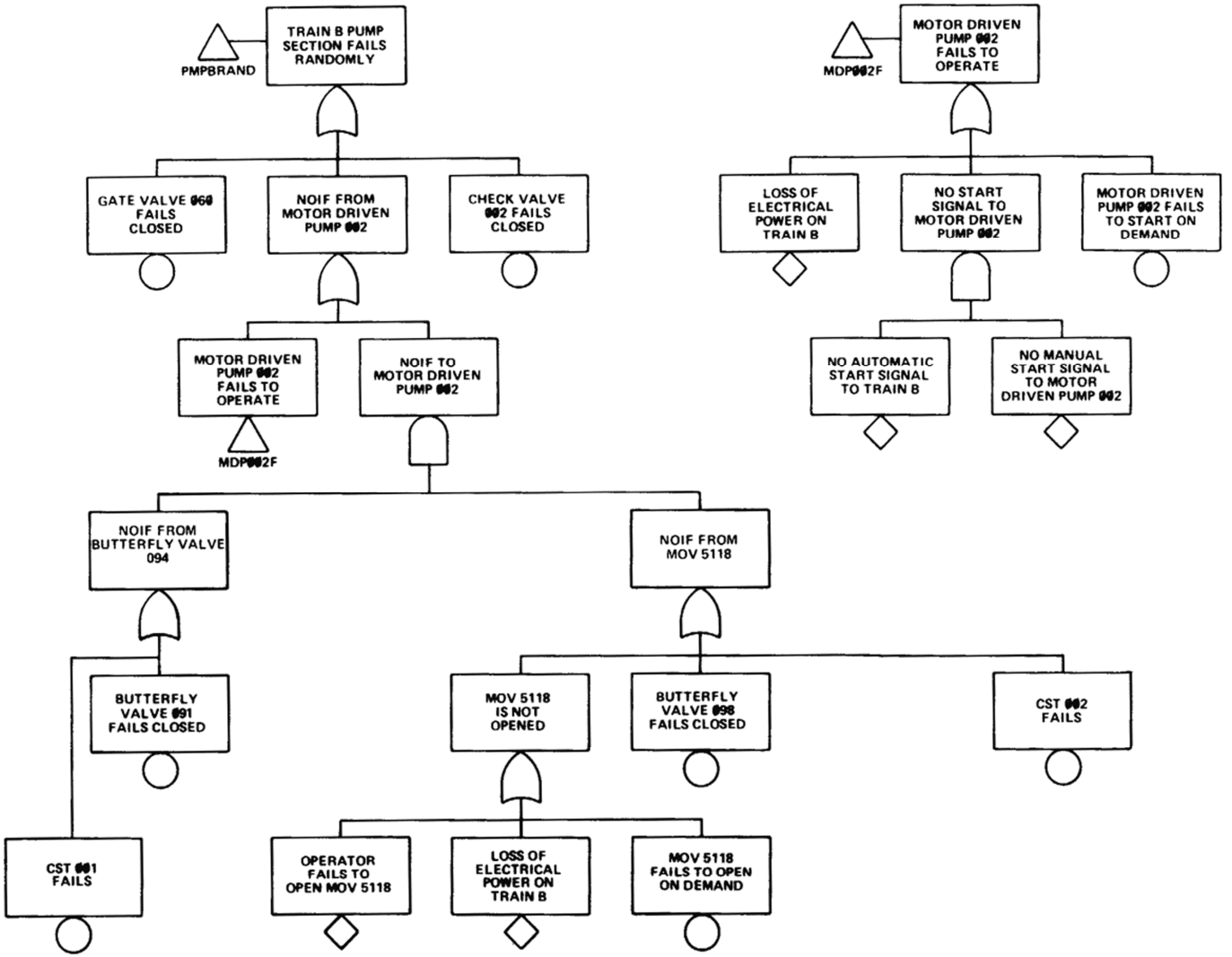
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**UNIT 1 AND UNIT 2**

UNIT 1 AFWS FAULT TREE MODEL

FIGURE 10A-7 (SHEET 16 OF 30)




NOIF = NO OR INSUFFICIENT FLOW

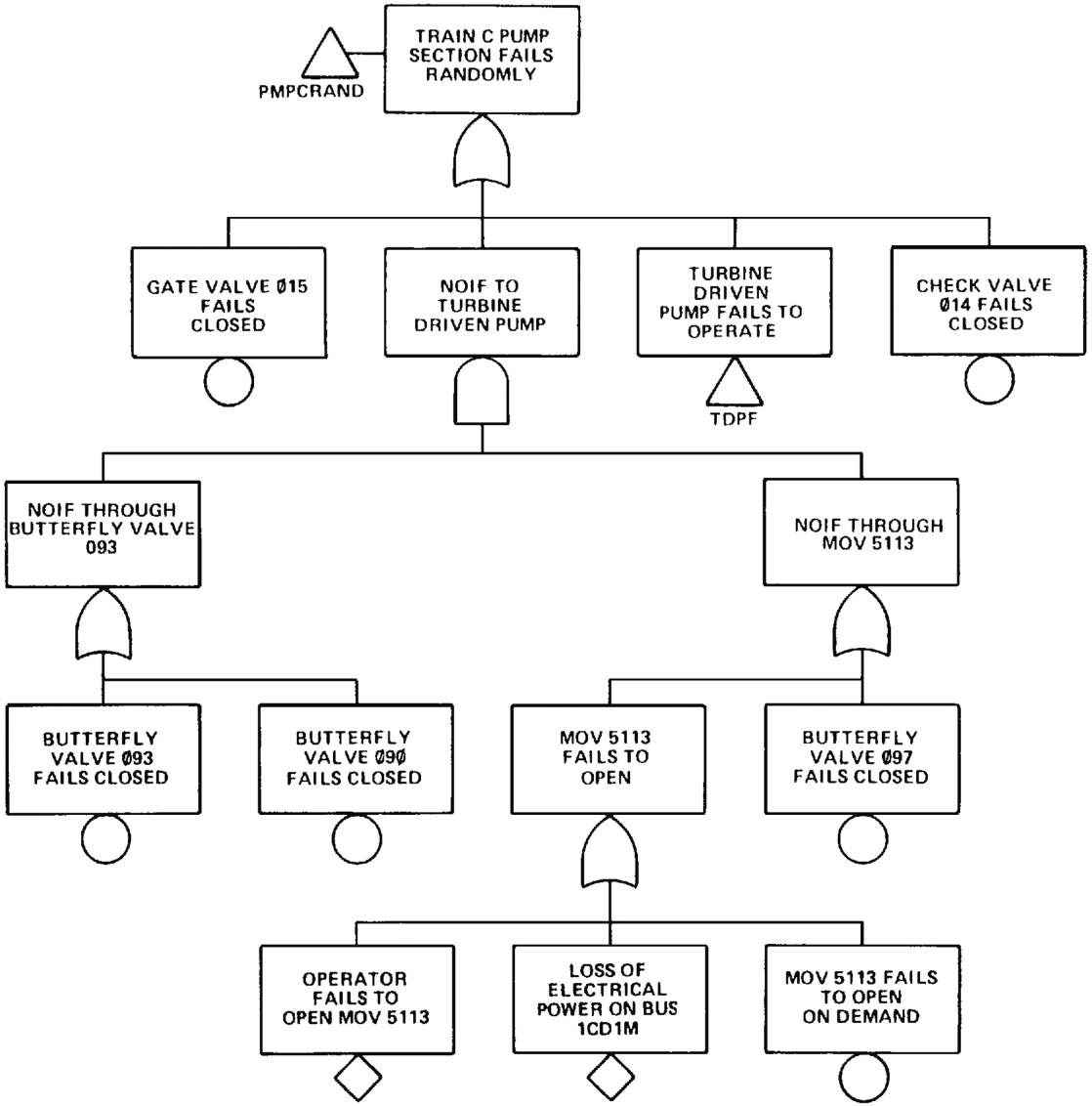
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**UNIT 1 AND UNIT 2**

UNIT 1 AFWS FAULT TREE MODEL

FIGURE 10A-7 (SHEET 17 OF 30)



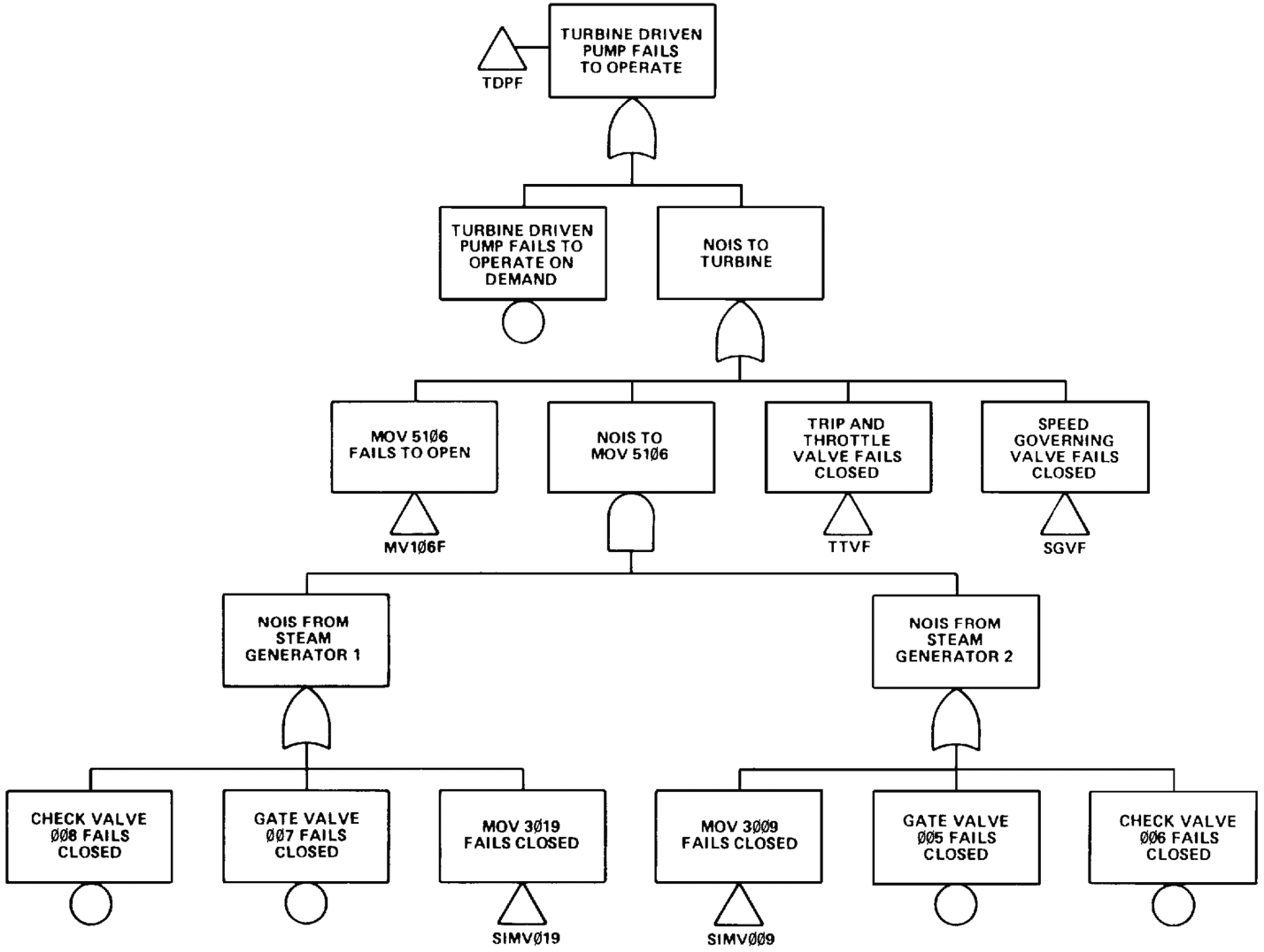

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**UNIT 1 AND UNIT 2**

UNIT 1 AFWS FAULT TREE MODEL

FIGURE 10A-7 (SHEET 18 OF 30)




NOIS = NO OR INSUFFICIENT STEAM

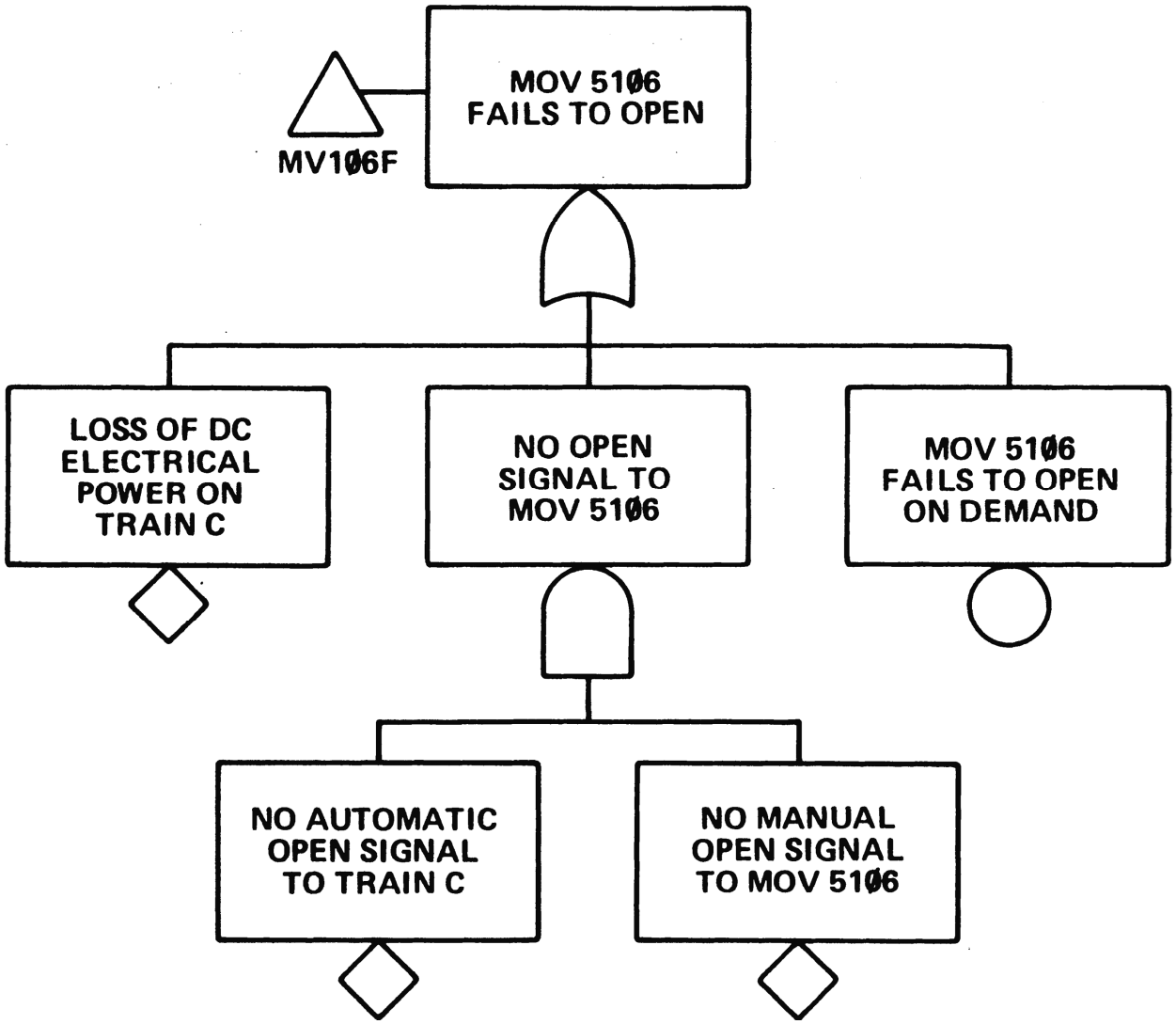
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**UNIT 1 AND UNIT 2**

UNIT 1 AFWS FAULT TREE MODEL

FIGURE 10A-7 (SHEET 19 OF 30)



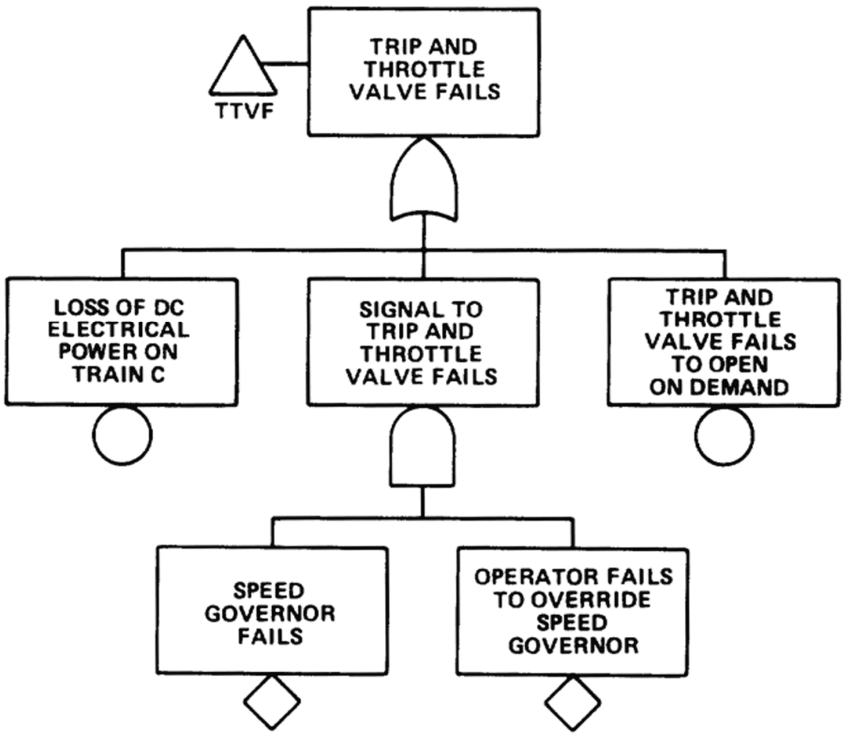

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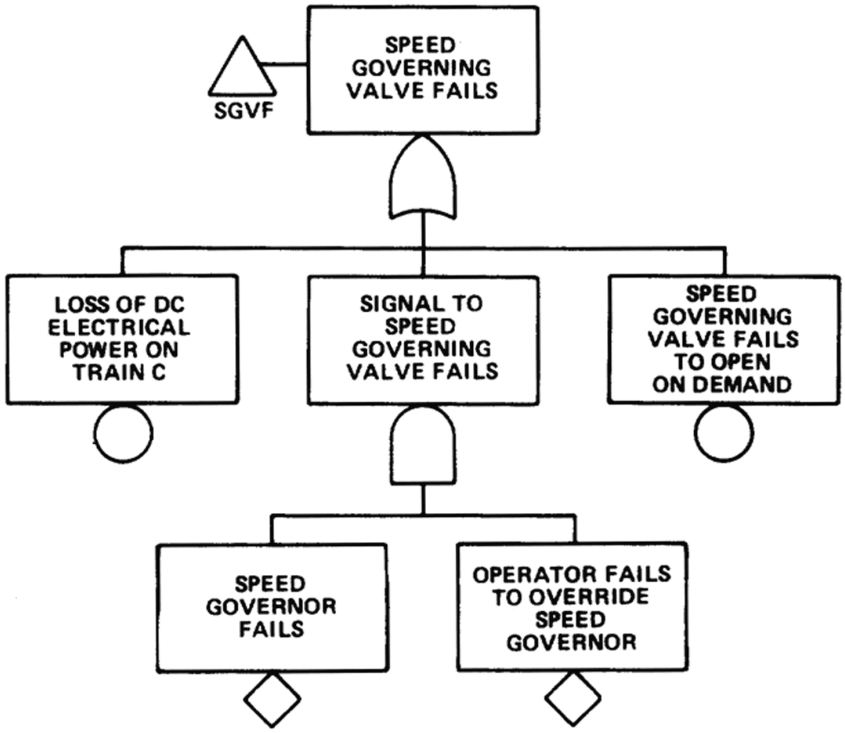
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UNIT 1 AND UNIT 2

UNIT 1 AFWS FAULT TREE MODEL

FIGURE 10A-7 (SHEET 20 OF 30)



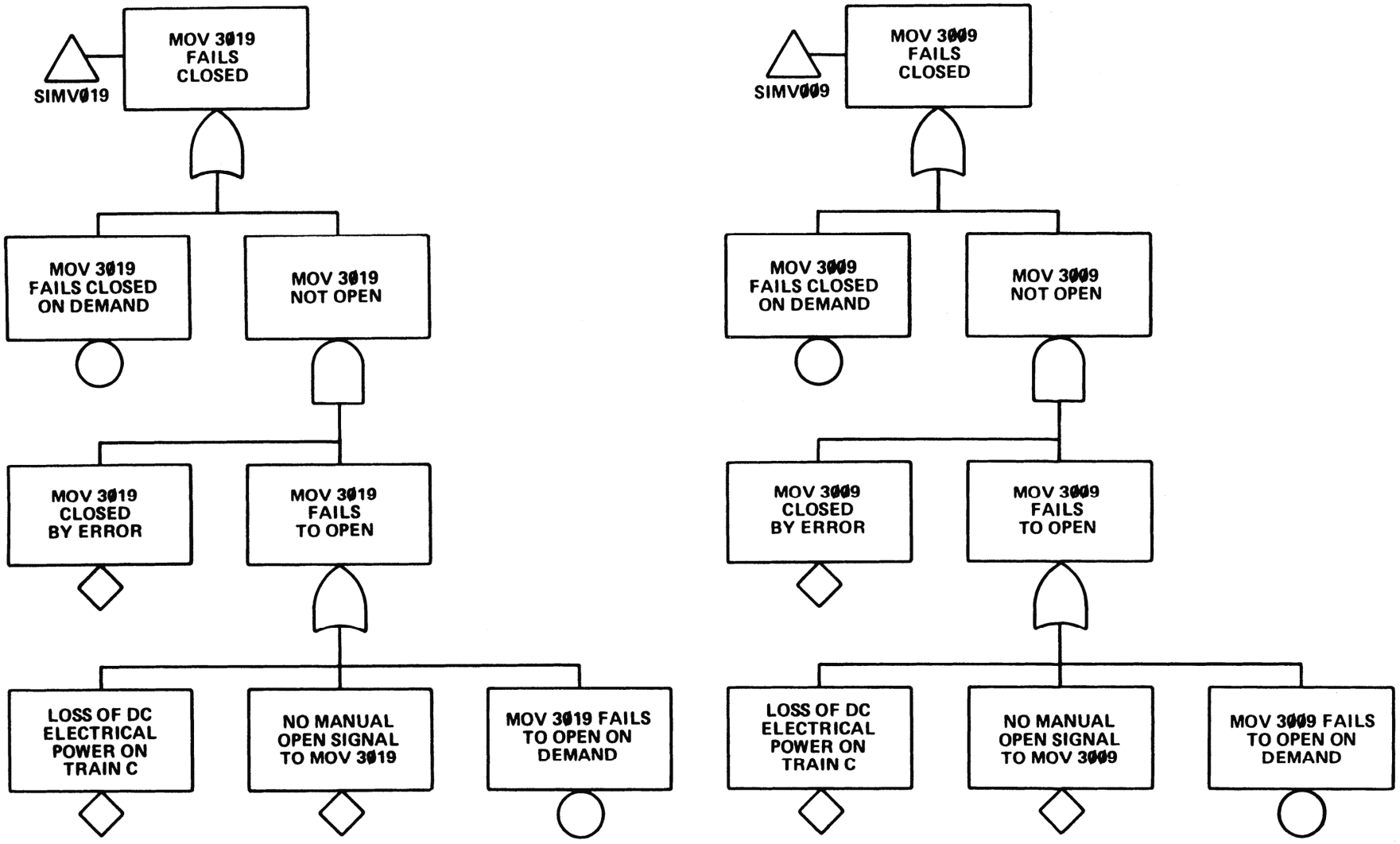

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**UNIT 1 AND UNIT 2**

UNIT 1 AFWS FAULT TREE MODEL

FIGURE 10A-7 (SHEET 21 OF 30)



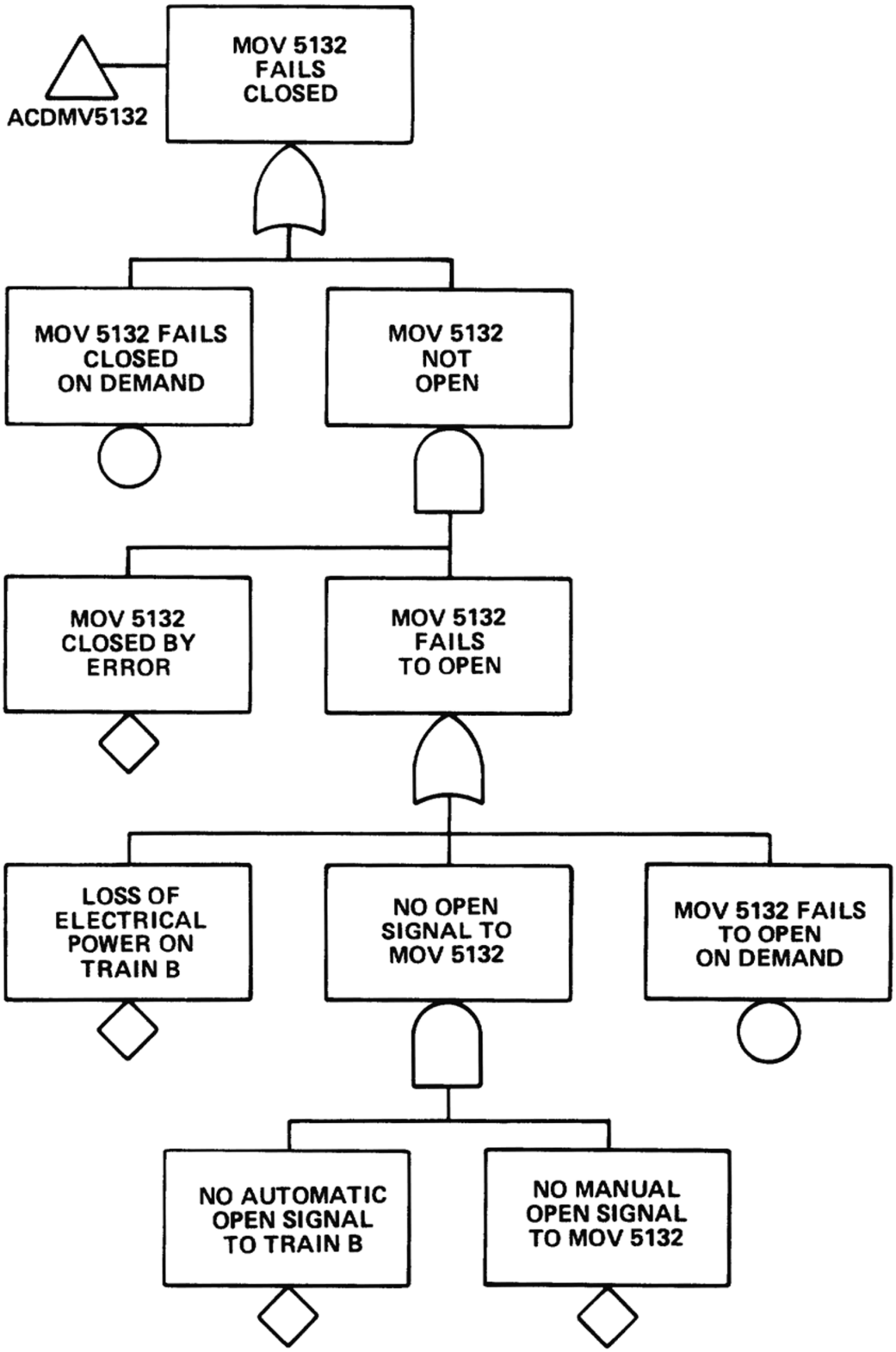

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UNIT 1 AND UNIT 2

UNIT 1 AFWS FAULT TREE MODEL

FIGURE 10A-7 (SHEET 22 OF 30)



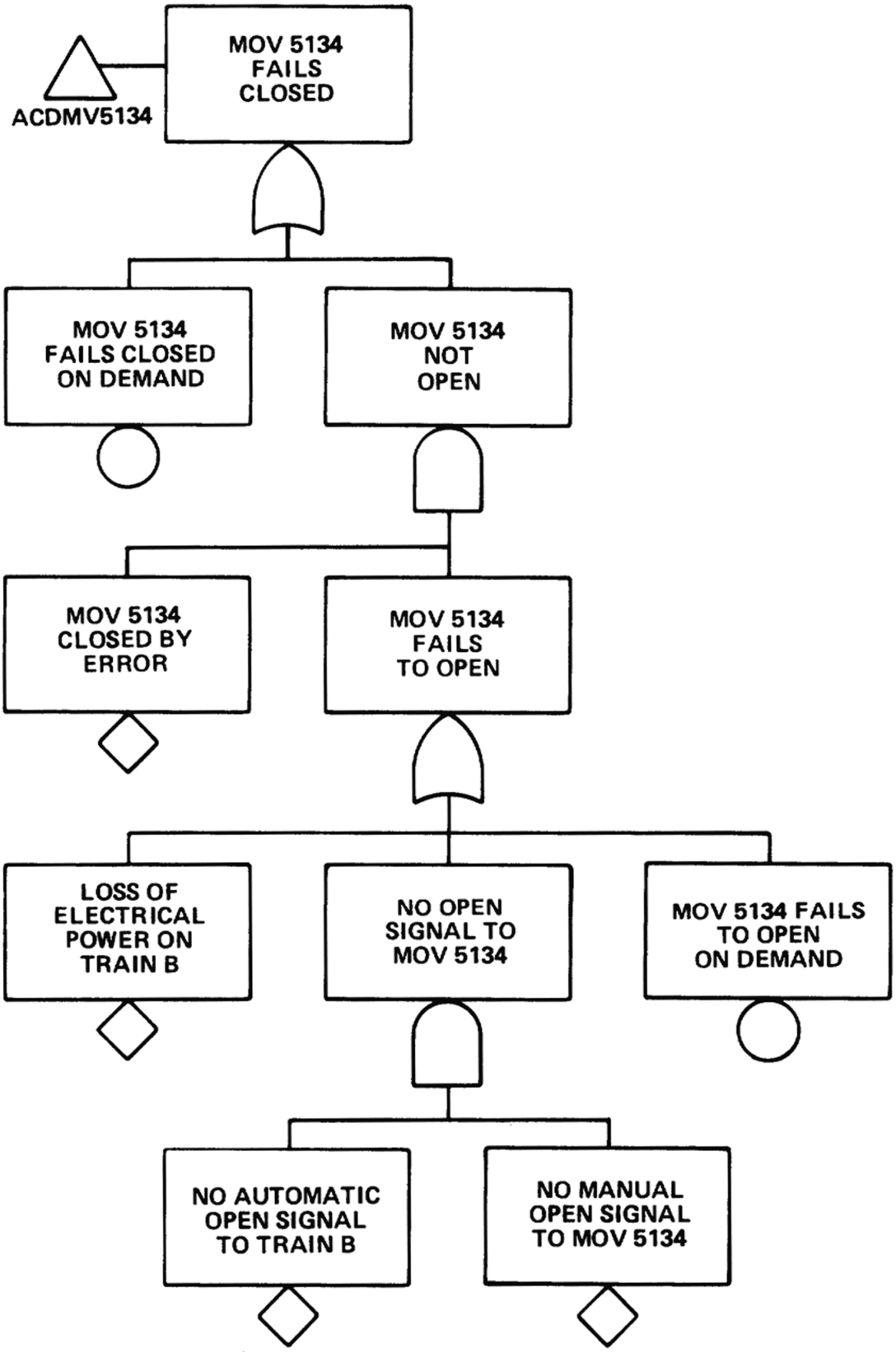

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**UNIT 1 AND UNIT 2**

UNIT 1 AFWS FAULT TREE MODEL

FIGURE 10A-7 (SHEET 23 OF 30)



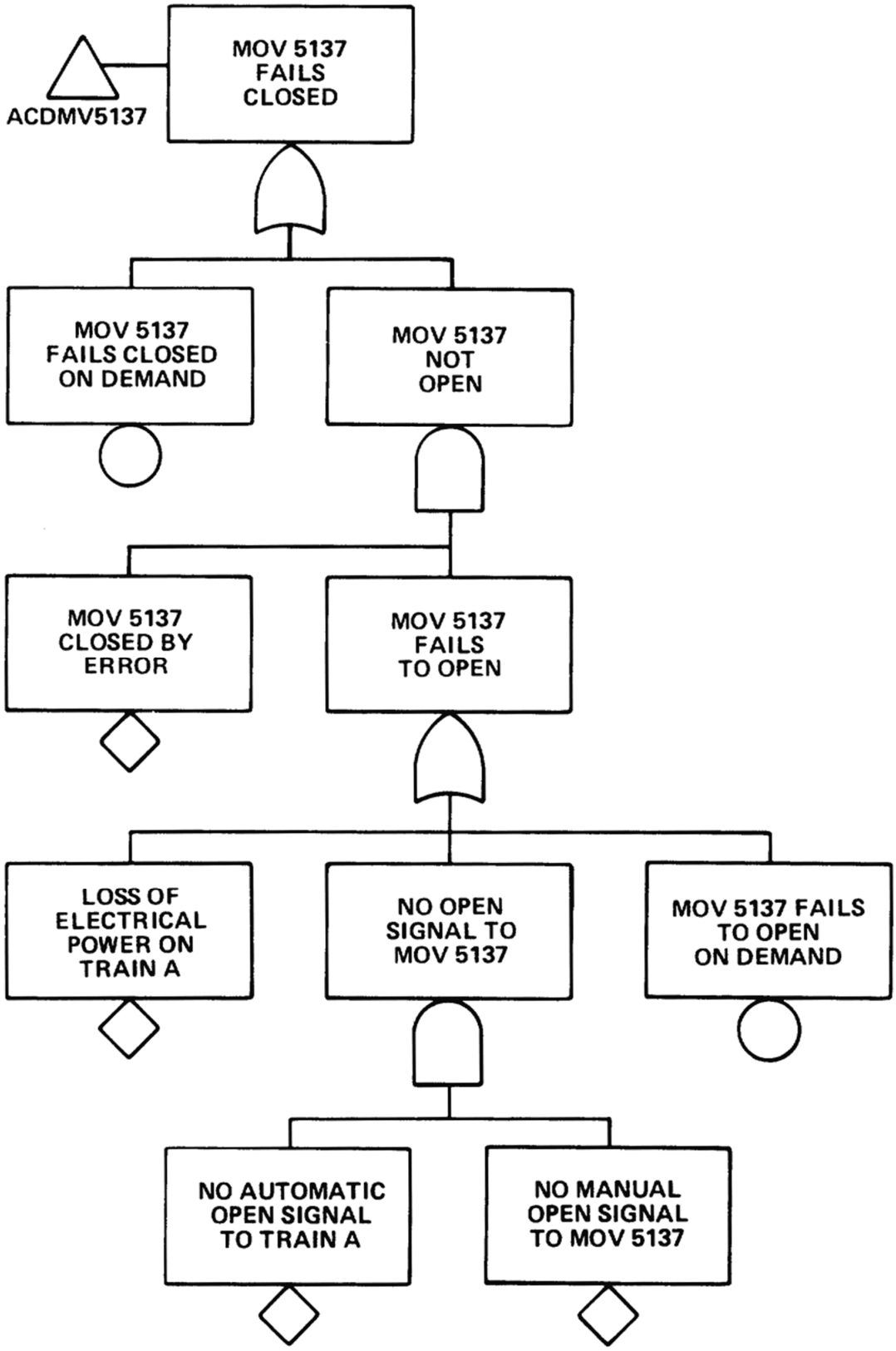

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**UNIT 1 AND UNIT 2**

UNIT 1 AFWS FAULT TREE MODEL

FIGURE 10A-7 (SHEET 24 OF 30)



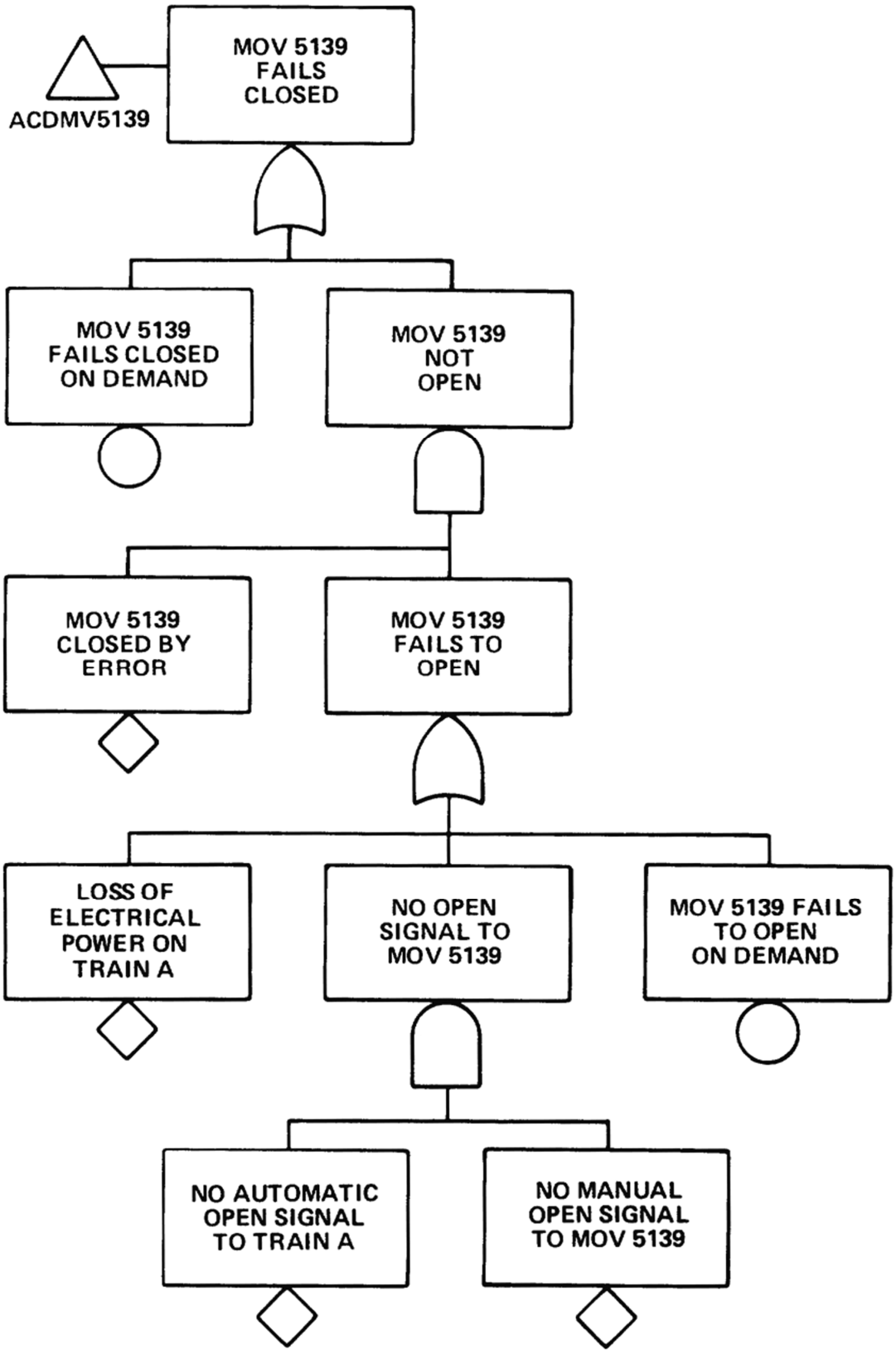

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**UNIT 1 AND UNIT 2**

UNIT 1 AFWS FAULT TREE MODEL

FIGURE 10A-7 (SHEET 25 OF 30)



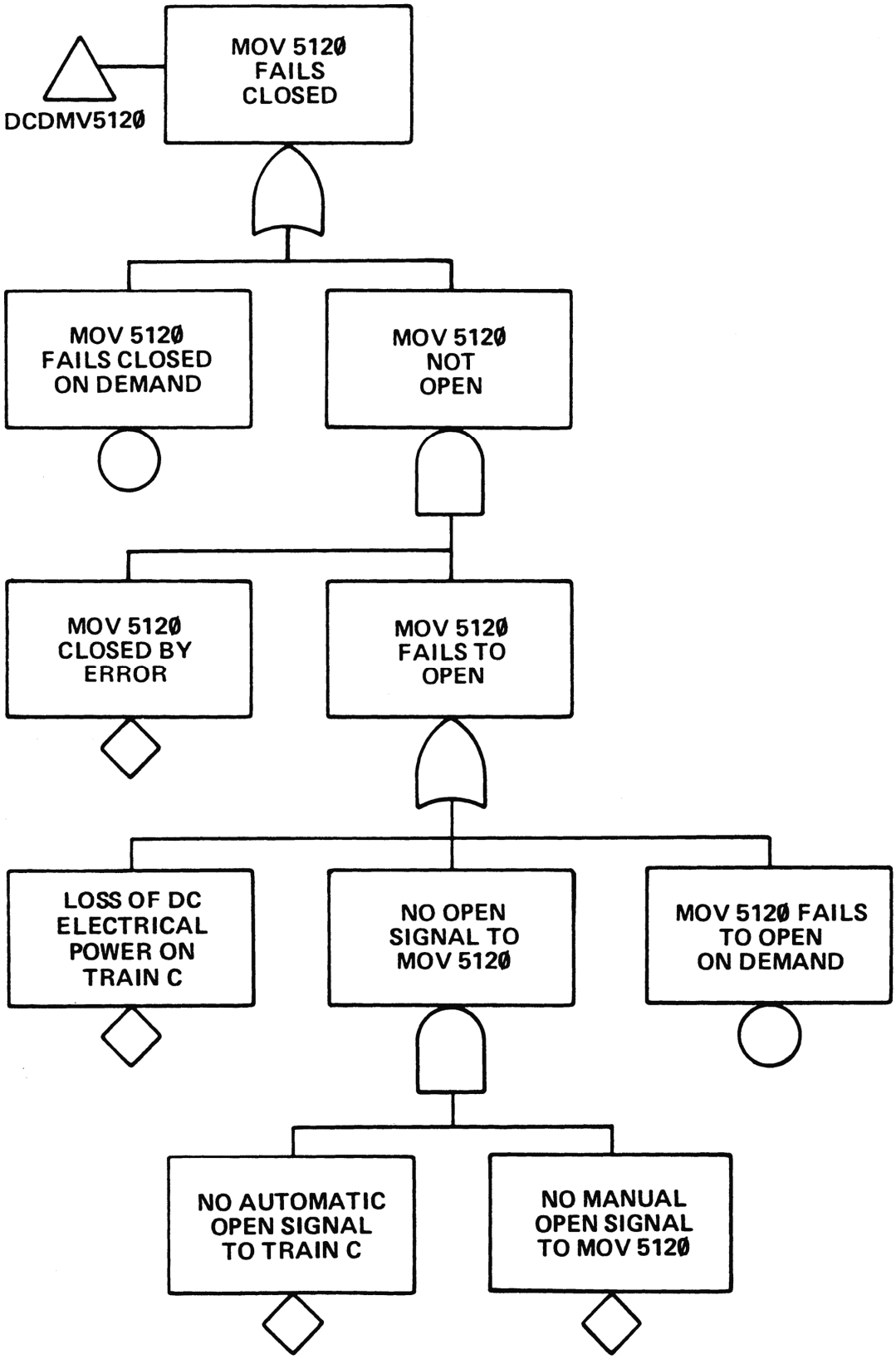

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**UNIT 1 AND UNIT 2**

UNIT 1 AFWS FAULT TREE MODEL

FIGURE 10A-7 (SHEET 26 OF 30)



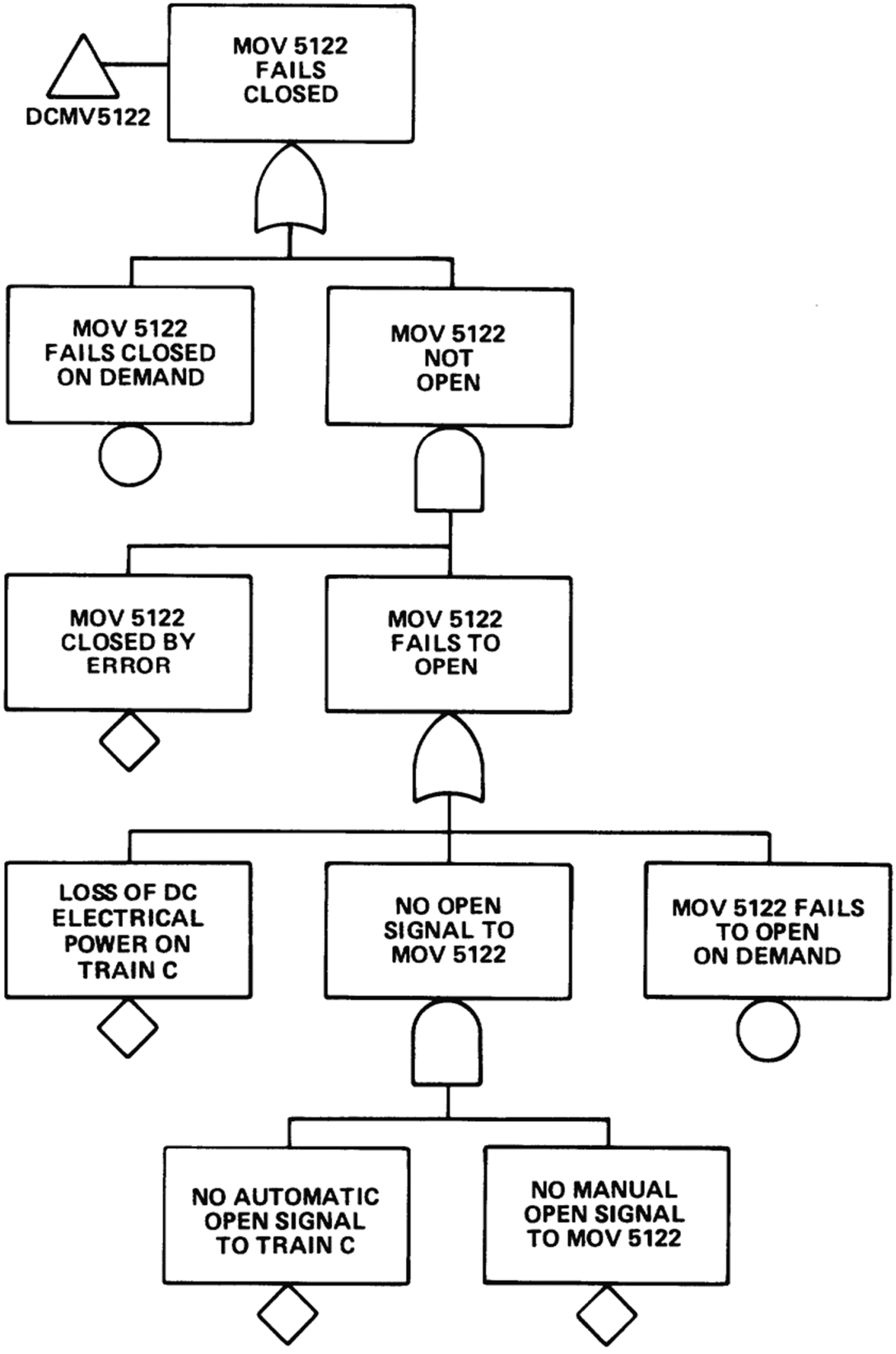

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UNIT 1 AND UNIT 2

UNIT 1 AFWS FAULT TREE MODEL

FIGURE 10A-7 (SHEET 27 OF 30)



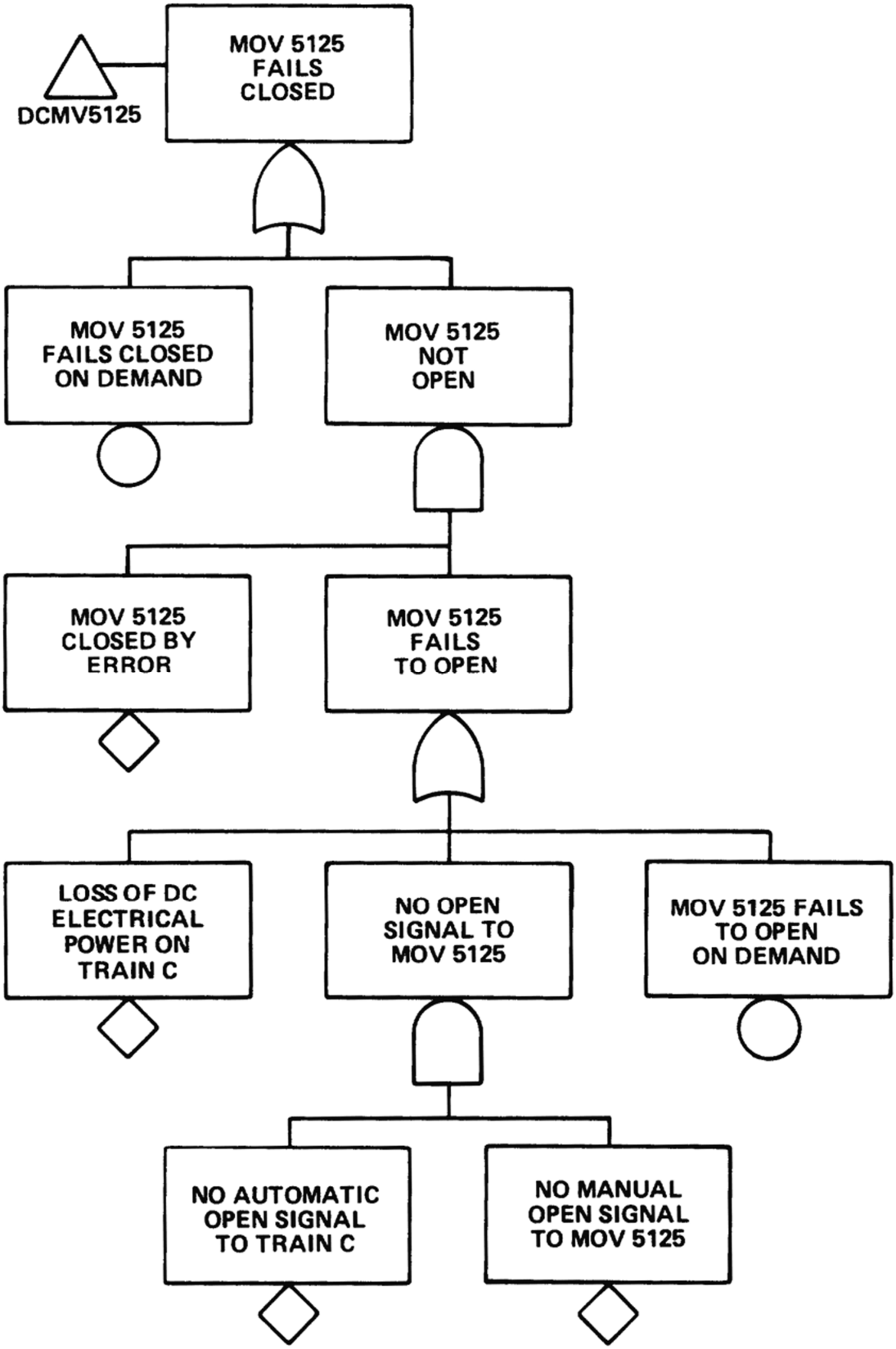

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**UNIT 1 AND UNIT 2**

UNIT 1 AFWS FAULT TREE MODEL

FIGURE 10A-7 (SHEET 28 OF 30)



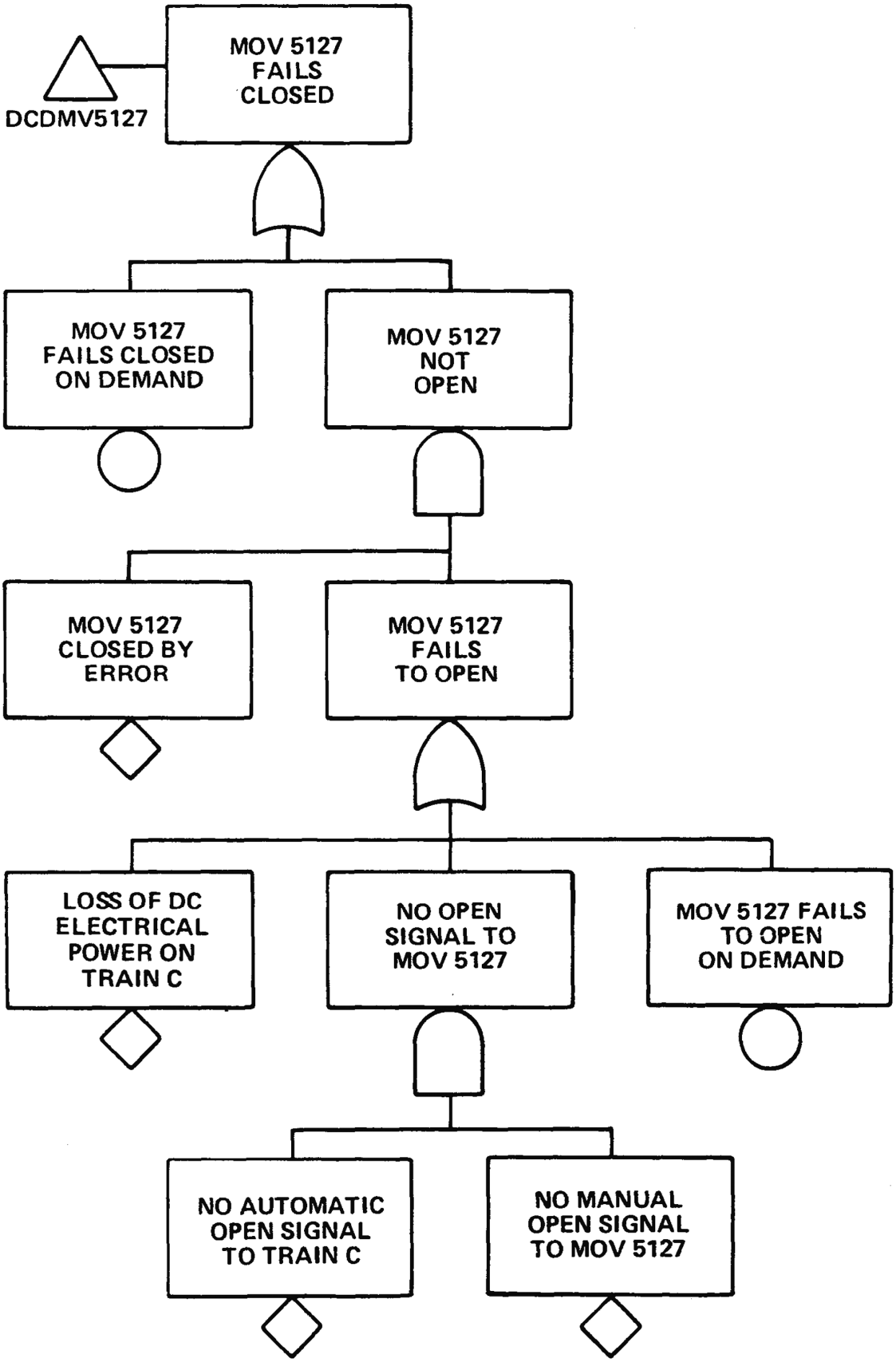

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**UNIT 1 AND UNIT 2**

UNIT 1 AFWS FAULT TREE MODEL

FIGURE 10A-7 (SHEET 29 OF 30)




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| | |
|---|---|
| SOUTHERN COMPANY *Energy to Serve Your World®* VOGTLE ELECTRIC GENERATING PLANT UNIT 1 AND UNIT 2 | UNIT 1 AFWS FAULT TREE MODEL |
| | FIGURE 10A-7 (SHEET 30 OF 30) |